Contents

Section 1

Marcus Kuo

Measure Spaces

1.1 Introduction

Example 1.1.1 (Non-measurable event).

Consider a uniform probability measure (measure of 1) on $[0, 2\pi)$. For example,

$$\mathbb{P}\left[[0,1]\right] = \frac{1}{2\pi}, \quad \mathbb{P}\left[[0,1] \cup [2,3]\right] = \frac{1}{\pi}.$$

Define the function T by

$$T(x) = \begin{cases} x+1, & x+1 < 2\pi \\ x+1-2\pi, & x+1 \geq 2\pi \end{cases}.$$

Thinking of a circle, the operator moves the point by one unit. With this operator, we can define an equivalence relation: we will say $x \sim y$ iff $T(x) = T(y)$ or $T(y) = T(x)$.
The equivalence relation leads to a partition $(A_\lambda : \lambda)$; i.e.,

$$A_\lambda \cap A'_\lambda = \varnothing, \ \lambda \neq \lambda', \quad \cup_\lambda A_\lambda = [0, 2\pi)$$

such that $x, y \in A_\lambda \iff x \sim y$.
For every λ, select an element $x_\lambda \in A_\lambda$ and collect them in $X = \{x_\lambda : \lambda\}$. Define the translation

$$X + i = \{x_\lambda + i : \lambda\}.$$

Note that $\{X + i \bmod 2\pi : i \in \mathbb{Z}\}$ is a partition of $[0, 2\pi)$.
We know that

$$\begin{aligned} 1 = \mathbb{P}\left[[0, 2\pi)\right] &= \sum_{i \in \mathbb{Z}} \mathbb{P}\left[(X+i)\right] \\ &= \sum_{i \in \mathbb{Z}} \mathbb{P}\left[X\right]. \end{aligned}$$

Then what is $\mathbb{P}\left[X\right]$? It is not well-defined; X is not a measurable event.

[1][2]

We need to restrict ourselves to measurable subsets to have our notion of probability be well-defined; hence measure theory is necessary.

[1] X can be called a representative set.
[2] Here we use the Axiom of Choice.

Definition 1.1.2 (σ-algebra).

Let Ω denote the sample space and $2^\Omega := \{\text{subsets of } \Omega\}$.
Given Ω, $\mathscr{F} \subseteq 2^\Omega$ is a σ-field (σ-algebra) if

(i) $\varnothing \in \mathscr{F}$

(ii) If $A \in \mathscr{F}$, then $A^c \in \mathscr{F}$

(iii) If $A_n \in \mathscr{F}$, then $\cup_n A_n \in \mathscr{F}$ (countable union).

Trivial examples include 2^Ω and $\{\varnothing, \Omega\}$.

Definition 1.1.3 (Generated σ-field).

Given $\mathscr{E} \subseteq 2^\Omega$, define $\sigma(\mathscr{E})$ to be the smallest σ-field containing $\mathscr{E}$. In other words, for any σ-field $\mathscr{F} \supseteq \mathscr{E}$, then $\sigma(\mathscr{E}) \subseteq \mathscr{F}$.
Note that $\sigma(\mathscr{E})$ is well-defined, because we can put

$$\sigma(\mathscr{E}) = \bigcap \{\mathscr{F} \subseteq 2^\Omega : \mathscr{F} \text{ is a } \sigma\text{-field and } \mathscr{F} \supseteq \mathscr{E}\}.$$

We use the following lemma, whose proof is immediate.

Lemma 1.1.4.

If $\mathscr{F}_\lambda$ is a σ-field for each λ, then $\cap_\lambda \mathscr{F}_\lambda$ is a σ-field.

Definition 1.1.5 (Monotone sequences of sets and their limits).

We say that A_n is an increasing sequence ($A_n \uparrow$) if

$$A_1 \subseteq A_2 \subseteq \dots$$

and similarly for decreasing sequences of sets.
If $A_n \uparrow$, define $\lim\limits_{n\to\infty} A_n := \cup_n A_n$, and if $A_n \downarrow$, define $\lim\limits_{n\to\infty} A_n := \cap_n A_n$.

We can define the limit of an arbitrary, non-monotone sequence.

Definition 1.1.6 ($\liminf$ and $\limsup$ of sequences).

For arbitrary A_n, define

$$\liminf_{n\to\infty} A_n = \cup_k \cap_{n\geq k} A_n, \quad \limsup_{n\to\infty} A_n = \cap_k \cup_{n\geq k} A_n.$$

If $\liminf\limits_{n\to\infty} A_n = \limsup\limits_{n\to\infty} A_n$, then the limit exists and is the common value.

Remark 1.1.7. Thinking hard, these definitions make sense. Alternatively,

$$\begin{aligned} x \in \liminf_{n\to\infty} A_n &\iff \exists\, k \text{ s.t. } \forall\, n \geq k,\ x \in A_n \\ &\iff x \text{ belongs to all but finitely many } A_n \\ x \in \limsup_{n\to\infty} A_n &\iff \forall\, k, \exists\, n \geq k \text{ s.t. } x \in A_n \\ &\iff x \text{ belongs to infinitely many } A_n. \end{aligned}$$

Furthermore, $\liminf\limits_{n\to\infty} A_n \subseteq \limsup\limits_{n\to\infty} A_n$.

Definition 1.1.8 (π-system, monotone class, λ-system, field).

We say that $\mathscr{P}$ is a π-system if it is closed under intersection; i.e., for any $A, B \in \mathscr{P}$, $A \cap B \in \mathscr{P}$.
We say that $\mathscr{M}$ is a monotone class if for any monotone sequence (A_n), $\lim\limits_{n\to\infty} A_n \in \mathscr{M}$. Monotone classes are closed under the limit operation.
We say that $\mathscr{L}$ is a λ-system if

(i) $\Omega \in \mathscr{L}$,

(ii) If $A, B \in \mathscr{L}$ such that $A \supseteq B$, then $A \setminus B \in \mathscr{L}$, and

(iii) If (A_n) is increasing sequence in $\mathscr{L}$, then $\lim\limits_{n\to\infty} A_n \in \mathscr{L}$. $\mathscr{L}$ is closed under countable unions, but only for increasing sequences.

$\mathscr{A}$ is a field (algebra) if

(i) $\varnothing \in \mathscr{A}$,

(ii) $\forall\, A \in \mathscr{A}$, $A^c \in \mathscr{A}$, and

(iii) $\forall\, A, B \in \mathscr{A}$, $A \cup B \in \mathscr{A}$.

[3]

The first and fourth definitions deal with finite operations; the second and third include limits.

Theorem 1.1.9.

(Monotone class.) If $\mathscr{A}$ is a field, then $m(\mathscr{A}) = \sigma(\mathscr{A})$.

[4]

Proof of Theorem 1.1.9. The inclusion $m(\mathscr{A}) \subseteq \sigma(\mathscr{A})$ is straightforward. By definition, $\sigma(\mathscr{A})$ contains $\mathscr{A}$. Also, $\sigma(\mathscr{A})$ is a monotone class. Since $m(\mathscr{A})$ is the smallest monotone class containing $\mathscr{A}$, we get $m(\mathscr{A}) \subseteq \sigma(\mathscr{A})$.

For the reverse inclusion, it suffices to show that $m(\mathscr{A})$ is a σ-field containing $\mathscr{A}$. It contains $\mathscr{A}$ by definition.

(i) Since $\mathscr{A}$ is a field, $\varnothing \in \mathscr{A} \subseteq m(\mathscr{A})$.

(ii) (Closure under complements.) Define

$$M_0 := \{A \in m(\mathscr{A}) : A^c \in m(\mathscr{A})\}.$$

We claim M_0 is a monotone class containing $\mathscr{A}$.

(a) $\mathscr{A} \subseteq M_0$. If $A \in \mathscr{A}$, then $A^c \in \mathscr{A} \subseteq m(\mathscr{A})$, hence $A \in M_0$.

[3] This assumption is weaker than the one in the definition of a σ-field.
[4] $m(\mathscr{A})$ is defined to be the smallest monotone class containing $\mathscr{A}$.

(b) M_0 is a monotone class. Suppose $A_n \uparrow A$ with each $A_n \in M_0$. Then $A_n^c \downarrow A^c$, and since each $A_n^c \in m(\mathscr{A})$ and $m(\mathscr{A})$ is a monotone class, we have $A^c \in m(\mathscr{A})$. Also $A \in m(\mathscr{A})$ because $A_n \uparrow A$. Hence $A \in M_0$. The argument for $A_n \downarrow A$ is identical (then $A_n^c \uparrow A^c$).

Thus M_0 is a monotone class containing $\mathscr{A}$, so by minimality $m(\mathscr{A}) \subseteq M_0$. In particular, $m(\mathscr{A})$ is closed under complements.

(iii) (Closure under countable unions.) First we show closure under finite unions.

Define

$$M_1 := \{A \in m(\mathscr{A}) : \forall\, B \in \mathscr{A},\ A \cup B \in m(\mathscr{A})\}.$$

Clearly $M_1 \subseteq m(\mathscr{A})$. We check that M_1 is a monotone class containing $\mathscr{A}$.

(a) $\mathscr{A} \subseteq M_1$. If $A \in \mathscr{A}$ and $B \in \mathscr{A}$, then $A \cup B \in \mathscr{A} \subseteq m(\mathscr{A})$ (since $\mathscr{A}$ is a field). Hence $A \in M_1$, so $\mathscr{A} \subseteq M_1$.

(b) M_1 is a monotone class. Suppose $A_n \uparrow A$ with $A_n \in M_1$ and fix a $B \in \mathscr{A}$. Then $A_n \cup B \uparrow A \cup B$. Since each $A_n \cup B \in m(\mathscr{A})$ and $m(\mathscr{A})$ is a monotone class, $A \cup B \in m(\mathscr{A})$. B was arbitrary in $\mathscr{A}$, so $A \in M_1$. The decreasing case is the same.

Therefore M_1 is a monotone class containing $\mathscr{A}$, hence $m(\mathscr{A}) \subseteq M_1$. Equivalently,

$$\forall\, A \in m(\mathscr{A}),\ \forall\, B \in \mathscr{A}, \qquad A \cup B \in m(\mathscr{A}).$$

Now define

$$M_2 := \{A \in m(\mathscr{A}) : \forall\, B \in m(\mathscr{A}),\ A \cup B \in m(\mathscr{A})\}.$$

Again $M_2 \subseteq m(\mathscr{A})$. We show M_2 is a monotone class containing $\mathscr{A}$.

(a) $\mathscr{A} \subseteq M_2$. Fix $A \in \mathscr{A}$ and $B \in m(\mathscr{A})$. From the conclusion about M_1 (with roles swapped), $B \cup A \in m(\mathscr{A})$, hence $A \cup B \in m(\mathscr{A})$. Thus $A \in M_2$.

(b) M_2 is a monotone class. If $A_n \uparrow A$ with $A_n \in M_2$ and $B \in m(\mathscr{A})$, then $A_n \cup B \uparrow A \cup B$. Since each $A_n \cup B \in m(\mathscr{A})$ and $m(\mathscr{A})$ is a monotone class, $A \cup B \in m(\mathscr{A})$, so $A \in M_2$. The decreasing case is identical.

Thus M_2 is a monotone class containing $\mathscr{A}$, so $m(\mathscr{A}) \subseteq M_2$. In particular, $m(\mathscr{A})$ is closed under finite unions.

Finally, let $(A_n)_{n\geq 1} \subseteq m(\mathscr{A})$ and set $U_m := \bigcup_{n=1}^m A_n$. By finite-union closure, $U_m \in m(\mathscr{A})$ for each m, and $U_m \uparrow \bigcup_{n=1}^\infty A_n$. Since $m(\mathscr{A})$ is a monotone class, $\bigcup_{n=1}^\infty A_n \in m(\mathscr{A})$.

We have shown $m(\mathscr{A})$ is a σ-field containing $\mathscr{A}$, hence $\sigma(\mathscr{A}) \subseteq m(\mathscr{A})$. Combined with the first inclusion, $m(\mathscr{A}) = \sigma(\mathscr{A})$. □

[5]

Theorem 1.1.10.

(π-λ.) If $\mathscr{P}$ is a π-system, then

$$l(\mathscr{P}) = \sigma(\mathscr{P}).$$

1.2 σ-fields

Definition 1.2.1 (Topology).

Let Ω be a set. A *topology* on Ω is a collection $\mathcal{T} \subseteq 2^\Omega$ such that:

(i) $\varnothing \in \mathcal{T}$ and $\Omega \in \mathcal{T}$,

(ii) if $U, V \in \mathcal{T}$, then $U \cap V \in \mathcal{T}$,

[5] This and the previous theorem give us two different ways to generate a σ-field. We can start with something more complicated (field) or less complicated (a π-system). In this course, the monotone class theorem will be more commonly used. The choice comes down to convenience.

(iii) if $\{U_i\}_{i\in I} \subseteq \mathcal{T}$, then $\bigcup_{i\in I} U_i \in \mathcal{T}$.

The sets in $\mathcal{T}$ are called *open sets*, and $(\Omega, \mathcal{T})$ is a *topological space*.

Definition 1.2.2 (Measurable space).

A *measurable space* is a pair $(\Omega, \mathscr{F})$ where Ω is a set and $\mathscr{F} \subseteq 2^\Omega$ is a σ-algebra on Ω, i.e.:

(i) $\Omega \in \mathscr{F}$,

(ii) if $A \in \mathscr{F}$ then $A^c \in \mathscr{F}$,

(iii) if $A_1, A_2, \ldots \in \mathscr{F}$ then $\bigcup_{n=1}^\infty A_n \in \mathscr{F}$.

[6]

Definition 1.2.3 (Borel σ-algebra).

If $(\Omega, \mathcal{T})$ is a topological space, the *Borel σ-algebra* on Ω is

$$\mathscr{B}(\Omega) \coloneqq \sigma(\mathcal{T}),$$

the smallest σ-algebra containing all open sets. In particular, for $\mathbb{R}$ with its usual topology,

$$\mathscr{B}_\mathbb{R} \coloneqq \mathscr{B}(\mathbb{R}) = \sigma\big(\{U \subseteq \mathbb{R} : U \text{ is open}\}\big) = \sigma\big(\{[a, \infty) : a \in \mathbb{R}\}\big).$$

Definition 1.2.4 (Measure).

Suppose $\mathscr{E}$ is a set collection such that $\varnothing \in \mathscr{E} \subseteq 2^\Omega$. We say $\mu : \mathscr{E} \to [0, \infty]$ is a measure if:

(i) $\mu(\varnothing) = 0$

(ii) For disjoint $A_n \in \mathscr{E}$ such that $\cup_n A_n \in \mathscr{E}$,

$$\mu(\cup_n A_n) = \sum_n \mu(A_n).$$

In particular, if $\mu(\Omega) < \infty$, μ is called a finite measure. If $\mu(\Omega) = 1$, then it is a probability measure. If for any event $A \in \mathscr{E}$, there exists $A_n \in \mathscr{E}$ such that $\cup_n A_n \supseteq A$ and $\mu(A_n) < \infty$ for each n, we call μ is a σ-finite measure. Any set in a σ-finite measure is the countable union of sets each of finite measure.

Definition 1.2.5 (Probability space).

We call $(\Omega, \mathscr{F}, \mu)$ a probability space if $\mu(\Omega) = 1$.

Remark 1.2.6 (Properties of measure). Suppose μ is a measure on a σ-field $\mathscr{E}$.[7] Then

(i) (Monotonicity.) If $A \subseteq B$, then $\mu(A) \leq \mu(B)$.

[6] A measure space is a measurable space equipped with a measure.

[7] Note now μ is not a premeasure on an arbitrary set collection; it is defined on a σ-field for these properties.

(ii) (Continuity from below.) If (A_n) is an increasing sequence ($A_n \subseteq A_{n+1}$), then

$$\lim_{n\to\infty} \mu(A_n) = \mu\Big(\lim_{n\to\infty} A_n\Big) \qquad \text{where} \quad \lim_{n\to\infty} A_n := \bigcup_{n=1}^{\infty} A_n.$$

(iii) (Continuity from above.) If (A_n) is a decreasing sequence ($A_{n+1} \subseteq A_n$) and $\mu(A_1) < \infty$, then

$$\mu\Big(\lim_{n\to\infty} A_n\Big) = \lim_{n\to\infty} \mu(A_n) \qquad \text{where} \quad \lim_{n\to\infty} A_n := \bigcap_{n=1}^{\infty} A_n.$$

Proof of Theorem 1.1.10. (i) If $A \subseteq B$, then $B = A \amalg (B \setminus A)$, so

$$\mu(B) = \mu(A) + \mu(B \setminus A) \geq \mu(A)$$

by nonnegativity.

(ii) Assume $A_n \uparrow$ and set $A := \bigcup_{n=1}^{\infty} A_n$. Define

$$D_1 := A_1, \qquad D_m := A_m \setminus A_{m-1} \quad (m \geq 2),$$

(with the convention $A_0 := \varnothing$). Then the D_m are pairwise disjoint and for each n,

$$A_n = \amalg_{m=1}^{n} D_m, \qquad \text{and} \qquad A = \amalg_{m=1}^{\infty} D_m.$$

By countable additivity,

$$\mu(A) = \sum_{m=1}^{\infty} \mu(D_m).$$

On the other hand, for each n,

$$\mu(A_n) = \mu(\amalg_{m=1}^{n} D_m) = \sum_{m=1}^{n} \mu(D_m),$$

so taking $n \to \infty$ gives

$$\lim_{n\to\infty} \mu(A_n) = \lim_{n\to\infty} \sum_{m=1}^{n} \mu(D_m) = \sum_{m=1}^{\infty} \mu(D_m) = \mu(A) = \mu\left(\bigcup_{n=1}^{\infty} A_n\right).$$

(iii) Assume $A_n \downarrow$ and $\mu(A_1) < \infty$. Consider $B_n := A_1 \setminus A_n$. Then (B_n) is increasing, so by (ii),

$$\mu\left(\bigcup_{n=1}^{\infty} B_n\right) = \lim_{n\to\infty} \mu(B_n).$$

But

$$\bigcup_{n=1}^{\infty} B_n = \bigcup_{n=1}^{\infty} (A_1 \setminus A_n) = A_1 \cap \bigcup_{n=1}^{\infty} A_n^c = A_1 \cap \left(\bigcap_{n=1}^{\infty} A_n\right)^c = A_1 \setminus \bigcap_{n=1}^{\infty} A_n.$$

Also, since $A_n \subseteq A_1$ and $\mu(A_1) < \infty$,

$$\mu(B_n) = \mu(A_1 \setminus A_n) = \mu(A_1) - \mu(A_n).$$

Therefore,

$$\mu\left(A_1 \setminus \bigcap_{n=1}^{\infty} A_n\right) = \lim_{n\to\infty} (\mu(A_1) - \mu(A_n)) = \mu(A_1) - \lim_{n\to\infty} \mu(A_n).$$

Finally, since

$$\mu\left(A_1 \setminus \bigcap_{n=1}^{\infty} A_n\right) = \mu(A_1) - \mu\left(\bigcap_{n=1}^{\infty} A_n\right),$$

again using $\mu(A_1) < \infty$, we conclude

$$\mu\left(\bigcap_{n=1}^{\infty} A_n\right) = \lim_{n\to\infty} \mu(A_n).$$

□

1.3 Constructing the Uniform Measure

Given a premeasure[8] on a small family of sets (a finite algebra/field), how can we get an extension to a useful σ-field?

We will first construct a measure μ on a field $\mathscr{A}$. Then we will extend μ from $\mathscr{A}$ to 2^Ω through outer measure as an approximation of μ. Finally, we will restrict the domain to $\mathscr{F}_\tau \subseteq 2^\Omega$ so that $\mathscr{F}_\tau$ is a σ-field and μ is a measure on that field.

Ultimately, we want to construct the uniform measure on $\left([0,1], \mathscr{B}_{[0,1]}\right)$.

Example 1.3.1.

With $\Omega = [0,1]$, put $\mathscr{A} = \{\text{finite unions of intervals}\}$. It is easy to construct a measure on $\mathscr{A}$ by considering lengths of disjoint intervals in its elements.

How do we extend the measure from $\mathscr{A}$ to $\sigma(\mathscr{A})$? If we are able to do so, it is unique? We first answer the second question.

Theorem 1.3.2.

Suppose $\mathscr{A}$ is a field and μ and ν are two finite measures on $\sigma(\mathscr{A})$ such that for every $A \in \mathscr{A}$, $\mu(A) = \nu(A)$. Then for every $B \in \sigma(\mathscr{A})$,

$$\mu(B) = \nu(B).$$

Proof of Theorem 1.3.2. Define

$$\mathscr{H} = \{A \in \sigma(\mathscr{A}) : \mu(A) = \nu(A)\}.$$

We know that $\mathscr{A} \subseteq \mathscr{H} \subseteq \sigma(\mathscr{A})$. Suppose we can show that $\mathscr{H}$ is a monotone class. Then it must contain the monotone class generated by $\mathscr{A}$, which is $\sigma(\mathscr{A})$ by the monotone class theorem. This would show equality. Let's prove it!

Suppose we have a monotone sequence A_n in $\mathscr{H}$. Then by definition of $\mathscr{H}$,[9]

$$\begin{aligned} \mu(A_n) &= \nu(A_n) \\ \implies \lim_{n\to\infty} \mu(A_n) &= \lim_{n\to\infty} \nu(A_n) \\ \mu(\lim_{n\to\infty} A_n) &= \nu\left(\lim_{n\to\infty} A_n\right) \\ \implies \lim_{n\to\infty} A_n &\in \mathscr{H}, \end{aligned}$$

[8] For example, a finitely/countably additive set function like one involving volumes of rectangles.

[9] Note the existence of $\lim_{n\to\infty} \mu(A_n), \nu(A_n)$ is guaranteed in $[0,\infty]$ by the sequence being monotone.

so $\mathscr{H}$ is a monotone class. □

Definition 1.3.3 (Outer measure).

We say that $\tau : 2^\Omega \to [0, \infty]$ is an outer measure if:

(i) $\tau(\varnothing) = 0$,

(ii) If $A \subseteq B$, then $\tau(A) \leq \tau(B)$, and

(iii) $\tau(\cup_n A_n) \leq \sum_n \tau(A_n)$.

[10]

Theorem 1.3.4 (Construction of outer measure).

Suppose $\varnothing \in \mathscr{E} \subseteq 2^\Omega$ and $\mu : \mathscr{E} \to [0, \infty]$ is such that $\mu(\varnothing) = 0$. For every $A \in 2^\Omega$, define

$$\tau(A) = \inf\{\sum_n \mu(A_n) : A_n \in \mathscr{E},\ A \subseteq \cup_n A_n\}.$$

Then τ is an outer measure.

[11][12]

Proof of Theorem 1.3.4. (i) Since $m(\varnothing) = 0$ and $\varnothing \subseteq \varnothing \cup \varnothing \cup \ldots$,

$$\tau(\varnothing) \leq \sum_{n=1}^{\infty} m(\varnothing) = 0.$$

τ is an inf over nonnegative numbers, so $\tau(\varnothing) = 0$.

(ii) Any covering of B is also a covering of A.

(iii) For any $\epsilon > 0$ and A_n, there exists $\{B_{n,k}\}_k$ such that $B_{n,k} \in \mathscr{E}$, $\cup_k B_{n,k} \supseteq A_n$, and

$$\tau(A_n) + \frac{\epsilon}{2^n} \geq \sum_k \mu(B_{n,k}).$$

We know

$$\bigcup_n A_n \subseteq \bigcup_n \bigcup_k B_{n,k},$$

so

$$\tau\left(\bigcup_n A_n\right) \leq \sum_n \sum_k \mu(B_{n,k}) \leq \sum_n \tau(A_n) + \epsilon.$$

We can send $\epsilon \to 0$ to conclude.

□

Now that we have an outer measure on 2^Ω, how do we restrict the domain to where we have countable additivity of disjoint sets?

[10] The definition of outer measure differs from that of measure in three ways: (i) outer measure is defined on 2^Ω instead of a subset; (ii) outer measure must explicitly be monotone, while measure does not; (iii) outer measure is only required to be countably *sub*additive.

[11] Think of $\mathscr{E}$ as a field, even though the theorem does not require this. For instance, $\mathscr{E}$ might be all finite unions of rectangles.

[12] If there does not exist a countable covering of A, convention is to put $\tau(A) = \infty$.

Theorem 1.3.5 (Restriction to Carathéodory-measurable sets).

Let τ be an outer measure. Define

$$\mathscr{F}_\tau := \{A \subseteq \Omega : \forall\ B \subseteq \Omega, \tau(B) = \tau(B \cap A) + \tau(B \cap A^c)\}.$$

Then $(\Omega, \mathscr{F}_\tau, \tau)$ is a measure space.

Thinking of A as a knife that splits any set B into $A \cap B$ and $A^c \cap B$, we may think that A is a good knife if it divides all sets in Ω well.

Proof of Theorem 1.3.5. Step I ($\mathscr{F}_\tau$ is a σ-field). We see that $\varnothing \in \mathscr{F}_\tau$, as for any $B \supseteq \Omega$,

$$\tau(B) = \underbrace{\tau(B \cap \varnothing)}_{=\tau(\varnothing)=0} + \underbrace{\tau(B \cap \varnothing^c)}_{=\tau(B)}.$$

Furthermore, because if $\tau(B) = \tau(A \cap B) + \tau(A^c \cap B) = \tau(A^c \cap B) + \tau(A \cap B)$, $\mathscr{F}_\tau$ is closed under complementation.

We will now show that if $A_1, A_2 \in \mathscr{F}_\tau$, $A_1 \cap A_2 \in \mathscr{F}_\tau$.[13] For any $B \in \Omega$,

$$\begin{aligned}\tau(B) &= \tau(B \cap A_1) + \tau(B \cap A_1^c) \\ &= \tau(B \cap A_1 \cap A_2) + \underbrace{\tau(B \cap A_1 \cap A_2^c) + \tau(B \cap A_1^c \cap A_2) + \tau(B \cap A_1^c \cap A_2^c)}_{\text{goal: make into } B \cap (A_1 \cap A_2)^c}.\end{aligned}$$

Showing the union of these three terms is equal to $B \cap (A_1 \cap A_2)^c$ is a bit difficult, so let's try decomposing the single set.

$$\begin{aligned}\tau\left(B \cap (A_1 \cap A_2)^c\right) &= \tau\left(B \cap (A_1^c \cup A_2^c)\right) \\ &= \underbrace{\tau\left(B \cap (A_2^c \cap A_1)\right)}_{\text{one of goal terms}} + \tau\left(B \cap (A_1^c)\right) \\ &= \tau\left(B \cap (A_2^c \cap A_1)\right) + \tau(B \cap A_1^c \cap A_2) + \tau(B \cap A_1^c \cap A_2^c),\end{aligned}$$

where we cut with A_1 for the second equality and by A_2 for the third.

Thus $\mathscr{F}_\tau$ is closed under finite intersection. Because we already showed it is closed under complementation, it is closed under finite union as well.

It just remains to show that $\mathscr{F}_\tau$ is closed under countable union. Suppose $A_n \in \mathscr{F}_\tau$. WLOG, we can assume that $\{A_n\}$ is pairwise disjoint. We know from $\mathscr{F}_\tau$ being closed under finite union that for any m, $\cup_{n=1}^m A_n \in \mathscr{F}_\tau$. For any $B \subseteq \Omega$,

$$\begin{aligned}\tau(B) &= \tau\left(B \cap \left(\bigcup_{n=1}^m A_n\right)\right) + \tau\left(B \cap \left(\bigcup_{n=1}^m A_n\right)^c\right) \\ &= \tau(B \cap A_1) + \tau\left(B \cap \left(\bigcup_{n=2}^m A_n\right)\right) + \tau\left(B \cap \left(\bigcup_{n=1}^m A_n\right)^c\right),\end{aligned}$$

since we can cut the first term by A_1 and the A_ns are disjoint. Iterate on this. Then by monotonicity of outer

[13] There is no good reason for proving closure under intersection instead of union. The proof for showing closure under finite union is analogous, so it might be a good idea to do it as an exercise.

measure,

$$\begin{aligned}
\tau(B) &= \sum_{n=1}^{m} \tau(B \cap A_n) + \tau\left(B \cap \left(\bigcup_{n=1}^{m} A_n\right)^c\right) \\
&\geq \sum_{n=1}^{m} \tau(B \cap A_n) + \tau\left(B \cap \underbrace{\left(\bigcup_{n=1}^{\infty} A_n\right)^c}_{\text{now smaller}}\right) \\
&\geq \sum_{n=1}^{\infty} \tau(B \cap A_n) + \tau\left(B \cap \left(\bigcup_{n=1}^{\infty} A_n\right)^c\right) \\
&\geq \tau\left(B \cap \left(\bigcup_{n=1}^{\infty} A_n\right)\right) + \tau\left(B \cap \left(\bigcup_{n=1}^{\infty} A_n\right)^c\right),
\end{aligned}$$

where the last inequality follows from countable subadditivity of τ. However, also by countable (in this case, finite) subadditivity,

$$\tau(B) \leq \tau\left(B \cap \left(\bigcup_{n=1}^{\infty} A_n\right)^c\right) + \tau\left(B \cap \left(\bigcup_{n=1}^{\infty} A_n\right)^c\right),$$

so $\cup_{n=1}^{\infty} A_n$ cuts arbitrary $B \subseteq \Omega$ well, so $\cup_{n=1}^{\infty} A_n \in \mathscr{F}_\tau$ and $\mathscr{F}_\tau$ is closed under countable union.

Step II (τ is a measure on $\mathscr{F}_\tau$). We check $\tau(\varnothing) = 0$ because it is an outer measure. The only other condition for τ being a measure is countable additivity on disjoint sets.

Again by τ as an outer measure, for disjoint $\{A_n\}$,

$$\tau\left(\cup_n A_n\right) \leq \sum_n \tau(A_n).$$

For the other direction, take $B = \cup_n A_n$, so

$$\begin{aligned}
\tau(B) = \tau(\cup_n A_n) &\leq \sum_{n=1}^{\infty} \tau(A_n) \\
&= \sum_{n=1}^{\infty} \tau(A_n \cap B) + \tau\left(A_n \cap B^c\right) \\
&\leq \sum_{n=1}^{\infty} \tau(A_n) + \tau(0) \\
&= \sum_{n=1}^{\infty} \tau(A_n).
\end{aligned}$$

$\mathscr{F}_\tau$ being a σ-field on Ω and τ being a valid measure on $\mathscr{F}_\tau$ satisfies the definition of $(\Omega, \mathscr{F}_\tau, \tau)$ being a measure space, so we are done. $\square$

Theorem 1.3.6 (Unique extension of measure).

Suppose μ is a finite measure on a field $\mathscr{A}$. Then there exists a unique extension of μ from $\mathscr{A}$ to $\sigma(\mathscr{A})$.

Proof of Theorem 1.3.6. For every $A \subseteq \Omega$, define

$$\tau(A) = \inf\{\sum_n \mu(B_n) : B_n \in \mathscr{A},\ \cup_n B_n \supseteq A\}.$$

By our earlier theorem, τ is an outer measure.

Defining $\mathscr{F}_\tau$ according to the Caratheodory criterion, we know $(\Omega, \mathscr{F}_\tau, \tau)$ is a measure space.

Since we already showed that a measure on a field is unique on a σ-algebra generated by that field, we just need to show that τ is actually an extension of μ to $\sigma(\mathscr{A})$[14]. This entails showing that for any $A \in \mathscr{A}$, $\mu(A) = \tau(A)$ and that $\mathscr{F}_\tau \supseteq \sigma(\mathscr{A})$.

Fix $A \in \mathscr{A}$. Since $A \subseteq A \cup \varnothing \cup \varnothing \cup \ldots$,

$$\tau(A) \leq \mu(A) + \varnothing + \varnothing + \cdots = \mu(A).$$

For the other direction of the inequality, note that for any covering B_n of A,

$$\begin{aligned}\mu(A) &= \mu\left(\cup_n (A \cap B_n)\right) \\ &\leq \sum_n \mu(A \cap B_n) \\ &\leq \sum_n \mu(B_n),\end{aligned}$$

where the second inequality follows from monotonicity of μ. Taking an inf yields

$$\mu(A) \leq \tau(A).$$

Since $\mathscr{F}_\tau$ is a σ-field, in order to show that $\mathscr{F}_\tau \supseteq \sigma(\mathscr{A})$, we just need to prove that $A \subseteq \mathscr{F}_\tau$. Hence, we want to show for every $A \in \mathscr{A}$ and every $B \subseteq \Omega$, $\tau(B) = \tau(B \cap A) + \tau(B \cap A^c)$. The $\leq$ direction is given by the subadditivity of τ with $(B \cap A) \cup (B \cap A^c)$

Fix $\epsilon > 0$. There exists $B_n \in \mathscr{A}$ such that $\cup_n B_n \supseteq B$ and

$$\tau(B) + \epsilon \geq \sum_n \mu(B_n).$$

Then[15]

$$\begin{aligned}\tau(B) + \epsilon &\geq \sum_n \mu(B_n) \\ &= \sum_n \left[\mu(B_n \cap A) + \mu(B_n \cap A^c)\right] \text{ by additivity of measure} \\ &= \sum_n \left[\tau(B_n \cap A) + \tau(B_n \cap A^c)\right] \text{ since } B_n, A, A^c \in \mathscr{A}\text{, where } \mu, \tau \text{ agree} \\ &\geq \tau\left((\cup_n B_n) \cap A\right) + \tau\left((\cup_n B_n) \cap A^c\right) \text{ by subadditivity of outer measure} \\ &\geq \tau\left(B \cap A\right) + \tau(B \cap A^c) \text{ by monotonicity of outer measure.}\end{aligned}$$

Sending $\epsilon \to 0$,

$$\tau(B) \geq \tau(B \cap A) + \tau\left(B \cap A^c\right),$$

so we have shown

$$\tau(B) = \tau(B \cap A) + \tau(B \cap A^c)$$

for arbitrary $A \in \mathscr{A}$ and $B \subseteq \Omega$, so $\mathscr{A} \subseteq \mathscr{F}_\tau$, and hence $\sigma(\mathscr{A}) \subseteq \mathscr{F}_\tau$. $\square$

Now we finally consider $\left([0,1], \mathscr{B}_{[0,1]}\right)$ and $\mathscr{A}$ as the set of finite unions of intervals and μ the length of an interval. By the theorem, we can extend μ to be a measure on $\mathscr{F} \supseteq \sigma(\mathscr{A}) = \mathscr{B}_{[0,1]}$.

[14] This is why we need μ to be a measure on a *field*.

[15] The last two inequalities use only outer-measure properties. At this point we do not yet know, for example, that $B \cap A \in \mathscr{F}_\tau$.

Definition 1.3.7 (Complete measure space).

A measure space $(\Omega, \mathscr{F}, \mu)$ is complete if

$$A \in \mathscr{F} \text{ and } \mu(A) = 0 \implies \forall\, B \subseteq A, B \in \mathscr{F}, .$$

i.e. if the σ-field contains all subsets of measure zero sets.

Theorem 1.3.8.

Given an outer measure τ, the measure space $(\Omega, \mathscr{F}_\tau, \tau)$ constructed from the Caratheodory criterion is a complete measure space.

Proof of Theorem 1.3.8. We already showed that $(\Omega, \mathscr{F}_\tau, \tau)$ is a measure space. Suppose $\tau(A) = 0$. For any $B \in \Omega$,

$$\tau(B) \leq \tau(B \cap A) + \tau(B \cap A^c)$$

is true by subadditivity.

By monotonicity of outer measure,

$$\tau(B \cap A^c) \leq \tau(B).$$

Also, $\tau(B \cap A) \leq \tau(A) = 0$, so

$$\tau(B) \geq \tau(B \cap A) + \tau(B \cap A^c).$$

Thus $\tau(B) = \tau(B \cap A) + \tau(B \cap A^c)$, for every $B \in \Omega$, so $A \in \mathscr{F}_\tau$. □

We receive the fact that $\left|\mathscr{B}_{[0,1]}\right| = |[0,1]|$, which allows us to claim that $\left([0,1], \mathscr{B}_{[0,1]}, \mu\right)$, where μ is the uniform measure, is not a complete measure space.[16]

Definition 1.3.9 (Cantor set).

Define $C_0 = [0,1]$, $C_1 = [0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$, and recursively $C_n = \frac{C_{n-1}}{3} \cup \left(\frac{2}{3} + \frac{C_{n-1}}{3}\right)$. Then define the Cantor set $C = \cap_{n=0}^{\infty} C_n$.

Immediately, $C \in \mathscr{B}_{[0,1]}$, and $\mu(C) = \lim_{n\to\infty} \left(\frac{2}{3}\right)^n = 0$. However, $|C| = |[0,1]|$ because C bijects with binary expansions of numbers in $[0,1]$. Thus[17]

$$\left|2^C\right| = \left|2^{[0,1]}\right| > |[0,1]| = \left|\mathscr{B}_{[0,1]}\right|.$$

Because this inequality is strict, there exists $A \in 2^C$ such that $A \notin \mathscr{B}_{[0,1]}$.

What is the connection between the result $\mathscr{F}_\tau$ of Caratheodory extension and $\mathscr{B}_{[0,1]}$?

Theorem 1.3.10 (Completion).

Let $(\Omega, \mathscr{F}, \mu)$ be a measure space. Define

$$\tilde{\mathscr{F}} = \{A \cup N : A \in \mathscr{F} \text{ and } \exists\, B \in \mathscr{F} \text{ s.t. } \mu(B) = 0 \text{ and } B \supseteq N\}.$$

[16] This implies that the measure space we constructed via Caratheodory is strictly larger than this Borel measure space.

[17] The inequality is because for any set S, $\left|2^S\right| > |S|$.

$\mathscr{F}$ is a σ-field. For any $A \cup N \in \tilde{\mathscr{F}}$, defining

$$\tilde{\mu}(A \cup N) = \mu(A),$$

$\left(\Omega, \tilde{\mathscr{F}}, \tilde{\mu}\right)$ is a complete and well-defined measure space. Moreover, for any $A \in \mathscr{F}$, $\tilde{\mu}(A) = \mu(A)$; the complete measure space is an extension of the original measure space.
We call $\tilde{\mathscr{F}}$ the *completion* of $\mathscr{F}$.

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Proof of Theorem 1.3.10. We need to check that $\tilde{\mathscr{F}}$ is a σ-field.

(i) Since $\varnothing = \varnothing \cup \varnothing$, where $\varnothing \in \mathscr{F}$ and $\varnothing \subseteq \varnothing$, $\varnothing \in \tilde{\mathscr{F}}$

(ii) Suppose $A \in \mathscr{F}$ and $N \subseteq B \in \mathscr{F}$ such that $m(B) = 0$.[20] We need to show $(A \cup N)^c \in \tilde{F}$. WE want to write $(A \cup N)^c$ as the union of something in $\mathscr{F}$ and a subset of a set of measure 0.

We can write[21]

$$(A \cup N)^c = (A \cup B)^c \cup (B \setminus (A \cup N)).$$

Since $A, B \in \mathscr{F} \implies (A \cup B)^c \in \mathscr{F}$ and $B \setminus (A \cup N) \subseteq B$ with $\mu(B) = 0$, we are done.

(iii) Suppose $A_n \in \mathscr{F}$, $N_n \subseteq B_n \in \mathscr{F}$, and $\mu(B_n) = 0$. We want to show $\cup_n (A_n \cup N_n) \in \tilde{\mathscr{F}}$.

Writing

$$\cup_n (A_n \cup N_n) = (\cup_n A_n) \cup (\cup_n N_n),$$

since $\cup_n N_n \subseteq \cup_n B_n$ and $\mu(\cup_n B_n) = 0$, we are done.

We want to show $\tilde{\mu}$ is well-defined. Suppose $A_1, A_2 \in \mathscr{F}$, $N_1 \subseteq B_1 \in \mathscr{F}$, $N_2 \subseteq B_2 \in \mathscr{F}$, $\mu(B_1) = \mu(B_2) = 0$, and $A_1 \cup N_1 = A_2 \cup N_2$. Can we show $\tilde{\mu}(A_1 \cup N_1) = \tilde{\mu}(A_2 \cup N_2)$?

By definition,[22]

$$\begin{aligned}
\tilde{\mu}(A_1 \cup B_1) &= \mu(A_1) \\
&= \mu(A_1 \cup B_1) \text{ because we can union with measure zero set} \\
&\geq \mu(A_2) = \tilde{\mu}(A_2 \cup N_2).
\end{aligned}$$

By symmetry, the other direction is true, so $\tilde{\mu}$ is well-defined.

Now we show $\tilde{\mu}$ is a measure on $\tilde{\mathscr{F}}$.

(i) $\tilde{\mu}(\varnothing) = \tilde{\mu}(\varnothing \cup \varnothing) = \mu(\varnothing) = 0$

(ii) Suppose $A_n \in \mathscr{F}$, $N_n \subseteq B_n \in \mathscr{F}$, $\mu(B_n) = 0$, and $A_n \cup N_n$ are pairwise disjoint.

[18] $\tilde{\mathscr{F}}$ just adds all subsets of sets in $\mathscr{F}$ with measure 0.
[19] $\tilde{\mathscr{F}}$ is the smallest completion of $\mathscr{F}$.
[20] With this sentence, we are supposing $A \cup N \in \tilde{\mathscr{F}}$.
[21] draw
[22] Adding in B_1 comes from: (i) monotonicity giving $\mu(A_1) \leq \mu(A_1 \cup B_1)$ and (ii) subadditivity giving the other direction.

Then

$$\begin{aligned}\tilde{\mu}\left(\cup_n\left(A_n \cup N_n\right)\right) &= \tilde{\mu}\left(\left(\cup_n A_n\right) \cup \left(\cup_n N_n\right)\right) \\ &= \mu\left(\cup_n A_n\right) \\ &= \sum_n \mu(A_n) \text{ by disjointness} \\ &= \sum_n \tilde{\mu}(A_n \cup N_n),\end{aligned}$$

so $\tilde{\mu}$ is a measure.

Since $\tilde{\mu}(A) = \tilde{\mu}\left(A \cup \varnothing\right) = \mu(A)$, $\tilde{\mu}$ is an extension of μ. □

Theorem 1.3.11.

Suppose μ is a finite measure on a field $\mathscr{A}$ of subsets of Ω and that τ is the outer measure constructed by μ. With $\mathscr{F}_\tau$ as the set of τ-measurable subsets of Ω according to the Caratheodory criterion, $(\Omega, \mathscr{F}_\tau, \tau)$ is the completion of $(\Omega, \sigma(\mathscr{A}), \tau)$.

[23]

We require the following lemma.

Lemma 1.3.12.

Under the same setting (finite measure μ with induced outer measure τ), for any $A \in \mathscr{F}_\tau$, there exists $B \in \sigma(\mathscr{A})$ such that $B \supseteq A$ and $\tau(B) = \tau(A)$. Elements in $\mathscr{F}_\tau$ can be approximated above by elements in $\sigma(\mathscr{A})$.

Proof of Lemma 1.3.12. By definition of outer measure induced by a premeasure,

$$\tau(A) = \inf\{\sum_n \mu(B_n) : B_n \in \mathscr{A}, \cup_n B_n \subseteq A\}.$$

For every $m \in \mathbb{N}$, there exists a sequence $\{B_{m,n}\}_n \subseteq \mathscr{A}$ such that $\cup_n B_{m,n} \supseteq A$ and

$$\tau(A) + \frac{1}{n} \geq \sum_n \mu(B_{m,n}).$$

Write

$$B = \cap_m \cup_n B_{m,n} \supseteq A.$$

Then $B \in \sigma(\mathscr{A})$ because each $B_{m,n} \in \mathscr{A}$. We see

$$\begin{aligned}\tau(A) + \frac{1}{m} &\geq \sum_n \mu(B_{m,n}) \\ &= \sum_n \tau(B_{m,n}) \\ &\geq \tau\left(\cup_n B_{m,n}\right) \\ &\geq \tau(B).\end{aligned}$$

Sending $m \to \infty$, $\tau(A) \geq \tau(B)$. Since $A \subseteq B$, $\tau(A) \leq \tau(B)$. □

[23] One example is taking $\mathscr{A}$ as all finite unions of intervals in the unit interval and $\mathscr{F}_\tau$ as the *completion* of the Borel σ-field.

Proof. (Of the theorem.) Write $\mathscr{F} = \sigma(\mathscr{A})$. We want to show that $\tilde{\mathscr{F}} = \mathscr{F}_\tau$, where

$$\tilde{\mathscr{F}} = \{A \cup N : A \in \mathscr{F}, \exists\ B \in \mathscr{F} \text{ s.t. } B \supseteq N, \mu(B) = 0\}.$$

We know $\tilde{\mathscr{F}} \subseteq \mathscr{F}_\tau$ since $\tilde{\mathscr{F}}$ is the smallest completion of $\mathscr{F}$ and $\mathscr{F}_\tau$ is a completion of $\mathscr{F}$.

For $\mathscr{F}_\tau \subseteq \tilde{\mathscr{F}}$, fix $A \in \mathscr{F}_\tau$. We want to write A as the union of something in $\mathscr{F}$ and a subset of a measure zero set.

By the lemma, there exists $B \in \mathscr{F}$ such that $B \supseteq A$ and $\tau(B) = \tau(A)$. By finiteness of τ, this implies $\tau(B \setminus A) = 0$.

Apply the lemma again to $B \setminus A$: there exists $C \in \mathscr{F}$ such that $C \subseteq B \setminus A$ and $\tau(C) = \tau(B \setminus A) = 0$.

Draw, then write

$$A = (A \cap C^c) \cup (A \cap C)\,.$$

We don't know that $A \cap C^c \in \mathscr{F}$, because we just know $A \in \mathscr{F}_\tau$. However, we can write

$$A = (B \cap C^c) \cup (A \cap C).$$

$B, C \in \mathscr{F}$, so $B \cap C^c \in \mathscr{F}$. Furthermore, $A \cap C \subseteq C$, where $\tau(C) = 0$, so we are done. $\square$

Section 2

Marcus Kuo

Measurable Functions

2.1 Preliminaries

Definition 2.1.1 (Preimage).

Let $f : X \to Y$ be a function, and let $\mathscr{G}$ be a collection of subsets of Y (e.g., a σ-algebra on Y). For any $B \in \mathscr{G}$, define the preimage of B under f by

$$f^{-1}(B) := \{x \in X : f(x) \in B\}.$$

Definition 2.1.2 (Measurable function).

We say $f : (X, \mathscr{F}) \to (Y, \mathscr{G})$ is measurable if $f^{-1}\mathscr{G} \subseteq \mathscr{F}$; i.e. the preimage of any measurable set in the second measurable space is measurable.
Alternatively, for any $B \in \mathscr{G}$, $f^{-1}B \in \mathscr{F}$.

[1]

Remark 2.1.3 (Rules of applying preimage). Some of these are shown on homework. We have that

- $f^{-1}\varnothing = \varnothing,\ f^{-1}Y = X$
- $f^{-1}(B^c) = (f^{-1}B)^c$
- $A \subseteq B \implies f^{-1}A \subseteq f^{-1}B$
- $f^{-1}(\cup_\lambda A_\lambda) = \cup_\lambda f^{-1}A_\lambda$
- $f^{-1}(\cap_\lambda A_\lambda) = \cap_\lambda f^{-1}A_\lambda$

Definition 2.1.4 (Random variable).

$X : (\Omega, \mathscr{F}) \to (\mathbb{R}, \mathscr{B}_\mathbb{R})$ is a random variable.

[1] f maps (sample space, σ-field) to (sample space, σ-field).

2.2 Properties of Measurable Functions

Lemma 2.2.1.

Suppose $f : (X, \mathscr{F}) \to (Y, \sigma(\mathscr{E}))$. f is measurable if $f^{-1}\mathscr{E} \subseteq \mathscr{F}$.

Proof of Lemma 2.2.1. Suppose $f^{-1}\mathscr{E} \subseteq \mathscr{F}$. We know that because $\mathscr{F}$ is a σ-field, $\sigma\left(f^{-1}\mathscr{E}\right) \subseteq \mathscr{F}$. We want to show that $f^{-1}\sigma(\mathscr{E}) \subseteq \mathscr{F}$. So, it suffices to show $\sigma\left(f^{-1}\mathscr{E}\right) = f^{-1}\sigma(\mathscr{E})$.

Since $\mathscr{E} \subseteq \sigma(\mathscr{E})$, it can be shown that $f^{-1}\sigma(\mathscr{E})$ is a σ-field containing $f^{-1}\mathscr{E}$. Since $\sigma\left(f^{-1}\mathscr{E}\right)$ is the smallest such σ-field, we have that

$$\sigma\left(f^{-1}\mathscr{E}\right) \subseteq f^{-1}\sigma\left(\mathscr{E}\right).$$

For the other direction, write $\mathscr{H} = \{B \subseteq Y : f^{-1}B \in \sigma\left(f^{-1}\mathscr{E}\right)\}$. Since $f^{-1}\mathscr{E} \subseteq \sigma(f^{-1}\mathscr{E})$, we have $\mathscr{E} \subseteq \mathscr{H}$. We can check that $\mathscr{H}$ is actually a σ-field, so $\sigma(\mathscr{E}) \subseteq \mathscr{H}$ and thus

$$f^{-1}\sigma(\mathscr{E}) \subseteq \sigma\left(f^{-1}\mathscr{E}\right).$$

□

Since $\mathscr{B}_{\mathbb{R}} = \sigma\left(\{(a, \infty) : a \in \mathbb{R}\}\right)$, a random variable $X : (\Omega, \mathscr{F}) \to (\mathbb{R}, \mathscr{B}_{\mathbb{R}})$ is measurable if $\{X > a\}$ is measurable for all $a \in \mathbb{R}$ (equivalently, if $\{X \geq a\}$ is measurable for all $a \in \mathbb{R}$).

Proposition 2.2.2.

Applying a continuous operator to a measurable function produces a measurable function. For instance, for f, g measurable, $f + g$ is measurable. $\limsup$, $\liminf$ and such preserve measurability.

Proof of Proposition 2.2.2. We see that $\{f + g > a\}$ is equivalent to the statement that there exists some t such that $\{f > t\} \cap \{g > a - t\}$. We can take t to be rational such that $t \in (a - g, f)$, so

$$\{f + g > a\} = \bigcup_{t \in \mathbb{Q}} \left[\{f > t\} \cap \{g > a - t\}\right] \in \mathscr{F}.$$

Similarly,

$$\{\sup_k f_k \leq a\} = \bigcap_{k=1}^{\infty} \{f_k \leq a\} \in \mathscr{F}.$$

□

Lemma 2.2.3.

Suppose $f : (X, \mathscr{F}) \to (Y, \mathscr{G})$ and $g : (Y, \mathscr{G}) \to (Z, \mathscr{H})$ are measurable. Then $g \circ f$ is measurable.

Proof of Lemma 2.2.3. Take two preimages. □

2.3 Integration

Definition 2.3.1 (Indicator function).

Define the indicator function on A as

$$\mathbb{1}_A(\omega) = \begin{cases} 1, & \omega \in A \\ 0, & \omega \notin A \end{cases} .$$

Because

$$\{\mathbb{1}_A \leq a\} = \begin{cases} \Omega, & a \geq 1 \\ A^c, & 0 \leq a < 1 \\ \varnothing, & a < 0 \end{cases},$$

the indicator function $\mathbb{1}_A$ is measurable iff A is measurable.

Definition 2.3.2 (Simple function).

Suppose $\{A_1, \ldots, A_n\}$ is a finite partition of Ω. A simple function f is a linear combination of indicator functions on sets of the partition:

$$f = \sum_{i=1}^{n} a_i \mathbb{1}_{A_i}, \quad a_i \in \mathbb{R}.$$

From simple functions, we can approximate measurable nonnegative functions, which we can use to get a general measurable function f as the difference of two nonnegative measurable functions: $f = f^+ - f^-$, where

$$\forall\, x \in \mathbb{R}, x^+ = \max(x, 0), \quad x^- = (-x)^+, \quad |x| = x^+ + x^-.$$

Lemma 2.3.3 (Approximation of nonnegative measurable functions).

For any nonnegative measurable function f, there exists a sequence (f_n) of nonnegative simple functions such that $f_n \uparrow f$ pointwise.
If f is bounded, then this convergence is uniform, i.e.

$$\sup_{\omega \in \Omega} |f_n(\omega) - f(\omega)| \to 0.$$

Proof of Lemma 2.3.3. Define

$$f_n = \sum_{k=0}^{n2^n - 1} \frac{k}{2^n} \mathbb{1}_{\frac{k}{2^n} \leq f < \frac{k+1}{2^n}} + n\mathbb{1}_{f \geq n}.$$

f_n divides the section where $f \leq n$ into $n2^n - 1$ intervals. Equivalently, we can write[2]

$$f_n = \max\{\frac{k}{2^n} : \frac{k}{2^n} \leq f, k = 0, 1, 2, \ldots, n2^n - 1\} \wedge n.$$

We see that $f_n \leq f_{n+1}$ because $\frac{2k}{2^{n+1}}, \frac{2k+1}{2^{n+1}} \geq \frac{k}{2^n}$ and $n + 1 \geq n$.

If $f(\omega) \leq n$, then $|f_n(\omega) - f(\omega)| < \frac{1}{2^n}$, so we have pointwise convergence. If f is bounded, then eventually, we get a uniform bound on the approximation error. $\square$

[2] $\wedge$ acts as a min operator.

Remark 2.3.4 (Lebesgue integral). Suppose $f : (\Omega, \mathscr{F}, \mu) \to (\mathbb{R}, \mathscr{B}_{\mathbb{R}})$ and that everything in these definitions is measurable when required. We will follow a similar process for constructing the Lebesgue integral.

(i) (Indicator.) Define $\int \mathbb{1}_A d\mu = \mu(A)$.

(ii) (Nonnegative simple function.) For $f = \sum_{i=1}^n a_i \mathbb{1}_{A_i}$, where $a_i \geq 0$ and A_i are measurable, define

$$\int f d\mu = \sum_{i=1}^n a_i \mu\left(A_i\right).$$

(iii) (Nonnegative function.) For $f \geq 0$, define

$$\int f d\mu = \sup\{\int g d\mu : g \leq f, g \text{ is a nonnegative simple function}\}.$$

(iv) (General function.) Define

$$\int f d\mu = \int f^+ d\mu - \int f^- d\mu$$

when at least one of the integrals is finite.

Is the integral of a simple function well-defined? Since simple functions have multiple representations (consider $f = a$ on $A_1 \cup A_2$), we want to make sure $\int f d\mu = \sum a_i \mu(A_i)$ is well-defined. We can write

$$f = \sum_{i=1}^n a_i \mathbb{1}_{A_i} = \sum_{j=1}^n b_j \mathbb{1}_{f=b_j}, \text{where } b_i \neq b_j,\ m \leq n,$$

which is a unique representation of the simple function.

For each i, j, either $A_i \subseteq \{f = b_j\}$ or $A_i \cap \{f = b_j\} = \varnothing$. Using the additivity of measure, write

$$\begin{aligned}
\int f d\mu &= \sum_{i=1}^n a_i \mu(A_i) \\
&= \sum_{i=1}^n a_j \sum_{j=1}^m \mu\left(A_i \cap \{f = b_j\}\right) \\
&= \sum_{j=1}^m \sum_{i=1}^n a_i \mu\left(A_i \cap \{f = b_j\}\right) \\
&= \sum_{j=1}^m \sum_{i:\ A_i \subseteq \{f=b_j\}} a_i \mu\left(A_i \cap \{f = b_j\}\right) \\
&= \sum_{j=1}^m \sum_{i: A_i \subseteq \{f=b_j\}} b_j \mu\left(A_i \cap \{f = b_j\}\right) \\
&= \sum_{j=1}^m \sum_{i=1}^n b_j \mu\left(A_i \cap \{f = b_j\}\right) \\
&= \sum_{j=1}^m b_j \mu\left(f = b_j\right).
\end{aligned}$$

Now that we know the integral of a simple function is well-defined, let's get basic properties.

Lemma 2.3.5.

Suppose $(f_n), f, g$ are nonnegative simple functions. Then

(i) $\int f d\mu \geq 0$

(ii) $\int (af)\, d\mu = a \int f d\mu$, for $a \geq 0$

(iii) $\int (f+g) d\mu = \int f d\mu + \int g d\mu$

(iv) $f \leq g \implies \int f d\mu \leq \int g d\mu$

(v) If $(f_n) \uparrow$ and $\lim_{n\to\infty} f_n \geq g$, then

$$\lim_{n\to\infty} \int f_n d\mu \geq \int g d\mu.$$

[3]

Proof of Lemma 2.3.5. (iii) For $f = \sum a_i \mathbb{1}_{A_i}$ and $g = \sum b_j \mathbb{1}_{B_j}$, we can write

$$f + g = \sum_{i=1}^{n} \sum_{j=1}^{m} (a_i + b_j) \mathbb{1}_{A_i \cap B_j},$$

so

$$\begin{aligned}
\int (f+g)\, d\mu &= \sum_{i=1}^{n} \sum_{j=1}^{m} (a_i + b_j) \mu\left(A_i \cap B_j\right) \\
&= \sum_{i=1}^{n} a_i \sum_{j=1}^{m} \mu\left(A_i \cap B_j\right) + \sum_{j=1}^{m} b_j \sum_{i=1}^{n} \mu\left(A_i \cap B_j\right) \\
&= \sum_{i=1}^{n} a_i \mu\left(A_i\right) + \sum_{j=1}^{m} b_j \mu(B_j) \\
&= \int f d\mu + \int g d\mu.
\end{aligned}$$

(iv) Write

$$\int g d\mu = \int \left((g-f) + f\right) d\mu,$$

where $g - f$ is a nonnegative simple function because $g \geq f$. We can then use property (iii) and property (i).

(v) Introduce for each $\alpha \in (0,1)$ the set

$$A_n(\alpha) = \{f_n \geq \alpha g\}.$$

Then $A_n(\alpha) \uparrow \Omega$[4].

[3] Note that even if the limit of (f_n) takes on a value of ∞, we are never actually evaluating the integral of the limit, so there is no cause for concern.

[4] If $\alpha = 1$, some example like $f_n = 1 - \frac{1}{n} \uparrow 1 = g$, shows that $A_n(\alpha) \uparrow \Omega$ would not hold.

Write $g = \sum_{j=1}^m b_j \mathbb{1}_{B_j}$. Then by property (iv), we get the first and second inequalities in

$$\begin{aligned}
\int f_n d\mu &\geq \int f_n \mathbb{1}_{A_n(\alpha)} d\mu \\
&\geq \int (\alpha g \mathbb{1}_{A_n(\alpha)}) d\mu \\
&= \alpha \int g \mathbb{1}_{A_n(\alpha)} d\mu \\
&= \sum_{j=1}^m b_j \mathbb{1}_{B_j \cap A_n(\alpha)} d\mu \\
&= \sum_{j=1}^m b_j \mu\left(B_j \cap A_n(\alpha)\right) \\
\implies \lim_{n\to\infty} \int f_n d\mu &\geq \lim_{n\to\infty} \alpha \sum_{j=1}^m b_j \mu\left(B_j \cap A_n(\alpha)\right) \\
&= \alpha \sum_{j=1}^m b_j \lim_{n\to\infty} \mu\left(B_j \cap A_n(\alpha)\right) \\
&= \alpha \sum_{j=1}^m b_j \mu\left(B_j\right) = \alpha \int g d\mu,
\end{aligned}$$

where the second-to-last equality follows from continuity of measure from below. Sending $\alpha \to 1^-$, we are done. The introduction of α allows us to reduce the switching of a limit and integral to the switching of a limit and a finite sum.

□

Is the integral of a nonnegative function well-defined? Recall that for nonnegative f, we define

$$\int f d\mu = \sup\{\int g d\mu : g \leq f, g \text{ nonnegative simple}\}.$$

If f itself is nonnegative and simple, does this definition coincide with our previous one?

Writing $f = \sum_{i=1}^n a_i \mathbb{1}_{A_i}$ for a partition $\{A_i\}$, define

$$T(f) = \sum_{i=1}^n a_i \mu(A_i), \quad S(f) = \sup\{T(g) : g \leq f, g \text{ nonnegative simple }\}.$$

Is $T(f) = S(f)$?

For any $g \leq f$ where g is nonnegative and simple, we see $T(g) \leq T(f)$ by monotonicity of the integral of a nonnegative simple function. Taking a sup on both sides over all nonnegative simple minorants g of f, we get $S(f) \leq T(f)$. Furthermore, since $f \leq f$, $S(f) \geq T(f)$ and the integral of a nonnegative function extends the integral of nonnegative simple functions.

Lemma 2.3.6.

Assume f, g are nonnegative measurable functions. Suppose we have a sequence (f_n) of nonnegative simple functions that increases to f.

(i) Then

$$\int f d\mu = \lim_{n\to\infty} \int f_n d\mu$$

(ii) We can write

$$\int f d\mu = \lim_{n\to\infty} \sum_{k=1}^{n2^n-1} \frac{k}{2^n} \mu\left(\frac{k}{2^n} \le f < \frac{k+1}{2^n}\right) + n\mu\left(f \ge n\right).$$

(iii) For all $a \ge 0$,

$$\int (af) d\mu = a \int f d\mu.$$

(iv)

$$\int (f+g) d\mu = \int f d\mu + \int g d\mu.$$

(v) The integral of nonnegative functions is monotone:

$$f \le g \implies \int f d\mu \le \int g d\mu.$$

[5]

Proof of Lemma 2.3.6. (i) By definition of the integral of f (not monotonicity),

$$\int f d\mu \ge \int f_n d\mu.$$

Taking a limit on both sides, we get

$$\int f d\mu \ge \lim_{n\to\infty} \int f_n d\mu.$$

By property (v) of the integral of nonnegative simple functions, for $g \le f$ where g is nonnegative and simple, we have

$$\lim_{n\to\infty} \int f_n d\mu \ge \int g d\mu.$$

Taking a sup over all nonnegative simple minorants g of f, we show

$$\lim_{n\to\infty} \int f_n d\mu \ge \int f d\mu.$$

(ii) Trivial.

(iii) There exists a sequence of simple $f_n \uparrow f$, so $af_n \uparrow af$. Then by repeatedly applying property (ii) of nonnegative simple functions,

$$\begin{aligned}\int (af) d\mu &= \lim_{n\to\infty} \int (af_n)\, d\mu \\ &= a \lim_{n\to\infty} \int f_n d\mu \\ &= a \int f d\mu.\end{aligned}$$

(iv) There exist simple $f_n \uparrow f$ and $g_n \uparrow g$, so $f_n + g_n \uparrow f + g$. Then by property (iii) of nonnegative simple

[5] This shows linearity of the integral of nonnegative functions, which is not immediate from the definition.

functions,

$$\begin{aligned}\int (f+g)d\mu &= \lim_{n\to\infty} \int f_n + g_n d\mu \\ &= \lim_{n\to\infty} \int f_n d\mu + \lim_{n\to\infty} \int g_n d\mu \\ &= \int f d\mu + \int g d\mu.\end{aligned}$$

(v) Trivial, since any simple function below f is also below g.

□

How do we extend the definition to an integral of a general measurable function? Decompose $f = f^+ - f^-$. If $\min\{\int f^+ d\mu, \int f^- d\mu\} < \infty$, then we say the integral of f exists and define $\int f d\mu = \int f^+ d\mu - \int f^- d\mu$ Both integrals are finite iff $\int |f|\, d\mu < \infty$, since $|f| = f^+ + f^-$. In this case (the integral of f is finite), we call f integrable.

From now on, we only consider f that are integrable.[6]

Definition 2.3.7 (Definite integral).

For any measurable A, define

$$\int_A f d\mu = \int (f \mathbb{1}_A)\, d\mu.$$

Definition 2.3.8 (Expectation of a random variable).

For a random variable $X : (\Omega, \mathscr{F}, \mathbb{P}) \to (\mathbb{R}, \mathscr{B}_{\mathbb{R}})$, we define the expectation of X as

$$\mathbb{E}[X] := \int X d\mathbb{P}.$$

We say the expectation exists if $\mathbb{E}[|X|] < \infty$ (X is integrable).[a]

[a] Note that the expectation exists if X is integrable. The integral of X existing is not sufficient; it must be finite as well.

Lemma 2.3.9.

Suppose f, g are integrable.

(i) If $A \in \mathscr{F}$ such that $\mu(A) = 0$,

$$\int_A f d\mu = 0.$$

(ii) If $f \le g$ almost everywhere ($\mu(f > g) = 0$), then

$$\int f d\mu \le \int g d\mu.$$

(iii) If $f = g$ almost everywhere, then

$$\int f d\mu = \int g d\mu.$$

[6] The integral of x^3 does not exist and x^3 is not integrable on the real line.

Proof of Lemma 2.3.9. (i) First, consider $f = \sum_i a_i \mathbb{1}_{A_i}$ to be a nonnegative simple function. Then

$$\int_A f d\mu = \int \left(\sum_i a_i \mathbb{1}_{A_i \cap A} \right) d\mu = \sum_i a_i \mu \left(A_i \cap A \right) = 0,$$

since $\mu(A_i \cap A) \leq \mu(A) = 0$.

Now consider f nonnegative. Then there exists nonnegative simple function sequence $f_n \uparrow f$, so $f_n \mathbb{1}_A \uparrow f \mathbb{1}_A$. Then

$$\int_A f d\mu = \lim_{n \to \infty} \int_A f_n d\mu = 0.$$

For a general f, put $f = f^+ - f^-$, so

$$f \mathbb{1}_A = f^+ \mathbb{1}_A - f^- \mathbb{1}_A.$$

Taking an integral on both sides with linearity of the integral on nonnegative functions, we see the right hand side is 0.

(ii) Suppose $f \leq g$ almost everywhere. Let

$$A = \{x : f \leq g\}, \quad A^c = \{x : f > g\}.$$

Then $\mu\left(A^c\right) = 0$. Suppose for now that $f, g \geq 0$.

We can write

$$\begin{aligned} \int f d\mu &= \int_A f d\mu + \int_{A^c} f d\mu \\ &= \int f \mathbb{1}_A d\mu + \int f \mathbb{1}_{A^c} d\mu \\ &\leq \int_A g d\mu + \int_{A^c} g d\mu \\ &= \int g d\mu. \end{aligned}$$

where the inequality follows from monotonicity of integrals of nonnegative functions.

For general f, g, write $f = f^+ - f^-$ and $g = g^+ - g^-$. If $f \leq g$, then $f^+ \leq g^+$ and $-f \geq -g$, so

$$(-f)^+ \geq (-g)^+ .$$

then $f^- \geq g^-$.

Using monotonicity for the integral of nonnegative functions,

$$\int f^+ d\mu \leq \int g^+ d\mu, \quad \int f^- d\mu \geq \int g^- d\mu.$$

we conclude by subtracting the second inequality from the first.

(iii) Follows from symmetry with (ii).

□

Lemma 2.3.10 (Linearity of integral).

Suppose f, g are integrable. Then

(i) $\int (af) d\mu = a \int f d\mu$, for all $a \in \mathbb{R}$.

(ii) $\int (f+g)d\mu = \int f d\mu + \int g d\mu$

Proof of Lemma 2.3.10. (i) If $a \geq 0$, we see that

$$(af)^+ = \max\{0, af\} = a\max\{0, f\} = af^+.$$

Similarly, $(af)^- = af^-$. This gives us the second equality in the following:

$$\begin{aligned}\int (af)d\mu &= \int (af)^+ d\mu - \int (af)^- d\mu \\ &= a\left(\int f^+ d\mu - \int f^- d\mu\right) \\ &= a\int f d\mu.\end{aligned}$$

If $a < 0$, then $(af)^+ = ((-a)(-f))^+ = -a(-f)^+ = -af^-$. Similarly, $(af)^- = -af^+$. We show linearity in the same way.

(ii) Note the three decompositions we have:

- $f = f^+ - f^-$
- $g = g^+ - g^-$
- $(f+g) = (f+g)^+ - (f+g)^-$.

We also know that

$$\begin{aligned}(f+g)^+ - (f+g)^- &= f^+ - f^- + g^+ - g^- \\ \text{Rearrange: } (f+g)^+ + f^- + g^- &= (f+g)^- + f^+ + g^+.\end{aligned}$$

We have a nonnegative function on either side; integrate, apply linearity of integral of nonnegative functions, and rearrange again to get

$$\int (f+g)^+ d\mu - \int (f+g)^- d\mu = \int f^+ d\mu - \int f^- d\mu + \int g^+ d\mu - \int g^- d\mu.$$

Use linearity of integrals of nonnegative functions again to conclude!

□

2.4 Convergence Theorems

Theorem 2.4.1 (Levi's monotone convergence).

Suppose $(f_n), f \geq 0$ and $f_n \uparrow f$ almost everywhere. Then

$$\lim_{n\to\infty} \int f_n d\mu = \int f d\mu.$$

Proof of Theorem 2.4.1. [7]We have shown the theorem but with f_n as simple functions. We want to approximate each f_n by a sequence of nonnegative simple functions.

[7]For the proof, consider $f_n \uparrow f$ everywhere. At the end, we can add in the measure zero set.

For each f_n, there exist nonnegative simple functions $(f_{n,k})_k$ such that $f_{n,k} \uparrow f_n$ as $k \to \infty$.

Define $g_k(x) := \max_{1 \le n \le k} f_{n,k}(x)$. We see that each g_k is a nonnegative simple function. Furthermore, because $f_{n,k} \le f_{n,k+1}$, (g_k) is increasing[8].

We see that $g_k \le f_k$, so $\lim\limits_{k\to\infty} g_k \le \lim\limits_{k\to\infty} f_k = f$.

Fix $i \in \mathbb{N}$. For every $k \ge i$, since

$$g_k(x) := \max_{1 \le n \le k} f_{n,k}(x),$$

we have $g_k(x) \ge f_{i,k}(x)$ for all x. Taking $k \to \infty$ and using $f_{i,k} \uparrow f_i$ gives

$$\lim_{k\to\infty} g_k(x) \ge \lim_{k\to\infty} f_{i,k}(x) = f_i(x), \quad \forall x.$$

Hence $\lim_{k\to\infty} g_k \ge f_i$ pointwise for every i. Therefore, pointwise,

$$\lim_{k\to\infty} g_k \ge \sup_{i\in\mathbb{N}} f_i = f,$$

and we have equality in $\lim\limits_{k\to\infty} g_k = f$.

Since (g_k) is increasing, by the monotone convergence theorem for simple functions in Lemma (**??**),

$$\lim_{k\to\infty} \int g_k d\mu = \int f d\mu.$$

By monotonicity of the integral of nonnegative functions, $\int f_n d\mu \le \int f d\mu$. We can take a limit to see $\lim_{n\to\infty} \int f_n d\mu \le \int f d\mu$. Because $g_n \le f_n$, we see

$$\lim_{n\to\infty} \int g_n d\mu \le \lim_{n\to\infty} \int f_n d\mu \le \int f d\mu.$$

The outer terms are equal, so

$$\lim_{n\to\infty} \int f_n d\mu = \int f d\mu.$$

□

Theorem 2.4.2 (Fatou's lemma).

We can take lim inf outside integrals of nonnegative functions, up to an inequality:

$$\int \left(\liminf_{n\to\infty} f_n\right) d\mu \le \liminf_{n\to\infty} \int f_n d\mu.$$

Proof of Theorem 2.4.2. Write

$$\begin{aligned}\liminf_{n\to\infty} f_n &= \lim_{k\to\infty} \left(\inf_{n\ge k} f_n\right) \\ &= \lim_{k\to\infty} g_k.\end{aligned}$$

Note that g_k is a monotone increasing sequence of nonnegative functions, so Levi's monotone convergence theorem applies and we get

$$\lim_{k\to\infty} \int \inf_{n\ge k} f_n d\mu = \lim_{k\to\infty} \int g_k d\mu = \int \lim_{k\to\infty} g_k d\mu = \int \liminf_{n\to\infty} f_n d\mu.$$

[8] $f_{n,k+1}$ is not considered in g_k, so the addition of $f_{n+1,k+1}$ poses no problem.

Since $\inf_{j\geq k} f_j \leq f_n$ for all $n \geq k$, we get

$$\int \inf_{n\geq k} f_n d\mu \leq \inf_{n\geq k} \int f_n d\mu, \forall\ n \geq k.$$

Taking a limit,

$$\lim_{k\to\infty} \int \inf_{n\geq k} f_n d\mu \leq \liminf_{n\to\infty} \int f_n d\mu.$$

Making the substitution with the result from Levi's MCT, we have

$$\int \liminf_{n\to\infty} f_n d\mu \leq \liminf_{n\to\infty} \int f_n d\mu.$$

□

The following theorem is the most general of the convergence theorems.

Theorem 2.4.3 (Lebesgue's Dominated Convergence Theorem).

Suppose $f_n \to f$ and there exists an integrable function g such that $|f_n| \leq g$ for all $n \in \mathbb{N}$. Then

$$\lim_{n\to\infty} \int f_n d\mu = \int f d\mu.$$

Finding the *control function* g is often possible, but difficult.

Proof of Theorem 2.4.3. By hypothesis, $-g \leq f_n \leq g$ for every $n \in \mathbb{N}$.

Since $f_n + g \geq 0$, by Fatou's lemma,

$$\begin{aligned}
\int \liminf_{n\to\infty} f_n + g d\mu &\leq \liminf_{n\to\infty} \int (f_n + g) d\mu \\
\int \liminf_{n\to\infty} f_n d\mu + \int g d\mu &\leq \liminf_{n\to\infty} \int f_n d\mu + \int g d\mu \text{ by linearity} \\
\int \liminf_{n\to\infty} f_n d\mu &\leq \liminf_{n\to\infty} \int f_n d\mu \text{ since } \int g d\mu < \infty.
\end{aligned}$$

We symmetrically derive the other desired inequality, using the fact that $g - f_n \geq 0$:

$$\begin{aligned}
\int \liminf_{n\to\infty} g - f_n d\mu &\leq \liminf_{n\to\infty} \int g - f_n d\mu \\
\int g d\mu + \int \liminf_{n\to\infty} -f_n &\leq \int g d\mu + \liminf_{n\to\infty} \int -f_n d\mu \\
\int \limsup_{n\to\infty} f_n &\geq \int \limsup_{n\to\infty} \int f_n d\mu.
\end{aligned}$$

Chaining together these two inequalities,

$$\limsup_{n\to\infty} \int f_n d\mu \leq \int \limsup_{n\to\infty} f_n d\mu = \int f d\mu = \int \liminf_{n\to\infty} f_n d\mu \leq \liminf_{n\to\infty} \int f_n d\mu.$$

Since $\liminf_{n\to\infty} \int f_n d\mu \leq \limsup_{n\to\infty} \int f_n d\mu$, we see that equality holds and thus the limit exists:

$$\lim_{n\to\infty} \int f_n d\mu = \int f d\mu.$$

□

Corollary 2.4.4 (Bounded Convergence Theorem).

Suppose $f_n \to f$, $|f_n| \leq M < \infty$ almost everywhere, and $\mu(\Omega) < \infty$. Then

$$\lim_{n\to\infty} \int f_n d\mu = \int f d\mu.$$

Proof of Corollary 2.4.4. Take $g = M$, which is integrable by finiteness of $\mu(\Omega)$. $\square$

Section 3

Marcus Kuo

Independence

3.1 Preliminaries

Consider a measure space $(\Omega, \mathscr{F}, \mathbb{P})$.

Notation 3.1.1.

For a random variable $X : (\Omega, \mathscr{F}, \mathbb{P}) \to (\mathbb{R}, \mathscr{B}_{\mathbb{R}})$, when we write $\{X \in B\}$, we are really referring to the set as

$$\{X \in B\} = \{\omega \in \Omega : X(\omega) \in B\}.$$

Similarly,

$$\{X \leq a\} = \{\omega \in \Omega : X(\omega) \leq a\},$$
$$\{X \in A, Y \in B\} = \{\omega \in \Omega : X(\omega) \in A, Y(\omega) \in B\},$$

and so on.
Perhaps most importantly,

$$\mathbb{P}[X \in B] = \mathbb{P}[\omega \in \Omega : X(\omega) \in B].$$

Definition 3.1.2 (Independent events).

We say that the events $A_1, \ldots, A_n \in \mathscr{F}$ are independent if for every $\Lambda \subseteq \{1, \ldots, n\}$,

$$\mathbb{P}\left[\bigcap_{i \in \Lambda} A_i\right] = \prod_{i \in \Lambda} \mathbb{P}[A_i].$$

We say $\mathscr{E}_1, \ldots, \mathscr{E}_n \subseteq \mathscr{F}$ are independent if for any $A_1 \in \mathscr{E}_1, \ldots, A_n \in \mathscr{E}_n$, $(A_1, \ldots, A_n)$ are independent.

Definition 3.1.3 (Generated σ-field (of a random variable)).

Let $X : (\Omega, \mathscr{F}, \mathbb{P}) \to (\mathbb{R}, \mathscr{B}_{\mathbb{R}})$ be a random variable. We define $\sigma(X)$ to be the smallest σ-field such that

$$X : (\Omega, \sigma(X), \mathbb{P}) \to (\mathbb{R}, \mathscr{B}_{\mathbb{R}})$$

is measurable.
Since $X^{-1}\mathscr{B}_{\mathbb{R}}$ is a σ-field containing X (preimage of a σ-field is also a σ-field),

$$\sigma(X) \subseteq X^{-1}\mathscr{B}_{\mathbb{R}} = X^{-1}\sigma(\{\{x > a\} : a \in \mathbb{Q}\}).$$

Conversely, suppose

$$X : (\Omega, \sigma(X), \mathbb{P}) \to (\mathbb{R}, \mathscr{B}_{\mathbb{R}})$$

is measurable. Then $X^{-1}(B) \in \sigma(X)$ for every $B \in \mathscr{B}_{\mathbb{R}}$. Hence,

$$X^{-1}(\mathscr{B}_{\mathbb{R}}) \subseteq \sigma(X).$$

Therefore,

$$\sigma(X) = X^{-1}\mathscr{B}_{\mathbb{R}}.$$

Example 3.1.4.

Suppose $A \in \mathscr{F}$ and $X = \mathbb{1}_A$. Then

$$\sigma(X) = \{\varnothing, \Omega, A, A^c\}.$$

Definition 3.1.5 (Independent random variables).

We say that $X_1, \ldots, X_n : (\Omega, \mathscr{F}, \mathbb{P}) \to (\mathbb{R}, \mathscr{B}_{\mathbb{R}})$ are independent if $\sigma(X_1), \ldots, \sigma(X_n)$ are independent.

Why does this definition make sense? For $X \perp\!\!\!\perp Y$ and for any $A, B \in \mathscr{B}_{\mathbb{R}}$, we want

$$\mathbb{P}[X \in A, Y \in B] = \mathbb{P}[X \in A]\,\mathbb{P}[Y \in B].$$

We see that

$$\begin{aligned} \mathbb{P}[X \in A] &= \mathbb{P}\left[X^{-1}A\right], \\ \mathbb{P}[Y \in B] &= \mathbb{P}\left[Y^{-1}B\right], \\ \mathbb{P}[X \in A, Y \in B] &= \mathbb{P}\left[X^{-1}A \cap Y^{-1}B\right] \end{aligned}.$$

We need $\mathbb{P}$ to be defined on and satisfy the independence condition on sets of this last type.

Lemma 3.1.6.

If $X \perp\!\!\!\perp Y$ and $\mathbb{E}|X|, \mathbb{E}|Y| < \infty$, then

$$\mathbb{E}[XY] = \mathbb{E}[X]\,\mathbb{E}[Y].$$

Proof of Lemma 3.1.6. (i) If $X = \mathbb{1}_A, Y = \mathbb{1}_B$, then $XY = \mathbb{1}_{A\cap B}$. Hence

$$\begin{aligned} \mathbb{E}[XY] &= \mathbb{E}[\mathbb{1}_{A\cap B}] \\ &= \mathbb{P}[A \cap B] \\ &= \mathbb{P}[A]\,\mathbb{P}[B] \text{ by independence} \\ &= \mathbb{E}[X]\,\mathbb{E}[Y]. \end{aligned}$$

(ii) For $\{A_i\}, \{B_j\}$ partitions of Ω, if $X = \sum_i a_i \mathbb{1}_{A_i}$ and $Y = \sum_j b_j \mathbb{1}_{B_j}$, then we are tempted to write

$$\begin{aligned} XY &= \sum_{i=1}^{n}\sum_{j=1}^{m} a_i b_j \mathbb{1}_{A_i \cap B_j} \\ \mathbb{E}[XY] &= \sum_{i=1}^{n}\sum_{j=1}^{m} a_i b_j \mathbb{P}[A_i \cap B_j] \\ &= \sum_{i=1}^{n}\sum_{j=1}^{m} a_i b_j \mathbb{P}[A_i]\,\mathbb{P}[B_j] \end{aligned}$$

and continue, but we don't know A_i and B_j are independent, since this requires $A_i \in \sigma(X)$ and $B_j \in \sigma(Y)$. This is not guaranteed if the a_i, b_j are not unique. (For example, if $X(\omega) = 1$ for $\omega \in A_1$ and $\omega \in A_2$, then $A_1 \notin \sigma(X)$, since A_1 being in $\sigma(X)$ is not necessary for X to be measurable — only $A_1 \cup A_2 \in \sigma(X)$ is required.

Instead, we should decompose as

$$X = \sum_{i=1}^{n} a_i \mathbb{1}_{X=a_i}, \quad Y = \sum_{j=1}^{m} b_j \mathbb{1}_{Y=b_j}.$$

Then we can actually continue to get

$$\begin{aligned} \mathbb{E}[XY] &= \sum_{i=1}^{n}\sum_{j=1}^{m} a_i b_j \mathbb{P}[X = a_i, Y = b_j] \\ &= \sum_{i=1}^{n}\sum_{j=1}^{m} a_i b_j \mathbb{P}[X = a_i]\,\mathbb{P}[Y = b_j] \\ &= \left(\sum_{i=1}^{n} a_i \mathbb{P}[X = a_i]\right)\left(\sum_{j=1}^{m} b_j \mathbb{P}[Y = b_j]\right) \\ &= \mathbb{E}[X]\,\mathbb{E}[Y]. \end{aligned}$$

(iii) Now consider $X, Y \geq 0$. Write

$$\begin{aligned} X_n &= \sum_{k=0}^{n2^n-1} \frac{k}{2^n} \mathbb{1}_{\frac{k}{2^n} \leq X < \frac{k+1}{2^n}} + n\mathbb{1}_{X \geq n} \\ Y_n &= \sum_{k=0}^{n2^n-1} \frac{k}{2^n} \mathbb{1}_{\frac{k}{2^n} \leq Y < \frac{k+1}{2^n}} + n\mathbb{1}_{Y \geq n}. \end{aligned}$$

Then $X_n \uparrow X$, $Y_n \uparrow Y$. Since $\sigma(X_n) \subseteq \sigma(X)$[1] and $\sigma(Y_n) \subseteq \sigma(Y)$ imply that $X_n \perp\!\!\!\perp Y_n$,

$$\mathbb{E}[X_n Y_n] = \mathbb{E}[X_n]\,\mathbb{E}[Y_n].$$

Taking $n \to \infty$ concludes.

(iv) For general X, Y, put $X = X^+ - X^-$ and $Y = Y^+ - Y^-$, so

$$XY = X^+Y^+ + X^-Y^- - X^+Y^- - X^-Y^+.$$

[1] Consider $X = 3\mathbb{1}_{A_1} + 4\mathbb{1}_{A_2} - \mathbb{1}_{A_3}$ for disjoint A_1, A_2, A_3, and find $\sigma(X)$. This provides intuition for what $\sigma(X)$ generally is.

Since $\sigma(X^+) \subseteq \sigma(X)$ and so on, we see that each term in the sum satisfies independence; we can take the expectation termwise. Using linearity of the integral to recombine X^+ with X^- and similarly for Y, we get independence of X and Y.

$\square$

Definition 3.1.7 (Identically distributed random variables).

We say that $X_1, \ldots, X_n$ are identically distributed if

$$\mathbb{P}[X_j \in A] = \mathbb{P}[X_i \in A]$$

for all i, j and $A \in \mathscr{F}$.

Section 4

Marcus Kuo

Law of Large Numbers

4.1 Preliminaries

Lemma 4.1.1 (Borel-Cantelli).

If $\sum_n \mathbb{P}[A_n] < \infty$, then

$$\mathbb{P}[A_n \text{ infinitely often}] = \mathbb{P}\left[\bigcap_{m=1}^{\infty} \bigcup_{n=m}^{\infty} A_n\right] = \mathbb{P}\left[\limsup_{n\to\infty} A_n\right] = 0.$$

Proof of Lemma 4.1.1. By finiteness of the entire sum, for any $\epsilon > 0$, there exists $m \in \mathbb{N}$ such that

$$\sum_{n=m}^{\infty} \mathbb{P}[A_n] < \epsilon.$$

Write

$$\begin{aligned}\mathbb{P}[A_n \text{ i.o.}] &= \mathbb{P}\left[\bigcap_{m=1}^{\infty} \bigcup_{n=m}^{\infty} A_n\right] \\ &\leq \mathbb{P}\left[\bigcup_{n=m}^{\infty} A_n\right] \\ &\leq \sum_{n=m}^{\infty} \mathbb{P}[A_n] < \epsilon.\end{aligned}$$

Sending $\epsilon \to 0$ yields the result. □

We need the following lemma to bound the second moments in one formulation of the LLN.

Lemma 4.1.2 (Existence of lower moments).

Suppose $\mathbb{E}[|X|^r] < \infty$. Then $\mathbb{E}[|X|^s] < \infty$ for all $s < r$.

Proof of Lemma 4.1.2. For all $t \geq 0$,

$$t^s \leq \max\{t^r, 1\} \leq t^r + 1.$$

So

$$\mathbb{E}[|X|^s] \leq \mathbb{E}[|X|^r] + 1 < \infty.$$

□

We also need the following lemma.

Lemma 4.1.3 (Maximal inequality).

Suppose $Z_1, \ldots, Z_N$ are independent. Then

$$\mathbb{P}\left[\max_{i \le N} |Z_1 + \cdots + Z_i| > \epsilon_1 + \epsilon_2\right] \le \beta \mathbb{P}\left[|Z_1 + \cdots + Z_N| > \epsilon_1\right],$$

where $\beta = \frac{1}{\min_{i \le N} \mathbb{P}[|Z_{i+1} + \cdots + Z_N| \le \epsilon_2]}$

Proof of Lemma 4.1.3. Define $S_i = Z_1 + \cdots + Z_i$ and $T_i = Z_{i+1} + \cdots + Z_N$. We see that

$$\mathbb{P}\left[\max_{i \le N} |S_i| > t\right] \le \sum_{i \le N} \mathbb{P}\left[|S_i| > t\right].$$

This is not helpful, because applying union bound is best when the terms are independent — if they are highly dependent, then union bound throws out a lot of information (think of $A_1 = A_2 = A_3 = \cdots$.)

Define the stopping time

$$\tau = \begin{cases} \text{first time } |S_i| > \epsilon_1 + \epsilon_2 \\ N \text{ if } \max_{i \le N} |S_i| \le \epsilon_1 + \epsilon_2 \end{cases}.$$

Then we can write [1]

$$\begin{aligned}
\mathbb{P}\left[\max_{i \le N} |S_i| > \epsilon_1 + \epsilon_2\right] &= \mathbb{P}\left[|S_\tau| > \epsilon_1 + \epsilon_2\right] \\
&= \sum_{i=1}^{N} \mathbb{P}\left[|S_\tau| > \epsilon_1 + \epsilon_2, \tau = i\right] \\
&= \sum_{i=1}^{N} \mathbb{P}\left[|S_i| > \epsilon_1 + \epsilon_2, \tau = i\right] \\
&= \sum_{i=1}^{N} \frac{\mathbb{P}\left[|S_i| > \epsilon_1 + \epsilon_2, \tau = i, |T_i| \le \epsilon_2\right]}{\mathbb{P}\left[|T_i| \le \epsilon_2\right]}.
\end{aligned}$$

This last equality is allowed because $\{|S_i| > \epsilon_1 + \epsilon_2, \tau = i\}$ is independent from $|T_i| \le \epsilon_2$.[2]

Continuing,

$$\begin{aligned}
\mathbb{P}\left[\max_{i \le N} |S_i| > \epsilon_1 + \epsilon_2\right] &\le \frac{1}{\min_i \mathbb{P}\left[|T_i| \le \epsilon_2\right]} \sum_{i=1}^{N} \mathbb{P}\left[|S_i| > \epsilon_1 + \epsilon_2, \tau = i, |T_i| \le \epsilon_2\right] \\
&\le \frac{1}{\min_i \mathbb{P}\left[|T_i| \le \epsilon_2\right]} \sum_{i=1}^{N} \mathbb{P}\left[|S_N| > \epsilon_1, \tau = i\right] \\
&= \frac{1}{\min_i \mathbb{P}\left[|T_i| \le \epsilon_2\right]} \mathbb{P}\left[|S_N| > \epsilon_1\right].
\end{aligned}$$

□

[1] Note that τ is random and S_i is random, so S_τ is very random! That is why we decouple in the second equality.

[2] Note the nuance with the independence of $\tau = i$ and $|T_i| \le \epsilon_2$. Also, we know that these events are independent because they are measurable functions of independent random variables (see the Appendix).

4.2 Strong and Stronger Laws

Theorem 4.2.1 (Strong law with uniformly bounded second moments).

Suppose $\{X_i\}$ is a sequence of independent random variables with $\mathbb{E}[X_i] = 0$ for each i and

$$\sup_i \mathbb{E}\left[X_i^2\right] < \infty.$$

Then

$$\frac{X_1 + \cdots + X_n}{n} \to 0$$

almost surely.

Proof of Theorem 4.2.1. First, consider the stronger condition

$$\sup_i \mathbb{E}\left[X_i^4\right] < \infty.$$

Write $S_n = \sum_{i=1}^n X_i$. It suffices to show that

$$\sum_n \mathbb{P}\left[\frac{|S_n|}{n} > \epsilon\right] < \infty$$

since by Borel-Cantelli we would have

$$\mathbb{P}\left[\frac{|S_n|}{n} > \epsilon \text{ i.o.}\right] = 0,$$

which implies

$$\mathbb{P}\left[\limsup_{n\to\infty} \frac{|S_n|}{n} \leq \epsilon\right] = 1,$$

from which we can send ϵ to 0.

[3]By the previous lemma, $\sup_i \mathbb{E}\left[X_i^2\right] < \infty$ as well. We have by Chebyshev's inequality that

$$\mathbb{P}\left[\frac{|S_n|}{n} > \epsilon\right] \leq \frac{\mathbb{E}\left[S_n^2\right]}{n^2\epsilon^2}.$$

Writing

$$\begin{aligned}
\mathbb{E}\left[S_n^2\right] &= \mathbb{E}\left[(X_1 + \ldots + X_n)(X_1 + \ldots + X_n)\right] \\
&= \mathbb{E}\left[X_1^2 + X_2^2 + \ldots + X_n^2 + \text{ terms in } X_iX_j\right] \\
&= \sum_{i=1}^n \mathbb{E}\left[X_i^2\right],
\end{aligned}$$

where the last equality follows from independence of X_i and X_j for $i \neq j$. Thus,

$$\mathbb{P}\left[\frac{|S_n|}{n} > \epsilon\right] \leq \frac{\sum_{i=1}^n \mathbb{E}\left[X_i^2\right]}{n^2\epsilon^2} \leq \frac{\sup_i \mathbb{E}\left[X_i^2\right]}{\epsilon^2} \cdot \frac{1}{n}.$$

[3]This part of the proof provides a reason for why the easiest case is with a uniform bound on the fourth moments, rather than on the second moments.

Summing across n, we do *not* get convergence. If we try to use the third power in Chebyshev's inequality, we cannot decompose $\mathbb{E}\left[|X_i|^3\right]$.

We compute

$$\begin{aligned} S_n^4 = X_1^4 + X_2^4 + \ldots + X_n^4 \\ &+ \text{terms in } X_i^3 X_j,\ i \neq j \\ &+ \underbrace{\text{terms of the form } X_i^2 X_j^2}_{\binom{n}{2}\binom{4}{2} \text{ terms}}. \end{aligned}$$

By assumption, there exists $M > 0$ such that $\mathbb{E}\left[X_i^4\right] < M$ for every i. We see by Cauchy–Schwarz that[4]

$$\mathbb{E}\left[X_1^2 X_2^2\right] \leq \sqrt{\mathbb{E}\left[X_1^4\right]}\sqrt{\mathbb{E}\left[X_2^4\right]} \leq M.$$

Note that the "odd" terms vanish from independence giving $\mathbb{E}\left[X_i^3 X_j\right] = \mathbb{E}\left[X_i^3\right] \cdot 0 = 0$. Computing $\binom{n}{2} = \frac{n!}{2!(n-2)!} = \frac{n(n-1)}{2}$, we have the bound

$$\begin{aligned} \mathbb{E}\left[S_n^4\right] &\leq nM + \binom{4}{2}\binom{n}{2}M \\ &\leq 100000 n^2 M. \end{aligned}$$

By Chebyshev's inequality,

$$\mathbb{P}\left[|S_n| > n\epsilon\right] \leq \frac{\mathbb{E}\left[S_n^4\right]}{n^4\epsilon^4} \leq \frac{100000M}{\epsilon^4 n^2}.$$

Summing across n,

$$\sum_{n=1}^{\infty} \mathbb{P}\left[|S_n| > n\epsilon\right] \leq \frac{100000M}{\epsilon^4} \sum_{n=1}^{\infty} \frac{1}{n^2} < \infty.$$

We have our result to which we can apply Borel-Cantelli.

Now suppose the weaker condition $\sup_i \mathbb{E}\left[X_i^2\right] = \sigma^2 < \infty$. Define blocks $B_1, B_2, \ldots$ such that

$$B_k = \{i \in \mathbb{N} : n_{k-1} < i \leq n_k\}$$

for $n_k = 2^k$ (this particular choice of n_k is not important to the proof).

We want to show that

$$\max_{n \in B_k} \frac{|S_n|}{n} \xrightarrow{k \to \infty} 0$$

almost surely. We can get this via Borel-Cantelli by showing for any $\epsilon > 0$,

$$\sum_{k=1}^{\infty} \mathbb{P}\left[\max_{n \in B_k} \frac{|S_n|}{n} > \epsilon\right] < \infty.$$

[4]Here, the inner product is defined

$$\langle X_i, X_j \rangle = \mathbb{E}\left[X_i X_j\right].$$

Cauchy–Schwarz gives

$$\mathbb{E}\left[X_i X_j\right] \leq \sqrt{\mathbb{E}\left[X_i^2\right]}\sqrt{\mathbb{E}\left[X_j^2\right]}.$$

We know thanks to our foresight in Lemma (**??**) that

$$\begin{aligned}
\sum_{k=1}^{\infty} \mathbb{P}\left[\max_{n\in B_k} \frac{|S_n|}{n} > \epsilon\right] &\leq \sum_{k=1}^{\infty} \mathbb{P}\left[\frac{1}{n_{k-1}} \max_{n\in B_k} |S_n| > \epsilon\right] \\
&\leq \sum_{k=1}^{\infty} \mathbb{P}\left[\max_{n\leq n_k} |S_n| > n_{k-1}\epsilon\right] \\
&\leq \sum_{k=1}^{\infty} \beta_k \mathbb{P}\left[|S_{n_k}| > \frac{n_{k-1}\epsilon}{2}\right], \\
\beta_k &= \frac{1}{\min_{n\leq n_k} \mathbb{P}\left[|S_{n_k} - S_n| \leq \frac{n_{k-1}\epsilon}{2}\right]}.
\end{aligned}$$

Analyzing each term of the product $\beta_k \mathbb{P}\left[|S_{n_k}| > \frac{n_{k-1}\epsilon}{2}\right]$ separately,

$$\begin{aligned}
\frac{1}{\beta_k} &= \min_{n\leq n_k} \mathbb{P}\left[|S_{n_k} - S_n| \leq \frac{n_{k-1}\epsilon}{2}\right] \\
&= \min_{n\leq k} \left(1 - \mathbb{P}\left[|S_{n_k} - S_n| > \frac{n_{k-1}\epsilon}{2}\right]\right) \\
&= 1 - \max_{n\leq n_k} \mathbb{P}\left[|S_{n_k} - S_n| > \frac{n_{k-1}\epsilon}{2}\right] \\
&\geq 1 - \frac{4\mathbb{E}\left[|S_{n_k} - S_n|^2\right]}{n_{k-1}^2\epsilon^2} \\
&\geq 1 - \max_{n\leq n_k} \frac{4(n_k - n)\sigma^2}{n_{k-1}^2\epsilon^2} \\
&\geq 1 - \frac{4n_k\sigma^2}{n_{k-1}^2\epsilon^2} \\
&\geq \frac{1}{2} \text{ for } k \geq K
\end{aligned}$$

and

$$\mathbb{P}\left[|S_{n_k}| > \frac{n_{k-1}\epsilon}{2}\right] \leq \frac{4\mathbb{E}\left[S_{n_k}^2\right]}{n_{k-1}^2\epsilon^2} \leq \frac{4n_k\sigma^2}{n_{k-1}^2\epsilon^2}.$$

Hence,

$$\mathbb{P}\left[\max_{n\leq n_k} |S_n| > n_{k-1}\epsilon\right] \leq 2\frac{4n_k\sigma^2}{n_{k-1}^2\epsilon^2} = \frac{8n_k\sigma^2}{n_{k-1}^2\epsilon^2}$$

for $k \geq K$.

Since the probabilities for $k < K$ are uniformly bounded by 1,

$$\begin{aligned}
\sum_{k=1}^{\infty} \mathbb{P}\left[\max_{n\leq n_k} |S_n| > n_{k-1}\epsilon\right] &\leq K - 1 + \sum_{k\geq K} \frac{8n_k\sigma^2}{n_{k-1}^2\epsilon^2} \\
&= K - 1 + O\left(\sum_k \frac{1}{2^k}\right) < \infty.
\end{aligned}$$

□

Theorem 4.2.2 (Kolmogorov's Law of Large Numbers).

Suppose $\{X_i\}$ is a sequence of independent random variables, $\mathbb{E}[X_i] = 0$ for each X_i, and $\sum_i \frac{\mathbb{E}[X_i^2]}{i^2} < \infty$. Then $\frac{X_1+\cdots+X_n}{n} \to 0$ almost surely.

Proof of Theorem 4.2.2. Left to homework. □

Theorem 4.2.3 (Strong Law of Large Numbers).

Suppose $\{X_i\}$ is a sequence of independent and identically distributed random variables with $\mathbb{E}[|X_1|] < \infty$. Then

$$\frac{X_1 + \cdots + X_n}{n} \to \mathbb{E}[X_1]$$

almost surely.

Proof of Theorem 4.2.3. By replacing X_i with $X_i - \mathbb{E}[X_1]$, it suffices to prove the centered case, so assume $\mathbb{E}[X_i] = 0$.

We proceed by truncation, splitting the sum $\frac{1}{n}\sum_{i=1}^n X_n$ into the difference of bad and good behavior, the difference of good behavior and its expectation, and the expectation of good behavior. Define

$$Y_i = X_i \mathbb{1}_{|X_i| \le i}.$$

Then

$$\begin{aligned}
\mu_i := \mathbb{E}[Y_i] &= \mathbb{E}\left[X_i \mathbb{1}_{|X_i|\le i}\right] \\
&= \mathbb{E}\left[X \mathbb{1}_{|X|\le i}\right] \text{ by identical distribution} \\
&= -\mathbb{E}\left[X \mathbb{1}_{|X|> i}\right] \text{ by } \mathbb{E}[X_i] = 0.
\end{aligned}$$

Decompose

$$\frac{1}{n}\sum_{i=1}^n X_i = \underbrace{\frac{1}{n}\sum_{i=1}^n (X_i - Y_i)}_{(I)} + \underbrace{\frac{1}{n}\sum_{i=1}^n (Y_i - \mu_i)}_{(II)} + \underbrace{\frac{1}{n}\sum_{i=1}^n \mu_i}_{(III)}.$$

Starting with the last term,

$$\begin{aligned}
|(III)| = \left|\frac{1}{n}\sum_{i=1}^n \mu_i\right| &= \left|\frac{1}{n}\sum_{i=1}^n \mathbb{E}\left[X \mathbb{1}_{|X|>i}\right]\right| \\
&\le \frac{1}{n}\sum_{i=1}^n \mathbb{E}\left[|X| \mathbb{1}_{|X|>i}\right] \\
&= \mathbb{E}\left[|X| \frac{1}{n}\sum_{i=1}^n \mathbb{1}_{|X|>i}\right].
\end{aligned}$$

The indicator sum is a counting function:

$$\begin{aligned}
\sum_{i=1}^{n} \mathbb{1}_{|X|>i} &\leq \sum_{i=1}^{\infty} \mathbb{1}_{|X|>i} \\
&= \sum_{i=1}^{\infty} \sum_{j=i}^{\infty} \mathbb{1}_{j<|X|\leq j+1} \\
&= \sum_{j=1}^{\infty} \sum_{i=1}^{j} \mathbb{1}_{j<|X|\leq j+1} \\
&= \sum_{j=1}^{\infty} j \mathbb{1}_{j<|X|\leq j+1} \\
&\leq |X| \sum_{j=1}^{\infty} \mathbb{1}_{j<|X|\leq j+1} = |X|,
\end{aligned}$$

Naively, we may try to plug this bound on the count to the bound on $|(III)|$ and get

$$|(III)| \leq \mathbb{E}\left[|X| \frac{1}{n} |X|\right],$$

which may be infinite because we don't have control over the second moment. However, noting that $\frac{1}{n} \sum_{j=1}^{n} \mathbb{1}_{|X|>i} \leq 1$, we get

$$|(III)| \leq \mathbb{E}\left[|X| \min\{\frac{1}{n} |X|, 1\}\right].$$

Because $|X|$ dominates $|X| \min\{\frac{1}{n} |X|, 1\}$ and $\mathbb{E}[|X|] < \infty$, we apply the DCT to see

$$\begin{aligned}
\lim_{n\to\infty} |(III)| &\leq \lim_{n\to\infty} \mathbb{E}\left[|X| \min\left\{\frac{1}{n} |X|, 1\right\}\right] \\
&= \mathbb{E}\left[\lim_{n\to\infty} |X| \min\left\{\frac{1}{n} |X|, 1\right\}\right]. \\
&\xrightarrow{n\to\infty} 0
\end{aligned}$$

For (I), consider with Borel-Cantelli in mind that

$$\begin{aligned}
\sum_{i=1}^{\infty} \mathbb{P}[X_i \neq Y_i] &\leq \sum_{i=1}^{\infty} \mathbb{P}[|X| > i] \\
&= \mathbb{E}\left[\sum_{i=1}^{\infty} \mathbb{1}_{|X|>i}\right] \\
&\leq \mathbb{E}[|X|] < \infty,
\end{aligned}$$

so

$$\mathbb{P}[X_i \neq Y_i \text{ i.o.}] = 0.$$

Then there exists $\Omega_0 \subseteq \Omega$ such that $\mathbb{P}[\Omega_0] = 1$ and for every $\omega \in \Omega_0$, there exists $i(\omega)$ such that

$$X_i(\omega) = Y_i(\omega)$$

for all $i > i(\omega)$. For each $\omega \in \Omega_0$,

$$(I) = \frac{1}{n}\sum_{i=1}^{n}(X_i(\omega) - Y_i(\omega)) = \frac{1}{n}\sum_{i=1}^{i(\omega)}(X_i(\omega) - Y_i(\omega)) \to 0,$$

where the convergence follows from the sum being fixed.

Since $\mathbb{P}[\Omega_0] = 1$,

$$(I) = \frac{1}{n}\sum_{i=1}^{n}(X_i - Y_i) \to 0$$

almost surely.

Examining (II), we want to use Kolmogorov's Law of Large Numbers. Note that $Y_i^2 = (Y_i - \mu_i + \mu_i)^2 = (Y_i - \mu_i)^2 + \mu_i^2 + 2($
The third term vanishes in expectation. This gives us the Kolmogorov condition: [5]

$$\begin{aligned}
\sum_{i=1}^{\infty}\frac{\mathbb{E}\left[(Y_i-\mu_i)^2\right]}{i^2} &= \sum_{k=1}^{\infty}\frac{\mathbb{E}\left[Y_i^2\right]}{i^2} - \frac{\mathbb{E}\left[\mu_i^2\right]}{i^2} \\
&\le \sum_{i=1}^{\infty}\frac{\mathbb{E}\left[Y_i^2\right]}{i^2} \\
&= \sum_{i=1}^{\infty}\frac{1}{i^2}\mathbb{E}\left[X^2\mathbb{1}_{|X|\le i}\right] \\
&= \sum_{i=1}^{\infty}\frac{1}{i^2}\mathbb{E}\left[X^2\sum_{j=1}^{i}\mathbb{1}_{j-1<|X|\le j}\right] \\
&= \sum_{j=1}^{\infty}\sum_{i=j}^{\infty}\frac{1}{i^2}\mathbb{E}\left[X^2\mathbb{1}_{j-1<|X|\le j}\right] \\
&\le C\sum_{j=1}^{\infty}\frac{1}{j}\mathbb{E}\left[X^2\mathbb{1}_{j-1<|X|\le j}\right] \\
&\le C\sum_{j=1}^{\infty}\mathbb{E}\left[|X|\,\mathbb{1}_{j-1<|X|\le j}\right] \\
&\le C\mathbb{E}\left[|X|\right] < \infty,
\end{aligned}$$

so we can apply Kolmogorov's LLN to get

$$\frac{1}{n}\sum_{i=1}^{n}(Y_i - \mu_i) \to 0$$

almost surely.

All three terms converge to 0 almost surely, so their sum does as well. □

4.3 Proof Techniques

When proving almost sure convergence for averages of partial sums, it helps to decide early which obstruction is preventing a direct Borel–Cantelli argument.

[5] The second inequality holds since $\sum_{i=j}^{\infty}\frac{1}{i^2} \le C\frac{1}{j}$ for some absolute constant C.

1. **Can I sum the tail probabilities directly?** If you can show

$$\sum_{n=1}^{\infty} \mathbb{P}\left[|S_n| > \epsilon a_n\right] < \infty$$

for every $\epsilon > 0$, then Borel–Cantelli already gives $\frac{S_n}{a_n} \to 0$ almost surely.

2. **Do I only have good estimates at selected times?** If the direct sum diverges but a sparse subsequence is manageable, choose block endpoints n_k and try to prove

$$\sum_{k=1}^{\infty} \mathbb{P}\left[|S_{n_k}| > \epsilon a_{n_k}\right] < \infty.$$

Geometric choices such as $n_k = 2^k$ are common because they often turn borderline series into summable ones.

3. **Do I need control between block endpoints?** If estimates at the endpoints are not enough, pass to block maxima. For

$$B_k = \{n_{k-1} < n \leq n_k\},$$

one tries to bound

$$\mathbb{P}\left[\max_{n \in B_k} \frac{|S_n|}{a_n} > \epsilon\right].$$

When $a_n = n$, blocking is especially natural because

$$\max_{n \in B_k} \frac{|S_n|}{n} \leq \frac{1}{n_{k-1}} \max_{n \in B_k} |S_n|.$$

This is the standard signal that a maximal inequality should be used.

4. **Are large jumps ruining the moment estimates?** If the original variables do not have enough finite moments, truncate:

$$Y_i = X_i \mathbb{1}_{|X_i| \leq b_i}.$$

Then split the average into three parts:

$$\frac{1}{n}\sum_{i=1}^{n} X_i = \frac{1}{n}\sum_{i=1}^{n}(X_i - Y_i) + \frac{1}{n}\sum_{i=1}^{n}(Y_i - \mathbb{E}[Y_i]) + \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[Y_i].$$

The first term handles rare large jumps, the second is the centered "good" part, and the third is the bias introduced by truncation.

5. **Does the centered truncated part fit Kolmogorov's LLN?** After truncation, set

$$Z_i = Y_i - \mathbb{E}[Y_i].$$

If you can verify

$$\sum_{i=1}^{\infty} \frac{\mathbb{E}\left[Z_i^2\right]}{i^2} < \infty,$$

then Kolmogorov's LLN applies to $\frac{1}{n}\sum_{i=1}^{n} Z_i$. This is often the reason for choosing the truncation threshold b_i to depend on i.

6. **When should I expect blocking and maximal inequalities together?** If the theorem concerns all n,

but your estimates naturally control either S_N or a sparse subsequence, then the usual upgrade is:

$$\text{endpoint estimate} \implies \text{block maximum} \implies \text{maximal inequality} \implies \text{all } n.$$

As a quick rule of thumb: use truncation when moments are too weak, use Kolmogorov's LLN for the centered truncated variables, and use blocking plus a maximal inequality when you need to pass from control at selected times to control for every n.

Section 5

Marcus Kuo

Central Limit Theorem

5.1 Weak Convergence

We will eventually prove the following.

Theorem 5.1.1 (Central Limit Theorem).

Let $\{X_i\}$ be iid with $\mathbb{E}[X_i] = 0$ and $\mathrm{Var}(X_i) = 1$. Then

$$\sqrt{n}\frac{X_1 + \cdots + X_n}{n} \xrightarrow{d} N(0,1).$$

But what does $\xrightarrow{d}$ mean? It means $\rightsquigarrow$.
Consider a metric space $(\mathscr{X}, d)$ and a random variable

$$X : (\Omega, \mathscr{F}, \mathbb{P}) \to (\mathscr{X}, \mathscr{B}(\mathscr{X})).$$

For every $B \in \mathscr{B}(\mathscr{X})$, define

$$P(B) = \mathbb{P}\left[X^{-1}(B)\right] = \mathbb{P}[X \in B].$$

We showed on the midterm that P is a probability measure on $(\mathscr{X}, \mathscr{B}(\mathscr{X}))$. For $f : (\mathscr{X}, \mathscr{B}(\mathscr{X})) \to (\mathbb{R}, \mathscr{B}_{\mathbb{R}})$, we equivalently write all of the following

$$Pf = \int f dP = \int_{\mathscr{X}} f(x) dP(x) = \int_{\Omega} f(X(\omega)) d\mathbb{P}(\omega) = \int f(X) d\mathbb{P} = \mathbb{P}f = \mathbb{E}[f(X)].$$

We say $X \sim P$, where P is the distribution of X.

Definition 5.1.2 (Bounded Lipschitz function).

Define the class of bounded Lipschitz functions on $\mathscr{X}$ as

$$\mathrm{BL}(\mathscr{X}) = \left\{ f : \mathscr{X} \to \mathbb{R} : \sup_x |f(x)| < \infty, \sup_{x \neq y} \frac{|f(x) - f(y)|}{d(x,y)} < \infty \right\}.$$

Definition 5.1.3 (Weak convergence).

Suppose P_n, P are probability measures on $(\mathscr{X}, \mathscr{B}(\mathscr{X}))$. We say $P_n \to P$ weakly if $P_n l \to Pl$ for all $l \in \mathrm{BL}(\mathscr{X})$.

[1]

[1] Equivalently, $\mathbb{E}[l(X_n)] \to \mathbb{E}[l(X)]$ for all $X_n \sim P_n$ and $X \sim P$.

Note that $X_1, X_2, \ldots$ can all live on different probability spaces, but they all must map to the same measure space. Each X_n then induces its own probability measure on the common measure space. Essentially,

$$\begin{aligned} X_1 : (\Omega_1, \mathscr{F}_1, \mathbb{P}_1) &\xrightarrow{X_1} (\mathscr{X}, \mathscr{B}(\mathscr{X})), \quad X_1 \sim P_1 = \mathbb{P} \circ X_1^{-1} \\ X_2 : (\Omega_2, \mathscr{F}_2, \mathbb{P}_2) &\xrightarrow{X_2} (\mathscr{X}, \mathscr{B}(\mathscr{X})), \quad X_2 \sim P_2 = \mathbb{P} \circ X_2^{-1} \\ &\vdots \end{aligned}$$

Definition 5.1.4 (Lower semi-continuity, upper semi-continuity).

We say that $g : \mathscr{X} \to \mathbb{R}$ is lower semi-continuous if

$$\{x \in \mathscr{X} : g(x) > t\} = \{g > t\}$$

is open in the topology of $\mathscr{X}$ for every $t \in \mathbb{R}$.
$f : \mathscr{X} \to \mathbb{R}$ is upper semi-continuous if $-f$ is lower semi-continuous (equivalently, $\{f < t\}$ is open for every $t \in \mathbb{R}$).

Example 5.1.5.

$g = \mathbb{1}_{(1,2)}$ is lower semi-continuous, but $h = \mathbb{1}_{[1,2]}$ is not. All continuous g are lower semi-continuous. Any indicator function of an open set is LSC.

Lemma 5.1.6.

Suppose g is lower semi-continuous and $g \geq -M$. Then there exist $l_n \in \mathrm{BL}(\mathscr{X})$ such that $l_n \uparrow g$.

Proof of Lemma 5.1.6. WLOG, assume $g \geq 0$. Define $h_t = t\mathbb{1}_{g>t}$. Then for each $x \in \mathscr{X}$, $\sup_t h_t(x) = g(x)$, as

$$\sup_t h_t(x) = \sup_t t\mathbb{1}_{g(x)>t} = g(x).$$

These step functions are not Lipschitz.

Define $l_{k,t}(x) = t \wedge (k \cdot d(x, \{g \leq t\}))$ with the metric between a point x and set B defined as

$$d(x, B) = \inf_{y \in B} d(x, y)$$

using the metric d on $\mathscr{X}$. We see that $l_{k,t} \in \mathrm{BL}(\mathscr{X})$ and

$$\begin{aligned} \sup_k l_{k,t}(x) &= t\mathbb{1}_{d(x,\{g\leq t\})>0} \\ &= t\mathbb{1}_{x \notin \{g \leq t\}} \\ &= t\mathbb{1}_{g(x)>t} = h_t(x). \end{aligned}$$

Combining our two results,

$$\sup_{k,t} l_{k,t} = g.$$

We have approximated g with the supremum of bounded Lipschitz functions, but not rigorously.

Because $\mathbb{Q}_+$ is countable and dense in $[0, \infty)$, the countable family $\{l_{k,t} : k \in \mathbb{N},\ t \in \mathbb{Q}_+\}$ still has supremum g. Enumerate it as $f_1, f_2, \ldots$, and replace f_n by $\max_{1 \leq i \leq n} f_i$; finite maxima preserve the bounded Lipschitz property, so we obtain $l_n \uparrow g$. $\square$

Lemma 5.1.7.

Suppose $P_n \to P$. Then

(i) For any lower semi-continuous g that is bounded below,

$$\liminf_{n\to\infty} P_n g \geq Pg.$$

(ii) For any upper semi-continuous f that is bounded above,

$$\limsup_{n\to\infty} P_n f \leq Pf.$$

Proof of Lemma 5.1.7. Note that it suffices to prove (i), since (ii) follows from the definition of upper semi-continuity. We know by our approximation lemma that there exists $\{l_i\} \subseteq \mathrm{BL}(\mathscr{X})$ such that $l_i \uparrow g$. For every $i \in \mathbb{N}$,

$$\begin{aligned}\liminf_{n\to\infty} P_n g &\geq \liminf_{n\to\infty} P_n l_i \\ &= P l_i\end{aligned}$$

by definition of weak convergence of $P_n \to P$. Sending $i \to \infty$, we get by the monotone convergence theorem that $Pl_i \to Pg$. □

Example 5.1.8.

While the directions of the inequalities seem counterintuitive when we look at where the inf and sup are, the direction is determined by the LSC/USC property of f and g.
We can see a simple example of the inequalities holding strictly. Let $X_n : \frac{1}{n}$ and $X := 0$ almost surely, and let P_n, P denote their distributions. Then $P_n \to P$ weakly since for every bounded continuous l,

$$\mathbb{E}\left[l(X_n)\right] = l\left(\frac{1}{n}\right) \to l(0) = \mathbb{E}\left[l(X)\right].$$

Now take

$$g = \mathbb{1}_{(0,\infty)}, \qquad f = \mathbb{1}_{\{0\}}.$$

Then g is lower semi-continuous and bounded below, while f is upper semi-continuous and bounded above. We see

$$P_n g = \mathbb{P}\left(X_n > 0\right) = 1, \qquad Pg = \mathbb{P}\left(X > 0\right) = 0,$$

so $\limsup_{n\to\infty} P_n g \leq Pg$ is false; and

$$P_n f = \mathbb{P}\left(X_n = 0\right) = 0, \qquad Pf = \mathbb{P}\left(X = 0\right) = 1,$$

so $\liminf_{n\to\infty} P_n f \geq Pf$ is false.

Corollary 5.1.9.

If $P_n \to P$, then

(i) For any open B, $\liminf_{n\to\infty} P_n(B) \geq P\left(B\right)$

(ii) For any closed G, $\limsup_{n\to\infty} P_n(G) \leq P(G)$.

Proof of Corollary 5.1.9. For (i), consider $f := \mathbb{1}_B$, which is LSC from B being open. Then

$$\liminf_{n\to\infty} P_n(B) = \liminf_{n\to\infty} P_n f \geq Pf = P(B).$$

For (ii), we can consider the lemma with $g := \mathbb{1}_G$, which is USC from G being closed. □

Notation 5.1.10 (Lower semi-continuous minorant).

We write

$$\mathring{f} := \sup\{g \leq f : g \text{ is LSC}\} = \sup_{g\in\mathscr{G}} g$$

and

$$\overline{f} = \inf\{g \geq f : g \text{ is USC}\}.$$

Note that $\mathring{f}$ itself is LSC since

$$\left\{\mathring{f} > t\right\} = \left\{\sup_{g\in\mathscr{G}} g > t\right\} = \bigcup_{g\in\mathscr{G}} \{g > t\}$$

is open. Hence $\mathring{f}$ is LSC.[2]

Similarly, $\overline{f}$ is USC. On the homework, we show that $\mathring{\mathbb{1}}_B = \mathbb{1}_{\mathring{B}}$ and also have $\overline{\mathbb{1}}_B = \mathbb{1}_{\overline{B}}$. This justifies the notation.

Theorem 5.1.11.

Suppose $|f| \leq M$. Then

$$\mathring{f}(x) = \overline{f}(x) \iff f \text{ is continuous at } x.$$

Proof of Theorem 5.1.11. Recall f is continuous at x if for every $\epsilon > 0$, there exists open $U \ni x$ such that for every $y \in U$, $|f(x) - f(y)| < \epsilon$.

($\Longrightarrow$) From $\mathring{f}(x) = f(x)$, $U = \left\{y : \mathring{f}(y) > f(x) - \epsilon\right\}$ is open. Then $\mathring{f}(x) = f(x) > f(x) - \epsilon$, so $x \in U$. For every $y \in U$, $f(y) \geq \mathring{f}(y) > f(x) - \epsilon$.

From $\overline{f}(x) = f(x)$, put

$$V = \left\{y : \overline{f}(y) < f(x) + \epsilon\right\}.$$

This set is open and contains x. For every $y \in V$, $f(y) \leq \overline{f}(y) < f(x) + \epsilon$.

Define $W = U \cap V$, which is the intersection of open sets that contain x, so $W \ni x$ is open. For every $y \in W$, $|f(x) - f(y)| < \epsilon$, so f is continuous at x.

($\Longleftarrow$) We want to find for any $\epsilon \in (0,1)$ a LSC $g \leq f$ and USC $h \geq f$ such that $h(x) - g(x) \leq 2\epsilon$, since then we can let $\epsilon \downarrow 0$ and conclude. We can expand the inequality given to us by continuity to see

$$f(x) - \epsilon \leq f(y) \leq f(x) + \epsilon$$

for all $y \in U$, where $U \ni x$ is open. We can upgrade this inequality to all of $\mathscr{X}$ from U by fixing some U then seeing

$$g := (f(x) - \epsilon)\,\mathbb{1}_U - (M+1)\mathbb{1}_{U^c} \leq f(y) \leq (f(x) + \epsilon)\,\mathbb{1}_U + (M+1)\,\mathbb{1}_{U^c} =: h.$$

[2] Since the interior of a set is the largest open subset, $\mathring{f}$ is the function analogue: the largest lower semi-continuous function not exceeding f.

Since $M \geq |f|$ and $1 \geq \epsilon$, the upper bound is upper semi-continuous and the lower bound is lower semi-continuous. Also,

$$g(x) = f(x) - \epsilon, \qquad h(x) = f(x) + \epsilon.$$

Hence

$$f(x) - \epsilon = g(x) \leq \mathring{f}(x) \leq \overline{f}(x) \leq h(x) = f(x) + \epsilon.$$

Sending $\epsilon \to 0$ gives

$$\mathring{f}(x) = \overline{f}(x).$$

□

Note that the $\implies$ direction does not require boundedness of f.

Remark 5.1.12. From this, we can write the points of $\mathscr{X}$ where f is continuous as

$$\begin{aligned} C_f &= \left\{x : \mathring{f}(x) = \overline{f}(x)\right\} \\ &= \left\{\mathring{f} = \overline{f}\right\}. \end{aligned}$$

Because $\mathring{f}$ and $\overline{f}$ are measurable (their definitions characterize the preimage of open sets), $\mathring{f} - \overline{f}$ is measurable. Hence, the preimage of $\{0\}$ under $\mathring{f} - \overline{f}$ is measurable; i.e., $C_f \in \mathscr{B}(\mathscr{X})$.

We also see that

$$P(C_f) = 1 \iff P\overline{f} = P\mathring{f} \iff P\left(\mathring{f} - \overline{f}\right) = 0.$$

Theorem 5.1.13.

Suppose $P_n \to P$ weakly. Then

(i) $P_n f \to Pf$ for every f such that $|f| \leq M$ and $P(C_f) = 1$

(ii) In particular, applying (i) to indicator functions yields that $P_n B \to PB$ for every $B \in \mathscr{B}(\mathscr{X})$ such that $P(\partial B) = 0$.

Proof of Theorem 5.1.13. Recall from Lemma (**??**) that

$$\begin{aligned} P\overline{f} &\geq \limsup_{n\to\infty} P_n f \\ &\geq \liminf_{n\to\infty} P_n f \\ &\geq P\mathring{f}. \end{aligned}$$

Now applying Theorem (**??**) yields $P\overline{f} = P\mathring{f}$, so all of the inequalities hold with equality. In particular,

$$\limsup_{n\to\infty} P_n f = \liminf_{n\to\infty} P_n f = \lim_{n\to\infty} P_n f = P\overline{f} = P\mathring{f} = Pf.$$

With $B \in \mathscr{B}(\mathscr{X})$ and $f := \mathbb{1}_B$, we see that f is discontinuous at ∂B. If $P(\partial B) = 0$, then $P(C_f) = 1$, so

$$P_n(B) = P_n \mathbb{1}_B \to P\mathbb{1}_B = P(B).$$

□

Theorem 5.1.14 (Portmanteau).

The following are equivalent:

(i) $P_n l \to Pl$ for every $l \in \mathrm{BL}(\mathscr{X})$

(ii) $\liminf_{n\to\infty} P_n g \geq Pg$ for every lower semi-continuous $g \geq -M$

(iii) $\limsup_{n\to\infty} P_n f \leq Pf$ for every upper semi-continuous $f \leq M$

(iv) $\liminf_{n\to\infty} P_n B \geq PB$ for every open B

(v) $\limsup_{n\to\infty} P_n G \leq PG$ for every closed G

(vi) $P_n f \to Pf$ for every $|f| \leq M$ such that $P(C_f) = 1$

(vii) $P_n B \to PB$ for every $B \in \mathscr{B}(\mathscr{X})$ such that $P(\partial B) = 0$.

Proof of Theorem 5.1.14. The implications proved above already showed that (i) implies (ii) and (iii), (iv) and (v), and (vi) and (vii), so we only need to show (vii) $\implies$ (i).

Proving the desired result for every bounded continuous f is more than enough, since Lipschitz continuous functions are a subclass of bounded continuous functions.

By splitting into positive and negative parts, we may assume $0 \leq f \leq 1$. Note that

$$\begin{aligned} f(x) &= \int_0^1 \mathbb{1}_{f(x)>t}\, dt \\ Pf = \int_{\mathscr{X}} f(x)\, dP(x) &= \int_{\mathscr{X}} \int_0^1 \mathbb{1}_{f(x)>t} dt dP(x) \\ &= \int_0^1 \int_{\mathscr{X}} \mathbb{1}_{f(x)>t} dP(x) dt \\ &= \int_0^1 P(f > t) dt \end{aligned}$$

by Fubini.

Similarly,

$$P_n f = \int_0^1 P_n(f > t) dt.$$

We can relate functions to sets by noting

$$\begin{aligned} \partial \{f > t\} &= \overline{\{f > t\}} \setminus \{f \overset{\circ}{>} t\} \\ &= \overline{\{f > t\}} \setminus \{f > t\} \\ &\subseteq \{f \geq t\} \setminus \{f > t\} \\ &= \{f = t\}. \end{aligned}$$

Hence, if $P(f = t) = 0$, then $P(\partial \{f > t\}) = 0$, so by (vii), we have

$$P_n(f > t) \to P(f > t).$$

Now define $T = \{t : P(f = t) > 0\}$. We claim that T is countable. To see why, write

$$\begin{aligned} T &= \{t : P(f = t) > 0\} \\ &= \bigcup_n \left\{ t : P(f = t) \geq \frac{1}{n} \right\}. \end{aligned}$$

Since the cardinality of $\left\{ t : P(f = t) \geq \frac{1}{n} \right\}$ is at most n, T is countable. Thus, the integrals of $P(f > t)$ and $P_n(f > t)$ over T are zero.

This lets us write

$$\begin{aligned} Pf &= \int_0^1 P(f > t)dt = \int_{T^c} P(f > t)dt \\ P_n f &= \int_0^1 P_n(f > t)dt = \int_{T^c} P_n(f > t)dt. \end{aligned}$$

For every $t \in T^c$, we have convergence of the integrands, and they are all bounded by 1. By the DCT,

$$P_n f \to Pf.$$

□

Definition 5.1.15 (Products).

For $(\Omega_1, \mathscr{F}_1, \mu_1)$ and $(\Omega_2, \mathscr{F}_2, \mu_2)$ where μ_1, μ_2 are finite measures, define

$$\begin{aligned} \Omega_1 \times \Omega_2 &= \{(\omega_1, \omega_2) : \omega_1 \in \Omega_1, \omega_2 \in \Omega_2\} \\ \mathscr{F}_1 \times \mathscr{F}_2 &= \sigma\left(\{\underbrace{A_1 \times A_2}_{\text{rectangle}}\} : A_1 \in \mathscr{F}_1, A_2 \in \mathscr{F}_2 \right) \end{aligned}$$

.

[3]

Theorem 5.1.16 (Fubini).

Suppose μ_1, μ_2 are finite.

(i) There exists a unique μ on $(\Omega_1 \times \Omega_2, \mathscr{F}_1 \times \mathscr{F}_2)$ such that for every $A_1 \in \mathscr{F}_1$ and $A_2 \in \mathscr{F}_2$,

$$\mu(A_1 \times A_2) = \mu_1(A_1)\mu_2(A_2).$$

(ii) Suppose $f : (\Omega_1 \times \Omega_2, \mathscr{F}_1 \times \mathscr{F}_2) \to (\mathbb{R}, \mathscr{B}_{\mathbb{R}})$ is measurable and $\int |f| \, d\mu < \infty$. Then

$$\begin{aligned} \int_{\Omega_1 \times \Omega_2} f d\mu &= \int_{\Omega_2} \int_{\Omega_1} f(\omega_1, \omega_2) d\mu_1(\omega_1) d\mu_2(\omega_2) \\ &= \int_{\Omega_2} \int_{\Omega_1} f(\omega_1, \omega_2) d\mu_2(\omega_2) d\mu_1(\omega_1). \end{aligned}$$

The Portmanteau theorem (**??**) above gave us several equivalent characteriztions of weak convergence of measures defined on an arbitrary measurable space $(\mathscr{X}, \mathscr{B}_{\mathscr{X}})$. We have the following equivalent characterizations

[3] Note that the product measure is first defined on the field of rectangles, then extended to the generated σ-field.

when the measurable space is $(\mathbb{R}, \mathscr{B}_{\mathbb{R}})$, the second of which is the usual notion of convergence in distribution.

Theorem 5.1.17 (Weak Convergence on $\mathbb{R}$).

The following are equivalent:

(i) $P_n l \to Pl$ for every $l \in \mathrm{BL}(\mathbb{R})$

(ii) $P_n(-\infty, t] \to P(-\infty, t]$ for every t such that $P(\{t\}) = 0$

(iii) $P_n f \to Pf$ for every $f \in C^\infty(\mathbb{R})$, where

$$C^\infty(\mathbb{R}) = \left\{ f : \mathbb{R} \to \mathbb{R} : \|f\|_\infty < \infty, \left\| f^{(k)} \right\|_\infty < \infty, \ \forall\ k \right\}.$$

Proof of Theorem 5.1.17. Homework is to show equivalence of (i) and (ii). (i) $\implies$ (iii) because $C^\infty(\mathbb{R}) \subseteq \mathrm{BL}(\mathbb{R})$, while (iii) $\implies$ (i) requires more work.

Fix $l \in \mathrm{BL}(\mathbb{R})$. We know l is continuous, but do not know it is infinitely differentiable, nor do we have control over the derivatives, if they exist. Define

$$l_\sigma(x) = \mathbb{E}\left[l(x + \sigma Z)\right], \quad Z \sim N(0,1).$$

We can check that $l_\sigma \in C^\infty(\mathbb{R})$ with induction.

We have that[4]

$$\begin{aligned} |l_\sigma(x) - l(x)| &= |\mathbb{E}\left[l(x+\sigma Z)\right] - l(x)| \\ &\le \mathbb{E}\left[|l(x+\sigma Z) - l(x)|\right] \\ &\le K\mathbb{E}\left[\sigma |Z|\right] \\ &= \sigma K \mathbb{E}\left[|Z|\right]. \end{aligned}$$

For every $\epsilon > 0$, there exists $\sigma > 0$ such that

$$\sup_x |l_\sigma(x) - l(x)| \le \epsilon,$$

which yields the uniform bounds

$$P_n\left(l - l_\sigma\right) \le \epsilon, \quad P\left(l - l_\sigma\right) \le \epsilon.$$

By the triangle inequality,

$$|P_n l - Pl| \le |P_n l_\sigma - P l_\sigma| + 2\epsilon \to 0 + 2\epsilon,$$

so

$$\limsup_{n\to\infty} |P_n l - Pl| \le 2\epsilon.$$

We send $\epsilon \to 0$ to conclude. □

5.2 Proving the Central Limit Theorem

We will use Lindeberg swapping, then see a secondary proof using Stein's method.

Because linear combinations of Gaussian random variables are Gaussian, we know that if $Z_i \sim N(0,1)$,

$$\frac{Z_1 + \cdots + Z_n}{\sqrt{n}} \sim N(0,1),$$

[4] l_σ is the Gaussian smoothing of l, i.e. convolution with the $N(0, \sigma^2)$ density.

as the first and second moments match. Lindeberg swapping will try to get us from $\frac{X_1+\cdots+X_n}{\sqrt{n}}$ to $\frac{Z_1+\cdots+Z_n}{\sqrt{n}}$.

First, we provide intuition for the specific statement of the CLT.

Aside 5.2.1. Consider $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Bernoulli}(p)$. Then the CLT tells us

$$\frac{\sum_{i=1}^{n}(X_i - p)}{\sqrt{np(1-p)}} \rightsquigarrow N(0,1).$$

But what if p depends on n, for example $p_n \to 0$? Then the previous iid statement no longer applies verbatim. If instead $np_n \to \lambda$, then $\sum_{i=1}^{n} X_i \rightsquigarrow \text{Poisson}(\lambda)$. What really happens is that we have

$$\begin{aligned}
n = 1 &: X_{11} && \sim \text{Bernoulli}(p_1) \\
n = 2 &: X_{21}, X_{22} && \overset{\text{iid}}{\sim} \text{Bernoulli}(p_2) \\
&\vdots \\
n &: X_{n_1}, \ldots, X_{nn} && \overset{\text{iid}}{\sim} \text{Bernoulli}(p_n).
\end{aligned}$$

We see that we need to consider this triangular array in our statement of the CLT. We want to say

$$\frac{\sum_{i=1}^{n}(X_{ni} - p_n)}{\sqrt{np_n(1-p_n)}} \sim P_n, \quad P_n \rightsquigarrow N(0,1).$$

Definition 5.2.2 (Triangular array).

An array is triangular if it is of the form

$$\begin{aligned}
&\xi_{1,1}, \xi_{1,2}, \ldots, \xi_{1,k(1)} \\
&\xi_{2,1}, \xi_{2,2}, \ldots, \xi_{2,k(2)} \\
&\vdots \\
&\xi_{n,1}, \xi_{n,2}, \ldots, \xi_{n,k(n)}
\end{aligned}$$

with $k(n) \to \infty$ as $n \to \infty$. We have independence within each row and do not care about dependence across rows.

Theorem 5.2.3 (Central Limit).

Suppose we have a triangular array of random variables $\xi_{n,i}$ such that

(i) $\sum_{i=1}^{k(n)} \mathbb{E}[\xi_{n,i}] \to \mu$,

(ii) $\sum_{i=1}^{k(n)} \text{Var}(\xi_{n,i}) \to \sigma^2$, and

(iii) $\sum_{i=1}^{k(n)} \mathbb{E}\left[|\xi_{n,i}|^3\right] \to 0$.

Then

$$\sum_{i=1}^{k(n)} \xi_{n,i} \rightsquigarrow N\left(\mu, \sigma^2\right).$$

[5]

Lemma 5.2.4.

Let $\xi_1, \ldots, \xi_k$ and $\eta_1, \ldots, \eta_k$ be independent random variables such that each η_i is Gaussian and, for every i,

$$\mathbb{E}\left[\xi_i\right] = \mathbb{E}\left[\eta_i\right], \qquad \mathbb{E}\left[\xi_i^2\right] = \mathbb{E}\left[\eta_i^2\right], \qquad \mathbb{E}\left[|\xi_i|^3\right] < \infty.$$

Then for every $f \in C^3(\mathbb{R})$ with $\|f'''\|_\infty < \infty$,

$$\left|\mathbb{E}\left[f\left(\sum_{i=1}^k \xi_i\right)\right] - \mathbb{E}\left[f\left(\sum_{i=1}^k \eta_i\right)\right]\right| \leq C\|f'''\|_\infty \sum_{i=1}^k \mathbb{E}\left[|\xi_i|^3\right],$$

where $C > 0$ is a universal constant.

Proof of Lemma 5.2.4. Defining

$$\begin{aligned} X_i &= \eta_1 + \eta_2 + \cdots + \eta_{i-1} + \xi_{i+1} + \cdots + \xi_k, \\ Y_i &= \xi_i, \\ Z_i &= \eta_i, \end{aligned}$$

the ith swap replaces $X_i + Y_i$ with $X_i + Z_i$. By the triangle inequality, it suffices to control one swap and then sum over i.

Let $M = \frac{1}{6}\|f'''\|_\infty$. Taylor's theorem gives

$$\begin{aligned} f(X_i + Y_i) &= f(X_i) + f'(X_i)Y_i + \frac{1}{2}f''(X_i)Y_i^2 + R(X_i, Y_i), \\ R(X_i, Y_i) &= \frac{1}{6}f'''(x^*)Y_i^3, \quad x^* \text{ between } X_i \text{ and } X_i + Y_i, \end{aligned}$$

so $|R(X_i, Y_i)| \leq M|Y_i|^3$. The same bound holds with Z_i in place of Y_i.

By independence of X_i from both Y_i and Z_i,

$$\begin{aligned} \mathbb{E}\left[f(X_i + Y_i)\right] &= \mathbb{E}\left[f(X_i)\right] + \mathbb{E}\left[f'(X_i)\right]\mathbb{E}\left[Y_i\right] \\ &\quad + \frac{1}{2}\mathbb{E}\left[f''(X_i)\right]\mathbb{E}\left[Y_i^2\right] + \mathbb{E}\left[R(X_i, Y_i)\right], \\ \mathbb{E}\left[f(X_i + Z_i)\right] &= \mathbb{E}\left[f(X_i)\right] + \mathbb{E}\left[f'(X_i)\right]\mathbb{E}\left[Z_i\right] \\ &\quad + \frac{1}{2}\mathbb{E}\left[f''(X_i)\right]\mathbb{E}\left[Z_i^2\right] + \mathbb{E}\left[R(X_i, Z_i)\right]. \end{aligned}$$

Since $\mathbb{E}\left[Y_i\right] = \mathbb{E}\left[Z_i\right]$ and $\mathbb{E}\left[Y_i^2\right] = \mathbb{E}\left[Z_i^2\right]$, the first three Taylor terms cancel, and therefore

$$\begin{aligned} \left|\mathbb{E}\left[f(X_i + Y_i)\right] - \mathbb{E}\left[f(X_i + Z_i)\right]\right| &\leq \left|\mathbb{E}\left[R(X_i, Y_i)\right]\right| + \left|\mathbb{E}\left[R(X_i, Z_i)\right]\right| \\ &\leq M\mathbb{E}\left[|Y_i|^3\right] + M\mathbb{E}\left[|Z_i|^3\right]. \end{aligned}$$

[5] Note that the summations are defined on each row, so the limit takes us *down* the array.

Write $Z_i \sim N(\mu, \sigma^2) = \mu + \sigma X$ for $X \sim N(0,1)$. We have the bound

$$\begin{aligned}
\mathbb{E}\left[|Z_i|^3\right] &= \mathbb{E}\left[|\mu + \sigma X|^3\right] \\
&\leq 4\left(|\mu|^3 + \sigma^3 |X|^3\right) \\
&\leq C_1\left(|\mu|^3 + \sigma^3\right) \\
&\leq C_2\left(\mathbb{E}\left[|Y_i|\right]^3 + \mathbb{E}\left[Y_i^2\right]^{\frac{3}{2}}\right) \\
&\leq C_3 \mathbb{E}\left[|Y_i|^3\right],
\end{aligned}$$

where the second to last step moves the absolute value inside the expectation and the last step follows from Jensen's inequality with convex $f(x) = x^{\frac{3}{2}}$. Hence

$$\left|\mathbb{E}\left[f(X_i + Y_i)\right] - \mathbb{E}\left[f(X_i + Z_i)\right]\right| \leq C \left\|f'''\right\|_\infty \mathbb{E}\left[|\xi_i|^3\right].$$

Summing over $i = 1, \ldots, k$ yields

$$\left|\mathbb{E}\left[f\left(\sum_{i=1}^k \xi_i\right)\right] - \mathbb{E}\left[f\left(\sum_{i=1}^k \eta_i\right)\right]\right| \leq C \left\|f'''\right\|_\infty \sum_{i=1}^k \mathbb{E}\left[|\xi_i|^3\right].$$

□

*Proof of Theorem (**??**).* Define

$$\begin{aligned}
\mu_n &= \sum_{i=1}^{k(n)} \mathbb{E}\left[\xi_{n,i}\right] \\
\sigma_n^2 &= \sum_{i=1}^{k(n)} \mathrm{Var}\left(\xi_{n,i}\right).
\end{aligned}$$

For every $f \in C^\infty(\mathbb{R})$ with $\left\|f'''\right\|_\infty < \infty$,

$$\left|\mathbb{E}\left[f\left(\sum_{i=1}^{k(n)} \xi_{n,i}\right)\right] - \mathbb{E}\left[f\left(N(\mu, \sigma^2)\right)\right]\right| \leq (\mathrm{I}) + (\mathrm{II}),$$

where

$$\begin{aligned}
(\mathrm{I}) &= \left|\mathbb{E}\left[f\left(\sum_{i=1}^{k(n)} \xi_{n,i}\right)\right] - \mathbb{E}\left[f\left(N\left(\mu_n, \sigma_n^2\right)\right)\right]\right|, \\
(\mathrm{II}) &= \left|\mathbb{E}\left[f\left(N\left(\mu_n, \sigma_n^2\right)\right)\right] - \mathbb{E}\left[f\left(N\left(\mu, \sigma^2\right)\right)\right]\right|.
\end{aligned}$$

By moment-matching, there exist independent $y_{n,1}, \ldots, y_{n,k(n)}$ that are Gaussian and match the first two moments of $\xi_{n,1}, \ldots, \xi_{n,k(n)}$, so[6]

$$\sum_{i=1}^{k(n)} y_{n,i} \sim N\left(\mu_n, \sigma_n^2\right).$$

[6] Because the sum of normal random variables has mean equal to the sum of the means and variance equal to the sum of the variances. We also use that the sum of independent normal random variables is again normal.

By assumption (iii) and Lemma (**??**), we have

$$(\mathrm{I}) \leq C \left\| f''' \right\|_\infty \sum_{i=1}^{k(n)} \mathbb{E}\left[\left| \xi_{n,i} \right|^3 \right] \to 0.$$

We can write

$$\begin{aligned}
(\mathrm{II}) &= \left| \mathbb{E}\left[f\left(\mu_n + \sigma_n Z\right) \right] - \mathbb{E}\left[f\left(\mu + \sigma Z\right) \right] \right|, \quad Z \sim N(0,1), \\
&\leq L\mathbb{E}\left[\left|\mu_n - \mu\right| + \left|\sigma_n - \sigma\right| \left|Z\right| \right] \\
&= L\left|\mu_n - \mu\right| + C_1 \mathbb{E}\left[\left|Z\right|\right] \left|\sigma_n - \sigma\right| \\
&\to 0,
\end{aligned}$$

where the convergence follows from assumptions (i) and (ii). □

Example 5.2.5.

Returning to our example that motivated the triangular matrix consideration, suppose

$$X_{n,1}, \ldots, X_{n,n} \overset{\text{iid}}{\sim} \text{Bernoulli}(p_n).$$

We want to show that

$$\frac{\sum_{i=1}^{n} \left(X_{n,i} - p_n\right)}{\sqrt{np_n(1-p_n)}} \rightsquigarrow N(0,1).$$

Define

$$\xi_{n,i} = \frac{X_{n,i} - p_n}{\sqrt{np_n(1-p_n)}}.$$

We see that the first two conditions are satisfied and the third when $np_n(1-p_n) \to \infty$:

(i) $\sum_{i=1}^{n} \mathbb{E}\left[\xi_{n,i}\right] = n \cdot 0 = 0$

(ii) $\sum_{i=1}^{n} \mathrm{Var}\left(\xi_{n,i}\right) = 1$

(iii) We see

$$\begin{aligned}
\sum_{i=1}^{n} \mathbb{E}\left[\left| \frac{\left|X_{n,i} - p_n\right|}{\sqrt{np_n(1-p_n)}} \right|^3 \right] &= n \frac{p_n(1-p_n)^3 + (1-p_n)p_n^3}{\left(np_n(1-p_n)\right)^{1.5}} \\
&= n \frac{p_n(1-p_n)}{n^{1.5}} \frac{(1-p_n)^2 + p_n^2}{\left(p_n(1-p_n)\right)^{1.5}}, \\
&\asymp \frac{1}{\sqrt{np_n(1-p_n)}}
\end{aligned}$$

which goes to 0 if $np_n(1-p_n) \to \infty$

Corollary 5.2.6.

Suppose $X_i \overset{\text{iid}}{\sim} \mathbb{P}$ where $\mathbb{P}$ does not depend on n, $\mathbb{E}[X_i] = 0$, and $\operatorname{Var}(X_i) = 1$. Then

$$\frac{X_1 + \cdots + X_n}{\sqrt{n}} \rightsquigarrow N(0,1).$$

In particular, we do not require a finite third moment.

Proof of Corollary 5.2.6. We proceed by truncation. Put $\xi_{n,i} = \frac{X_i}{\sqrt{n}} \mathbb{1}_{|X_i| \le \sqrt{n}}$. Then

$$\begin{aligned}
\left| \sum_{i=1}^n \mathbb{E}[\xi_{n,i}] \right| &= \left| \sqrt{n} \mathbb{E}\left[X \mathbb{1}_{|X| \le \sqrt{n}} \right] \right| \\
&= \left| \sqrt{n} \mathbb{E}\left[X \mathbb{1}_{|X| > \sqrt{n}} \right] \right| \\
&\le \mathbb{E}\left[X^2 \mathbb{1}_{|X| > \sqrt{n}} \right] \\
&\to 0,
\end{aligned}$$

where in the convergence we use a dominating function of X^2.

To check the second condition, consider

$$\begin{aligned}
\sum_{i=1}^n \operatorname{Var}(\xi_{n,i}) &= n \operatorname{Var}\left(\frac{X}{\sqrt{n}} \mathbb{1}_{|X| \le \sqrt{n}} \right) \\
&= \operatorname{Var}\left(X \mathbb{1}_{|X| \le \sqrt{n}} \right) \\
&= \mathbb{E}\left[X^2 \mathbb{1}_{|X| \le \sqrt{n}} \right] - \left(\mathbb{E}\left[X \mathbb{1}_{|X| \le \sqrt{n}} \right] \right)^2.
\end{aligned}$$

We analyze these terms separately. We see from our previous work that

$$\begin{aligned}
\mathbb{E}\left[X^2 \mathbb{1}_{|X| \le \sqrt{n}} \right] &= \mathbb{E}\left[X^2 \right] - \mathbb{E}\left[X^2 \mathbb{1}_{|X| > \sqrt{n}} \right] \\
&\to 1 + 0 = 1,
\end{aligned}$$

while

$$\begin{aligned}
\left| \mathbb{E}\left[X \mathbb{1}_{|X| \le \sqrt{n}} \right] \right| &= \mathbb{E}\left[X \mathbb{1}_{|X| > \sqrt{n}} \right] \\
&\le \mathbb{E}\left[|X| \, \mathbb{1}_{|X| > \sqrt{n}} \right] \to 0
\end{aligned}$$

by the DCT with a dominating function of $|X|$. Hence, the second condition is satisfied.

For the third condition, we examine

$$\begin{aligned}\sum_{i=1}^{n}\mathbb{E}\left[\left|\xi_{n,i}\right|^3\right] &= n\mathbb{E}\left[\left(\frac{|X|}{\sqrt{n}}\right)^3 \mathbb{1}_{|X|\le\sqrt{n}}\right] \\ &= \mathbb{E}\left[\frac{|X|^3}{\sqrt{n}}\mathbb{1}_{|X|\le\sqrt{n}}\right] \\ &= \mathbb{E}\left[X^2\frac{|X|}{\sqrt{n}}\mathbb{1}_{|X|\le\sqrt{n}}\right] \\ &\le \mathbb{E}\left[X^2\min\left\{\frac{|X|}{\sqrt{n}},1\right\}\right] \to 0\end{aligned}$$

by X^2 being a dominating function.

Therefore, we can invoke the CLT to get $\sum\limits_{i=1}^{n}\xi_{n,i} \rightsquigarrow N(0,1)$.

For any $f \in C^\infty(\mathbb{R})$, we want to show that

$$\left|\mathbb{E}\left[f\left(\sum_{i=1}^{n}\xi_{n,i}\right)\right] - \mathbb{E}\left[f\left(\sum_{i=1}^{n}\frac{X_i}{\sqrt{n}}\right)\right]\right| \to 0,$$

as we can use the triangle inequality with our CLT result.

Put

$$S_n = \sum_{i=1}^{n}\xi_{n,i}, \qquad T_n = \sum_{i=1}^{n}\frac{X_i}{\sqrt{n}}.$$

Then

$$\begin{aligned}\left|\mathbb{E}\left[f\left(S_n\right)\right] - \mathbb{E}\left[f\left(T_n\right)\right]\right| &= \left|\mathbb{E}\left[\left(f\left(S_n\right) - f\left(T_n\right)\right)\mathbb{1}_{S_n\neq T_n}\right]\right| \\ &\le 2\left\|f\right\|_\infty \mathbb{P}\left[S_n \neq T_n\right] \\ &\le 2\left\|f\right\|_\infty \sum_{i=1}^{n}\mathbb{P}\left[\xi_{n,i} \neq \frac{X_i}{\sqrt{n}}\right] \\ &\le 2\left\|f\right\|_\infty \sum_{i=1}^{n}\mathbb{P}\left[|X_i| > \sqrt{n}\right] \\ &= 2\left\|f\right\|_\infty \mathbb{E}\left[n\mathbb{1}_{|X|>\sqrt{n}}\right] \\ &\le 2\left\|f\right\|_\infty \mathbb{E}\left[X^2\mathbb{1}_{|X|>\sqrt{n}}\right] \to 0.\end{aligned}$$

where the convergence follows from X^2 being a dominating function and the indicator giving us pointwise convergence to 0. Consequently,

$$\begin{aligned}\left|\mathbb{E}\left[f\left(T_n\right)\right] - \mathbb{E}\left[f\left(N(0,1)\right)\right]\right| \le &\left|\mathbb{E}\left[f\left(T_n\right)\right] - \mathbb{E}\left[f\left(S_n\right)\right]\right| \\ &+ \left|\mathbb{E}\left[f\left(S_n\right)\right] - \mathbb{E}\left[f\left(N(0,1)\right)\right]\right| \to 0.\end{aligned}$$

□

Theorem 5.2.7 (Lindeberg Central Limit Theorem).

Suppose we have a triangular array $\{X_{n,i}\}$ such that

(i) $\mathbb{E}\left[X_{n,i}\right] = 0$

(ii) $\sum_{i=1}^{k(n)} \operatorname{Var}(X_{n,i}) = 1$

(iii) (Lindeberg condition) For every $\epsilon > 0$,

$$L_n(\epsilon) := \sum_{i=1}^{k(n)} \mathbb{E}\left[X_{n,i}^2 \mathbb{1}_{|X_{n,i}|>\epsilon}\right] \xrightarrow{n\to\infty} 0.$$

Then

$$\sum_{i=1}^{k(n)} X_{n,i} \rightsquigarrow N(0,1).$$

[7]

We require a lemma.

Lemma 5.2.8.

Suppose for every $\epsilon > 0$,

$$H_n(\epsilon) = \frac{L_n(\epsilon)}{\epsilon^2} \to 0.$$

Then there exists a sequence $(\epsilon_n) \to 0$ such that $H_n(\epsilon_n) \to 0$.

Proof of Lemma 5.2.8. For any $k \in \mathbb{N}$, there exists n_k such that if $n \geq n_k$,

$$H_n(\frac{1}{k}) \leq \frac{1}{k}.$$

From this, we can pick an increasing subsequence (n_k).

Putting $\epsilon_n = \frac{1}{k}$ for $n_k \leq n < n_{k+1}$, we get that

$$H_n(\epsilon_n) \leq \frac{1}{k} \to 0.$$

[8] □

*Proof of Theorem (**??**).* Define $H_n(\epsilon) = \frac{L_n(\epsilon)}{\epsilon^2}$. For any fixed $\epsilon > 0$, we see $H_n(\epsilon) \to 0$. By Lemma (**??**), there exists a sequence $\epsilon_n \to 0$ such that $H_n(\epsilon_n) \to 0$.

Define

$$\xi_{n,i} = X_{n,i} \mathbb{1}_{|X_{n,i}|\leq\epsilon_n}.$$

Our process is similar to the proof of the previous CLT corollary.

[7] Note that the first two conditions are exact equalities, while in the general CLT, we only require convergence. This is what we give up for the weakening of the third condition.

[8] We cannot simply put $\epsilon_n = \frac{1}{n}$: the hypothesis says $H_n(\epsilon) \to 0$ for each *fixed* $\epsilon > 0$, so we cannot substitute a sequence that changes with n directly.

We check the first condition:

$$\begin{aligned}\left|\sum_{i=1}^{k(n)} \mathbb{E}\left[\xi_{n,i}\right]\right| &= \left|\sum_i \mathbb{E}\left[X_{n,i}\mathbb{1}_{|X_{n,i}|>\epsilon_n}\right]\right| \\ &\leq \frac{1}{\epsilon_n}\left|\sum_i \mathbb{E}\left[X_{n,i}^2\mathbb{1}_{|X_{n,i}|>\epsilon_n}\right]\right| \\ &= \epsilon_n H_n(\epsilon_n) \to 0,\end{aligned}$$

since both ϵ_n and $H_n \to 0$.

For the variance condition, note

$$\begin{aligned}\sum_{i=1}^{k(n)} \mathrm{Var}\left(\xi_{n,i}\right) &= \sum_i \mathbb{E}\left[\xi_{n,i}^2\right] - \sum_i \left(\mathbb{E}\left[\xi_{n,i}\right]\right)^2 \\ \sum_i \mathbb{E}\left[\xi_{n,i}^2\right] &= \sum_i \mathbb{E}\left[X_{n,i}^2\mathbb{1}_{|X_{n,i}|\leq\epsilon_n}\right] \\ &= \sum_i \mathbb{E}\left[X_{n,i}^2\right] - \sum_i \mathbb{E}\left[X_{n,i}^2\mathbb{1}_{|X_{n,i}|>\epsilon_n}\right] \\ &= 1 - L_n(\epsilon_n) = 1 - \epsilon_n^2 H_n(\epsilon_n) \to 1 - 0 = 1 \\ \sum_i \mathbb{E}\left[\xi_{n,i}\right]^2 &= \sum_i \left(\mathbb{E}\left[X_{n,i}\mathbb{1}_{|X_{n,i}|\leq\epsilon_n}\right]\right)^2 \\ &= \sum_i \left(\mathbb{E}\left[X_{n,i}\mathbb{1}_{|X_{n,i}|>\epsilon_n}\right]\right)^2 \text{ by } \mathbb{E}\left[X_{n,i}\right] = 0,\ a^2 = (-a)^2 \\ &\leq \sum_i \mathbb{E}\left[X_{n,i}^2\mathbb{1}_{|X_{n,i}|>\epsilon_n}\right] = L_n(\epsilon_n) \to 0.\end{aligned}$$

Hence the sum of the variances goes to 1.

For the third moment condition, we see

$$\begin{aligned}\sum_i \mathbb{E}\left[\left|\xi_{n,i}\right|^3\right] &= \sum_i \mathbb{E}\left[\left|X_{n,i}\right|^3 \mathbb{1}_{|X_{n,i}|\leq\epsilon_n}\right] \\ &\leq \epsilon_n \sum_i \mathbb{E}\left[X_{n,i}^2\mathbb{1}_{|X_{n,i}|\leq\epsilon_n}\right] \\ &= \epsilon_n\left(1 - L_n(\epsilon_n)\right) \leq \epsilon_n \to 0,\end{aligned}$$

so we apply the CLT to get $\sum_i \xi_{n,i} \rightsquigarrow N(0,1)$.

As in the previous proof, we write

$$\begin{aligned}\left|\mathbb{E}\left[f\left(\sum_i \xi_{n,i}\right)\right]-\mathbb{E}\left[f\left(\sum_i X_{n,i}\right)\right]\right| &\leq 2\|f\|_\infty \mathbb{P}\left[\sum_i \xi_{n,i} \neq \sum_i X_{n,i}\right] \\ &\leq 2\|f\|_\infty \sum_i \mathbb{P}\left[|X_{n,i}|>\epsilon_n\right] \\ &= 2\|f\|_\infty \mathbb{E}\left[\sum_i \mathbb{1}_{|X_{n,i}|>\epsilon_n}\right] \\ &\leq 2\|f\|_\infty \mathbb{E}\left[\sum_i \frac{X_{n,i}^2}{\epsilon_n^2}\mathbb{1}_{|X_{n,i}|>\epsilon_n}\right] \\ &= 2\|f\|_\infty H_n(\epsilon_n) \to 0.\end{aligned}$$

By the triangle inequality, we see that

$$\left|\mathbb{E}\left[f\left(\sum_i X_{n,i}\right)\right]-\mathbb{E}\left[f\left(N(0,1)\right)\right]\right| \to 0,$$

so

$$\sum_{i=1}^{k(n)} X_{n,i} \rightsquigarrow N(0,1).$$

□

Theorem 5.2.9 (Lindeberg CLT (arbitrary variance)).

Suppose for each n the random variables $X_{n,1},\ldots,X_{n,k(n)}$ are independent and satisfy $\mathbb{E}\left[X_{n,i}\right]=0$. Let

$$S_n^2 := \sum_{i=1}^{k(n)} \mathbb{E}\left[X_{n,i}^2\right],$$

and assume $S_n>0$ for all sufficiently large n. If for every $\epsilon>0$,

$$\frac{1}{S_n^2}\sum_{i=1}^{k(n)} \mathbb{E}\left[X_{n,i}^2 \mathbb{1}_{|X_{n,i}|>\epsilon S_n}\right] \to 0,$$

then

$$\frac{\sum_{i=1}^{k(n)} X_{n,i}}{S_n} \rightsquigarrow N(0,1).$$

Proof of Theorem 5.2.9. We standardize the array by setting

$$Y_{n,i} = \frac{X_{n,i}}{S_n}.$$

Then $\mathbb{E}\left[Y_{n,i}\right]=0$ and $\sum_i \mathrm{Var}(Y_{n,i})=1$. The Lindeberg condition for $\{Y_{n,i}\}$ follows directly from the hypothesis, with cutoff ϵS_n. The conclusion follows from Theorem (**??**).

Note that there is no problem with (i) losing independence within the array's rows by dividing by S_n nor (ii) having cutoff ϵS_n instead of ϵ, because the term S_n is a deterministic value not dependent on realized values of $X_{n,i}$. □

Theorem 5.2.10 (Lyapunov CLT).

Suppose $X_{n,1}, \ldots, X_{n,k(n)}$ are independent such that $\mathbb{E}[X_{n,i}] = 0$ and for $S_n = \sqrt{\sum_{i=1}^{k(n)} \mathbb{E}\left[X_{n,i}^2\right]}$, there exists $\delta > 0$ such that

$$\frac{1}{S_n^{2+\delta}} \sum_{i=1}^{k(n)} \mathbb{E}\left[|X_{n,i}|^{2+\delta}\right] \to 0,$$

then

$$\frac{X_{n,1} + \cdots + X_{n,k(n)}}{S_n} \rightsquigarrow N(0,1).$$

[9]

Proof of Theorem 5.2.10. For any $\epsilon > 0$, we see that

$$\begin{aligned} X_{n,i}^2 \mathbb{1}_{|X_{n,i}| > \epsilon S_n} &= X_{n,i}^2 \mathbb{1}_{|X_{n,i}|^\delta > \epsilon^\delta S_n^\delta} \\ &\leq \frac{|X_{n,i}|^{2+\delta}}{\epsilon^\delta S_n^\delta} \end{aligned}$$

$$\frac{1}{S_n^2} \sum_{i=1}^{k(n)} \mathbb{E}\left[X_{n,i}^2 \mathbb{1}_{|X_{n,i}| > \epsilon S_n}\right] \leq \frac{1}{\epsilon^\delta} \frac{1}{S_n^{2+\delta}} \sum_{i=1}^{k(n)} \mathbb{E}\left[|X_{n,i}|^{2+\delta}\right] \to 0.$$

The previous theorem's condition is satisfied. □

Remark 5.2.11. The Lindeberg condition is *sufficient* for the CLT conclusion, but not in general *necessary*: a triangular array can satisfy the CLT without satisfying the Lindeberg condition. However, if the individual variances are uniformly negligible relative to the total variance, the two are equivalent.

Example 5.2.12 (Lindeberg's condition is not necessary).

Consider the triangular array with $k(n) = 1$ and $X_{n,1} \sim N(0,1)$ for all n. Since $X_{n,1}$ is already standard normal,

$$\sum_{i=1}^{k(n)} X_{n,i} = X_{n,1} \sim N(0,1) \rightsquigarrow N(0,1),$$

so the CLT conclusion holds trivially.
However, the Lindeberg condition fails. Since $\mathbb{E}[X_{n,1}] = 0$ and $\mathrm{Var}(X_{n,1}) = 1$, conditions (i) and (ii) of Theorem (**??**) are satisfied. But for any fixed $\epsilon > 0$,

$$L_n(\epsilon) = \mathbb{E}\left[X_{n,1}^2 \mathbb{1}_{|X_{n,1}| > \epsilon}\right] = \mathbb{E}\left[Z^2 \mathbb{1}_{|Z| > \epsilon}\right] > 0$$

is a positive constant independent of n, so $L_n(\epsilon) \not\to 0$.
Note also that $\mathbb{E}\left[|X_{n,1}|^3\right] = \mathbb{E}\left[|Z|^3\right]$ is a positive constant, so condition (iii) of the general triangular CLT (Theorem **??**) fails as well; the CLT holds here because the distributions are already Gaussian, not because any of our sufficient conditions are met.
The key reason Lindeberg fails is that the uniform infinitesimality condition $\max_{k=1,\ldots,k(n)} \frac{\sigma_k^2}{s_n^2} \to 0$ is violated: with $k(n) = 1$ the only term contributes all of $s_n^2 = 1$ to itself, so $\frac{\sigma_1^2}{s_n^2} = 1 \not\to 0$.

[9] This is a rescaled version of Theorem (**??**).

[10]

Theorem 5.2.13 (Necessity of Lindeberg's condition under uniform infinitesimality).

Suppose for each n the random variables $X_{n,1}, \ldots, X_{n,k(n)}$ are independent with $\mathbb{E}[X_{n,i}] = 0$, and let $\sigma_{n,i}^2 = \mathrm{Var}(X_{n,i})$ and $s_n^2 = \sum_{i=1}^{k(n)} \sigma_{n,i}^2 > 0$. If

$$\max_{i=1,\ldots,k(n)} \frac{\sigma_{n,i}^2}{s_n^2} \to 0 \quad \text{as } n \to \infty,$$

then the Lindeberg condition

$$\frac{1}{s_n^2} \sum_{i=1}^{k(n)} \mathbb{E}\left[X_{n,i}^2 \mathbb{1}_{|X_{n,i}| > \epsilon s_n}\right] \to 0 \quad \text{for every } \epsilon > 0$$

holds if and only if

$$\frac{\sum_{i=1}^{k(n)} X_{n,i}}{s_n} \rightsquigarrow N(0,1).$$

Proof of Theorem 5.2.13. The sufficiency direction is Theorem (**??**). We prove necessity.

By normalizing, set $\tilde{X}_{n,i} = X_{n,i}/s_n$, so $\mathbb{E}\left[\tilde{X}_{n,i}\right] = 0$, $\sum_i \tilde{\sigma}_{n,i}^2 = 1$ (where $\tilde{\sigma}_{n,i}^2 = \sigma_{n,i}^2/s_n^2$), $\max_i \tilde{\sigma}_{n,i}^2 \to 0$, and $Y_n = \sum_i \tilde{X}_{n,i} \rightsquigarrow N(0,1)$.

Denote $\phi_{n,i}(t) = \mathbb{E}\left[e^{it\tilde{X}_{n,i}}\right]$. By the continuity theorem, $\prod_i \phi_{n,i}(t) \to e^{-t^2/2}$ for every t.

Step 1. *Under uniform infinitesimality,* $\sum_i \mathbb{E}\left[1 - \cos(t\tilde{X}_{n,i})\right] \to \frac{t^2}{2}$.

Using $|e^{ix} - 1 - ix| \le x^2/2$, we get $|\phi_{n,i}(t) - 1| \le \frac{t^2}{2}\tilde{\sigma}_{n,i}^2$, so $|\phi_{n,i}(t) - 1| \to 0$ uniformly in i. For large n, each $|\phi_{n,i}(t) - 1| \le \frac{1}{2}$, and using $|\log(1+z) - z| \le |z|^2$ for $|z| \le \frac{1}{2}$:

$$\begin{aligned}
\left|\sum_i \log \phi_{n,i}(t) - \sum_i (\phi_{n,i}(t) - 1)\right| &\le \sum_i |\phi_{n,i}(t) - 1|^2 \\
&\le \max_i |\phi_{n,i}(t) - 1| \cdot \sum_i |\phi_{n,i}(t) - 1| \\
&\le \frac{t^2 \max_i \tilde{\sigma}_{n,i}^2}{2} \cdot \frac{t^2}{2} \sum_i \tilde{\sigma}_{n,i}^2 \\
&= \frac{t^4 \max_i \tilde{\sigma}_{n,i}^2}{4} \to 0.
\end{aligned}$$

Hence $\sum_i (\phi_{n,i}(t) - 1) \to -t^2/2$.

Since $\mathbb{E}\left[\tilde{X}_{n,i}\right] = 0$, we write $\phi_{n,i}(t) - 1 = \mathbb{E}\left[e^{it\tilde{X}_{n,i}} - 1 - it\tilde{X}_{n,i}\right]$. Then

$$\sum_i (\phi_{n,i}(t) - 1) = \sum_i \mathbb{E}\left[e^{it\tilde{X}_{n,i}} - 1 - it\tilde{X}_{n,i} + \frac{t^2 \tilde{X}_{n,i}^2}{2}\right] - \frac{t^2}{2}.$$

[10] Theorem (**??**) was not done in class, nor was Example (**??**). After the preceeding remark was made in class, I found answers to the resulting question.

Since $\sum_i(\phi_{n,i}(t)-1) \to -t^2/2$, taking real parts and using $\mathrm{Re}(e^{ix}-1-ix+x^2/2) = \cos(x)-1+x^2/2$ yields

$$\sum_i \mathbb{E}\left[1-\cos(t\tilde{X}_{n,i})\right] = \frac{t^2}{2} + o(1).$$

Step 2. *Conclude $L_n(\epsilon) \to 0$.*
Fix $\epsilon > 0$. Split the expectation at ϵ:

$$\frac{t^2}{2} + o(1) = \sum_i \mathbb{E}\left[(1-\cos(t\tilde{X}_{n,i}))\mathbb{1}_{|\tilde{X}_{n,i}|\le\epsilon}\right] + \sum_i \mathbb{E}\left[(1-\cos(t\tilde{X}_{n,i}))\mathbb{1}_{|\tilde{X}_{n,i}|>\epsilon}\right].$$

For the first sum, we use $1-\cos(u) \le u^2/2$:

$$\sum_i \mathbb{E}\left[(1-\cos(t\tilde{X}_{n,i}))\mathbb{1}_{|\tilde{X}_{n,i}|\le\epsilon}\right] \le \frac{t^2}{2}\sum_i \mathbb{E}\left[\tilde{X}_{n,i}^2\mathbb{1}_{|\tilde{X}_{n,i}|\le\epsilon}\right] = \frac{t^2}{2}(1-L_n(\epsilon)).$$

For the second sum, we use $1-\cos(u) \le 2$ and Markov's inequality:

$$\sum_i \mathbb{E}\left[(1-\cos(t\tilde{X}_{n,i}))\mathbb{1}_{|\tilde{X}_{n,i}|>\epsilon}\right] \le 2\sum_i \mathbb{P}\left(|\tilde{X}_{n,i}|>\epsilon\right) \le \frac{2}{\epsilon^2}\sum_i \mathbb{E}\left[\tilde{X}_{n,i}^2\mathbb{1}_{|\tilde{X}_{n,i}|>\epsilon}\right] = \frac{2L_n(\epsilon)}{\epsilon^2}.$$

Combining,

$$\frac{t^2}{2} + o(1) \le \frac{t^2}{2}(1-L_n(\epsilon)) + \frac{2L_n(\epsilon)}{\epsilon^2} = \frac{t^2}{2} - L_n(\epsilon)\left(\frac{t^2}{2} - \frac{2}{\epsilon^2}\right).$$

Rearranging,

$$L_n(\epsilon)\left(\frac{t^2}{2} - \frac{2}{\epsilon^2}\right) \le o(1).$$

Choosing $t > 2/\epsilon$ makes the prefactor $\frac{t^2}{2} - \frac{2}{\epsilon^2} > 0$, so $L_n(\epsilon) \to 0$. □

5.3 Stein's Method

This proof technique provides an alternative to characteristic functions and Lindeberg swapping.

Lemma 5.3.1 (Stein's identity).

Suppose $Z \sim N(0,1)$. If f is absolutely continuous such that

$$\mathbb{E}\left[f(Z)\right] < \infty, \quad \mathbb{E}\left[Zf(Z)\right] < \infty,$$

we have

$$\mathbb{E}\left[f'(Z)\right] = \mathbb{E}\left[Zf(Z)\right].$$

[11]

Proof of Lemma 5.3.1. Let $\phi(t) = \frac{1}{\sqrt{2\pi}}e^{-t^2/2}$ denote the standard normal density, so $\mathbb{E}[g(Z)] = \int_{-\infty}^{\infty} g(t)\phi(t)\,dt$ for any measurable g. We begin with

$$\mathbb{E}\left[f'(Z)\right] = \int_{-\infty}^{\infty} f'(t)\phi(t)\,dt$$

[11] This identity is the defining identity of $N(0,1)$.

Since $\phi(-\infty) = 0$, the FTC gives $\phi(t) = \int_{-\infty}^{t} \phi'(x)\,dx$. Substituting:

$$= \int_{-\infty}^{\infty} f'(t) \int_{-\infty}^{t} \phi'(x)\,dx\,dt$$

Apply Fubini on the region $\{(x,t) : x < t\}$:

$$= \int_{-\infty}^{\infty} \phi'(x) \int_{x}^{\infty} f'(t)\,dt\,dx.$$

We would like to apply the FTC to the inner integral: $\int_x^\infty f'(t)\,dt = f(\infty) - f(x)$. However, $f(\infty)$ may not be finite, so this fails.

The fix is to split at $t = 0$ and use a *different* representation of ϕ on each piece, chosen so that after Fubini the FTC only runs over a finite interval. For $t < 0$, anchor at $-\infty$: $\phi(t) = \int_{-\infty}^{t} \phi'(x)\,dx$. For $t > 0$, anchor at $+\infty$: since $\phi(+\infty) = 0$, the FTC gives $\phi(t) = -\int_t^\infty \phi'(x)\,dx$. In each case, after Fubini the inner integral only reaches 0.

$$\mathbb{E}\left[f'(Z)\right] = \int_{-\infty}^{0} f'(t)\phi(t)\,dt + \int_{0}^{\infty} f'(t)\phi(t)\,dt$$

Substitute the appropriate representation of $\phi(t)$ on each piece:

$$= \int_{-\infty}^{0} f'(t) \int_{-\infty}^{t} \phi'(x)\,dx\,dt + \int_{0}^{\infty} f'(t) \int_{t}^{\infty} -\phi'(x)\,dx\,dt$$

Apply Fubini to each piece. For the left piece the region $\{-\infty < x < t < 0\}$ becomes $\{-\infty < x < 0,\ x < t < 0\}$; for the right piece the region $\{0 < t < x < \infty\}$ becomes $\{0 < x < \infty,\ 0 < t < x\}$:

$$= \int_{-\infty}^{0} \phi'(x) \int_{x}^{0} f'(t)\,dt\,dx + \int_{0}^{\infty} (-\phi'(x)) \int_{0}^{x} f'(t)\,dt\,dx$$

Apply the FTC (valid since f is absolutely continuous) to the inner integrals:

$$\begin{aligned}
&= \int_{-\infty}^{0} \phi'(x)\left(f(0) - f(x)\right)dx - \int_{0}^{\infty} \phi'(x)\left(f(x) - f(0)\right)dx \\
&= f(0)\left(\int_{-\infty}^{0} \phi'(x)dx + \int_{0}^{\infty} \phi'(x)dx\right) \\
&\quad - \int_{-\infty}^{0} \phi'(x)f(x)dx - \int_{0}^{\infty} \phi'(x)f(x)dx
\end{aligned}$$

Since $\phi'(x) = -x\phi(x)$ and $\mathbb{E}[Z] = 0$, the first term vanishes:

$$\begin{aligned}
&= f(0)\int_{-\infty}^{\infty} -x\phi(x)\,dx + \int_{-\infty}^{\infty} x\phi(x)f(x)\,dx \\
&= \int_{-\infty}^{\infty} x\phi(x)f(x)\,dx \\
&= \mathbb{E}\left[Zf(Z)\right].
\end{aligned}$$

□

Aside 5.3.2. Alternatively, we can derive Stein's identity via integration by parts if we don't see the clever

representation trick. Recall $\phi'(t) = -t\phi(t)$. Then

$$\begin{aligned}
\mathbb{E}\left[Zf(Z)\right] &= \int tf(t)\phi(t)\,dt \\
&= -\int f(t)\phi'(t)\,dt \quad \text{since } t\phi(t) = -\phi'(t) \\
&= -\left(f(t)\phi(t)|_{-\infty}^{\infty} - \int f'(t)\phi(t)\,dt\right) \quad \text{(integration by parts)} \\
&= \mathbb{E}\left[f'(Z)\right],
\end{aligned}$$

where the boundary term vanishes because $\phi(t) \to 0$ exponentially fast and $\mathbb{E}\left[f(Z)\right] < \infty$ ensures $f(t)\phi(t) \to 0$ as $t \to \pm\infty$.

[12]

Proposition 5.3.3 (Converse of Stein's identity).

Suppose W is a random variable such that for every absolutely continuous f with

$$\mathbb{E}\left[f(W)\right] < \infty, \quad \mathbb{E}\left[|Wf(W)|\right] < \infty$$

and

$$\mathbb{E}\left[f'(W)\right] = \mathbb{E}\left[Wf(W)\right],$$

we have $W \sim N(0,1)$.

Proof of Proposition 5.3.3. Consider *Stein's equation*:

$$f'(w) - wf(w) = \mathbb{1}_{w \le t} - \Phi(t).$$

If this differential equation has a solution for each t, then $W \sim N(0,1)$. To see why, note that for $W \sim P$, we can take an expectation of Stein's equation and get

$$\mathbb{E}\left[f'(W) - Wf(W)\right] = P\left[W \le t\right] - \Phi(t),$$

so if $\mathbb{E}\left[f'(W)\right] = \mathbb{E}\left[Wf(W)\right]$ (our hypotheses), we have

$$P\left[W \le t\right] = \Phi(t),$$

so $W \sim N(0,1)$. (Since agreement on the half-intervals $(-\infty, t]$, which generate $\mathscr{B}(\mathbb{R})$, implies agreement on all of $\mathscr{B}(\mathbb{R})$.)

Using the integrating factor $e^{-\frac{w^2}{2}}$, we can indeed solve the differential equation:

$$\begin{aligned}
e^{-\frac{w^2}{2}} f'(w) - we^{-\frac{w^2}{2}} f(w) &= e^{-\frac{w^2}{2}}\left(\mathbb{1}_{w \le t} - \Phi(t)\right) \\
\frac{d}{dw}\left(e^{-\frac{w^2}{2}} f(w)\right) &= e^{-\frac{w^2}{2}}\left(\mathbb{1}_{w \le t} - \Phi(t)\right) \\
\implies e^{-\frac{w^2}{2}} f(w) &= \int_{-\infty}^{w} e^{-\frac{x^2}{2}}\left(\mathbb{1}_{x \le t} - \Phi(t)\right) dx + C \\
f(w) &= e^{\frac{w^2}{2}} \int_{-\infty}^{w} e^{-\frac{x^2}{2}}\left(\mathbb{1}_{x \le t} - \Phi(t)\right) dx + Ce^{\frac{w^2}{2}}.
\end{aligned}$$

[12]This proof technique generalizes integration by parts: the key step is multiplying by the integrating factor $e^{-w^2/2}$ to convert Stein's ODE into an exact derivative.

Taking $C = 0$ gives the standard admissible solution used in Stein's method. $\square$

Remark 5.3.4. In our proof, where did we use that Φ was the Gaussian CDF? Couldn't we have replaced Φ by a different CDF G, then gotten that $W \sim G$?

No. A Gaussian RV is the only one that will satisfy $\mathbb{E}[f'(Z)] = \mathbb{E}[Zf(Z)]$ for all admissible f. After setting $C = 0$, we require

$$\begin{aligned}\infty &> \mathbb{E}[Zf(Z)] \\ &= \int_{-\infty}^{\infty} w \frac{1}{\sqrt{2\pi}} e^{-\frac{w^2}{2}} f(w) dw \\ &= \int_{-\infty}^{\infty} w \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{w} e^{-\frac{x^2}{2}} \left(\mathbb{1}_{x \le t} - G(t)\right) dx\, dw.\end{aligned}$$

This necessitates that $\lim_{w\to\infty} \int_{-\infty}^{w} e^{-\frac{x^2}{2}} \left(\mathbb{1}_{x\le t} - G(t)\right) dx = 0$. (Otherwise, we would have linear growth coming from the w term in the outer integral.)

Therefore,

$$G(t) = \frac{\int_{-\infty}^{\infty} \mathbb{1}_{x\le t} e^{-\frac{x^2}{2}} dx}{\int_{-\infty}^{\infty} e^{-\frac{x^2}{2}} dx} = \Phi(t).$$

Aside 5.3.5. The converse implies that we can use discrepancy in Stein's equality as a criterion for convergence to $N(0,1)$: a random variable W is standard normal if and only if $\mathbb{E}[f'(W)] - \mathbb{E}[Wf(W)] = 0$ for all admissible f. So to show $W_n = \frac{1}{\sqrt{n}} \sum X_i \rightsquigarrow N(0,1)$, it suffices to show this discrepancy vanishes as $n \to \infty$.

This gives a different strategy from Lindeberg swapping. In Lindeberg swapping, we bounded the direct distance between distributions:

$$\left| \mathbb{E}\left[f\left(\sum_i \frac{X_i}{\sqrt{n}} \right) \right] - \mathbb{E}\left[f\left(N(0,1)\right) \right] \right|.$$

With Stein's method, we instead show W_n satisfies the Gaussian characterization by bounding

$$\left| \mathbb{E}\left[f'(W_n)\right] - \mathbb{E}\left[W_n f(W_n)\right] \right|.$$

For Theorem (**??**), we require the following lemma and definition.

Lemma 5.3.6.

We have that $\|f_h\|_\infty$, $\|f_h'\|_\infty$, $\|f_h''\|_\infty \le 2\|h'\|_\infty$.

Proof of Lemma 5.3.6. We just prove that $\|f_h\|_\infty \le 2\,\mathrm{Lip}(h)$; the derivative bounds are standard and follow by differentiating Stein's equation and using the same Gaussian-tail estimates. Recall

$$f_h(w) = e^{\frac{w^2}{2}} \int_{-\infty}^{w} e^{-\frac{x^2}{2}} \left(h(x) - Nh\right) dx.$$

When $w > 0$, the identity

$$\int_{-\infty}^{\infty} e^{-\frac{x^2}{2}} \left(h(x) - Nh\right) dx = 0$$

lets us rewrite

$$f_h(w) = -e^{\frac{w^2}{2}} \int_{w}^{\infty} e^{-\frac{x^2}{2}} \left(h(x) - Nh\right) dx.$$

Hence

$$\begin{aligned}|f_h(w)| &\le e^{\frac{w^2}{2}} \int_w^\infty e^{-\frac{x^2}{2}} |h(x) - h(0)| \, dx + e^{\frac{w^2}{2}} |h(0) - Nh| \int_w^\infty e^{-\frac{x^2}{2}} dx \\ &\le \mathrm{Lip}(h) e^{\frac{w^2}{2}} \int_w^\infty x e^{-\frac{x^2}{2}} dx + \mathrm{Lip}(h) \mathbb{E}\left[|Z|\right] e^{\frac{w^2}{2}} \int_w^\infty e^{-\frac{x^2}{2}} dx.\end{aligned}$$

Using

$$\int_w^\infty x e^{-\frac{x^2}{2}} dx = e^{-\frac{w^2}{2}}$$

and the standard Gaussian-tail bound

$$e^{\frac{w^2}{2}} \int_w^\infty e^{-\frac{x^2}{2}} dx \le \sqrt{\frac{\pi}{2}},$$

we obtain

$$|f_h(w)| \le \mathrm{Lip}(h) \left(1 + \sqrt{\frac{\pi}{2}} \mathbb{E}\left[|Z|\right]\right) = 2\,\mathrm{Lip}(h).$$

By symmetry, the same bound holds for $w < 0$, hence $\|f_h\|_\infty \le 2\,\mathrm{Lip}(h)$. □

Definition 5.3.7 (Wasserstein distance).

Suppose we have two distributions $W \sim P$ and $Z \sim Q$. The Wasserstein distance is defined between distributions (but often notated as the distance between random variables) and is

$$(\mathscr{W}(W, Z) =) \mathscr{W}(P, Q) := \sup_{\mathrm{Lip}(h) \le 1} \left|\mathbb{E}\left[h(W)\right] - \mathbb{E}\left[h(Z)\right]\right|.$$

We see that convergence in Wasserstein distance implies weak convergence.

We can prove the CLT with Stein's method instead of Lindeberg swapping. The key step is obtaining an explicit bound on the Wasserstein distance $\mathscr{W}(W_n, N(0,1))$ where $W_n = (X_1 + \cdots + X_n)/\sqrt{n}$; the CLT then follows by a triangular array argument.

Theorem 5.3.8 (Wasserstein distance bound; Wasserstein CLT).

Suppose $X_1, \ldots, X_n$ are independent with $\mathbb{E}\left[X_i\right] = 0$ and $\mathbb{E}\left[X_i^2\right] = 1$. Then

$$\mathscr{W}\left(\frac{X_1 + \cdots + X_n}{\sqrt{n}}, N(0,1)\right) \le \frac{3}{\sqrt{n}} \frac{1}{n} \sum_{i=1}^n \mathbb{E}\left[|X_i|^3\right].$$

In particular, if $\frac{1}{n^{\frac{3}{2}}} \sum_{i=1}^n \mathbb{E}\left[|X_i|^3\right] \to 0$, then

$$\frac{\sum_{i=1}^n X_i}{\sqrt{n}} \rightsquigarrow N(0,1).$$

Proof of Theorem 5.3.8. Consider the more general form of Stein's equation, where we replace the indicator with some function $h(\cdot)$ of w:

$$f'(w) - w f(w) = h(w) - Nh,$$

where $Nh := \mathbb{E}\left[h(Z)\right]$ for $Z \sim N(0,1)$.

We can follow the previous derivation to get a solution (dependent on h) as

$$f_h(w) = e^{\frac{w^2}{2}} \int_{-\infty}^{w} e^{-\frac{x^2}{2}} \left(h(x) - Nh\right) dx.$$

The definition of Wasserstein distance between W and Z is

$$\mathscr{W}(W, Z) = \sup_{\mathrm{Lip}(h) \le 1} \left|\mathbb{E}\left[h(W)\right] - \mathbb{E}\left[h(Z)\right]\right|.$$

Since Stein's equation $f_h'(w) - w f_h(w) = h(w) - Nh$ holds pointwise for every w, we may take expectations to get

$$\mathbb{E}\left[f_h'(W) - W f_h(W)\right] = \mathbb{E}\left[h(W) - Nh\right] = \mathbb{E}\left[h(W)\right] - \mathbb{E}\left[h(Z)\right].$$

Therefore

$$\begin{aligned}
\mathscr{W}(W, Z) &= \sup_{\mathrm{Lip}(h) \le 1} \left|\mathbb{E}\left[f_h'(W) - W f_h(W)\right]\right| \\
&\le \sup_{\substack{\|f\|_\infty \le 2 \\ \|f'\|_\infty \le 2 \\ \|f''\|_\infty \le 2}} \left|\mathbb{E}\left[f'(W)\right] - \mathbb{E}\left[W f(W)\right]\right|.
\end{aligned}$$

Suppose $X_1, \ldots, X_n$ are independent[13] with $\mathbb{E}\left[X_i\right] = 0$, $\mathrm{Var}\left(X_i\right) = 1$, and $\mathbb{E}\left[|X_i|^3\right] < \infty$. Define

$$W = \frac{X_1 + \cdots + X_n}{\sqrt{n}}, \quad W_i = W - \frac{X_i}{\sqrt{n}},$$

which gives $X_i \perp\!\!\!\perp W_i$.

Then

$$\begin{aligned}
\mathbb{E}\left[W f(W)\right] &= \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \mathbb{E}\left[X_i f(W)\right] \\
\mathbb{E}\left[X_i f(W)\right] &= \mathbb{E}\left[X_i \left(f(W) - f(W_i)\right)\right] + \underbrace{\mathbb{E}\left[X_i f(W_i)\right]}_{=\mathbb{E}[X_i] f(W_i) = 0} \\
&= \mathbb{E}\left[X_i \left(f(W) - f(W_i)\right)\right] \\
&= \mathbb{E}\left[X_i \left(f(W) - f(W_i) - f'(W_i)(W - W_i)\right)\right] + \mathbb{E}\left[X_i f'(W_i)(W - W_i)\right] \\
&= \frac{1}{2}\mathbb{E}\left[X_i f''(\xi_i)\left(W - W_i\right)^2\right] + \frac{1}{\sqrt{n}}\mathbb{E}\left[X_i^2 f'(W_i)\right] \\
&= \frac{1}{2n}\mathbb{E}\left[X_i^3 f''(\xi_i)\right] + \frac{1}{\sqrt{n}}\mathbb{E}\left[f'(W_i)\right]
\end{aligned}$$

by independence of X_i and W_i and $(W - W_i) = X_i$.

Put

$$A_n = \frac{1}{2n\sqrt{n}} \sum_{i=1}^{n} \mathbb{E}\left[X_i^3 f''(\xi_i)\right], \qquad B_n = \frac{1}{n} \sum_{i=1}^{n} \mathbb{E}\left[f'(W_i)\right].$$

Then

$$\mathbb{E}\left[W f(W)\right] = A_n + B_n.$$

[13] Notably, not necessarily identically distributed.

Putting

$$m_1 = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\left[|X_i|\right], \qquad m_3 = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\left[|X_i|^3\right],$$

we bound A_n and $\mathbb{E}[f'(W)] - B_n$ separately. For A_n, using $\|f''\|_\infty \le 2$:

$$|A_n| \le \frac{\|f''\|_\infty}{2n\sqrt{n}}\sum_{i=1}^{n}\mathbb{E}\left[|X_i|^3\right] \le \frac{m_3}{\sqrt{n}}.$$

For $\mathbb{E}[f'(W)]-B_n$, the mean value theorem and $W-W_i = X_i/\sqrt{n}$ give $|f'(W) - f'(W_i)| \le \|f''\|_\infty |W - W_i| = 2|X_i|/\sqrt{n}$, so

$$\left|\mathbb{E}\left[f'(W)\right] - B_n\right| \le \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\left|f'(W) - f'(W_i)\right| \le \frac{2}{n\sqrt{n}}\sum_{i=1}^{n}\mathbb{E}\left|X_i\right| = \frac{2m_1}{\sqrt{n}}.$$

Combining via the triangle inequality:

$$\left|\mathbb{E}\left[f'(W)\right] - \mathbb{E}\left[Wf(W)\right]\right| \le \frac{1}{\sqrt{n}}\left(2m_1 + m_3\right) \le \frac{3m_3}{\sqrt{n}}.$$

Note that in bounding the first moment by the third moment of X_i, we use

$$\mathbb{E}\left[|X_i|\right] \le \left(\mathbb{E}\left[|X_i|^3\right]\right)^{\frac{1}{3}},$$

which follows by Jensen's inequality, since $t \mapsto t^3$ is convex on $[0,\infty)$. By assumption,

$$\mathbb{E}[X_i^2] = 1 \le \left(\mathbb{E}\left[|X_i|^3\right]\right)^{2/3},$$

so $\mathbb{E}\left[|X_i|^3\right] \ge 1$; plugging this back into the previous equation shows $\mathbb{E}\left[|X_i|\right] \le \mathbb{E}\left[|X_i|^3\right]$.

Substituting m_3 into the bound on the distance,

$$\mathscr{W}\left(\frac{X_1 + \cdots + X_n}{\sqrt{n}}, N(0,1)\right) \le \frac{3}{\sqrt{n}} \cdot \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\left[|X_i|^3\right].$$

This bound holds for every fixed n and is non-asymptotic.

To deduce the CLT, put the problem into triangular-array notation: for each n, let $X_{n,1}, \ldots, X_{n,n}$ be independent with $\mathbb{E}[X_{n,i}] = 0$, $\mathbb{E}[X_{n,i}^2] = 1$, and set $W_n = n^{-1/2}\sum_{i=1}^{n} X_{n,i}$. Note that

$$S_n = \sqrt{\sum_{i=1}^{n}\mathbb{E}\left[X_{n,i}^2\right]} = \sqrt{\sum_{i=1}^{n} 1} = \sqrt{n},$$

so for $\delta = 1$,

$$\frac{1}{S_n^{2+\delta}}\sum_{i=1}^{n}\mathbb{E}\left[|X_{n,i}|^{2+\delta}\right] = \frac{1}{n^{\frac{3}{2}}}\sum_{i=1}^{k}\mathbb{E}\left[|X_{n,i}|^3\right].$$

If this expression goes to 0 as $n \to \infty$, then the Wassterstein bound gives

$$\mathscr{W}\left(W_n, N(0,1)\right) \to 0,$$

[14]and since Wasserstein convergence implies weak convergence, $W_n \xrightarrow{d} N(0,1)$.

[14]The Lyapunov condition with $\delta = 1$ is satisfied. We only presented this as a Lyapunov condition to relate this way of proving the CLT to a way that we already know. The convergence condition required for the Wasserstein bound to go to 0 is the same as a

In the classical i.i.d. case with $X_1, X_2, \ldots$ i.i.d., $\mathbb{E}[X_i] = 0$, $\mathbb{E}[X_i^2] = 1$, $\mathbb{E}[|X_i|^3] < \infty$: set $X_{n,i} = X_i$ for all n, so $L_n = \mathbb{E}[|X_1|^3]/\sqrt{n} \to 0$, giving $\mathscr{W}(W_n, N(0,1)) \leq 3\mathbb{E}[|X_1|^3]/\sqrt{n} \to 0$. □

Definition 5.3.9 (Kolmogorov-Smirnov distance).

Suppose $W \sim P$ and $Z \sim Q$. Then we define the KS distance as

$$\begin{aligned} \mathrm{KS}(P,Q) &:= \sup_t |P(-\infty, t] - Q(-\infty, t]| \\ &= \sup_t |\mathbb{P}[W \leq t] - \mathbb{P}[Z \leq t]| \end{aligned}.$$

Proposition 5.3.10.

Suppose $Z \sim N(0,1)$. Then for every $\alpha > 0$,

$$\mathrm{KS}(W,Z) = \sup_t |\mathbb{P}[W \leq t] - \mathbb{P}[Z \leq t]| \leq \frac{1}{\alpha}\mathscr{W}(W,Z) + \alpha.$$

Hence,

$$\mathrm{KS}(W,Z) \leq 2\sqrt{\mathscr{W}(W,Z)}.$$

The KS distance can be bounded by twice the square root of the Wasserstein distance. (Details to follow in next lecture.)

Proof of Proposition 5.3.10. Fix $t \in \mathbb{R}$, and define $h_{t,\alpha}$ by

$$h_{t,\alpha}(w) = \begin{cases} 1, & w \leq t \\ 1 - \frac{w-t}{\alpha}, & t < w < t + \alpha \\ 0, & w \geq t + \alpha \end{cases}.$$

Then $\mathrm{Lip}(h_{t,\alpha}) = \frac{1}{\alpha}$ and

$$\mathbb{1}_{w \leq t} \leq h_{t,\alpha}(w) \leq \mathbb{1}_{w \leq t+\alpha}.$$

Hence

$$\begin{aligned} \mathbb{P}[W \leq t] - \mathbb{P}[Z \leq t] &\leq \mathbb{E}[h_{t,\alpha}(W)] - \mathbb{E}[h_{t,\alpha}(Z)] + \mathbb{P}[t < Z \leq t + \alpha] \\ &\leq \frac{1}{\alpha}\mathscr{W}(W,Z) + \alpha. \end{aligned}$$

Since the same bound holds with W and Z interchanged, taking absolute values and then the supremum over t yields the first claim. The second follows by choosing $\alpha = \sqrt{\mathscr{W}(W,Z)}$. □

[15]

Lyapunov convergence condition.

[15]The Berry–Esseen bound is uniform across all t. Furthermore, the existence of third moment is necessary for this uniform convergence instead of just weak convergence, which does not require existence of the third moment.

Corollary 5.3.11.

Under the same setting,

$$\mathrm{KS}\left(\frac{X_1 + \cdots + X_n}{\sqrt{n}}, N(0,1)\right) \leq 2\sqrt{\frac{3}{\sqrt{n}}\frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\left[|X_i|^3\right]}.$$

We can get a sharper bound on the KS distance instead of the previous bound, which used Stein's method.

Proposition 5.3.12 (Berry–Esseen).

We have

$$\mathrm{KS}\left(\frac{X_1 + \cdots + X_n}{\sqrt{n}}, N(0,1)\right) \leq \frac{C}{\sqrt{n}}\frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\left[|X_i|^3\right].$$

Section 6

Marcus Kuo

Poisson Limit Theorem

We eventually want to show that

$$\mathrm{Bin}\,(n,p) \rightsquigarrow \mathrm{Poi}(\lambda)$$

when $np \to \lambda$.

6.1 Coupling

We introduce the notion of *coupling*. Suppose

$$X : (\Omega_1, \mathscr{F}_1, \mathbb{P}_1) \to (\mathbb{R}, \mathscr{B}(\mathbb{R}))$$
$$Y : (\Omega_2, \mathscr{F}_2, \mathbb{P}_2) \to (\mathbb{R}, \mathscr{B}_{\mathbb{R}}).$$

It would be nice if X and Y lived on the same space.

Definition 6.1.1 (Coupling).

The pair $\left(\hat{X}, \hat{Y}\right)$ is a coupling of (X, Y) if $\hat{X}$ and $\hat{Y}$ are on the same space and

$$\hat{X} \overset{\mathrm{d}}{=} X, \quad \hat{Y} \overset{\mathrm{d}}{=} Y.$$

Example 6.1.2.

Suppose $X \sim N(0,1)$ and $Y \sim N(1,1)$. One coupling is

$$\hat{X} = X, \quad \hat{Y} = X + 1.$$

There are many possible couplings for the above example. Are any couplings more notable than others?

Definition 6.1.3 (Maximal coupling).

We say $(\hat{X}, \hat{Y})$ is a maximal coupling of (X, Y) if

$$\mathbb{P}\left[\hat{X} = \hat{Y}\right]$$

is maximized.

Clearly, the previous example has $\mathbb{P}\left[\hat{X} = \hat{Y}\right] = 0$. How do we construct the maximal coupling?

Theorem 6.1.4 (Construction of maximal coupling, discrete RVs).

Suppose X, Y are both discrete and notate

$$\mathbb{P}[X = i] = p_i, \quad \mathbb{P}[Y = i] = q_k.$$

The maxmimal coupling satisfies

$$\mathbb{P}\left[\hat{X} \neq \hat{Y}\right] = \sum_i p_i \wedge q_i.$$

[12]

Proof of Theorem 6.1.4. Step I (Showing an upper bound). Define $\alpha = \sum_i p_i \wedge q_i$. For any coupling $(\hat{X}, \hat{Y})$ of (X, Y), define

$$A = \{i : p_i < q_i\}.$$

Then we can write

$$\begin{aligned}
\mathbb{P}\left[\hat{X} = \hat{Y}\right] &= \mathbb{P}\left[\hat{X} = \hat{Y} \in A\right] + \mathbb{P}\left[\hat{X} = \hat{Y} \in A^c\right] \\
&\leq \mathbb{P}\left[\hat{X} \in A\right] + \mathbb{P}\left[\hat{Y} \in A^c\right] \\
&= \mathbb{P}[X \in A] + \mathbb{P}[Y \in A^c] \\
&= \sum_{i \in A} p_i + \sum_{i \in A^c} q_i \\
&= \sum_i p_i \wedge q_i = \alpha.
\end{aligned}$$

Step II (Showing the upper bound is achieved). Define

$$\begin{cases} b_i &= \frac{p_i \wedge q_i}{\alpha} \\ c_i &= \frac{p_i - p_i \wedge q_i}{1-\alpha} \\ d_i &= \frac{q_i - p_i \wedge q_i}{1-\alpha} \end{cases}.$$

We see that all three of b_i, c_i, d_i are probabilities because they are nonnegative and individually sum to 1.[3]

Generate independent random variables B, C, D such that

$$\mathbb{P}[B = i] = b_i, \quad \mathbb{P}[C = i] = c_i, \quad \mathbb{P}[D = i] = d_i.$$

Then generate a binary indicator $I \sim \text{Bernoulli}(\alpha)$.

If $I = 1$, then construct $\hat{X} = \hat{Y} = B$. If $I = 0$, then put $\hat{X} = C, \hat{Y} = D$.[4] We have constructed $\hat{X}, \hat{Y}$ on the same space. We need to check that it is a coupling of (X, Y) and that it is indeed maximal.[5]

[1] The same argument applied to continuous RVs if we work with densities; take 383.

[2] The LHS is called total variation affinity/distance.

[3] The denominators are scaling factors.

[4] If $I = 1$, which happens with probability α, $\hat{X}$ and $\hat{Y}$ agree. Otherwise, with probability $1 - \alpha$, they disagree, as if the probability of $C = i$, c_i is nonzero, the probability od $D = i$, d_i is zero.

[5] When making a maximal coupling, we begin with marginal distributions of X and Y and want to find the joint distribution such that they are as dependent as possible.

We can write

$$\begin{aligned}\mathbb{P}\left[\hat{X}=i\right] &= \mathbb{P}\left[\hat{X}=i \mid I=1\right]\mathbb{P}\left[I=1\right]+\mathbb{P}\left[\hat{X}=i \mid I=0\right]\mathbb{P}\left[I=0\right]\\ &= \mathbb{P}\left[B=i\right]\mathbb{P}\left[I=1\right]+\mathbb{P}\left[C=i\right]\mathbb{P}\left[I=0\right]\\ &= \frac{p_i\wedge q_i}{\alpha}\alpha+\frac{p_i-p_i\wedge q_i}{1-\alpha}(1-\alpha)\\ &= p_i=\mathbb{P}\left[X=i\right].\end{aligned}$$

Similarly,

$$\mathbb{P}\left[\hat{Y}=i\right]=q_i=\mathbb{P}\left[Y=i\right].$$

Therefore,

$$\mathbb{P}\left[\hat{X}=\hat{Y}\right]\geq\mathbb{P}\left[I=1\right]=\alpha,$$

so $(\hat{X},\hat{Y})$ is maximal. □

6.2 LeCam and Stein–Chen Poisson Limit Theorems

Definition 6.2.1 (Total variation).

For random variables P, Q on the same measure space $(\Omega,\mathscr{F})$, define

$$\mathrm{TV}(P,Q)=\sup_{B\in\mathscr{F}}\left|P(B)-Q(B)\right|.$$

Theorem 6.2.2.

Suppose P, Q are discrete random variables on $(\Omega,\mathscr{F})$. The maximial coupling $(\hat{X},\hat{Y})$ of (P,Q) satisfies

$$\mathbb{P}\left[\hat{X}\neq\hat{Y}\right]=\mathrm{TV}(P,Q).$$

Proof of Theorem 6.2.2. It suffices to show $\mathrm{TV}(P,Q)=1-\underbrace{\sum_i p_i\wedge q_i}_{\text{total variation affinity}}$.

Again, let $A=\{i:p_i<q_i\}$. Since $A\in\mathscr{F}$, the definition of total variation gives

$$\mathrm{TV}(P,Q)\geq|P(A)-Q(A)|=Q(A)-P(A).$$

For every B,

$$\begin{aligned}
|P(B) - Q(B)| &= \left|\sum_{i\in B}(p_i - q_i)\right| \\
&= \left|\sum_{i\in B\cap A}\underbrace{(p_i - q_i)}_{\leq 0} + \sum_{i\in B\cap A^c}\underbrace{(p_i - q_i)}_{\geq 0}\right| \\
&= \left|\sum_{i\in B\cap A^c}(p_i - q_i) - \sum_{i\in B\cap A}(q_i - p_i)\right| \\
&\leq \max\left\{\sum_{i\in B\cap A^c}(p_i - q_i), \sum_{i\in B\cap A}(q_i - p_i)\right\} \\
&\leq \max\left\{\sum_{i\in A}(p_i - q_i), \sum_{i\in A} q_i - p_i\right\} \\
&= \max\{1 - P(A) - (1 - Q(A)), Q(A) - P(A)\} \\
&= Q(A) - P(A).
\end{aligned}$$

Taking a sup over all $B \in \mathscr{F}$, we have

$$\mathrm{TV}(P, Q) \leq Q(A) - P(A).$$

Combining with the other direction we already showed, we have equality.

Since

$$\begin{aligned}
\mathrm{TV}(P, Q) &= Q(A) - P(A) \\
\sum_i p_i \wedge q_i &= \sum_{i\in A} p_i + \sum_{i\in A^c} q_i \\
&= P(A) + Q(A^c)
\end{aligned}$$

sum to 1, we see that

$$\mathrm{TV}(P, Q) + \sum_i p_i \wedge q_i = 1,$$

and subtract $\sum_i p_i \wedge q_i$ to conclude. □

Theorem 6.2.3 (LeCam Poisson Limit Theorem).

Suppose $X_i \sim \mathrm{Bernoulli}(p_i)$ independently for all $i = 1, \ldots, n$ and define $\lambda = \sum_{i=1}^n p_i$. Then

$$\mathrm{TV}\left(\sum_{i=1}^n X_i, \mathrm{Poi}(\lambda)\right) \leq \sum_{i=1}^n p_i^2.$$

[6]

[6] Like other notions of distance, the total variation between two random variables is really the total variation between their distributions.

Example 6.2.4.

Suppose $p_i = \frac{\lambda}{n}$. Then we have

$$\mathrm{TV}\left(\mathrm{Bin}(n, \frac{\lambda}{n}), \mathrm{Poi}(\lambda)\right) \leq \frac{\lambda^2}{n} \to 0.$$

The strength of this result is that it only relies on independence. We allow p_i to vary, have an explicit bound on the total variation, and have that total variation goes to zero, which implies weak convergence.[a]

[a] Total variation is the distance in measures across all sets, including half intervals. So we have convergence in CDF and thus weak convergence.

Proof of Theorem 6.2.3. Define $Z_i \sim \mathrm{Poi}(p_i)$ independently for all i.

Then

$$\sum_{i=1}^{n} Z_i \sim \mathrm{Poi}(\sum_{i=1}^{n} p_i) = \mathrm{Poi}(\lambda).$$

So it suffices to bound the total variation between $\sum_i X_i$ and $\sum_i Z_i$. By Theorem (**??**), this is equal to the probability that the maximal coupling variables are not equal.

Let $\left(\hat{X}_i, \hat{Z}_i\right)$ be the maximal coupling of (X_i, Z_i). Then by our theorem,

$$\begin{aligned}
\mathbb{P}\left[\hat{X}_i = \hat{Z}_i\right] &= 1 - TV\left(X_i, Z_i\right) \\
&= \sum_j \mathbb{P}\left[X_i = k\right] \wedge \mathbb{P}\left[Z_i = k\right] \\
\mathbb{P}\left[Z_i = k\right] = \frac{p_i^k}{k!} e^{-p_i}, \quad \mathbb{P}\left[X_i = k\right] &= \begin{cases} p_i, & k = 1 \\ 1 - p_i, & k = 0 \end{cases} \\
\implies \mathbb{P}\left[\hat{X}_i = \hat{Z}_i\right] &= \min\left\{1 - p_i, e^{-p_i}\right\} + \min\left\{p_i, p_i e^{-p_i}\right\} \\
&= 1 - p_i + p_i e^{-p_i} \\
&\geq 1 - p_i + p_i(1 - p_i) \\
&= 1 - p_i^2.
\end{aligned}$$

Note that

$$\left(\sum_{i=1}^{n} \hat{X}_i, \sum_{i=1}^{n} \hat{Z}_i\right)$$

is a coupling of

$$\left(\sum_{i=1}^{n} X_i, \sum_{i=1}^{n} Z_i\right).$$

It need not be maximal, but it suffices for the first inequality in

$$\begin{aligned}
\mathrm{TV}\left(\sum_{i=1}^n X_i, \mathrm{Poi}(\lambda)\right) &= \mathrm{TV}\left(\sum_{i=1}^n X_i, \sum_{i=1}^n Z_i\right) \\
&= \mathbb{P}\left[\widehat{\sum_{i=1}^n X_i} \neq \widehat{\sum_{i=1}^n Z_i}\right] \\
&\leq \mathbb{P}\left[\sum_{i=1}^n \hat{X}_i \neq \sum_{i=1}^n \hat{Z}_i\right] \\
&\leq \sum_{i=1}^n \mathbb{P}\left[\hat{X}_i \neq \hat{Z}_i\right] \\
&\leq \sum_{i=1}^n p_i^2.
\end{aligned}$$

□

Lemma 6.2.5 (Stein-Chen).

We have that

$$Z \sim \mathrm{Poi}(\lambda) \iff \forall\ f, \mathbb{E}\left[Zf(Z)\right] = \lambda \mathbb{E}\left[f(Z+1)\right].$$

[7]

Proof of Lemma 6.2.5. ($\Longrightarrow$) Since we have for every $k = 0, 1, 2, \ldots$

$$\mathbb{P}\left[Z = k\right] = \frac{\lambda^k}{k!} e^{-\lambda},$$

we know

$$\begin{aligned}
\mathbb{E}\left[Zf(Z)\right] &= \sum_{k=0}^\infty k f(k) \frac{\lambda^k}{k!} e^{-\lambda} \\
&= \sum_{k=1}^\infty k f(k) \frac{\lambda^k}{k!} e^{-\lambda} \\
&= \lambda \sum_{k=1}^\infty f(k) \frac{\lambda^{k-1}}{(k-1)!} e^{-\lambda} \\
&= \lambda \sum_{k=0}^\infty f(k+1) \frac{\lambda^k}{k!} e^{-\lambda} = \lambda \mathbb{E}\left[f(Z+1)\right].
\end{aligned}$$

($\Longleftarrow$) We use a difference equation instead of the differential equation in Stein's method. Consider for each $k = 0, 1, 2, \ldots$ and the equation

$$\lambda f(k+1) - k f(k) = \mathbb{1}_{k \in A} - \mathbb{P}\left[Z \in A\right] \tag{6.1}$$

for some set A. If[8] we can show that this equation has a solution f_A, then we know for a random variable W with

[7] The forward direction lets f be any function such that the expectations exist. A minimal assumption $|Zf(Z)| < \infty$. The backwards direction requires the identity hold for all bounded $f : \mathbb{N}_0 \to \mathbb{R}$.

[8] If W satisfies the Stein–Chen identity for every f, then plugging in the solution f_A forces $\mathbb{P}(W \in A) = \mathbb{P}(Z \in A)$ for every set A, hence $W \sim \mathrm{Poi}(\lambda)$.

distribution satisfying the Stein–Chen identity for all f, we can plug in $f = f_A$ to get

$$\mathbb{P}[W \in A] = \mathbb{P}[Z \in A],$$

so $W \sim \text{Poi}(\lambda)$.

We would like to get equation (**??**) into a form of $g(k+1) - g(k) = [\cdots]$, since then we can sum the right hand side to get g. Multiplying each side of (**??**) by $\frac{\lambda^k}{k!}$, we have that for $k = 1, 2, \ldots$,

$$\frac{\lambda^{k+1}}{k!} f(k+1) - \frac{\lambda^k}{(k-1)!} f(k) = \frac{\lambda^k}{k!} \left(\mathbb{1}_{k \in A} - \mathbb{P}[Z \in A]\right), \tag{6.2}$$

and for $k = 0$,

$$\lambda f(1) = \mathbb{1}_{0 \in A} - \mathbb{P}[Z \in A].$$

For $g(k) = \frac{\lambda^k}{(k-1)!} f(k)$, we have that

$$\begin{cases} g(k+1) - g(k) = \frac{\lambda^k}{k!} \left(\mathbb{1}_{k \in A} - \mathbb{P}[Z \in A]\right), & k = 1, 2, \ldots \\ g(1) = \mathbb{1}_{0 \in A} - \mathbb{P}[Z \in A] \end{cases},$$

so we sum to get

$$g(k+1) = \sum_{j=0}^{k} \frac{\lambda^j}{j!} \left(\mathbb{1}_{j \in A} - \mathbb{P}[Z \in A]\right), \quad k = 0, 1, 2, \ldots.$$

Therefore, our solution to the original difference equation is

$$f_A(k+1) = \frac{k!}{\lambda^{k+1}} \sum_{j=0}^{k} \frac{\lambda^j}{j!} \left(\mathbb{1}_{j \in A} - \mathbb{P}[Z \in A]\right), \quad k = 0, 1, 2, \ldots.$$

Note that we don't actually need the solution $f(0)$, since the right hand side of the goal equation does not involve $f(0)$ and the left hand side multiplies it by 0.[9]

Let $Z \sim \text{Poi}(\lambda)$. Suppose that for some random variable W, for any f,

$$\mathbb{E}[W f(W)] = \lambda \mathbb{E}[f(W+1)].$$

Plugging in our f_A, we know

$$\begin{aligned} 0 &= \mathbb{E}[\lambda f_A(W+1) - W f_A(W)] \\ &= \mathbb{E}[\mathbb{1}_{W \in A} - \mathbb{P}[Z \in A]] \\ &= \mathbb{P}[W \in A] - \mathbb{P}[Z \in A]. \end{aligned}$$

Since A is arbitrary, we know $W \sim \text{Poi}(\lambda)$. □

Lemma 6.2.6.

The solution f_A to the Stein-Chen equation is

$$f_A(k+1) = \frac{k!}{\lambda^{k+1}} \sum_{j=0}^{k} \frac{\lambda^j}{j!} \left(\mathbb{1}_{j \in A} - \mathbb{P}[Z \in A]\right),$$

[9] Review why the term $\mathbb{1}_{j \in A} - \mathbb{P}[Z \in A]$ is necessary (the sum must be 0).

and it satisfies, for integers k, l,

$$|f_A(k) - f_A(l)| \leq \min\left\{1, \frac{1}{\lambda}\right\} |k - l|.$$

Proof of Lemma 6.2.6. The formula for f_A was established in the proof of the ($\Longleftarrow$) direction of the Stein–Chen lemma above. It remains to prove the Lipschitz bound.

Write $f_A(k+1) = \sum_{i \in A} f_i(k+1)$, where $f_i := f_{\{i\}}$. Substituting $A = \{i\}$ into the formula and using the Poisson probability $\mathbb{P}[Z = j] = \frac{\lambda^j}{j!} e^{-\lambda}$,

$$\begin{aligned} f_i(k+1) &= \frac{k!}{\lambda^{k+1}} \sum_{j=0}^{k} \frac{\lambda^j}{j!} (\mathbb{1}_{j=i} - \mathbb{P}[Z = i]) \\ &= \frac{k!}{\lambda^{k+1}} \cdot \frac{\lambda^i}{i!} (\mathbb{1}_{i \leq k} - \mathbb{P}[Z \leq k]) \\ &= \frac{\mathbb{P}[Z = i]}{\lambda \mathbb{P}[Z = k]} (\mathbb{1}_{i \leq k} - \mathbb{P}[Z \leq k]), \end{aligned}$$

where the second line uses $\sum_{j=0}^{k} \frac{\lambda^j}{j!} \mathbb{1}_{j=i} = \frac{\lambda^i}{i!} \mathbb{1}_{i \leq k}$ and

$$\mathbb{P}[Z = i] \sum_{j=0}^{k} \frac{\lambda^j}{j!} = \frac{\lambda^i}{i!} e^{-\lambda} \cdot e^{\lambda} \mathbb{P}[Z \leq k] = \frac{\lambda^i}{i!} \mathbb{P}[Z \leq k],$$

and the third uses

$$\frac{k!}{\lambda^{k+1}} \cdot \frac{\lambda^i}{i!} = \frac{e^{-\lambda}}{\lambda \mathbb{P}[Z = k]} \cdot \frac{\mathbb{P}[Z = i]}{e^{-\lambda}} = \frac{\mathbb{P}[Z = i]}{\lambda \mathbb{P}[Z = k]}.$$

We claim:

(i) $f_k(k+1) \geq f_k(k)$,

(ii) For $i \neq k$, $f_i(k+1) \leq f_i(k)$.

With these, we could bound $f_A(k+1) - f_A(k)$ from above: adding k to A increases the sum by (i), and each remaining $i \neq k$ term is non-positive by (ii), so

$$\begin{aligned} f_A(k+1) - f_A(k) &\leq f_{A \cup \{k\}}(k+1) - f_{A \cup \{k\}}(k) \\ &\leq f_k(k+1) - f_k(k). \end{aligned}$$

We evaluate using $\frac{\mathbb{P}[Z=k]}{\mathbb{P}[Z=k-1]} = \frac{\lambda}{k}$:

$$\begin{aligned} f_k(k+1) &= \frac{\mathbb{P}[Z > k]}{\lambda}, \\ f_k(k) &= -\frac{\mathbb{P}[Z = k]\mathbb{P}[Z \leq k-1]}{\lambda \mathbb{P}[Z = k-1]} = -\frac{\mathbb{P}[Z \leq k-1]}{k}, \end{aligned}$$

so

$$f_k(k+1) - f_k(k) = \frac{\mathbb{P}[Z > k]}{\lambda} + \frac{\mathbb{P}[Z \leq k-1]}{k}.$$

For each $0 \leq j \leq k-1$, the Poisson recurrence gives $\mathbb{P}[Z = j] = \frac{j+1}{\lambda} \mathbb{P}[Z = j+1]$, so $\frac{\mathbb{P}[Z=j]}{k} = \frac{j+1}{k} \cdot \frac{\mathbb{P}[Z=j+1]}{\lambda} \leq \frac{\mathbb{P}[Z=j+1]}{\lambda}$

since $j+1 \le k$. Summing,

$$\frac{\mathbb{P}[Z \le k-1]}{k} \le \frac{1}{\lambda}\sum_{j=0}^{k-1}\mathbb{P}[Z = j+1] = \frac{\mathbb{P}[1 \le Z \le k]}{\lambda}.$$

Therefore,

$$f_k(k+1) - f_k(k) \le \frac{\mathbb{P}[Z > k] + \mathbb{P}[1 \le Z \le k]}{\lambda} = \frac{\mathbb{P}[Z \ge 1]}{\lambda} = \frac{1-e^{-\lambda}}{\lambda}.$$

Since $1 - e^{-\lambda} \le 1$ and $e^{-\lambda} \ge 1 - \lambda$ gives $1 - e^{-\lambda} \le \lambda$,

$$f_k(k+1) - f_k(k) \le \frac{1-e^{-\lambda}}{\lambda} \le \min\left\{1, \frac{1}{\lambda}\right\}.$$

For the other direction, note that $\mathbb{P}[Z \in \mathbb{N}_0] = 1$, so applying the formula gives $f_{\mathbb{N}_0} \equiv 0$ and thus $f_{A^c} = -f_A$. Thus

$$f_A(k) - f_A(k+1) = f_{A^c}(k+1) - f_{A^c}(k) \le \min\left\{1, \frac{1}{\lambda}\right\},$$

by the same argument applied to A^c. Combining both directions,

$$\left|f_A(k+1) - f_A(k)\right| \le \min\left\{1, \frac{1}{\lambda}\right\},$$

and the triangle inequality extends this to $|f_A(k) - f_A(l)| \le \min\{1, \frac{1}{\lambda}\}|k-l|$.

It remains to prove (i) and (ii). From the formulas above, $f_k(k+1) = \frac{\mathbb{P}[Z>k]}{\lambda} \ge 0$ and $f_k(k) = -\frac{\mathbb{P}[Z \le k-1]}{k} \le 0$, so $f_k(k+1) \ge 0 \ge f_k(k)$, proving (i).

For (ii), when $i < k$ both numerators involve the tail:

$$f_i(k+1) = \frac{\mathbb{P}[Z=i]\mathbb{P}[Z>k]}{\lambda\mathbb{P}[Z=k]}, \qquad f_i(k) = \frac{\mathbb{P}[Z=i]\mathbb{P}[Z>k-1]}{\lambda\mathbb{P}[Z=k-1]},$$

so $f_i(k+1) \le f_i(k)$ iff $\frac{\mathbb{P}[Z>k]}{\mathbb{P}[Z=k]} \le \frac{\mathbb{P}[Z>k-1]}{\mathbb{P}[Z=k-1]}$. Writing

$$\frac{\mathbb{P}[Z>k]}{\mathbb{P}[Z=k]} = \sum_{l>0}\lambda^l\frac{k!}{(l+k)!},$$

each term $\frac{k!}{(l+k)!} = \frac{1}{(k+1)\cdots(k+l)}$ is decreasing in k, so the sum is decreasing in k, which gives $f_i(k+1) \le f_i(k)$.

When $i > k$ both numerators involve the CDF:

$$f_i(k+1) = -\frac{\mathbb{P}[Z=i]\mathbb{P}[Z \le k]}{\lambda\mathbb{P}[Z=k]}, \qquad f_i(k) = -\frac{\mathbb{P}[Z=i]\mathbb{P}[Z \le k-1]}{\lambda\mathbb{P}[Z=k-1]},$$

so $f_i(k+1) \le f_i(k)$ iff $\frac{\mathbb{P}[Z \le k]}{\mathbb{P}[Z=k]} \ge \frac{\mathbb{P}[Z \le k-1]}{\mathbb{P}[Z=k-1]}$. Writing

$$\frac{\mathbb{P}[Z \le k]}{\mathbb{P}[Z=k]} = \sum_{l=0}^{k}\lambda^{-l}\frac{k!}{(k-l)!},$$

both the number of terms and each factor $\frac{k!}{(k-l)!} = k(k-1)\cdots(k-l+1)$ increase in k, so the ratio is increasing in k, giving $f_i(k+1) \le f_i(k)$. $\square$

Proposition 6.2.7.

For $Z \sim \mathrm{Poi}(\lambda)$ and a random variable W, the total variation is

$$\begin{aligned}\mathrm{TV}(W, Z) &= \sup_A |\mathbb{P}[W \in A] - \mathbb{P}[Z \in A]| \\ &= \sup_A |\mathbb{E}[W f_A(W)] - \lambda \mathbb{E}[f_A(W+1)]| \\ &\leq \sup_{\mathrm{Lip}(f) \leq \min\left\{1, \frac{1}{\lambda}\right\}} |\mathbb{E}[W f(W) - \lambda \mathbb{E}[f(W+1)]]|.\end{aligned}$$

Proof of Proposition 6.2.7. The first equality is the definition of total variation, the second comes from the Stein–Chen identity, and the third from the Lipschitz bound we just derived. □

Theorem 6.2.8 (Stein-Chen).

Suppose we have independent $X_i \sim \mathrm{Bernoulli}(p_i)$ for $i = 1, \ldots, n$. Put $\lambda = \sum_{i=1}^n p_i$. Then we have

$$\mathrm{TV}\left(\sum_{i=1}^n X_i, \mathrm{Poi}(\lambda)\right) \leq \min\left\{1, \frac{1}{\lambda}\right\} \sum_{i=1}^n p_i^2.$$

We have a stronger result than LeCam's result because of the min term. When $\lambda \to \infty$ and $\sum_{i=1}^n p_i^2$ is controlled, we get that $\sum_{i=1}^n X_i \rightsquigarrow \mathrm{Poi}(\lambda)$.

Proof of Theorem 6.2.8. Let $W = X_1 + \cdots + X_n$, with $X_i \sim \mathrm{Bernoulli}(p_i)$ independent for $i = 1, \ldots, n$. Write $W_i = W - X_i$ and put $\lambda = \sum_{i=1}^n p_i$.

Step I (Decomposing the discrepancy). We decompose the Stein–Chen discrepancy as

$$\lambda \mathbb{E}[f(W+1)] - \mathbb{E}[W f(W)] = \sum_{i=1}^n p_i \mathbb{E}[f(W+1)] - \sum_{i=1}^n \mathbb{E}[X_i f(W_i + X_i)].$$

Since $X_i \in \{0, 1\}$ and $X_i \perp\!\!\!\perp W_i$, conditioning on W_i yields

$$\begin{aligned}\mathbb{E}[X_i f(W_i + X_i)] &= \mathbb{E}[\mathbb{E}[X_i f(W_i + X_i) \mid W_i]] \\ &= \mathbb{E}[p_i \cdot f(W_i + 1) + (1 - p_i) \cdot 0] \\ &= p_i \mathbb{E}[f(W_i + 1)],\end{aligned}$$

where the $X_i = 0$ term vanishes because $X_i \cdot f(W_i + X_i) = 0 \cdot f(W_i) = 0$ when $X_i = 0$. Substituting back,

$$\lambda \mathbb{E}[f(W+1)] - \mathbb{E}[W f(W)] = \sum_{i=1}^n p_i \left(\mathbb{E}[f(W+1)] - \mathbb{E}[f(W_i + 1)]\right).$$

Step II (Applying the Lipschitz bound). Since $W = W_i + X_i$ and $X_i \in \{0, 1\}$, we have $|W - W_i| = X_i$. The

Lipschitz bound on f_A from the lemma above then gives $|f(W+1) - f(W_i+1)| \leq \min\left\{1, \frac{1}{\lambda}\right\} X_i$, so

$$\begin{aligned}
|\lambda \mathbb{E}[f(W+1)] - \mathbb{E}[Wf(W)]| &\leq \sum_{i=1}^{n} p_i \mathbb{E}[|f(W+1) - f(W_i+1)|] \\
&\leq \min\left\{1, \frac{1}{\lambda}\right\} \sum_{i=1}^{n} p_i \mathbb{E}[X_i] \\
&= \min\left\{1, \frac{1}{\lambda}\right\} \sum_{i=1}^{n} p_i^2.
\end{aligned}$$

The total variation bound then follows from the proposition above. □

Example 6.2.9.

Consider the typical example $p_i = \frac{\lambda}{n}$. Then we get the Law of Small Numbers.

Proposition 6.2.10 (TV convergence implies weak convergence).

If $\mu_n \to \mu$ in total variation, i.e. $\mathrm{TV}(\mu_n, \mu) \to 0$, then for every bounded measurable function f,

$$\int f \, d\mu_n \to \int f \, d\mu.$$

In particular, $\mu_n \rightsquigarrow \mu$.

Proof of Proposition 6.2.10. For any set $A \in \mathscr{F}$,

$$\left| \int \mathbf{1}_A \, d\mu_n - \int \mathbf{1}_A \, d\mu \right| = |\mu_n(A) - \mu(A)| \leq \mathrm{TV}(\mu_n, \mu) \to 0,$$

so the result holds for indicators. By linearity it extends to simple functions. For general bounded measurable f, given $\varepsilon > 0$ choose a simple function s with $\|f - s\|_\infty < \varepsilon$. Then

$$\begin{aligned}
\left| \int f \, d\mu_n - \int f \, d\mu \right| &\leq \left| \int (f - s) \, d\mu_n \right| + \left| \int s \, d\mu_n - \int s \, d\mu \right| + \left| \int (s - f) \, d\mu \right| \\
&\leq 2\varepsilon + \left| \int s \, d\mu_n - \int s \, d\mu \right|.
\end{aligned}$$

The second term tends to 0 as $n \to \infty$. Since ε was arbitrary, $\int f \, d\mu_n \to \int f \, d\mu$. Weak convergence follows because bounded continuous functions are bounded and measurable. □

Example 6.2.11.

The boundedness assumption cannot be dropped. Let μ_n assign mass $\frac{1}{n}$ to $\{n\}$ and $1 - \frac{1}{n}$ to $\{0\}$, and let $\mu(x) = \mathbb{1}_0$. The only points with positive probability are 0 and n, so $\mathrm{TV}(\mu_n, \mu) = \frac{1}{n} \to 0$. However, taking $f(x) = x$ shows

$$\int f \, d\mu_n = n \cdot \frac{1}{n} + 0 \cdot \left(1 - \frac{1}{n}\right) = 1 \neq 0 = \int f \, d\mu$$

for every n, so $\int f \, d\mu_n \not\to \int f \, d\mu$.

6.3 Dependence

What if $X_1, \ldots, X_n$ are dependent?

Definition 6.3.1 (Dependency neighborhood).

For each $i \in \mathbb{N}$, consider $\{N_i\}$ where

$$i \in N_i \subseteq [n] := \{1, 2, \ldots, n\}$$

such that

$$X_i \perp\!\!\!\perp \{X_j\}_{j \notin N_i}.$$

We call N_i the dependency neighborhood of X_i.

Theorem 6.3.2 (Stein-Chen, dependent variables).

Suppose $X_i \sim \text{Bernoulli}(p)$ for $i = 1, \ldots, n$, where the X_is are not necessarily independent. Put $\lambda = \sum_{i=1}^n p_i$. Then for dependency neighborhoods N_i, we have the total variation bound

$$\text{TV}\left(\sum_{i=1}^n X_i, \text{Poi}(\lambda)\right) \leq \min\left\{1, \frac{1}{\lambda}\right\}\left(\sum_{i=1}^n \sum_{j \in N_i \setminus \{i\}} \mathbb{E}\left[X_i X_j\right] + \sum_{i=1}^n \sum_{j \in N_i} p_i p_j\right).$$

Proof of Theorem 6.3.2. Fix a Stein solution f with

$$|f(k+1) - f(k)| \leq \min\left\{1, \frac{1}{\lambda}\right\}.$$

For each i, write

$$\begin{aligned} W &= X_i + \sum_{j \in N_i \setminus \{i\}} X_j + \sum_{j \notin N_i} X_j \\ &=: X_i + \sum_{j \in N_i \setminus \{i\}} X_j + V_i. \end{aligned}$$

By assumption, $X_i \perp\!\!\!\perp V_i$.

Now decompose

$$\begin{aligned} \mathbb{E}\left[W f(W)\right] - \lambda \mathbb{E}\left[f(W+1)\right] &= \sum_{i=1}^n \left(\mathbb{E}\left[X_i f(W)\right] - p_i \mathbb{E}\left[f(W+1)\right]\right) \\ &= \text{(I)} + \text{(II)}, \end{aligned}$$

where

$$\begin{aligned} \text{(I)} &= \sum_{i=1}^n \left(\mathbb{E}\left[X_i f(W)\right] - \mathbb{E}\left[X_i f(V_i + 1)\right]\right), \\ \text{(II)} &= \sum_{i=1}^n \left(\mathbb{E}\left[X_i f(V_i + 1)\right] - p_i \mathbb{E}\left[f(W+1)\right]\right). \end{aligned}$$

Since $X_i \in \{0, 1\}$,

$$\mathbb{E}\left[X_i f(V_i + X_i)\right] = \mathbb{E}\left[X_i f(V_i + 1)\right] = p_i \mathbb{E}\left[f(V_i + 1)\right],$$

simplifying (II).

Writing $f(W) - f(V_i + 1) \leq |f(W) - f(V_i + 1)|$,

$$\begin{aligned}
|(\mathrm{I})| &\leq \sum_{i=1}^{n} \mathbb{E}\left[X_i \left|f(W) - f(V_i+1)\right|\right] \\
&\leq \min\left\{1, \frac{1}{\lambda}\right\} \sum_{i=1}^{n} \mathbb{E}\left[X_i \sum_{j \in N_i \setminus \{i\}} X_j\right] \\
&= \min\left\{1, \frac{1}{\lambda}\right\} \sum_{i=1}^{n} \sum_{j \in N_i \setminus \{i\}} \mathbb{E}\left[X_i X_j\right],
\end{aligned}$$

where the second inequality follows from the Lipschitz bound on f. Also,

$$\begin{aligned}
|(\mathrm{II})| &\leq \sum_{i=1}^{n} p_i \left|\mathbb{E}\left[f(V_i+1)\right] - \mathbb{E}\left[f(W+1)\right]\right| \\
&\leq \min\left\{1, \frac{1}{\lambda}\right\} \sum_{i=1}^{n} p_i \mathbb{E}\left[|W - V_i|\right] \\
&= \min\left\{1, \frac{1}{\lambda}\right\} \sum_{i=1}^{n} p_i \mathbb{E}\left[\sum_{j \in N_i} X_j\right] \\
&= \min\left\{1, \frac{1}{\lambda}\right\} \sum_{i=1}^{n} \sum_{j \in N_i} p_i p_j.
\end{aligned}$$

This gives the stated bound. □

Example 6.3.3.

Consider the undirected Erdös–Rényi graph with n nodes and edges $A_{ij} \in \{0, 1\}$ with $A_{ij} = A_{ji} \overset{\text{iid}}{\sim} \text{Bernoulli}(p)$. We can count the number of edges with the CLT (for large n) or the independent version of the PLT (small n). More interestingly, we can count the number of triangles:

$$T := \sum_{1 \leq i < j < k \leq n} A_{ij} A_{jk} A_{ki}.$$

We can use the dependency neighborhood condition. To make indexing easier, consider the dependency neighborhoods N_α, so $\alpha \in N_\alpha$ and $X_\alpha \perp\!\!\!\perp \{X_\beta\}_{\beta \notin N_\alpha}$. The dependency version of Stein-Chen PLT then says for $X_\alpha \sim \text{Bernoulli}(p_\alpha)$ and $\lambda = \sum_\alpha p_\alpha$, we have

$$\begin{aligned}
\mathrm{TV}\left(\sum_\alpha X_\alpha, \mathrm{Poi}(\lambda)\right) \leq \min\left\{1, \frac{1}{\lambda}\right\} \sum_\alpha \sum_{\beta \in N_\alpha \setminus \{\alpha\}} \mathbb{E}\left[X_\alpha X_\beta\right] \\
+ \min\left\{1, \frac{1}{\lambda}\right\} \sum_\alpha \sum_{\beta \in N_\alpha} p_\alpha p_\beta
\end{aligned}.$$

Write $\alpha = (i, j, k)$, so $X_\alpha = A_{ij} A_{jk} A_{ki}$. Since a triangle with a shared node shares exactly one edge, and each of the three edges of a triangle can be shared with triangles constructed by connecting the two nodes of the edge with $n - 3$ different nodes, we see that when N_α are as small as possible, $|N_\alpha| = 3(n-3) + 1$. (The dependency neighborhood of a triangle includes itself.)

We see

$$\begin{aligned}
\text{(II)} &= \sum_{\alpha} \sum_{\beta \in N_\alpha} p_\alpha p_\beta \\
&= O\left(n^3 \cdot n \cdot p^6\right) \\
&= O\left(n^4 p^6\right) \\
\text{(I)} &= \sum_{\alpha} \sum_{\beta \in N_\alpha \setminus \{\alpha\}} \mathbb{E}\left[X_\alpha X_\beta\right] \\
&= \sum_{\alpha} \sum_{\beta \in N_\alpha \setminus \{\alpha\}} \mathbb{E}\left[A_{ab} A_{bc} A_{cd} A_{de} A_{ea}\right] \\
&= O\left(n^4 p^5\right) \\
\lambda &= \binom{n}{3} p^3 = O\left(n^4 p^5, np^2\right).
\end{aligned}$$

Therefore,

$$\mathrm{TV}\left(T, \mathrm{Poi}\left(\binom{n}{3} p^3\right)\right) = O\left(\min\left\{n \cdot n^3 p^3, np^2\right\}\right),$$

which tends to 0 when $p \ll \frac{1}{\sqrt{n}}$.

Remark 6.3.4. We can prove this dependent PLT using the martingale CLT instead of Stein's method.

Section 7

Marcus Kuo

Conditional Expectation

We will talk about conditional expectation from the perspective of the Radom–Nikodym derivative.

7.1 Signed Measure and the Radom–Nikodym Derivative

Definition 7.1.1 (Signed measure).

Suppose we have a measurable space $(\Omega, \mathscr{F})$. ϕ is a signed measure on $(\Omega, \mathscr{F})$ if:

(i) $\phi(\varnothing) = 0$

(ii) For disjoint countable $\{A_n\} \subseteq \mathscr{F}$,

$$\phi\left(\bigcup_n A_n\right) = \sum_n \phi(A_n).$$

(Implicitly, the sum is absolutely convergent.)

The only difference between a measure and signed measure is that a signed measure can take any value, not just nonnegative ones.

With nonnegative measures, there was no trouble with sets having measure infinity. But there is a potential collision with signed measure if two sets have measures infinity and negative infinity.

However, the following proposition rules this out.

Proposition 7.1.2.

Either

$$-\infty < \phi(A) \leq \infty, \quad \forall\ A \in \mathscr{F}$$

or

$$-\infty \leq \phi(A) < \infty, \quad \forall\ A \in \mathscr{F}.$$

Proof of Proposition 7.1.2. Suppose not, so there are $A, B \in \mathscr{F}$ such that

$$\phi(A) = \infty, \quad \phi(B) = -\infty.$$

By countable additivity, we can write [1]

$$\phi(A \cup B) = \phi(A) + \phi(B \setminus A) = \infty$$
$$\phi(A \cup B) = \phi(B) + \phi(A \setminus B) = -\infty.$$

[1] The definition of a signed measure imposes that the measure of any set in $\mathscr{F}$ is well-defined.

□

Theorem 7.1.3 (Hahn decomposition).

Given a signed measure ϕ, there exist $\Omega^+, \Omega^- \in \mathscr{F}$ such that they partition Ω:

$$\Omega^+ \cup \Omega^- = \Omega, \quad \Omega^+ \cap \Omega^- = \varnothing$$

and for any $A \in \mathscr{F}$,

$$\phi(A \cap \Omega^-) \leq 0 \leq \phi(A \cap \Omega^+).$$

Furthermore, this partition is unique in the sense that if we have Hahn decompositions $\left(\Omega_1^+, \Omega_1^-\right)$ and $\left(\Omega_2^+, \Omega_1^-\right)$,

$$\begin{aligned} A \subseteq \Omega_1^+ \Delta \Omega_2^+ &\implies \phi(A) = 0, \quad \forall\, A \in \mathscr{F} \\ A \subseteq \Omega_1^- \Delta \Omega_2^- &\implies \phi(A) = 0. \end{aligned}$$

Proposition 7.1.4 (Jordan decomposition).

From a signed measure ϕ, we can recover measures ϕ^+, ϕ^- by putting

$$\begin{aligned} \phi^+(A) &= \phi(A \cap \Omega^+) \\ \phi^-(A) &= \phi(A \cap \Omega^-). \end{aligned}$$

Note that Jordan decomposition is unique; starting with different Hahn decompositions yields the same Jordan decomposition. There is a remark if the Jordan decomposition $\phi = \phi^+ - \phi^-$ imposes mutual singularity.

Proof of Proposition 7.1.4. Immediate (or maybe not). □

Example 7.1.5.

Given a measure space $(\Omega, \mathscr{F}, \mu)$, where $f : (\Omega, \mathscr{F}) \to (\mathbb{R}, \mathscr{B}_{\mathbb{R}})$, we have that

$$\phi(A) := \int_A f d\mu$$

is a signed measure with the Hahn decomposition

$$\Omega^+ = \{f \geq 0\}, \quad \Omega^- = \{f < 0\}.$$

Note if f is nonnegative, ϕ is a measure.
The Jordan decomposition is

$$\phi^{\pm}(A) = \int_A f^{\pm} d\mu.$$

Definition 7.1.6 (Absolute continuity).

Given a signed measure ϕ and measure μ on $(\Omega, \mathscr{F})$, $\phi \ll \mu$[a] if for any $A \in \mathscr{F}$ such that $\mu(A) = 0$, $\phi(A) = 0$.

[a] Read as ϕ is absolutely continuous with respect to μ.

Remark 7.1.7. The signed measure ϕ as an integral of f over μ is absolutely continuous with respect to μ.

Theorem 7.1.8 (Radom–Nikodym).

Suppose ϕ is a signed measure and μ is a σ-finite measure on $(\Omega, \mathscr{F})$. If $\phi \ll \mu$, then there exists a function f (unique up to a measure zero set on $(\Omega, \mathscr{F}, \mu)$) such that

$$\phi(A) = \int_A f d\mu$$

for every $A \in \mathscr{F}$.
f is called the Random–Nikodym derivative (or the derivative between the two measures) and is also notated

$$f = \frac{d\phi}{d\mu}.$$

Example 7.1.9.

Consider $(\mathbb{R}, \mathscr{B}_{\mathbb{R}})$ and $\mathbb{P} = N(0,1)$, with λ as Lebesgue measure. Then $\frac{d\mathbb{P}}{d\lambda}$ is the density of a standard Gaussian distribution.
For $\left(\mathbb{N}, 2^{\mathbb{N}}\right)$ with $\mathbb{P}$ as the Poisson distribution and λ the counting measure, $\frac{d\mathbb{P}}{d\lambda}$ is the probability mass function of the Poisson distribution.

7.2 Motivation of Conditional Expectation

For random variables X, Y, what should $\mathbb{E}[X \mid Y]$ be? In the simplest case, where $X = \mathbb{1}_A$ and $Y = \mathbb{1}_B$, we want

$$\mathbb{E}[X \mid Y] = \mathbb{P}[A \mid \mathbb{1}_B].$$

If $\mathbb{1}_B = 1$, then $\mathbb{P}[A \mid \mathbb{1}_B] = \mathbb{P}[A \mid B]$ and if $\mathbb{1}_B = 0$, then $\mathbb{P}[A \mid \mathbb{1}_B] = \mathbb{P}[A \mid B^c]$. So,

$$\mathbb{E}[X \mid Y] = \mathbb{P}[A \mid B]\mathbb{1}_B + \mathbb{P}[A \mid B^c]\mathbb{1}_{B^c}, \tag{7.1}$$

and the conditional expectation is measurable with respect to $\mathscr{G} = \{\varnothing, \Omega, B, B^c\}$.

What if $Z = a\mathbb{1}_B + b\mathbb{1}_{B^c}$ with $a \neq b$? Then

$$\mathbb{E}[X \mid Z] = \mathbb{E}[X \mid Y] = \mathbb{E}[X \mid \mathscr{G}],$$

as $\sigma(Z) = \{\varnothing, \Omega, B, B^c\} = \sigma(Y)$.

For any $D \in \mathscr{G} = \{\varnothing, \Omega, B, B^c\}$, we have that by equation (**??**),

$$\begin{aligned}
\mathbb{E}[\mathbb{E}[X \mid Y]\mathbb{1}_D] &= \mathbb{P}[A \mid B]\mathbb{P}[B \cap D] + \mathbb{P}[A \mid B^c]\mathbb{P}[B^c \cap D] \\
&= \begin{cases} 0, \quad D & = \varnothing \\ \mathbb{P}[A], \quad D & = \Omega \\ \mathbb{P}[A \cap B], \quad D & = B \\ \mathbb{P}[A \cap B^c], \quad D & = B^c \end{cases} \\
&= \mathbb{E}[X\mathbb{1}_D].
\end{aligned}$$

We have shown that the tower property holds *in this example.*

Proposition 7.2.1 (Tower Property).

For $D \in \mathscr{G}$ (D is measurable with respect to $\mathscr{G}$),

$$\mathbb{E}\left[\mathbb{E}\left[X \mid \mathscr{G}\right] \mathbb{1}_D\right] = \mathbb{E}\left[X \mathbb{1}_D\right].$$

Summarizing our three motivating examples, we see that $\mathbb{E}\left[X \mid Y\right]$ should:

(i) Equal $\mathbb{E}\left[X \mid \mathscr{G}\right]$ with $\mathscr{G} = \sigma(Y)$,

(ii) Be measurable with respect to $\mathscr{G}$, and

(iii) Satisfy the tower property.

It can be shown that there is a unique function satisfying all of these requirements. We call this conditional expectation.

Definition 7.2.2 (Conditional expectation with respect to a σ-field).

Suppose $(\Omega, \mathscr{F}, \mathbb{P})$ is a probability space, $\mathscr{G} \subseteq \mathscr{F}$ is a σ-field, and $X : (\Omega, \mathscr{F}) \to (\mathbb{R}, \mathscr{B}_{\mathbb{R}})$ is a random variable such that $\mathbb{E}\left[|X|\right] < \infty$.
$\mathbb{E}\left[X \mid \mathscr{G}\right]$ is defined as the unique[a] $\mathscr{G}$-measurable function such that for every $A \in \mathscr{G}$,

$$\int_A \mathbb{E}\left[X \mid \mathscr{G}\right] d\mathbb{P} = \int_A X d\mathbb{P}.$$

[a] Up to null sets in $\mathscr{G}$, *not* in $\mathscr{F}$.

7.3 Existence and Uniqueness

Why does $\mathbb{E}\left[X \mid \mathscr{G}\right]$ exist and why is it unique? Define

$$\phi(A) = \int_A X d\mathbb{P}$$

for every $A \in \mathscr{G}$. ϕ is a signed measure and $\mathbb{P}$ is a measure on the smaller measurable space $(\Omega, \mathscr{G})$ and $\phi \ll \mathbb{P}$. By the Radon–Nikodym theorem applied to $(\Omega, \mathscr{G})$, there exists a unique function $\frac{d\phi}{d\mathbb{P}}$ such that

$$\int_A X d\mathbb{P} = \phi(A) = \int_A \frac{d\phi}{d\mathbb{P}} d\mathbb{P},$$

which we define to be $\mathbb{E}\left[X \mid \mathscr{G}\right]$.

Example 7.3.1.

If $\mathscr{G} = \{\varnothing, \Omega\}$, then

$$\mathbb{E}\left[X \mid \mathscr{G}\right] = \mathbb{E}\left[X\right].$$

Returning to one of our motivating examples, if $X = \mathbb{1}_A$ and $\mathscr{G} = \{\varnothing, \Omega, B, B^c\}$, then

$$\begin{aligned}\mathbb{E}[X \mid G] &= a\mathbb{1}_B + b\mathbb{1}_{B^c} \\ &= \begin{cases} a\mathbb{P}[B] = \mathbb{P}[A \cap B], \quad D & = B \\ b\mathbb{P}[B^c] = \mathbb{P}[A \cap B^c], \quad D = B^c & \\ a\mathbb{P}[B] + b\mathbb{P}[B^c] = \mathbb{P}[A], \quad D & = \Omega \\ a \cdot 0 + b \cdot 0 = 0, \quad D & = \varnothing \end{cases}.\end{aligned}$$

The third and fourth cases are redundant. If $\mathbb{P}[B] \neq 0$, then we have the solution

$$a = \begin{cases} \frac{\mathbb{P}[A \cap B]}{\mathbb{P}[B]} = \mathbb{P}[A \mid B], & \mathbb{P}[B] \neq 0, \\ \text{arbitrary}, & \mathbb{P}[B] = 0, \end{cases}$$

$$b = \begin{cases} \mathbb{P}[A \mid B^c], \quad \mathbb{P}[B^c] & \neq 0 \\ \text{arbitrary}, \quad \mathbb{P}[B^c] & = 0 \end{cases}.$$

If $\mathscr{G} = \mathscr{F}$, then

$$\mathbb{E}[X \mid \mathscr{G}] = X.$$

Lemma 7.3.2 (Doob–Dynkin).

Suppose $Y : (\Omega, \mathscr{F}) \to (\mathbb{R}, \mathscr{B}_{\mathbb{R}})$ and $Z : (\Omega, \sigma(Y)) \to (\mathbb{R}, \mathscr{B}_{\mathbb{R}})$.
Then there exists $f : (\mathbb{R}, \mathscr{B}_{\mathbb{R}}) \to (\mathbb{R}, \mathscr{B}_{\mathbb{R}})$ such that $Z = f(Y)$.

[2]

Proof of Lemma 7.3.2. The proof is standard: first check the property for Z simple, then for nonnegative Z, and finally for general Z. We only check the simple-function case.

We can write $Z = \sum_{i=1}^n a_i \mathbb{1}_{A_i}$. Since $\sigma(Z) \subseteq \sigma(Y)$, we have

$$A_i \in \sigma(Y) = \left\{Y^{-1}B : B \in \mathscr{B}_{\mathbb{R}}\right\}.$$

Thus there exists $B_i \in \mathscr{B}_{\mathbb{R}}$ such that $A_i = Y^{-1}B_i$, which lets us write

$$\begin{aligned} Z(\omega) &= \sum_{i=1}^n a_i \mathbb{1}_{\omega \in A_i} \\ &= \sum_{i=1}^n a_i \mathbb{1}_{\omega \in Y^{-1}(B_i)} \\ &= \sum_{i=1}^n a_i \mathbb{1}_{Y(\omega) \in B_i} \\ &= f(Y), \end{aligned}$$

with $f(t) = \sum_{i=1}^n a_i \mathbb{1}_{t \in B_i}$. □

[2]The conditions of this lemma imply $\sigma(Z) \subseteq \sigma(Y)$.

Definition 7.3.3 (Conditional expectation with respect to a random variable).

For random variables X, Y define

$$\mathbb{E}[X \mid Y] = \mathbb{E}[X \mid \sigma(Y)].$$

By Doob–Dynkin, this is a function of Y.

Note that the statement

$$\mathbb{E}[(X - \mathbb{E}[X \mid \mathscr{G}])\mathbb{1}_A] = 0, \quad \forall\, A \in \mathscr{G}$$

is equivalent to

$$\mathbb{E}[(X - \mathbb{E}[X \mid \mathscr{G}])\, Y] = 0, \quad \forall\, Y \text{ s.t. } \sigma(Y) \subseteq \mathscr{G}.$$

This motivates some sort of orthogonality property — the conditional expectation $\mathbb{E}[X \mid \mathscr{G}]$ is a projection of X onto $\{Y : \sigma(Y) \subseteq \mathscr{G}\}$.

Similarly,[3] since $\mathbb{E}[X \mid Y] = \mathbb{E}[X \mid \sigma(Y)]$, we have

$$\mathbb{E}[(X - \mathbb{E}[X \mid Y])\, g(Y)] = 0, \forall \text{ functions } g \text{ of } Y.$$

We state these formally:

Proposition 7.3.4 (Conditional expectation as projection).

$$\mathbb{E}[X \mid \mathscr{G}] = \min_{Y:\sigma(Y)\subseteq\mathscr{G}} \mathbb{E}\left[(Y - X)^2\right],$$

while

$$\mathbb{E}[X \mid Y] = \min_{g} \mathbb{E}\left[(g(Y) - X)^2\right].$$

Remark 7.3.5. We have shown that if the second moment of X exists, we can define conditional expectation from the projection perspective. But the orthogonality condition is more general, as it does not require existence of the second moment.

7.4 Properties of Conditional Expectation

Proposition 7.4.1 (Properties of conditional expectation).

(i) If $\sigma(X) \subseteq \mathscr{G}$, then $\mathbb{E}[X \mid \mathscr{G}] = X$

(ii) If $\sigma(X) \perp\!\!\!\perp \mathscr{G}$, then $\mathbb{E}[X \mid \mathscr{G}] = \mathbb{E}[X]$

(iii) If $\mathscr{G}_0 \subseteq \mathscr{G}$, then

$$\mathbb{E}[\mathbb{E}[X \mid \mathscr{G}_0] \mid \mathscr{G}] = \mathbb{E}[\mathbb{E}[X \mid \mathscr{G}] \mid \mathscr{G}_0] = \mathbb{E}[X \mid \mathscr{G}_0].$$

(iv) If $X \leq Y$, then

$$\mathbb{E}[X \mid \mathscr{G}] \leq \mathbb{E}[Y \mid \mathscr{G}].$$

(v) $\mathbb{E}[aX + bY \mid \mathscr{G}] = a\mathbb{E}[X \mid \mathscr{G}] + b\mathbb{E}[Y \mid \mathscr{G}]$

[3] We have a condition (from where?) that $\mathbb{E}[(X - \mathbb{E}[X \mid Z])\, Z] = 0$ for all Z such that $\sigma(Z) \subseteq \sigma(Y)$. By Doob–Dynkin, this is equivalent to the proposition.

(vi) Suppose $\{X_n\}$ is a increasing sequence of nonnegative functions converging to X almost everywhere. Then

$$\mathbb{E}\left[X_n \mid \mathscr{G}\right] \uparrow \mathbb{E}\left[X \mid \mathscr{G}\right].$$

(vii) For $X_n \geq 0$,

$$\mathbb{E}\left[\liminf_{n\to\infty} X_n \mid \mathscr{G}\right] \leq \liminf_{n\to\infty} \mathbb{E}\left[X_n \mid \mathscr{G}\right].$$

(viii) Suppose $X_n \to X$ almost everywhere, $|X_n| \leq Y$ almost everywhere, and $\mathbb{E}[Y] < \infty$. Then

$$\mathbb{E}\left[X_n \mid \mathscr{G}\right] \to \mathbb{E}\left[X \mid \mathscr{G}\right].$$

(ix) If $\sigma(Y) \subseteq \mathscr{G}$, then

$$\mathbb{E}\left[XY \mid \mathscr{G}\right] = Y\mathbb{E}\left[X \mid \mathscr{G}\right].$$

Proof of Proposition 7.4.1. Left as exercises; these are all routine. □

Section A

Marcus Kuo

Appendix – Summary of Major Results: CLT and PLT

A.1 Statements of Theorems and Corollaries

Definition A.1.1 (Weak convergence).

Suppose P_n, P are probability measures on $(\mathscr{X}, \mathscr{B}(\mathscr{X}))$. We say $P_n \rightsquigarrow P$ if $P_n l \to Pl$ for all $l \in \mathrm{BL}(\mathscr{X})$.

Theorem A.1.2 (Portmanteau).

The following are equivalent:

(i) $P_n l \to Pl$ for every $l \in \mathrm{BL}(\mathscr{X})$

(ii) $\liminf_{n\to\infty} P_n g \geq Pg$ for every lower semicontinuous $g \geq -M$

(iii) $\limsup_{n\to\infty} P_n f \leq Pf$ for every upper semicontinuous $f \leq M$

(iv) $\liminf_{n\to\infty} P_n(B) \geq P(B)$ for every open B

(v) $\limsup_{n\to\infty} P_n(G) \leq P(G)$ for every closed G

(vi) $P_n f \to Pf$ for every $|f| \leq M$ such that $P(C_f) = 1$

(vii) $P_n B \to PB$ for every $B \in \mathscr{B}(\mathscr{X})$ with $P(\partial B) = 0$

Theorem A.1.3 (Weak convergence on $\mathbb{R}$).

The following are equivalent:

(i) $P_n l \to Pl$ for every $l \in \mathrm{BL}(\mathbb{R})$

(ii) $P_n(-\infty, t] \to P(-\infty, t]$ for every t such that $P(\{t\}) = 0$ (CDF convergence at continuity points)

(iii) $P_n f \to Pf$ for every $f \in C^\infty(\mathbb{R})$

Theorem A.1.4 (CLT for triangular arrays).

Suppose we have a triangular array of independent (within rows) random variables $\xi_{n,i}$, $i = 1, \ldots, k(n)$, such that

(i) $\sum_{i=1}^{k(n)} \mathbb{E}\left[\xi_{n,i}\right] \to \mu$,

(ii) $\sum_{i=1}^{k(n)} \mathrm{Var}\left(\xi_{n,i}\right) \to \sigma^2$, and

(iii) $\sum_{i=1}^{k(n)} \mathbb{E}\left[\left|\xi_{n,i}\right|^3\right] \to 0$.

Then $\sum_{i=1}^{k(n)} \xi_{n,i} \rightsquigarrow N(\mu, \sigma^2)$.

Corollary A.1.5 (Classical CLT).

If $X_1, X_2, \ldots$ are iid with $\mathbb{E}\left[X_i\right] = 0$ and $\mathrm{Var}(X_i) = 1$, then

$$\frac{X_1 + \cdots + X_n}{\sqrt{n}} \rightsquigarrow N(0,1).$$

In particular, existence of the second moment suffices (no third moment needed). The proof uses truncation to reduce to the triangular array CLT.

Theorem A.1.6 (Lindeberg CLT).

Suppose we have a triangular array $\{X_{n,i}\}$ with independence within rows such that

(i) $\mathbb{E}\left[X_{n,i}\right] = 0$ for all n, i,

(ii) $\sum_{i=1}^{k(n)} \mathrm{Var}\left(X_{n,i}\right) = 1$ for all n, and

(iii) (Lindeberg condition) for every $\epsilon > 0$,

$$L_n(\epsilon) := \sum_{i=1}^{k(n)} \mathbb{E}\left[X_{n,i}^2 \mathbb{1}_{|X_{n,i}|>\epsilon}\right] \to 0.$$

Then $\sum_i X_{n,i} \rightsquigarrow N(0,1)$.
Compared to the triangular array CLT, conditions (i) and (ii) are exact equalities rather than limits, but the third moment condition is replaced by the weaker Lindeberg condition.

Theorem A.1.7 (Lindeberg CLT, unnormalized).

Suppose for each n the random variables $X_{n,1}, \ldots, X_{n,k(n)}$ are independent with $\mathbb{E}\left[X_{n,i}\right] = 0$. Let $S_n^2 := \sum_{i=1}^{k(n)} \mathbb{E}\left[X_{n,i}^2\right]$ and assume $S_n > 0$ for all large n. If for every $\epsilon > 0$,

$$\frac{1}{S_n^2} \sum_{i=1}^{k(n)} \mathbb{E}\left[X_{n,i}^2 \mathbb{1}_{|X_{n,i}|>\epsilon S_n}\right] \to 0,$$

then $\frac{1}{S_n}\sum_{i=1}^{k(n)} X_{n,i} \rightsquigarrow N(0,1)$.

Theorem A.1.8 (Lyapunov CLT).

Suppose $X_{n,1},\dots,X_{n,n}$ are independent with $\mathbb{E}[X_{n,i}]=0$ and $S_n=\sqrt{\sum_{i=1}^n \mathbb{E}\left[X_{n,i}^2\right]}$. If there exists $\delta>0$ such that

$$\frac{1}{S_n^{2+\delta}}\sum_{i=1}^n \mathbb{E}\left[|X_{n,i}|^{2+\delta}\right]\to 0,$$

then $\frac{X_{n,1}+\cdots+X_{n,n}}{S_n}\rightsquigarrow N(0,1)$.
The Lyapunov condition implies the Lindeberg condition.

Lemma A.1.9 (Stein's identity).

$Z\sim N(0,1)$ if and only if for all absolutely continuous f with $\mathbb{E}[f(Z)]<\infty$ and $\mathbb{E}[Zf(Z)]<\infty$,

$$\mathbb{E}\left[f'(Z)\right]=\mathbb{E}\left[Zf(Z)\right].$$

Theorem A.1.10 (Wasserstein CLT).

Suppose $X_1,\dots,X_n$ are independent (not necessarily identically distributed) with $\mathbb{E}[X_i]=0$ and $\mathbb{E}\left[X_i^2\right]=1$. Then

$$\mathscr{W}\left(\frac{X_1+\cdots+X_n}{\sqrt{n}},N(0,1)\right)\le\frac{3}{\sqrt{n}}\frac{1}{n}\sum_{i=1}^n\mathbb{E}\left[|X_i|^3\right].$$

Proposition A.1.11 (Berry-Esseen).

Under the same conditions,

$$\mathrm{KS}\left(\frac{X_1+\cdots+X_n}{\sqrt{n}},N(0,1)\right)\le\frac{C}{\sqrt{n}}\frac{1}{n}\sum_{i=1}^n\mathbb{E}\left[|X_i|^3\right].$$

The bound is uniform across all t, and existence of the third moment is necessary for this uniform convergence.

Definition A.1.12 (Total variation).

For probability measures P,Q on $(\Omega,\mathscr{F})$,

$$\mathrm{TV}(P,Q)=\sup_{B\in\mathscr{F}}|P(B)-Q(B)|.$$

For discrete distributions, $\mathrm{TV}(P,Q)=1-\sum_i p_i\wedge q_i$.

Definition A.1.13 (Wasserstein distance).

For distributions P, Q,

$$\mathscr{W}(P,Q) := \sup_{\mathrm{Lip}(h)\leq 1} \left|\mathbb{E}_P[h] - \mathbb{E}_Q[h]\right|.$$

Convergence in Wasserstein distance implies weak convergence.

Definition A.1.14 (Kolmogorov-Smirnov distance).

For distributions P, Q,

$$\mathrm{KS}(P,Q) := \sup_t \left|P(-\infty,t] - Q(-\infty,t]\right|.$$

Proposition A.1.15 (Relationship between KS and Wasserstein).

$\mathrm{KS}(W,Z) \leq 2\sqrt{\mathscr{W}(W,Z)}$.

Definition A.1.16 (Coupling).

The pair $(\hat{X},\hat{Y})$ is a coupling of (X,Y) if $\hat{X}$ and $\hat{Y}$ live on the same probability space and $\hat{X} \overset{\mathrm{d}}{=} X$, $\hat{Y} \overset{\mathrm{d}}{=} Y$.

Definition A.1.17 (Maximal coupling).

$(\hat{X},\hat{Y})$ is a maximal coupling of (X,Y) if $\mathbb{P}\left[\hat{X}=\hat{Y}\right]$ is maximized. For discrete RVs, the maximal coupling satisfies $\mathbb{P}\left[\hat{X}\neq\hat{Y}\right] = \mathrm{TV}(P,Q)$.

Theorem A.1.18 (LeCam Poisson Limit Theorem).

Suppose $X_i \sim \mathrm{Bernoulli}(p_i)$ independently for $i = 1,\ldots,n$ and $\lambda = \sum_{i=1}^n p_i$. Then

$$\mathrm{TV}\left(\sum_{i=1}^n X_i, \mathrm{Poi}(\lambda)\right) \leq \sum_{i=1}^n p_i^2.$$

In particular, $\mathrm{Bin}(n,\frac{\lambda}{n}) \rightsquigarrow \mathrm{Poi}(\lambda)$.

Lemma A.1.19 (Stein-Chen).

$Z \sim \mathrm{Poi}(\lambda)$ if and only if for all f,

$$\mathbb{E}\left[Zf(Z)\right] = \lambda\mathbb{E}\left[f(Z+1)\right].$$

Theorem A.1.20 (Stein-Chen PLT, independent).

Suppose $X_i \sim \text{Bernoulli}(p_i)$ independently for $i = 1, \ldots, n$ and $\lambda = \sum_{i=1}^{n} p_i$. Then

$$\text{TV}\left(\sum_{i=1}^{n} X_i, \text{Poi}(\lambda)\right) \leq \min\left\{1, \frac{1}{\lambda}\right\} \sum_{i=1}^{n} p_i^2.$$

This strengthens LeCam's result by the $\min\left\{1, \frac{1}{\lambda}\right\}$ factor.

Theorem A.1.21 (Stein-Chen PLT, dependent).

Suppose $X_i \sim \text{Bernoulli}(p_i)$ for $i = 1, \ldots, n$ and $\lambda = \sum_{i=1}^{n} p_i$. Given dependency neighborhoods N_i with $i \in N_i$ and $X_i \perp\!\!\!\perp \{X_j\}_{j \notin N_i}$,

$$\begin{aligned}\text{TV}\left(\sum_{i=1}^{n} X_i, \text{Poi}(\lambda)\right) \leq \min\left\{1, \frac{1}{\lambda}\right\} \sum_{i=1}^{n} \sum_{j \in N_i \setminus \{i\}} \mathbb{E}\left[X_i X_j\right] \\ + \min\left\{1, \frac{1}{\lambda}\right\} \sum_{i=1}^{n} \sum_{j \in N_i} p_i p_j\end{aligned}.$$

A.2 Proof Techniques

Lindeberg replacement: compare to a Gaussian sum

The triangular-array CLT is proved by comparing the given sum to a Gaussian sum with the same first two moments and replacing the summands *one at a time*. This is useful because Gaussian sums are easy to identify, while general sums of independent variables are not. The replacement argument isolates the effect of each summand and shows that, once the first two moments are matched, the remaining error is controlled by a higher-moment term. Conceptually, this is why the triangular-array CLT is the basic engine behind several later CLTs: it gives a systematic way to reduce a difficult distributional comparison to a sequence of local comparisons.

Truncation: reduce to a setting with controlled tails

The classical iid CLT and the Lindeberg CLT are both reduced to a more manageable finite-third-moment setting by truncating large summands. The reason truncation appears is that the replacement/Taylor argument needs tail control, while the original assumptions may only give finite variance. Truncation separates the proof into two parts: first show that the truncated variables satisfy the hypotheses of the triangular-array CLT, then show that the discarded large terms are rare enough that they do not affect the limit. In other words, truncation is not the main approximation theorem; it is the device that puts us in a regime where the main theorem can be applied.

Stein's method for the normal law: turn approximation into an identity

For quantitative normal approximation, direct comparison with the standard normal is awkward. Stein's method avoids this by characterizing $Z \sim N(0, 1)$ through the identity

$$\mathbb{E}\left[f'(Z)\right] = \mathbb{E}\left[Zf(Z)\right].$$

Instead of proving W is close to Z head-on, we rewrite the difference $\mathbb{E}[h(W)] - \mathbb{E}[h(Z)]$ in terms of a Stein equation and then exploit the structure of W, typically $W = n^{-1/2} \sum X_i$. This is why Stein's method is used in the Wasserstein CLT and Berry–Esseen-type bounds: it converts a distributional question into an expectation that can be estimated using independence and moment bounds.

Stein–Chen: the Poisson version of the same idea

For Poisson approximation, the normal Stein identity is replaced by the Poisson identity

$$\mathbb{E}\left[Zf(Z)\right] = \lambda \mathbb{E}\left[f(Z+1)\right], \qquad Z \sim \mathrm{Poi}(\lambda).$$

This is useful when studying sums of rare indicators, because such sums already have the local "one event at a time" structure encoded by the Poisson distribution. In the independent case, and later with dependency neighborhoods, the Stein–Chen method gives explicit total-variation bounds by measuring how far the sum fails to satisfy the Poisson characterizing identity. So the role of Stein–Chen is not merely to prove convergence to Poisson, but to explain *why* sums of weakly dependent rare events should behave Poisson-like.

Coupling: realize two laws on the same probability space

Coupling is used when it is easier to compare two random variables after putting them on a common probability space. In the Poisson limit theorem, the key example is to couple each Bernoulli variable with a Poisson variable of the same mean. This is natural because when p_i is small, a Bernoulli(p_i) and a Poi(p_i) variable are usually equal, so the mismatch probability is of order p_i^2. Summing these local mismatch probabilities yields LeCam's bound. More broadly, coupling is useful because it turns an abstract distance between laws into an event whose probability we can estimate directly.

Comparing distances: move from the bound you can prove to the conclusion you want

The previous proofs produce bounds in different metrics for different reasons. Total variation is natural for coupling and Poisson approximation. Wasserstein distance is natural for Stein's method with Lipschitz test functions. Kolmogorov–Smirnov distance is natural when the conclusion is stated in terms of distribution functions. The comparison inequalities among these metrics explain why these proofs are enough: a strong bound in total variation or Wasserstein can be converted into weak convergence, and Wasserstein bounds can sometimes be turned into KS bounds. This is why the metric comparisons matter conceptually: they connect the output of the proof method to the exact form of the theorem statement.

Section B

Marcus Kuo

Appendix – Additional Notes

B.1 Measure Theory

Something that is often tediously required in measure theory is to show that some property holds for every set in a σ-algebra. The monotone class theorem and the π-λ lemma help us accomplish this by letting us check that the property holds on a subset.

Definition B.1.1 (Monotone class).

A monotone class $\mathscr{M}$ on Ω is a collection of subsets of Ω that is closed under countable monotone unions and intersections. For any sequence $A_n \in \mathscr{M}$,

(i) If $A_1 \subseteq A_2 \subseteq \cdots$, then $\cup_{n\geq 1} A_n \in \mathscr{M}$.

(ii) If $A_1 \supseteq A_2 \supseteq \cdots$, then $\cap_{n\geq 1} A_n \in \mathscr{M}$.

Theorem B.1.2 (Monotone class).

If $\mathscr{A}$ is an algebra, then the smallest monotone class $\mathscr{M}$ containing $\mathscr{A}$ is $\sigma(\mathscr{A})$.

Proof of Theorem B.1.2. Note that $\sigma(\mathscr{A})$ must itself be a monotone class, so $\mathscr{M} \subseteq \sigma(\mathscr{A})$.

To show the other direction, we can show that $\mathscr{M}$ is a σ-algebra. Because $\mathscr{M}$ contains $\mathscr{A}$, we will have $\mathscr{M} \supseteq \sigma(\mathscr{A})$. Note that $\varnothing \in \mathscr{A} \subseteq \mathscr{M}$, so now we need to show $\mathscr{M}$ is closed under complementation and countable unions and intersections.

Let $\mathscr{M}_0$ be the subset of $\mathscr{M}$ such that $F \in \mathscr{M} \implies F^c \in \mathscr{M}$. We see that since $\mathscr{A}$ is closed under complementation, $\mathscr{A} \subseteq \mathscr{M}_0$. Now we want to show $\mathscr{M}_0$ is a monotone class. Suppose we have a sequence $F_1 \subseteq F_2 \subseteq \cdots$ in $\mathscr{M}_0$. Then since

$$\left(\bigcup_{n\geq 1} F_n\right)^c = \bigcap_{n\geq 1} (F_n)^c$$

with $F_n^c \in \mathscr{M}$, since $\mathscr{M}$ is a monotone class, $(\cup_{n\geq 1} F_n)^c \in \mathscr{M}$. Hence $\mathscr{M}_0$ is closed under monotone unions, and similarly, we can show it is closed under monotone intersections. Therefore $\mathscr{M}_0$ is a monotone class containing $\mathscr{A}$, so $\mathscr{M} \subseteq \mathscr{M}_0$, and we have equality of $\mathscr{M} = \mathscr{M}_0$. $\mathscr{M}$ is closed under complements.

Now let $\mathscr{M}_1$ be the subset of all sets $F \in \mathscr{M}$ such that $A \in \mathscr{M} \implies F \cap A \in \mathscr{M}$. We see that $\mathscr{A} \subseteq \mathscr{M}_1$ because algebras are closed under finite intersection. For a monotone increasing sequence $\{F_n\} \subseteq \mathscr{M}_1$, we can write

$$\left(\bigcup F_n\right) \cap A = \bigcup (F_n \cap A)\,,$$

where $F_n \cap A \in \mathscr{M}$, so $(\cup F_n) \cap A \in \mathscr{M}$, which implies $\cup F_n \in \mathscr{M}_1$. Similarly, we can show that $\mathscr{M}_1$ is closed under monotone intersection, so it is a monotone class containing $\mathscr{A}$. $\mathscr{M}_1 \subseteq \mathscr{M}$ and thus $\mathscr{M}_1 = \mathscr{M}$, and so $\mathscr{M}$ is closed

under finite intersection.

Since $\mathscr{M}$ is closed under finite intersections and complements, it must be closed under finite unions. For a sequence $F_i \in \mathscr{M}$, we can define $N_n = \cup_{i=1}^n F_i$. Then $N_1 \subseteq N_2 \subseteq \cdots$, so $\cup_{i=1}^\infty F_i = \cup_n N_n \in \mathscr{M}$, and $\mathscr{M}$ is closed under countable unions.

We have shown that $\varnothing \in \mathscr{M}$ and $\mathscr{M}$ is closed under complementation and countable unions. Therefore, $\mathscr{M}$ is a σ-algebra. Since $\mathscr{M}$ contains $\mathscr{A}$, $\mathscr{M} \subseteq \sigma(\mathscr{A})$, and we have shown the reverse inclusion. □

Proposition B.1.3.

If μ and ν are probability measures on $\sigma(\mathscr{A})$ that agree on $\mathscr{A}$, an algebra of subsets of Ω, then $\mu = \nu$ on $\sigma(\mathscr{A})$.

Proof of Proposition B.1.3. We want to show that the set collection $\mathscr{H} \subseteq \sigma(\mathscr{A})$ of sets $H \in \sigma(\mathscr{A})$ such that $\mu(H) = \nu(H)$ is a monotone class that contains $\mathscr{A}$, since by the Monotone Class Theorem, we will have $\sigma(\mathscr{A}) \subseteq \mathscr{H}$, so $\sigma(\mathscr{A}) = \mathscr{H}$.

Note that we are given $\mathscr{A} \subseteq \mathscr{H}$.

Suppose $H_1 \subseteq H_2 \subseteq \cdots$ in $\mathscr{H}$. We want to show that $\cup H_n \in \mathscr{H}$, i.e. that $\mu(\cup_n H_n) = \nu(\cup_n H_n)$.

Since $\mu(H_n) = \nu(H_n)$ for each $n \in \mathbb{N}$, we know $\lim_{n\to\infty} \mu(H_n) = \lim_{n\to\infty} \nu(H_n)$. By continuity of measure from below,

$$\mu\left(\cup_n H_n\right) = \lim_{n\to\infty} \mu(H_n) = \lim_{n\to\infty} \nu(H_n) = \nu\left(\cup_n H_n\right).$$

Hence $\cup_n H_n \in \mathscr{H}$, so $\mathscr{H}$ is closed under monotone unions. We can similarly show it is closed under monontone intersections using continuity from above (since μ, ν being probability measures guarantee $\mu(H_1) = \nu(H_1) < \infty$).

$\mathscr{H}$ is a monotone class containing $\mathscr{A}$, so $\mathscr{H} \supseteq \sigma(\mathscr{A})$. By definition, $\mathscr{H} \subseteq \sigma(\mathscr{A})$, so we have equality; μ and ν agree on all of $\sigma(\mathscr{A})$. □

This was a typical use of the Monotone Class Theorem. In general, to show that a property holds on a σ-algebra, we want to show that the property holds on a monotone class containing a set collection that generates the σ-algebra.

For completeness, we provide an alternative to the Monotone Class Theorem.

Definition B.1.4 (π-system, λ-system).

A π-system Π is a collection of subsets of Ω that is closed under finite intersections.
A λ-system Λ is a collection of subsets of Ω such that

(i) $\Omega \in \Lambda$,

(ii) If $A, B \in \Lambda$, then $B \setminus A \in \Lambda$, and

(iii) If $A_n \in \Lambda$ and $A_n \cap A_m = \varnothing$ for $n \neq m$, then $\cup_{n=1}^\infty A_n \in \Lambda$.

Theorem B.1.5 (Dynkin's π-λ lemma).

Any λ-system that contains a π-system Π contains $\sigma(\Pi)$. In particular, the smallest λ-system containing Π is $\sigma(\Pi)$.

With the π-λ lemma, we can show that some property holds on a σ-algebra if we can show that the subset it holds on is a λ-system that contains a π-system Π that generates the σ-algebra.[1]

[1] π-systems are easy to work with; the sets of all intervals in $\mathbb{R}$ and rectangles in $\mathbb{R}^2$ are π-systems.

B.2 Measurable Transformations and Random Variables

Definition B.2.1 (Measurable transformation, random variable).

Let $(\mathscr{X}, \mathscr{F})$ and $(\mathscr{Y}, \mathscr{G})$ be measurable spaces and $T : \mathscr{X} \to \mathscr{Y}$. We call T a measurable transformation if for every $G \in \mathscr{G}$, then preimage $T^{-1}(G) \in \mathscr{F}$.
In the special case where $\mathscr{Y} = \mathbb{R}$ and $\mathscr{G} = \mathscr{B}_{\mathbb{R}}$, T is called a random variable.

To check that a transformation is measurable, we don't have to check that every set in the image's σ-algebra is measurable. We can just check that every element in a generating set for that σ-algebra is measurable.

Lemma B.2.2.

A transformation $T : \mathscr{X} \to \mathscr{Y}$ is measurable if there is a collection $\mathscr{K} \subseteq \mathscr{G}$ such that $\mathscr{G} = \sigma(\mathscr{K})$ and for every $K \in \mathscr{K}$,

$$T^{-1}(K) \in \mathscr{F}.$$

Equivalently (as written in class), $f : (\mathscr{X}, \mathscr{F}) \to (\mathscr{Y}, \sigma(\mathscr{K}))$ is measurable if $f^{-1}(\mathscr{K}) \subseteq \mathscr{F}$.

Proof of Lemma B.2.2. Suppose that $f^{-1}(\mathscr{K}) \subseteq \mathscr{F}$. Because $\mathscr{F}$ is a σ-field, we know that $\sigma\left(f^{-1}(\mathscr{K})\right) \subseteq \mathscr{F}$. Really, we want to show that $f^{-1}(\sigma(\mathscr{K})) \subseteq \mathscr{F}$. We conjecture that $\sigma\left(f^{-1}(\mathscr{K})\right) = f^{-1}(\sigma(\mathscr{K}))$ because pullbacks generally behave well. Showing this will provide the desired result.

($\subseteq$) Since $\mathscr{K} \subseteq \sigma(\mathscr{K})$, $f^{-1}(\mathscr{K}) \subseteq f^{-1}\sigma(\mathscr{K})$. We see that $f^{-1}\sigma(\mathscr{K})$ is a σ-field of subsets of $\mathscr{X}$ containing $f^{-1}(\mathscr{K})$:

(i) We know that $\mathscr{Y} \subseteq \sigma(\mathscr{K})$, so $\mathscr{X} = f^{-1}(\mathscr{Y}) \subseteq f^{-1}\sigma(\mathscr{K})$

(ii) Suppose $A \in f^{-1}(\sigma(\mathscr{K}))$. There is a $B \in \sigma(\mathscr{K})$ such that $f(A) = B$. Then $A^c = f^{-1}(B^c)$, and since $\sigma(\mathscr{K})$ is closed under complementation, $B^c \in \sigma(\mathscr{K})$. Hence, $A^c \in f^{-1}(\sigma(\mathscr{K}))$.

(iii) Suppose $A_n \in f^{-1}(\sigma(\mathscr{K}))$, so we have $B_n \in \sigma(\mathscr{K})$ such that $f(A_n) = B_n$. Note that $\cup_n A_n = \cup_n f^{-1}(B_n) = f^{-1}(\cup_n B$

$\sigma\left(f^{-1}(\mathscr{K})\right)$ is the smallest σ-field containing $f^{-1}(\mathscr{K})$, so we have shown that $\sigma\left(f^{-1}(\mathscr{K})\right) \subseteq f^{-1}\sigma(\mathscr{K})$.

($\supseteq$) We cannot directly use the same "smallest σ-field" trick. Instead, define

$$\mathscr{H} = \{B \subseteq \mathscr{Y} : f^{-1}(B) \in \sigma\left(f^{-1}(\mathscr{K})\right)\}$$

with the goal of showing $\sigma(\mathscr{K}) \subseteq \mathscr{H}$. Now we can do the same trick. $\mathscr{K} \subseteq \mathscr{H}$ because for any $K \in \mathscr{K}$, $f^{-1}(K) \in f^{-1}(\mathscr{K}) \subseteq \sigma\left(f^{-1}(\mathscr{K})\right)$. We show that $\mathscr{H}$ is a σ-field:

(i) $f^{-1}(\mathscr{Y}) = \mathscr{X} \in \sigma\left(f^{-1}(\mathscr{K})\right)$, where the inclusion follows from $\sigma\left(f^{-1}(\mathscr{K})\right)$ being a σ-field on $\mathscr{X}$.

(ii) Suppose $B \in \mathscr{H}$, so $f^{-1}(B) \in \sigma\left(f^{-1}(\mathscr{K})\right)$. We know $f^{-1}(B^c) = \left(f^{-1}(B)\right)^c \in \sigma\left(f^{-1}(\mathscr{K})\right)$, so $B^c \in \mathscr{H}$.

(iii) Suppose $B_n \in \mathscr{H}$, so $f^{-1}(B_n) \in \sigma\left(f^{-1}(\mathscr{K})\right)$. Then

$$f^{-1}(\cup_n B_n) = \cup_n f^{-1}(B_n) \in \sigma\left(f^{-1}(\mathscr{K})\right),$$

so $\cup_n B_n \in \mathscr{H}$.

$\mathscr{H}$ is a σ-field containing $\mathscr{K}$, so $\sigma(\mathscr{K}) \subseteq \mathscr{H}$. By how we defined $\mathscr{H}$, this means that $f^{-1}(\sigma(\mathscr{K})) \subseteq \sigma\left(f^{-1}(\mathscr{K})\right)$. $\square$

Proposition B.2.3.

If X_n is a sequence of random variables defined on a measurable space $(\Omega, \mathscr{F})$, then $\limsup_{n\to\infty} X_n$ is a random variable.

Proof of Proposition B.2.3. By the class lemma, we just need to check that for an arbitrary Borel set B,

$$\left(\limsup_{n\to\infty} X_n\right)^{-1}(B) = \left\{\omega : \lim_{k\to\infty} \sup_{n\geq k} X_n(\omega) \in B\right\} \in \mathscr{F}.$$

We claim that $\sup_{n\geq k} X_n \in \mathscr{F}$. For each $a \in \mathbb{R}$,

$$\left\{\omega : \sup_{n\geq k} X_n \leq a\right\} = \bigcap_{n=k}^{\infty} \{\omega : X_n(\omega) \leq a\},$$

which is a countable intersection of sets that are in $\mathscr{F}$ by measurability of the individual X_ns.

Now we claim that for measurable Y_n such that $Y_n \geq Y_{n+1}$, $\lim_{n\to\infty} Y_n$ is measurable. Since the sequence is decreasing, we have $\lim_{n\to\infty} Y_n = \inf_{n\geq 1} Y_n$. For each $a \in \mathbb{R}$,

$$\{\omega : \inf_{n\geq 1} Y_n(\omega) \geq a\} = \bigcap_{n=1}^{\infty} \{\omega : Y_n(\omega) \geq a\} \in \mathscr{F},$$

which is a countable intersection of measurable sets (by measurability of each Y_n). Hence $\inf_{n\geq 1} Y_n$ is measurable. Applying this with $Y_k = \sup_{n\geq k} X_n$ completes the proof. □

Proposition B.2.4.

Suppose (X, Σ_X), (Y, Σ_Y) measurable spaces, $f : X \to Y$ is measurable and μ a measure on (X, Σ_X). Define the pushforward measure $\nu := f_{\#}\mu$, where

$$\nu(A) := \mu(f^{-1}(A))$$

for $A \in \Sigma_Y$.
For any measurable $g : Y \to \overline{\mathbb{R}}$,

$$\int_X (g \circ f) d\mu \;=\; \int_Y g d\nu$$

whenever both sides are well-defined.

Proof of Proposition B.2.4. (i) Suppose $g = \mathbb{1}_A$ for $A \in \Sigma_Y$ Then

$$\begin{aligned}
\int_X (g \circ f) d\mu &= \int_X \mathbb{1}_{f^{-1}(A)} d\mu \\
&= \mu(\{x \in X : f(x) \in A\}) = \mu(f^{-1}(A)), \\
\int_Y g d\nu &= \int_Y \mathbb{1}_A d\nu \\
&= \nu(A) = \mu(f^{-1}(A))
\end{aligned}$$

(ii) Suppose $g = \sum_{i=1}^n a_i \mathbb{1}_{A_i}$ for $a_i \geq 0, A_i \in \Sigma_Y$. Without loss, we can take A_i to be disjoint. Then

$$g(y) = \sum_{i=1}^n a_i \mathbb{1}(y \in A_i), \qquad (g \circ f)(x) = \sum_{i=1}^n a_i \mathbb{1}(x \in f^{-1}(A_i)).$$

Then

$$\begin{aligned}
\int_X (g \circ f) d\mu &= \int_X \sum_{i=1}^n a_i \mathbb{1}_{f^{-1}(A_i)}(x) d\mu \\
&= \sum_{i=1}^n a_i \int_X \mathbb{1}_{f^{-1}(A_i)}(x) d\mu \qquad \text{by linearity} \\
&= \sum_{i=1}^n a_i \int_Y \mathbb{1}_{A_i}(y) d\nu \qquad \text{by (i)} \\
&= \int_Y \sum_{i=1}^n a_i \mathbb{1}_{A_i}(y) d\nu \qquad \text{by linearity} \\
&= \int_Y g d\nu.
\end{aligned}$$

(iii) Suppose $g \geq 0$. Then $\exists\, g_n \uparrow g$ such that g_n are simple functions. We have

$$g_n(f(x)) \uparrow g(f(x)).$$

Consequently,

$$\begin{aligned}
\int_X g \circ f d\mu &= \lim_{n\to\infty} \int_X (g_n \circ f) d\mu \qquad \text{by MCT} \\
&= \lim_{n\to\infty} \int_Y g_n d\nu \qquad \text{by (ii)} \\
&= \int_Y \lim_{n\to\infty} g_n d\nu \qquad \text{by MCT} \\
&= \int_Y g d\nu.
\end{aligned}$$

(iv) Suppose g is integrable. Then let $g^+ := \max\{0, g\}$, $\quad g^- := \max\{0, -g\}$, g^+ and g^- are nonnegative, and

$$\begin{aligned}
\int_X (g \circ f) d\mu &= \int_X (g \circ f)^+ d\mu - \int_X (g \circ f)^- d\mu \qquad \text{by definition of the general integral} \\
&= \int_X \left(g^+ \circ f\right) d\mu - \int_X \left(g^- \circ f\right) d\mu \\
&= \int_Y g^+ d\nu - \int_Y g^- d\nu \\
&= \int_Y g d\nu \qquad \text{by definition of the general integral}
\end{aligned}$$

□

Proposition B.2.5.

Convergence almost surely implies convergence in probability but the converse does not hold.

Proof of Proposition B.2.5. Assume $X_n \to X$ almost surely. Fix $\epsilon > 0$ and define

$$g_n(\omega) := \mathbb{1}_{\{|X_n(\omega) - X(\omega)| \geq \epsilon\}}.$$

Then each g_n is measurable and $0 \le g_n \le 1$. Let

$$g(\omega) := \lim_{n\to\infty} g_n(\omega).$$

Since g is a pointwise limit of measurable functions, it is measurable.

By $X_n \to X$ a.s., there exists a measurable set $N \subseteq \Omega$ with $\mathbb{P}(N) = 0$ such that for every $\omega \notin N$ we have $X_n(\omega) \to X(\omega)$. Fix $\omega \notin N$. Then there exists $N(\omega)$ such that for all $n \ge N(\omega)$, $|X_n(\omega) - X(\omega)| < \epsilon$, hence $g_n(\omega) = 0$ for all $n \ge N(\omega)$ and therefore $g(\omega) = 0$. Thus $g = 0$ on $\Omega \setminus N$.

Now split the integral of g:

$$\int_\Omega g\, d\mathbb{P} \;=\; \int_{\Omega\setminus N} g\, d\mathbb{P} \;+\; \int_N g\, d\mathbb{P}.$$

On $\Omega \setminus N$ we have $g = 0$, so $\int_{\Omega\setminus N} g\, d\mathbb{P} = 0$. On N we have $0 \le g \le 1$, hence

$$0 \le \int_N g\, d\mathbb{P} \le \int_N 1\, d\mathbb{P} = \mathbb{P}(N) = 0,$$

so $\int_N g\, d\mathbb{P} = 0$. Therefore $\int_\Omega g\, d\mathbb{P} = 0$.

Since $0 \le g_n \le 1$ and $\int_\Omega 1 d\mathbb{P} = \mathbb{P}(\Omega) = 1$, the Dominated Convergence Theorem yields

$$\lim_{n\to\infty} \int_\Omega g_n\, d\mathbb{P} = \int_\Omega \lim_{n\to\infty} g_n\, d\mathbb{P} = \int_\Omega g\, d\mathbb{P} = 0.$$

But $\int_\Omega g_n\, d\mathbb{P} = \mathbb{P}(|X_n - X| \ge \epsilon)$, so $\mathbb{P}(|X_n - X| \ge \epsilon) \to 0$. Since $\epsilon > 0$ was arbitrary, $X_n \to X$ in probability. $\square$

For the more complicated proof, consider

Proof. Suppose $X_n, X : (\Omega, \mathscr{F}, \mathbb{P}) \to (\mathbb{R}, \mathscr{B}_\mathbb{R})$ such that for almost every $\omega \in \Omega$, $X_n(\omega) \to X(\omega)$.

We write $X_n(\omega) \to X(\omega)$ if $\forall \epsilon > 0$, $\exists N \in \mathbb{N} : \forall n \ge N$, $|X_n(\omega) - X(\omega)| < \epsilon$.

$X_n(\omega) \nrightarrow X(\omega)$ if $\exists \epsilon > 0 : \forall N \in \mathbb{N}$, $\exists n \ge N : |X_n(\omega) - X(\omega)| \ge \epsilon$.

Consider $\{\omega \in \Omega : X_n(\omega) \nrightarrow X(\omega)\}$. Then since $\frac{1}{k} < \epsilon$ for some large k,

$$0 = \mathbb{P}\{\omega \in \Omega : X_n(\omega) \nrightarrow X(\omega)\} = \mathbb{P}\left\{\bigcup_{k=1}^\infty \bigcap_{N=1}^\infty \bigcup_{n\ge N} \left\{\omega \in \Omega : |X_n(\omega) - X(\omega)| \ge \frac{1}{k}\right\}\right\}.$$

By monotonicity, $\forall k \in \mathbb{N}$,

$$\mathbb{P}\left[\bigcap_{N=1}^\infty \bigcup_{n\ge N} \left\{\omega \in \Omega : |X_n(\omega) - X(\omega)| \ge \frac{1}{k}\right\}\right] = 0.$$

Fix $\epsilon > 0$ and pick K large enough so $\frac{1}{K} < \epsilon$. Then

$$\mathbb{P}\left[\bigcap_{N=1}^\infty \bigcup_{n\ge N} \left\{\omega \in \Omega : |X_n(\omega) - X(\omega)| \ge \frac{1}{K}\right\}\right] = 0.$$

Note that

$$\bigcup_{n\ge N} \left\{\omega \in \Omega : |X_n(\omega) - X(\omega)| \ge \frac{1}{K}\right\}$$

is a decreasing sequence in N. Since

$$\bigcup_{n\ge N} \left\{\omega \in \Omega : |X_n(\omega) - X(\omega)| \ge \frac{1}{K}\right\} \subseteq \Omega,$$

where $\mathbb{P}[\Omega] = 1$, we have by continuity of measure from above that

$$\lim_{N\to\infty} \mathbb{P}\left[\bigcup_{n\geq N}\left\{\omega \in \Omega : |X_n(\omega) - X(\omega)| \geq \frac{1}{K}\right\}\right] = 0.$$

By choice of K,

$$\bigcup_{n\geq N}\left\{\omega \in \Omega : |X_n(\omega) - X(\omega)| \geq \frac{1}{K}\right\} \supseteq \bigcup_{n\geq N}\{\omega \in \Omega : |X_n(\omega) - X(\omega)| \geq \epsilon\} \supseteq \{\omega \in \Omega : |X_N(\omega) - X(\omega)| \geq \epsilon\}.$$

$$\lim_{N\to\infty} \mathbb{P}\left[\{\omega \in \Omega : |X_N(\omega) - X(\omega)| \geq \epsilon\}\right] = 0,$$

and thus

$$X_n \xrightarrow{m} X.$$

To show the converse does not hold, put $X_1 = \mathbb{1}_{[0,\frac{1}{2}]}$, $X_2 = \mathbb{1}_{(\frac{1}{2},1]}$, $X_3 = \mathbb{1}_{[0,\frac{1}{3}]}$, $X_4 = \mathbb{1}_{[\frac{1}{3},\frac{2}{3}]}$, $X_5 = \mathbb{1}_{[\frac{2}{3},1]}$, $X_6 = \mathbb{1}_{[0,\frac{1}{4}]}$, and so on. We see that $X_n = 1$ on a set of measure less than $\frac{2}{n}$ and is 0 otherwise. With $X = 0$ defined on $[0,1]$ and $\epsilon \in (0,1)$, $\mathbb{P}[\omega \in [0,1] : |X_n(\omega) - X(\omega)| > \epsilon] = \frac{1}{m} \xrightarrow{m\to\infty} 0$ for m being the group number of X_n, but for each $\omega \in [0,1]$ $X_n(\omega) \not\to 0 = X(\omega)$. □

B.3 Independence

Recall that for a random variable $X : (\Omega, \mathscr{F}, \mathbb{P}) \to (\mathbb{R}, \mathscr{B}_{\mathbb{R}})$, we defined $\sigma(X)$ as the smallest σ-field such that $X : (\Omega, \sigma(X), \mathbb{P}) \to (\mathbb{R}, \mathscr{B}_{\mathbb{R}})$. It ends up that $\sigma(X) = X^{-1}(\mathscr{B}_{\mathbb{R}})$.

Proposition B.3.1.

If $A \in \sigma(X)$, then for every $\omega, \omega' \in \Omega$,

$$X(\omega) = X(\omega') \implies \mathbb{1}_A(\omega) = \mathbb{1}_A(\omega').$$

In other words, if the value of X is the same for ω, ω', then no set in $\sigma(X)$ can distinguish between the two (in membership).
If $X(\Omega)$ is countable, then the reverse implication holds; i.e., if for every $\omega, \omega' \in \Omega$, $X(\omega) = X(\omega') \implies \mathbb{1}_A(\omega) = \mathbb{1}_A(\omega')$, then $A \in \sigma(X)$.

Proof of Proposition B.3.1. Fix $A \in \sigma(X)$ and $\omega, \omega' \in \Omega$ such that $X(\omega) = X(\omega')$. By definition of $X : (\Omega, \sigma(X), \mathbb{P}) \to (\mathbb{R}, \mathscr{B}$ being the smallest σ-algebra making X measurable, we know there exists $B \in \mathscr{B}_{\mathbb{R}}$ such that $A = X^{-1}(B)$.

If $X(\omega) = X(\omega') \in B$, then $\omega, \omega' \in X^{-1}(B) = A$. If $X(\omega) = X(\omega') \notin B$, then $\omega, \omega' \notin X^{-1}(B) = A$. There are no other possibilities, so either ω, ω' are both in A or both not in A.

For the case where $X(\Omega)$ is countable, consider $A \subseteq \Omega$ such that it cannot distinguish ω, ω' who map to the same value under X. To show $A \in \sigma(X)$, we need to find $B \in \mathscr{B}_{\mathbb{R}}$ such that $A = X^{-1}(B)$. Defining $B = X(A) \subseteq X(\Omega)$, we see that B is the countable union of points, which are closed sets in $\mathbb{R}$. Therefore, $B \in \mathscr{B}_{\mathbb{R}}$, and we see that

$$X^{-1}(B) = X^{-1}(X(A)) = A,$$

so $A \in \sigma(X)$. □

Proposition B.3.2.

If $X_1, \ldots, X_n : (\Omega, \mathscr{F}, \mathbb{P}) \to (\mathbb{R}, \mathscr{B}_{\mathbb{R}})$ are independent random variables and $f : (\mathbb{R}, \mathscr{B}_{\mathbb{R}}) \to (\mathbb{R}, \mathscr{B}_{\mathbb{R}})$ be a measurable transformation. Then $f(X_1), \ldots, f(X_n)$ are independent.

Proof of Proposition B.3.2. First we prove the $n = 2$ case.

Consider $A_i \in \sigma(f(X_i))$ and $A_j \in \sigma(f(X_j))$. There exist some $B_i, B_j \in \mathscr{B}_{\mathbb{R}}$ such that $A_i = (f \circ X_i)^{-1}(B_i)$ and $A_j = (f \circ X_j)^{-1}(B_j)$. We want to show independence of A_i and A_j. We see that

$$\begin{aligned}
\mathbb{P}\left[(f \circ X_i)^{-1} B_i \cap (f \circ X_j)^{-1} B_j\right] &= \mathbb{P}\left[\omega \in \Omega : (f \circ X_i)(\omega) \in B_i, (f \circ X_j)(\omega) \in B_j\right] \\
&= \mathbb{P}\left[\omega \in \Omega : X_i(\omega) \in f^{-1}(B_i), X_j(\omega) \in f^{-1}(B_j)\right] \\
&= \mathbb{P}\left[X_i^{-1}\left(f^{-1}B_i\right) \cap X_j^{-1}\left(f^{-1}B_j\right)\right].
\end{aligned}$$

Recalling that $f : (\mathbb{R}, \mathscr{B}_{\mathbb{R}}) \to (\mathbb{R}, \mathscr{B}_{\mathbb{R}})$ is measurable, we see that $f^{-1}B_i, f^{-1}B_j \in \mathscr{B}_{\mathbb{R}}$. Because $X_i^{-1}(f^{-1}B_i) \in \sigma(X_i)$ and $X_j^{-1}(f^{-1}B_j) \in \sigma(X_j)$, by independence of X_i and X_j, we get

$$\begin{aligned}
\mathbb{P}\left[(f \circ X_i)^{-1}B_i \cap (f \circ X_j)^{-1}B_j\right] &= \mathbb{P}\left[X_i^{-1}(f^{-1}B_i)\right]\mathbb{P}\left[X_j^{-1}(f^{-1}B_j)\right] \\
&= \mathbb{P}\left[(f \circ X_i)^{-1} B_i\right]\mathbb{P}\left[(f \circ X_j)^{-1} B_j\right] \\
\mathbb{P}\left[A_i \cap A_j\right] &= \mathbb{P}\left[A_i\right]\mathbb{P}\left[A_j\right].
\end{aligned}$$

A_i and A_j were arbitrary in $\sigma(f(X_i))$ and $\sigma(f(X_j))$, so $f(X_i)$ and $f(X_j)$ are independent.

For $n > 2$, the proof follows exactly the same structure. □

Proof. Consider arbitrary sets $A_1 \in \sigma(f(X_1)), \ldots, A_n \in \sigma(f(X_n))$. By definition of the σ-algebra generated by a random variable, there exist sets $B_1, \ldots, B_n \in \mathscr{B}_{\mathbb{R}}$ such that

$$A_i = (f \circ X_i)^{-1}(B_i) \quad \text{for } i = 1, \ldots, n.$$

We want to show that

$$\mathbb{P}\left[A_1 \cap \cdots \cap A_n\right] = \mathbb{P}[A_1] \cdots \mathbb{P}[A_n].$$

We compute:

$$\begin{aligned}
\mathbb{P}\left[A_1 \cap \cdots \cap A_n\right] &= \mathbb{P}\left[(f \circ X_1)^{-1}(B_1) \cap \cdots \cap (f \circ X_n)^{-1}(B_n)\right] \\
&= \mathbb{P}\left[\{\omega \in \Omega : (f \circ X_1)(\omega) \in B_1, \ldots, (f \circ X_n)(\omega) \in B_n\}\right] \\
&= \mathbb{P}\left[\{\omega \in \Omega : X_1(\omega) \in f^{-1}(B_1), \ldots, X_n(\omega) \in f^{-1}(B_n)\}\right] \\
&= \mathbb{P}\left[X_1^{-1}\left(f^{-1}(B_1)\right) \cap \cdots \cap X_n^{-1}\left(f^{-1}(B_n)\right)\right].
\end{aligned}$$

Since $f : (\mathbb{R}, \mathscr{B}_{\mathbb{R}}) \to (\mathbb{R}, \mathscr{B}_{\mathbb{R}})$ is measurable and $B_i \in \mathscr{B}_{\mathbb{R}}$ for each i, we have $f^{-1}(B_i) \in \mathscr{B}_{\mathbb{R}}$ for $i = 1, \ldots, n$. Therefore, $X_i^{-1}(f^{-1}(B_i)) \in \sigma(X_i)$ for each $i = 1, \ldots, n$.

By independence of $X_1, \ldots, X_n$, we get:

$$\begin{aligned}
\mathbb{P}\left[X_1^{-1}(f^{-1}(B_1)) \cap \cdots \cap X_n^{-1}(f^{-1}(B_n))\right] &= \mathbb{P}\left[X_1^{-1}(f^{-1}(B_1))\right] \cdots \mathbb{P}\left[X_n^{-1}(f^{-1}(B_n))\right] \\
&= \mathbb{P}\left[(f \circ X_1)^{-1}(B_1)\right] \cdots \mathbb{P}\left[(f \circ X_n)^{-1}(B_n)\right] \\
&= \mathbb{P}[A_1] \cdots \mathbb{P}[A_n].
\end{aligned}$$

Since $A_1, \ldots, A_n$ were arbitrary in $\sigma(f(X_1)), \ldots, \sigma(f(X_n))$ respectively, we conclude that $f(X_1), \ldots, f(X_n)$ are independent. $\square$

Proposition B.3.3.

If $Z = g(X)$ for a random variable X and measurable $g : \mathbb{R} \to \mathbb{R}$, then $\sigma(Z) \subseteq \sigma(X)$.

Proof of Proposition B.3.3. For any Borel set $B \in \mathscr{B}_{\mathbb{R}}$, we see that

$$\begin{aligned} Z^{-1}(B) = (g \circ X)^{-1}(B) &= \{\omega : g(X(\omega)) \in B\} \\ &= \{\omega : X(\omega) \in g^{-1}(B)\} \\ &= \{\omega : \omega \in X(g^{-1}(B))\} \\ &= \{\omega : \omega \in X(B')\} \\ &= X^{-1}(B') \in X^{-1}(\mathscr{B}_{\mathbb{R}}) = \sigma(X) \end{aligned}$$

for some $B' \in \mathscr{B}_{\mathbb{R}}$ by measurability of g. Since B was arbitrary, we have

$$\sigma(Z) = Z^{-1}(\mathscr{B}_{\mathbb{R}}) \subseteq \sigma(X).$$

$\square$

B.4 Limits, Sets, and Quantifiers

Limsup and Liminf as Sets

For any sequence (Y_n) of measurable functions:

$$\{\limsup_{n\to\infty} Y_n > a\} = \bigcap_{k=1}^{\infty} \bigcup_{n=k}^{\infty} \{Y_n > a\} \quad \text{“}\forall k\, \exists n \geq k\text{”}$$

$$\{\limsup_{n\to\infty} Y_n < a\} = \bigcup_{k=1}^{\infty} \bigcap_{n=k}^{\infty} \{Y_n < a\} \quad \text{“}\exists k\, \forall n \geq k\text{”}$$

$$\{\liminf_{n\to\infty} Y_n < a\} = \bigcap_{k=1}^{\infty} \bigcup_{n=k}^{\infty} \{Y_n < a\} \quad \text{“}\forall k\, \exists n \geq k\text{”}$$

$$\{\liminf_{n\to\infty} Y_n > a\} = \bigcup_{k=1}^{\infty} \bigcap_{n=k}^{\infty} \{Y_n > a\} \quad \text{“}\exists k\, \forall n \geq k\text{”}$$

Sup and Inf as Sets

$$\{\sup_{n\geq k} Y_n > a\} = \bigcup_{n=k}^{\infty} \{Y_n > a\} \quad \text{“}\exists n \geq k\text{”}$$
$$\{\sup_{n\geq k} Y_n \leq a\} = \bigcap_{n=k}^{\infty} \{Y_n \leq a\} \quad \text{“}\forall n \geq k\text{”}$$

$$\{\inf_{n\geq k} Y_n < a\} = \bigcup_{n=k}^{\infty} \{Y_n < a\} \quad \text{“}\exists n \geq k\text{”}$$
$$\{\inf_{n\geq k} Y_n \geq a\} = \bigcap_{n=k}^{\infty} \{Y_n \geq a\} \quad \text{“}\forall n \geq k\text{”}$$
$$\{\sup_{n\geq k} Y_n < a\} = \bigcup_{r\in\mathbb{Q},\, r<a} \bigcap_{n=k}^{\infty} \{Y_n \leq r\}$$
$$\{\inf_{n\geq k} Y_n > a\} = \bigcup_{r\in\mathbb{Q},\, r>a} \bigcap_{n=k}^{\infty} \{Y_n \geq r\}$$

We can also write (if continuity from above holds)

$$\mathbb{P}(T \leq t) = \mathbb{P}\Big(\bigcap_{n=1}^{\infty} \{T < t + 1/n\}\Big) = \lim_{n\to\infty} \mathbb{P}(T < t + 1/n).$$

(See 2016 HW1.5.)

General Translation Rules

$$\forall \longleftrightarrow \bigcap$$
$$\exists \longleftrightarrow \bigcup$$
$$\sup \longleftrightarrow \bigcup \quad \text{(for strict inequalities)}$$
$$\inf \longleftrightarrow \bigcap \quad \text{(for non-strict inequalities)}$$

Monotone Sequences

If $Y_n \uparrow$ (increasing): $\lim_{n\to\infty} Y_n = \sup_{n\geq 1} Y_n$
If $Y_n \downarrow$ (decreasing): $\lim_{n\to\infty} Y_n = \inf_{n\geq 1} Y_n$

Key Identities

$$\limsup_{n\to\infty} Y_n = \inf_{k\geq 1} \sup_{n\geq k} Y_n = \lim_{k\to\infty} \sup_{n\geq k} Y_n$$
$$\liminf_{n\to\infty} Y_n = \sup_{k\geq 1} \inf_{n\geq k} Y_n = \lim_{k\to\infty} \inf_{n\geq k} Y_n$$

Section C

Marcus Kuo

Appendix - Exercises

C.1 Measure Theory

Proposition C.1.1 (Exercise 1.5).

Show that if $X = \mathbb{R}$ then the Borel σ-algebra $\mathscr{B}_{\mathbb{R}}$ is the σ-algebra generated by the collection of open half-lines (a, ∞) where $a \in \mathbb{Q}$.

Proof of Proposition C.1.1. Let

$$\mathscr{G} := \sigma\big(\{(q, \infty) : q \in \mathbb{Q}\}\big).$$

First, for any $a \in \mathbb{R}$ we have

$$(a, \infty) = \bigcup_{\substack{q \in \mathbb{Q} \\ q > a}} (q, \infty) \in \mathscr{G},$$

so *all* rays (a, ∞) (with real a) lie in $\mathscr{G}$.

Next, for any $b \in \mathbb{R}$,

$$[b, \infty) = \bigcap_{n=1}^{\infty} (b - \tfrac{1}{n}, \infty) \in \mathscr{G}, \qquad (-\infty, b) = \big([b, \infty)\big)^c \in \mathscr{G}.$$

Hence for any $a < b$,

$$(a, b) = (a, \infty) \cap (-\infty, b) \in \mathscr{G}.$$

Therefore every open interval is in $\mathscr{G}$, so every open set (a countable union of open intervals) is in $\mathscr{G}$. Thus

$$\mathscr{B}_{\mathbb{R}} = \sigma(\text{open sets}) \subseteq \mathscr{G}.$$

Conversely each (q, ∞) is open, so $\mathscr{G} \subseteq \mathscr{B}_{\mathbb{R}}$. Hence $\mathscr{G} = \mathscr{B}_{\mathbb{R}}$. □

Proposition C.1.2 (Exercise 1.7).

Define a metric on $\Omega = \{0, 1\}^{\mathbb{N}}$ by setting

$$d(x, y) = 2^{-n(x,y)},$$

where $n(x, y)$ is defined to be the maximum n such that $x_i = y_i$ for all $i \leq n$.

(a) Show that this is in fact a metric.

(b) Show that the metric space (Ω, d) is compact.

(c) Show that the Borel σ-algebra on the metric space (Ω, d) is the σ-algebra generated by the cylinder sets.

Proof of Proposition C.1.2. It is convenient to make the definition precise as follows: for $x \neq y$ define

$$n(x, y) := \max\{n \geq 0 : x_i = y_i \ \forall\, i \leq n\},$$

(with the convention that $n(x, y) = 0$ if $x_1 \neq y_1$), and set $d(x, x) = 0$; equivalently one may take the convention $2^{-\infty} = 0$.

(a) (Metric.) Clearly $d(x, y) \geq 0$, and $d(x, y) = 0$ iff $x = y$ by definition. Symmetry is immediate since $n(x, y) = n(y, x)$.

For the triangle inequality, we prove the stronger *ultrametric* inequality

$$d(x, z) \leq \max\{d(x, y), d(y, z)\}.$$

Let $m := \min\{n(x, y), n(y, z)\}$. Then $x_i = y_i = z_i$ for all $i \leq m$, hence $n(x, z) \geq m$. Therefore

$$d(x, z) = 2^{-n(x,z)} \leq 2^{-m} = \max\{2^{-n(x,y)}, 2^{-n(y,z)}\} = \max\{d(x, y), d(y, z)\}.$$

This implies the usual triangle inequality.

(b) (Compactness.) Since (Ω, d) is a metric space, it suffices to prove sequential compactness.

Take any sequence $(x^{(k)})_{k\geq 1}$ in Ω. Because each coordinate takes values in $\{0, 1\}$, we can choose a subsequence along which the first coordinate is constant. From that subsequence, choose a further subsequence along which the second coordinate is constant, and so on. This produces a nested family of subsequences; take the diagonal subsequence $(x^{(k_j)})$.

Define $x \in \Omega$ by setting x_i equal to the eventual constant value of the i-th coordinate along the diagonal subsequence. Fix N. For all sufficiently large j, the sequence $x^{(k_j)}$ agrees with x in the first N coordinates, so $n(x^{(k_j)}, x) \geq N$, hence

$$d(x^{(k_j)}, x) \leq 2^{-N}.$$

Since N was arbitrary, $d(x^{(k_j)}, x) \to 0$. Thus every sequence has a convergent subsequence, so (Ω, d) is compact.

(c) (Borel σ-algebra and cylinders.) For $x \in \Omega$ and $n \in \mathbb{N}$, define the cylinder set determined by the first n coordinates of x:

$$C(x; n) := \{y \in \Omega : y_i = x_i \ \forall\, i \leq n\}.$$

Observe that the open ball of radius 2^{-n} around x is exactly this cylinder:

$$B(x, 2^{-n}) = \{y : d(x, y) < 2^{-n}\} = \{y : n(x, y) \geq n\} = C(x; n).$$

Hence cylinder sets form a base for the topology induced by d (every open set is a union of such balls/cylinders). Therefore the Borel σ-algebra (generated by open sets) is the same as the σ-algebra generated by cylinder sets.

□

Proposition C.1.3 (Lemma 1.17).

The algebra $\mathscr{A}$ generated by the intervals $J \subseteq [0, 1]$ consists of all subsets of $[0, 1]$ that are finite unions of intervals.

Proof of Proposition C.1.3. Let $\mathscr{C}$ be the collection of all finite unions of intervals contained in $[0,1]$. Then:

- $\mathscr{C}$ contains every interval $J \subseteq [0,1]$ (take a union with one term).
- $\mathscr{C}$ is an algebra: it is closed under finite unions by construction, and if $E = \bigcup_{k=1}^m I_k$ with intervals I_k, then $[0,1] \setminus E$ is a finite union of intervals as well (after ordering endpoints and taking the complementary gaps). Hence $\mathscr{C}$ is closed under complements in $[0,1]$.

Since $\mathscr{A}$ is the *smallest* algebra containing all intervals, we have $\mathscr{A} \subseteq \mathscr{C}$.

Conversely, because $\mathscr{A}$ is an algebra containing all intervals, it is closed under finite unions; hence every finite union of intervals lies in $\mathscr{A}$, i.e. $\mathscr{C} \subseteq \mathscr{A}$.

Therefore $\mathscr{A} = \mathscr{C}$. □

Proposition C.1.4 (Exercise 1.26).

Define a d-dimensional rectangle to be a Cartesian product $A_1 \times \cdots \times A_d$, where each A_i is an interval of the real line. Let $\mathscr{A}_d$ be the algebra generated by all rectangles. Prove that there is a unique measure $\lambda = \lambda_d$ on $\mathscr{A}_d$ such that for any rectangle,

$$\lambda(A_1 \times \cdots \times A_d) = \prod_{i=1}^d |A_i|,$$

where $|A_i|$ is the length of the interval A_i.

Proof of Proposition C.1.4. (Outline that is usually sufficient on an exam; the only real work is well-definedness and countable additivity.)

Step 1: Disjoint rectangle decompositions. Let $\mathscr{R}$ be the collection of *half-open* rectangles

$$R = \prod_{i=1}^d (a_i, b_i] \qquad (a_i < b_i).$$

Every (finite) union of rectangles can be written as a finite union of half-open rectangles, and any finite union of half-open rectangles can be refined into a *finite disjoint* union of half-open rectangles by taking all endpoints in each coordinate and forming the corresponding finite grid. In particular, every $E \in \mathscr{A}_d$ admits a representation

$$E = \bigsqcup_{k=1}^m R_k, \qquad R_k \in \mathscr{R} \text{ pairwise disjoint.}$$

Step 2: Define λ and show it is well-defined. For $R = \prod_{i=1}^d (a_i, b_i] \in \mathscr{R}$ define

$$\lambda_0(R) := \prod_{i=1}^d (b_i - a_i).$$

For general $E \in \mathscr{A}_d$, choose a disjoint decomposition $E = \bigsqcup_{k=1}^m R_k$ with $R_k \in \mathscr{R}$ and set

$$\lambda(E) := \sum_{k=1}^m \lambda_0(R_k).$$

To see this does not depend on the chosen disjoint decomposition, take two such decompositions and refine both to a common grid (using the union of all endpoints appearing in either decomposition). On the common refinement, both decompositions become unions of the same collection of elementary grid rectangles, and the sum of λ_0 over those elementary rectangles is the same in both cases. Hence λ is well-defined.

By construction λ is finitely additive on $\mathscr{A}_d$, and it satisfies the desired rectangle formula.

Step 3: Countable additivity on $\mathscr{A}_d$. Let $(E_n)_{n\geq 1} \subseteq \mathscr{A}_d$ be pairwise disjoint and suppose $E := \bigsqcup_{n\geq 1} E_n \in \mathscr{A}_d$. Write E as a *finite disjoint* union of half-open rectangles:

$$E = \bigsqcup_{j=1}^{M} R_j, \qquad R_j \in \mathscr{R}.$$

Then for each fixed j,

$$R_j = \bigsqcup_{n\geq 1} (E_n \cap R_j)$$

is a disjoint union. Moreover, each $E_n \cap R_j \in \mathscr{A}_d$ and can be written as a finite disjoint union of rectangles in $\mathscr{R}$ contained in R_j.

A standard fact (proved by induction on d using the $d = 1$ case for interval length and Tonelli/monotone convergence for nonnegative sums of indicators) is that λ_0 is *countably additive on* $\mathscr{R}$: if $R \in \mathscr{R}$ and $R = \bigsqcup_{k\geq 1} R_k$ with disjoint $R_k \in \mathscr{R}$, then $\lambda_0(R) = \sum_k \lambda_0(R_k)$. Applying this inside each R_j and summing over $j = 1, \ldots, M$ yields

$$\lambda(E) = \sum_{j=1}^{M} \lambda_0(R_j) = \sum_{j=1}^{M} \sum_{n\geq 1} \lambda(E_n \cap R_j) = \sum_{n\geq 1} \sum_{j=1}^{M} \lambda(E_n \cap R_j) = \sum_{n\geq 1} \lambda(E_n),$$

where we used disjointness to justify the sums and the fact that each E_n is the disjoint union of $(E_n \cap R_j)_{j\leq M}$.

Thus λ is a measure on $\mathscr{A}_d$.

Step 4: Uniqueness. If $\tilde{\lambda}$ is another measure on $\mathscr{A}_d$ with the same rectangle formula, then for any $E \in \mathscr{A}_d$ with disjoint rectangle decomposition $E = \bigsqcup_{k=1}^{m} R_k$,

$$\tilde{\lambda}(E) = \sum_{k=1}^{m} \tilde{\lambda}(R_k) = \sum_{k=1}^{m} \lambda_0(R_k) = \lambda(E).$$

Hence $\tilde{\lambda} = \lambda$ on all of $\mathscr{A}_d$. □

Proposition C.1.5 (Exercise 1.33).

Let X_n be a sequence of random variables defined on a measurable space $(\Omega, \mathscr{F})$. Prove that $\limsup_{n\to\infty} X_n$ is a random variable.

Proof of Proposition C.1.5. Recall the identity

$$\limsup_{n\to\infty} X_n \;=\; \inf_{m\geq 1} \sup_{n\geq m} X_n.$$

Fix $a \in \mathbb{R}$. Then

$$\Big\{\limsup_{n\to\infty} X_n > a\Big\} = \bigcap_{m=1}^{\infty} \Big\{\sup_{n\geq m} X_n > a\Big\} = \bigcap_{m=1}^{\infty} \bigcup_{n=m}^{\infty} \{X_n > a\}.$$

Each set $\{X_n > a\}$ lies in $\mathscr{F}$ since X_n is measurable, and $\mathscr{F}$ is closed under countable unions and intersections. Hence the displayed set is in $\mathscr{F}$ for every a, so $\limsup_{n\to\infty} X_n$ is measurable (a random variable). □

Proposition C.1.6 (HW1.1(a) (Sequence space metric)).

Let $\Omega = \{0,1\}^{\mathbb{N}}$. For $x = (x_i)_{i\geq 1}, y = (y_i)_{i\geq 1} \in \Omega$, define

$$n(x,y) := \max\{n \in \mathbb{N}_0 : \; x_i = y_i \;\forall\, i \leq n\},$$

with the convention that $n(x,y) = \infty$ if $x = y$, and $\mathbb{N}_0 = \{0, 1, 2, \dots\}$. Define

$$d(x,y) := 2^{-n(x,y)} \quad \text{with the convention } 2^{-\infty} = 0.$$

Then d is a metric on Ω.

Proof of Proposition C.1.6. First, $d(x,y) \geq 0$ and $d(x,y) = d(y,x)$ are immediate from the definition.

If $x = y$, then $n(x,y) = \infty$ so $d(x,y) = 0$. Conversely, if $d(x,y) = 0$, then $2^{-n(x,y)} = 0$ forces $n(x,y) = \infty$, hence $x = y$. Thus $d(x,y) = 0 \iff x = y$.

For the triangle inequality, fix $x, y, z \in \Omega$ and set $m := n(x,y)$ and $\ell := n(y,z)$. Then x and y agree on coordinates $1, \dots, m$, and y and z agree on $1, \dots, \ell$. Hence x and z agree on $1, \dots, \min\{m, \ell\}$, i.e.

$$n(x,z) \geq \min\{m, \ell\}.$$

Therefore

$$d(x,z) = 2^{-n(x,z)} \leq 2^{-\min\{m,\ell\}} = \max\{2^{-m}, 2^{-\ell}\} = \max\{d(x,y), d(y,z)\} \leq d(x,y) + d(y,z).$$

So d satisfies the triangle inequality and is a metric. □

Proposition C.1.7 (HW1.1(b) (Compactness)).

The metric space (Ω, d) is compact.

Proof of Proposition C.1.7. It suffices to prove sequential compactness. Let $(x^{(k)})_{k\geq 1}$ be any sequence in Ω, where $x^{(k)} = (x_i^{(k)})_{i\geq 1}$.

Since each coordinate takes values in $\{0, 1\}$, there is an infinite subsequence along which the first coordinate is constant. From that subsequence, extract an infinite subsequence along which the second coordinate is constant; continue inductively. This yields a nested family of subsequences

$$(x^{(k_j^{(n)})})_{j\geq 1} \quad (n = 1, 2, \dots)$$

such that for each fixed n, the first n coordinates are constant along the subsequence $(x^{(k_j^{(n)})})_{j\geq 1}$.

Define the diagonal subsequence $y^{(n)} := x^{(k_n^{(n)})}$. Fix $N \geq 1$. For all $n \geq N$, $y^{(n)}$ belongs to the Nth subsequence, hence the first N coordinates of $y^{(n)}$ are constant for all $n \geq N$. Define $y \in \Omega$ by letting y_i be that eventual constant value of the ith coordinate.

Now fix $\varepsilon > 0$. Choose N such that $2^{-N} < \varepsilon$. For all $n \geq N$, $y^{(n)}$ agrees with y on coordinates $1, \dots, N$, so $n(y^{(n)}, y) \geq N$ and

$$d(y^{(n)}, y) = 2^{-n(y^{(n)},y)} \leq 2^{-N} < \varepsilon.$$

Thus $y^{(n)} \to y$ in (Ω, d), so every sequence has a convergent subsequence. Hence (Ω, d) is compact. □

Proposition C.1.8 (HW1.1(c) (Borel σ-algebra equals cylinder σ-algebra)).

Let $\mathcal{C}$ denote the collection of cylinder sets

$$C(a_1, \dots, a_n) := \{x \in \Omega : \ x_1 = a_1, \dots, x_n = a_n\}, \qquad n \geq 1, \ a_i \in \{0, 1\}.$$

Then the Borel σ-algebra $\mathcal{B}(\Omega)$ of (Ω, d) equals $\sigma(\mathcal{C})$.

Proof of Proposition C.1.8. First, every cylinder set is open. Indeed, fix a cylinder $C(a_1, \ldots, a_n)$ and $x \in C(a_1, \ldots, a_n)$. If $d(x, y) < 2^{-n}$, then $n(x, y) > n$, so in particular $y_i = x_i = a_i$ for $i = 1, \ldots, n$; hence $B(x, 2^{-n}) \subseteq C(a_1, \ldots, a_n)$, proving openness.

Conversely, cylinder sets form a basis for the topology. Let U be open and $x \in U$. Choose $r > 0$ such that $B(x, r) \subseteq U$. Pick n with $2^{-n} < r$ and consider the cylinder

$$C_n(x) := \{y \in \Omega : \ y_i = x_i \ \forall\, i = 1, \ldots, n\}.$$

For any $y \in C_n(x)$ we have $n(x, y) \geq n$, hence $d(x, y) \leq 2^{-n} < r$, so $C_n(x) \subseteq B(x, r) \subseteq U$. Thus every open set is a union of cylinder sets, so $\mathcal{B}(\Omega) \subseteq \sigma(\mathcal{C})$.

Since every cylinder set is open, $\sigma(\mathcal{C}) \subseteq \mathcal{B}(\Omega)$ as well. Therefore $\mathcal{B}(\Omega) = \sigma(\mathcal{C})$. □

Proposition C.1.9 (HW1.1(d) (Law of large numbers-type set is Borel)).

Define $S_n(x) := \sum_{i=1}^{n} x_i$ and $A_n(x) := \frac{1}{n} S_n(x)$. Then the set

$$E := \left\{ x \in \Omega : \ \lim_{n \to \infty} A_n(x) = \frac{1}{3} \right\}$$

is a Borel subset of (Ω, d).

Proof of Proposition C.1.9. Write the limit condition as a countable Boolean combination:

$$E = \bigcap_{k=1}^{\infty} \bigcup_{N=1}^{\infty} \bigcap_{n \geq N} \left\{ x \in \Omega : \ \left| A_n(x) - \frac{1}{3} \right| < \frac{1}{k} \right\}.$$

Fix n and k. The map $x \mapsto A_n(x)$ depends only on $(x_1, \ldots, x_n)$, hence is continuous in the metric d (if two sequences agree on the first n coordinates then A_n is identical). Therefore the set $\{|A_n - 1/3| < 1/k\}$ is open, hence Borel. Since E is formed from these sets using countable unions and intersections, E is Borel. □

Proposition C.1.10 (HW1.2 (A σ-algebra is not countably infinite)).

A σ-algebra cannot have countably infinite cardinality: it is either finite or uncountable.

Proof of Proposition C.1.10. Let $\mathscr{F}$ be a σ-algebra on Ω. If $\mathscr{F} = \{\varnothing, \Omega\}$ then it is finite. Assume $\mathscr{F}$ is infinite. We construct a strictly decreasing sequence

$$\Omega \supsetneq A_1 \supsetneq A_2 \supsetneq A_3 \supsetneq \cdots \quad \text{with each } A_n \in \mathscr{F}.$$

Start by choosing $A_1 \in \mathscr{F}$ with $\varnothing \neq A_1 \neq \Omega$ (otherwise $\mathscr{F}$ would be $\{\varnothing, \Omega\}$). Given A_n, consider the restricted σ-algebra on A_n:

$$\mathscr{F} \restriction_{A_n} := \{B \cap A_n : \ B \in \mathscr{F}\}.$$

If $\mathscr{F} \restriction_{A_n}$ were finite for both A_n and A_n^c, then $\mathscr{F}$ would be finite because every $B \in \mathscr{F}$ decomposes as

$$B = (B \cap A_n) \ \cup \ (B \cap A_n^c),$$

so $|\mathscr{F}| \leq |\mathscr{F} \restriction_{A_n}| \cdot |\mathscr{F} \restriction_{A_n^c}| < \infty$. Since $\mathscr{F}$ is infinite, at least one of these restrictions is infinite. Replace A_n by whichever of A_n or A_n^c has infinite restriction; in particular, we may assume $\mathscr{F} \restriction_{A_n}$ is infinite.

Because $\mathscr{F} \restriction_{A_n}$ is infinite, it contains some C with $\varnothing \neq C \neq A_n$. Write $C = B \cap A_n$ for some $B \in \mathscr{F}$, and set $A_{n+1} := C$. Then $A_{n+1} \in \mathscr{F}$ and $A_{n+1} \subsetneq A_n$.

Now define disjoint nonempty sets $D_n := A_n \setminus A_{n+1} \in \mathscr{F}$. For each subset $S \subseteq \mathbb{N}$ define

$$D_S := \bigcup_{n \in S} D_n \in \mathscr{F}$$

(a countable union). If $S \neq S'$, pick $m \in S \triangle S'$; then $D_m \subseteq D_S \triangle D_{S'}$, so $D_S \neq D_{S'}$. Hence the map $S \mapsto D_S$ is injective from $\mathcal{P}(\mathbb{N})$ into $\mathscr{F}$. Since $\mathcal{P}(\mathbb{N})$ is uncountable, $\mathscr{F}$ is uncountable. □

Proposition C.1.11 (HW1.3(A) (Union of nested σ-algebras is a field)).

Let $\mathscr{F}_1 \subseteq \mathscr{F}_2 \subseteq \cdots$ be σ-algebras on Ω and let

$$\mathscr{A} := \bigcup_{n=1}^{\infty} \mathscr{F}_n.$$

Then $\mathscr{A}$ is a field (algebra): it is closed under complements and finite unions.

Proof of Proposition C.1.11. Let $A \in \mathscr{A}$. Then $A \in \mathscr{F}_n$ for some n, so $A^c \in \mathscr{F}_n \subseteq \mathscr{A}$.

Let $A, B \in \mathscr{A}$. Choose m, n with $A \in \mathscr{F}_m$ and $B \in \mathscr{F}_n$. With $k := \max\{m, n\}$, the nesting gives $A, B \in \mathscr{F}_k$, hence $A \cup B \in \mathscr{F}_k \subseteq \mathscr{A}$. Thus $\mathscr{A}$ is a field. □

Proposition C.1.12 (HW1.3(B) (Union need not be a σ-algebra)).

There exist nested σ-algebras $\mathscr{F}_1 \subseteq \mathscr{F}_2 \subseteq \cdots$ such that $\bigcup_{n=1}^{\infty} \mathscr{F}_n$ is not a σ-algebra.

Proof of Proposition C.1.12. Let $\Omega = \mathbb{N}$. For each $n \geq 1$, let $\mathscr{F}_n := \sigma(\{1\}, \ldots, \{n\})$. Equivalently, $\mathscr{F}_n$ is the σ-algebra generated by the finite partition

$$\{1\}, \{2\}, \ldots, \{n\}, \ \{n+1, n+2, \ldots\}.$$

Hence every set in $\mathscr{F}_n$ is either finite (a union of some of $\{1\}, \ldots, \{n\}$) or cofinite (contains $\{n+1, n+2, \ldots\}$ and possibly some of $\{1\}, \ldots, \{n\}$). Therefore

$$\bigcup_{n=1}^{\infty} \mathscr{F}_n = \{\text{finite subsets of } \mathbb{N}\} \ \cup \ \{\text{cofinite subsets of } \mathbb{N}\}.$$

This is a field, but it is not a σ-algebra: each singleton $\{2k\}$ is finite, hence belongs to the union, but the countable union $\bigcup_{k=1}^{\infty} \{2k\} = \{2, 4, 6, \ldots\}$ is infinite and not cofinite, so it is not in the union. □

Proposition C.1.13 (HW1.4(a) (Almost sure equality is an equivalence relation)).

On a probability space $(\Omega, \mathscr{F}, \mathbb{P})$ define $d(A, B) := \mathbb{P}(A \triangle B)$. Define $A \sim A'$ iff $d(A, A') = 0$. Then $\sim$ is an equivalence relation on $\mathscr{F}$.

Proof of Proposition C.1.13. Reflexivity: $A \triangle A = \varnothing$, so $d(A, A) = \mathbb{P}(\varnothing) = 0$.

Symmetry: $A \triangle B = B \triangle A$, hence $d(A, B) = d(B, A)$.

Transitivity: If $A \sim B$ and $B \sim C$, then

$$A \triangle C \subseteq (A \triangle B) \cup (B \triangle C),$$

so by subadditivity

$$\mathbb{P}(A \triangle C) \leq \mathbb{P}(A \triangle B) + \mathbb{P}(B \triangle C) = 0.$$

Thus $A \sim C$. $\square$

Proposition C.1.14 (HW1.4(b) (Metric on equivalence classes)).

Let $\mathscr{F}^*$ be the set of equivalence classes under $\sim$. Define $d([A],[B]) := \mathbb{P}(A \triangle B)$. Then this is well-defined and makes $(\mathscr{F}^*, d)$ a metric space.

Proof of Proposition C.1.14. Well-defined: If $A \sim A'$ and $B \sim B'$, then

$$A' \triangle B' \subseteq (A' \triangle A) \cup (A \triangle B) \cup (B \triangle B'),$$

hence

$$\mathbb{P}(A' \triangle B') \leq \mathbb{P}(A' \triangle A) + \mathbb{P}(A \triangle B) + \mathbb{P}(B \triangle B') = \mathbb{P}(A \triangle B).$$

By symmetry the reverse inequality holds, so $\mathbb{P}(A' \triangle B') = \mathbb{P}(A \triangle B)$.

Metric axioms: nonnegativity and symmetry are immediate. If $d([A],[B]) = 0$, then $\mathbb{P}(A \triangle B) = 0$, i.e. $A \sim B$, so $[A] = [B]$. Triangle inequality follows from $A \triangle C \subseteq (A \triangle B) \cup (B \triangle C)$:

$$d([A],[C]) = \mathbb{P}(A \triangle C) \leq \mathbb{P}(A \triangle B) + \mathbb{P}(B \triangle C) = d([A],[B]) + d([B],[C]).$$

$\square$

Proposition C.1.15 (HW1.4(c) (Completeness)).

The metric space $(\mathscr{F}^*, d)$ is complete.

Proof of Proposition C.1.15. Let $([A_n])_{n \geq 1}$ be a Cauchy sequence in $(\mathscr{F}^*, d)$, and choose representatives $A_n \in \mathscr{F}$. Since it is Cauchy, we can pick a subsequence $(A_{n_k})_{k \geq 1}$ such that

$$\mathbb{P}(A_{n_{k+1}} \triangle A_{n_k}) < 2^{-k} \qquad (k \geq 1).$$

Let $B_k := A_{n_{k+1}} \triangle A_{n_k}$. Then $\sum_{k=1}^{\infty} \mathbb{P}(B_k) < \infty$. By Borel–Cantelli,

$$\mathbb{P}(\limsup_{k \to \infty} B_k) = 0.$$

Define

$$A := \liminf_{k \to \infty} A_{n_k} = \limsup_{k \to \infty} A_{n_k}$$

(the equality holds outside a null set because only finitely many B_k occur a.s., so membership in A_{n_k} stabilizes a.s.).

Fix $K \geq 1$. If $\omega \in A_{n_K} \triangle A$, then along the tail the membership changes at least once, so

$$A_{n_K} \triangle A \subseteq \bigcup_{k \geq K} (A_{n_{k+1}} \triangle A_{n_k}) = \bigcup_{k \geq K} B_k.$$

Therefore

$$\mathbb{P}(A_{n_K} \triangle A) \leq \sum_{k \geq K} \mathbb{P}(B_k) \xrightarrow[K \to \infty]{} 0.$$

So $[A_{n_K}] \to [A]$ in $(\mathscr{F}^*, d)$.

Finally, since $([A_n])$ is Cauchy, convergence of a subsequence implies convergence of the whole sequence to the same limit: given $\varepsilon > 0$, choose N such that $d([A_m],[A_n]) < \varepsilon/2$ for all $m, n \geq N$, and choose K such that $n_K \geq N$

and $d([A_{n_K}],[A]) < \varepsilon/2$. Then for $n \geq N$,

$$d([A_n],[A]) \leq d([A_n],[A_{n_K}]) + d([A_{n_K}],[A]) < \varepsilon.$$

Thus $(\mathscr{F}^*, d)$ is complete. □

Proposition C.1.16 (HW1.4(d) (Density of an algebra)).

Assume $\mathscr{F} = \sigma(\mathscr{A})$ where $\mathscr{A}$ is an algebra. Let $\mathscr{A}^* \subseteq \mathscr{F}^*$ be the set of equivalence classes that contain an element of $\mathscr{A}$. Then $\mathscr{A}^*$ is dense in $(\mathscr{F}^*, d)$.

Proof of Proposition C.1.16. Define

$$\mathcal{D} := \{F \in \mathscr{F} : \ \forall \varepsilon > 0 \ \exists A \in \mathscr{A} \text{ with } \mathbb{P}(F \triangle A) < \varepsilon\}.$$

Then $\mathscr{A} \subseteq \mathcal{D}$ trivially.

We claim $\mathcal{D}$ is a monotone class: (i) If $F_n \uparrow F$ with $F_n \in \mathcal{D}$, then for $\varepsilon > 0$ choose N with $\mathbb{P}(F \setminus F_N) < \varepsilon/2$ (using continuity from below of $\mathbb{P}$), and choose $A \in \mathscr{A}$ with $\mathbb{P}(F_N \triangle A) < \varepsilon/2$. Then

$$\mathbb{P}(F \triangle A) \leq \mathbb{P}(F \setminus F_N) + \mathbb{P}(F_N \triangle A) < \varepsilon.$$

So $F \in \mathcal{D}$. (ii) If $F_n \downarrow F$ with $F_n \in \mathcal{D}$, similarly choose N with $\mathbb{P}(F_N \setminus F) < \varepsilon/2$ (continuity from above, since $\mathbb{P}$ is finite), then approximate F_N by $A \in \mathscr{A}$ within $\varepsilon/2$ and conclude $F \in \mathcal{D}$.

By the monotone class theorem, the monotone class generated by the algebra $\mathscr{A}$ equals $\sigma(\mathscr{A}) = \mathscr{F}$. Since $\mathcal{D}$ is a monotone class containing $\mathscr{A}$, it contains $\mathscr{F}$. Thus for every $F \in \mathscr{F}$ and $\varepsilon > 0$ there exists $A \in \mathscr{A}$ with $\mathbb{P}(F \triangle A) < \varepsilon$. Equivalently, for every $[F] \in \mathscr{F}^*$ and $\varepsilon > 0$ there exists $[A] \in \mathscr{A}^*$ with $d([F],[A]) < \varepsilon$, so $\mathscr{A}^*$ is dense. □

Proposition C.1.17 (HW1.5(A) (Lebesgue measure on the unit interval is non-atomic)).

Lebesgue measure λ on $([0,1], \mathcal{B}([0,1]))$ is non-atomic: if F is measurable with $\lambda(F) > 0$, then there exists measurable $G \subseteq F$ with $0 < \lambda(G) < \lambda(F)$.

Proof of Proposition C.1.17. Fix measurable $F \subseteq [0,1]$ with $\lambda(F) > 0$. Define $\phi : [0,1] \to [0,\infty)$ by

$$\phi(t) := \lambda\big(F \cap [0,t]\big).$$

Then $\phi(0) = 0$ and $\phi(1) = \lambda(F) > 0$, and ϕ is continuous: for $s < t$,

$$0 \leq \phi(t) - \phi(s) = \lambda\big(F \cap (s,t]\big) \leq \lambda((s,t]) = t - s,$$

so $\phi(t) \to \phi(s)$ as $t \to s$.

By continuity and $\phi(0) = 0 < \lambda(F) = \phi(1)$, there exists $t \in (0,1)$ with $0 < \phi(t) < \phi(1)$. Let $G := F \cap [0,t]$. Then $G \subseteq F$ and $0 < \lambda(G) = \phi(t) < \phi(1) = \lambda(F)$. □

Proposition C.1.18 (HW1.5(B) (Exact splitting under non-atomicity)).

Let $(\Omega, \mathscr{F}, \mathbb{P})$ be non-atomic. If $F \in \mathscr{F}$ satisfies $\mathbb{P}(F) > 0$ and if $x \in (0, \mathbb{P}(F))$, then there exists $G \in \mathscr{F}$ with $G \subseteq F$ and $\mathbb{P}(G) = x$.

Proof of Proposition C.1.18. We first note: for any $E \in \mathscr{F}$ with $\mathbb{P}(E) > 0$ and any $\eta > 0$, there exists $H \subseteq E$ with $0 < \mathbb{P}(H) < \eta$. Indeed, by non-atomicity choose $H_1 \subseteq E$ with $0 < \mathbb{P}(H_1) < \mathbb{P}(E)$. If $\mathbb{P}(H_1) < \eta$ we are done.

Otherwise, since $\mathbb{P}(E \setminus H_1) = \mathbb{P}(E) - \mathbb{P}(H_1) > 0$, either $\mathbb{P}(H_1) \le \mathbb{P}(E)/2$ or $\mathbb{P}(E \setminus H_1) \le \mathbb{P}(E)/2$; in either case we have a subset of E of positive measure at most $\mathbb{P}(E)/2$. Iterating this splitting yields subsets of E with arbitrarily small positive measure.

Now fix F and $x \in (0, \mathbb{P}(F))$. We will construct a decreasing sequence $E_0 \supseteq E_1 \supseteq E_2 \supseteq \cdots$ of measurable subsets of F such that $\mathbb{P}(E_n) \downarrow x$. Set $E_0 := F$. Given E_{n-1} with $\mathbb{P}(E_{n-1}) > x$, let $\delta_n := \mathbb{P}(E_{n-1}) - x > 0$. Using the preliminary fact with $\eta := \min\{\delta_n, 2^{-n}\}$, choose $H_n \subseteq E_{n-1}$ with $0 < \mathbb{P}(H_n) < \min\{\delta_n, 2^{-n}\}$. Define $E_n := E_{n-1} \setminus H_n$. Then $E_n \subseteq E_{n-1}$ and

$$\mathbb{P}(E_n) = \mathbb{P}(E_{n-1}) - \mathbb{P}(H_n) \in (x,\ x + 2^{-n}),$$

so $\mathbb{P}(E_n) \downarrow x$.

Let $G := \bigcap_{n=1}^{\infty} E_n$. By continuity from above (since $\mathbb{P}(E_1) < \infty$),

$$\mathbb{P}(G) = \lim_{n\to\infty} \mathbb{P}(E_n) = x,$$

and $G \subseteq F$. □

Proposition C.1.19 (HW1.5(C) (Increasing family of sets with prescribed measure)).

Let $(\Omega, \mathscr{F}, \mathbb{P})$ be non-atomic. Then there exists a family $\{G_t\}_{t\in[0,1]} \subseteq \mathscr{F}$ such that (i) $s \le t \implies G_s \subseteq G_t$, and (ii) $\mathbb{P}(G_t) = t$ for all $t \in [0,1]$.

Proof of Proposition C.1.19. Let $\mathbb{D} := \{k/2^n :\ n \in \mathbb{N},\ 0 \le k \le 2^n\}$ be the dyadic rationals in $[0,1]$.

We first construct $\{G_q\}_{q\in\mathbb{D}}$ with $q \mapsto G_q$ increasing and $\mathbb{P}(G_q) = q$. Set $G_0 := \varnothing$ and $G_1 := \Omega$. Assume inductively that for some $n \ge 0$ we have defined $G_{k/2^n}$ for all $k = 0, \ldots, 2^n$, increasing in k, with $\mathbb{P}(G_{k/2^n}) = k/2^n$. For each $k = 0, \ldots, 2^n - 1$, apply (B) to the set

$$F_{k,n} := G_{(k+1)/2^n} \setminus G_{k/2^n},$$

which has measure $\mathbb{P}(F_{k,n}) = 2^{-n}$, with target value $x := 2^{-(n+1)}$. Thus we can choose $H_{k,n} \subseteq F_{k,n}$ with $\mathbb{P}(H_{k,n}) = 2^{-(n+1)}$, and define

$$G_{(2k+1)/2^{n+1}} := G_{k/2^n} \ \cup\ H_{k,n},$$

while keeping the already-defined even dyadics $G_{2k/2^{n+1}} := G_{k/2^n}$. This preserves monotonicity and enforces $\mathbb{P}(G_{m/2^{n+1}}) = m/2^{n+1}$.

Now extend to all $t \in [0,1]$ by

$$G_t := \bigcup_{\substack{q\in\mathbb{D}\\ q<t}} G_q, \qquad t \in (0,1], \quad \text{and } G_0 := \varnothing.$$

This is measurable because $\mathbb{D}$ is countable. If $s \le t$, then the index set $\{q \in \mathbb{D} : q < s\}$ is contained in $\{q \in \mathbb{D} : q < t\}$, hence $G_s \subseteq G_t$. Moreover, for each t, choose dyadics $q_n \uparrow t$. Then $G_{q_n} \uparrow G_t$ by construction, so by continuity from below,

$$\mathbb{P}(G_t) = \lim_{n\to\infty} \mathbb{P}(G_{q_n}) = \lim_{n\to\infty} q_n = t.$$

□

Proposition C.1.20 (HW1.5(D) (Uniform random variable exists)).

If $(\Omega, \mathscr{F}, \mathbb{P})$ is non-atomic, then there exists a random variable $T : \Omega \to [0,1]$ whose distribution is $\mathrm{Unif}[0,1]$.

Proof of Proposition C.1.20. Let $\{G_t\}_{t\in[0,1]}$ be as in (C), and define

$$T(\omega) := \inf\{t \in [0,1] : \ \omega \in G_t\},$$

with the convention $\inf \varnothing := 1$ (this is irrelevant since $G_1 = \Omega$).

For any $t \in [0,1]$, we claim $\{T < t\} = G_t$. Indeed: if $\omega \in G_t$, then by construction $G_t = \bigcup_{q<t,\ q\in\mathbb{D}} G_q$, so there exists $q < t$ with $\omega \in G_q$, hence $T(\omega) \le q < t$. Thus $\omega \in \{T < t\}$, so $G_t \subseteq \{T < t\}$. Conversely, if $T(\omega) < t$, then there exists $s < t$ with $\omega \in G_s$, and by monotonicity $G_s \subseteq G_t$, so $\omega \in G_t$. Hence $\{T < t\} \subseteq G_t$.

Therefore $\{T < t\} = G_t \in \mathscr{F}$ for each t, so T is measurable. Moreover,

$$\mathbb{P}(T < t) = \mathbb{P}(G_t) = t \qquad (t \in [0,1]).$$

Finally, for $t \in [0,1]$,

$$\mathbb{P}(T \le t) = \mathbb{P}\Big(\bigcap_{n=1}^{\infty}\{T < t + 1/n\}\Big) = \lim_{n\to\infty} \mathbb{P}(T < t + 1/n) = \lim_{n\to\infty}(t + 1/n) = t,$$

so T has the $\mathrm{Unif}[0,1]$ distribution. □

Proposition C.1.21 (HW1.6 (Existence/uniqueness of d-dimensional Lebesgue measure on the rectangle algebra)).

Let $\mathcal{R}_d$ be the collection of half-open rectangles in $\mathbb{R}^d$,

$$R = \prod_{i=1}^{d}(a_i, b_i], \qquad a_i < b_i,\ a_i, b_i \in \mathbb{R},$$

and let $\mathcal{A}_d$ be the algebra generated by (all) rectangles in $\mathbb{R}^d$ (equivalently: finite unions of rectangles). There exists a unique measure $\mu = \mu_d$ on $\mathcal{A}_d$ such that for every rectangle $A_1 \times \cdots \times A_d$,

$$\mu(A_1 \times \cdots \times A_d) = \prod_{i=1}^{d} |A_i|,$$

where $|A_i|$ denotes the interval length.

Proof of Proposition C.1.21. Step 1 (Define a pre-measure on a semiring). On the semiring $\mathcal{R}_d$ define

$$\mu_0\Big(\prod_{i=1}^{d}(a_i, b_i]\Big) := \prod_{i=1}^{d}(b_i - a_i).$$

(One checks $\mathcal{R}_d$ is a semiring: it is closed under intersections, and differences of two elements can be written as a finite disjoint union of elements of $\mathcal{R}_d$ by splitting along the finitely many coordinate endpoints.)

A standard induction on d (reducing to the $d = 1$ additivity of length on half-open intervals, and then slicing in the last coordinate) shows that μ_0 is a *pre-measure* on $\mathcal{R}_d$: if $R \in \mathcal{R}_d$ and $R = \bigsqcup_{n=1}^{\infty} R_n$ with pairwise disjoint $R_n \in \mathcal{R}_d$, then

$$\mu_0(R) = \sum_{n=1}^{\infty} \mu_0(R_n).$$

Step 2 (Extend to the algebra of finite unions). Any $E \in \mathcal{A}_d$ can be written as a finite disjoint union of half-open rectangles: there exist $R_1, \ldots, R_m \in \mathcal{R}_d$ disjoint such that $E = \bigsqcup_{j=1}^{m} R_j$ (obtain a common finite grid by collecting

all coordinate endpoints appearing in a finite rectangle representation of E and subdividing). Define

$$\mu(E) := \sum_{j=1}^{m} \mu_0(R_j).$$

This is well-defined: if E has two such disjoint decompositions, refine both by intersecting rectangles; the semiring property implies the refinement is a finite disjoint family of half-open rectangles, and finite additivity of μ_0 on such refinements forces the same total sum.

Thus μ is finitely additive on $\mathcal{A}_d$ and agrees with the desired formula on rectangles.

Step 3 (Countable additivity on $\mathcal{A}_d$). Let $(E_n)_{n\geq 1} \subseteq \mathcal{A}_d$ be pairwise disjoint and suppose $E := \bigsqcup_{n\geq 1} E_n \in \mathcal{A}_d$. Write $E = \bigsqcup_{j=1}^{m} R_j$ as a disjoint union of half-open rectangles $R_j \in \mathcal{R}_d$. Then for each j,

$$R_j = R_j \cap E = \bigsqcup_{n\geq 1} (R_j \cap E_n),$$

where the pieces are disjoint and each $R_j \cap E_n \in \mathcal{A}_d$ is a finite disjoint union of elements of $\mathcal{R}_d$. Using the pre-measure property of μ_0 on $\mathcal{R}_d$ and the definition of μ via disjoint unions of rectangles, we get

$$\mu(R_j) = \sum_{n\geq 1} \mu(R_j \cap E_n).$$

Summing over $j = 1, \ldots, m$ and using disjointness,

$$\mu(E) = \sum_{j=1}^{m} \mu(R_j) = \sum_{j=1}^{m} \sum_{n\geq 1} \mu(R_j \cap E_n) = \sum_{n\geq 1} \mu(E_n).$$

So μ is countably additive on $\mathcal{A}_d$, i.e. a measure.

Step 4 (Uniqueness on $\mathcal{A}_d$). If $\tilde{\mu}$ is any measure on $\mathcal{A}_d$ agreeing with the rectangle formula, then on a disjoint union $E = \bigsqcup_{j=1}^{m} R_j$ of half-open rectangles we have

$$\tilde{\mu}(E) = \sum_{j=1}^{m} \tilde{\mu}(R_j) = \sum_{j=1}^{m} \mu_0(R_j) = \mu(E),$$

so $\tilde{\mu} = \mu$ on $\mathcal{A}_d$.

This completes existence and uniqueness on $\mathcal{A}_d$. □

Proposition C.1.22 (HW2.1: Approximation by an algebra (Littlewood's third principle)).

Let $(\Omega, \mathscr{F}, \mathbb{P})$ be a probability space and let $\mathscr{A} \subseteq \mathscr{F}$ be an algebra with $\sigma(\mathscr{A}) = \mathscr{F}$. Then for every $F \in \mathscr{F}$ and every $\varepsilon > 0$ there exists $A \in \mathscr{A}$ such that $\mathbb{P}(A \triangle F) < \varepsilon$.

Proof of Proposition C.1.22. Define

$$\mathscr{D} := \Big\{ F \in \mathscr{F} : \ \forall \varepsilon > 0 \ \exists A \in \mathscr{A} \text{ with } \mathbb{P}(A \triangle F) < \varepsilon \Big\}.$$

Clearly $\mathscr{A} \subseteq \mathscr{D}$ (take $A = F$).

We claim $\mathscr{D}$ is a monotone class.

(Closure under increasing limits.) Let $F_n \uparrow F$ with each $F_n \in \mathscr{D}$. Fix $\varepsilon > 0$. By continuity from below, $\mathbb{P}(F \setminus F_N) \to 0$, so choose N with $\mathbb{P}(F \setminus F_N) < \varepsilon/2$. Since $F_N \in \mathscr{D}$, choose $A \in \mathscr{A}$ with $\mathbb{P}(A \triangle F_N) < \varepsilon/2$. Then

$$A \triangle F \subseteq (A \triangle F_N) \ \cup \ (F_N \triangle F) = (A \triangle F_N) \ \cup \ (F \setminus F_N),$$

so

$$\mathbb{P}(A\triangle F) \le \mathbb{P}(A\triangle F_N) + \mathbb{P}(F \setminus F_N) < \varepsilon.$$

Hence $F \in \mathscr{D}$.

(Closure under decreasing limits.) Let $F_n \downarrow F$ with each $F_n \in \mathscr{D}$. Since $\mathbb{P}(F_1) < \infty$, continuity from above gives $\mathbb{P}(F_n \setminus F) \to 0$. Fix $\varepsilon > 0$, choose N with $\mathbb{P}(F_N \setminus F) < \varepsilon/2$. Pick $A \in \mathscr{A}$ with $\mathbb{P}(A\triangle F_N) < \varepsilon/2$. Then

$$A\triangle F \subseteq (A\triangle F_N) \ \cup \ (F_N\triangle F) = (A\triangle F_N) \ \cup \ (F_N \setminus F),$$

so again $\mathbb{P}(A\triangle F) < \varepsilon$. Thus $F \in \mathscr{D}$.

Therefore $\mathscr{D}$ is a monotone class containing the algebra $\mathscr{A}$. By the monotone class theorem (the monotone class generated by an algebra equals the σ-algebra it generates), we have $\sigma(\mathscr{A}) \subseteq \mathscr{D}$. Since $\sigma(\mathscr{A}) = \mathscr{F}$, we conclude $\mathscr{D} = \mathscr{F}$, which is exactly the desired approximation statement. □

Proposition C.1.23 (HW2.2: Translates of Borel sets are Borel).

Let $B \in \mathscr{B}(\mathbb{R})$ and $x \in \mathbb{R}$. Then $B + x := \{b + x : b \in B\} \in \mathscr{B}(\mathbb{R})$.

Proof of Proposition C.1.23. Define the translation map $T_{-x} : \mathbb{R} \to \mathbb{R}$ by $T_{-x}(y) = y - x$, which is continuous. Observe that

$$B + x = \{y \in \mathbb{R} : \ y - x \in B\} = T_{-x}^{-1}(B).$$

Since T_{-x} is continuous (hence Borel measurable) and B is Borel, its preimage is Borel. Thus $B + x \in \mathscr{B}(\mathbb{R})$. □

Proposition C.1.24 (HW2.3: $\limsup$ of measurable maps is measurable).

Let $(\Omega, \mathscr{F})$ be a measurable space and let $X_n : \Omega \to \overline{\mathbb{R}}$ be measurable for each n. Then $\limsup_{n\to\infty} X_n$ is measurable as a map $\Omega \to \overline{\mathbb{R}}$.

Proof of Proposition C.1.24. Let

$$Y := \limsup_{n\to\infty} X_n = \inf_{N\ge 1} \sup_{n\ge N} X_n.$$

For each fixed N, the function $\sup_{n\ge N} X_n$ is measurable because for any $a \in \mathbb{R}$,

$$\left\{\sup_{n\ge N} X_n < a\right\} = \bigcap_{n\ge N} \{X_n < a\} \in \mathscr{F},$$

and similarly for other threshold events (using the Borel sets in $\overline{\mathbb{R}}$). Then Y is the infimum of a countable family of measurable functions, hence measurable, since for $a \in \mathbb{R}$,

$$\{Y < a\} = \bigcup_{N\ge 1} \left\{\sup_{n\ge N} X_n < a\right\} \in \mathscr{F}.$$

Therefore $\limsup_{n\to\infty} X_n$ is measurable. □

Proposition C.1.25 (HW2.4: Countably generated sigma-algebras are generated by one unit interval-valued r.v.).

Let $(\Omega, \mathscr{F}, \mathbb{P})$ be a probability space and let $\mathscr{G} \subseteq \mathscr{F}$ be a sub-σ-algebra. Then $\mathscr{G}$ is countably generated if and only if there exists a measurable $Y : \Omega \to [0, 1]$ such that $\mathscr{G} = \sigma(Y)$.

Proof of Proposition C.1.25. ($\Leftarrow$) Assume $\mathscr{G} = \sigma(Y)$ for some measurable $Y : \Omega \to [0,1]$. The Borel σ-algebra $\mathscr{B}([0,1])$ is countably generated, for example by the countable collection

$$\mathcal{I} := \{(a,b] \cap [0,1] : \ a, b \in \mathbb{Q}, \ 0 \le a < b \le 1\}.$$

Hence

$$\sigma(Y) = \sigma(\{Y^{-1}(I) : I \in \mathcal{I}\}),$$

which is generated by a countable family. Thus $\mathscr{G}$ is countably generated.

($\Rightarrow$) Assume $\mathscr{G}$ is countably generated, say $\mathscr{G} = \sigma(\{A_n\}_{n\ge1})$ for some $A_n \in \mathscr{G}$. Let $X_n := \mathbb{1}_{A_n}$, so each $X_n : \Omega \to \{0,1\}$ is $\mathscr{G}$-measurable and

$$\mathscr{G} = \sigma(\{A_n\}_{n\ge1}) = \sigma(\{X_n\}_{n\ge1}).$$

Now define

$$Y := \sum_{n=1}^{\infty} \frac{2X_n}{3^n}.$$

Then $0 \le Y \le \sum_{n\ge1} \frac{2}{3^n} = 1$, so $Y : \Omega \to [0,1]$ is well-defined and measurable as a pointwise limit of measurable partial sums; in particular Y is measurable with respect to $\sigma(\{X_n\}) = \mathscr{G}$, so $\sigma(Y) \subseteq \mathscr{G}$.

It remains to show $\mathscr{G} \subseteq \sigma(Y)$, i.e. each X_n is $\sigma(Y)$-measurable. Fix $n \ge 1$ and, for each $(d_1, \dots, d_{n-1}) \in \{0,2\}^{n-1}$, define the interval

$$I(d_1, \dots, d_{n-1}, 2) := \left[\sum_{k=1}^{n-1} \frac{d_k}{3^k} + \frac{2}{3^n}, \ \sum_{k=1}^{n-1} \frac{d_k}{3^k} + \frac{3}{3^n}\right].$$

(These are the stage-n Cantor subintervals corresponding to having digit 2 in position n.) Let

$$B_n := \bigcup_{(d_1,\dots,d_{n-1})\in\{0,2\}^{n-1}} I(d_1, \dots, d_{n-1}, 2) \ \subseteq [0,1].$$

This is a finite union of closed intervals, hence $B_n \in \mathscr{B}([0,1])$. By construction of Y (ternary expansion with digits 0 or 2), we have the identity of events

$$\{X_n = 1\} = \{Y \in B_n\}.$$

Therefore $\{X_n = 1\} \in \sigma(Y)$ for each n, so X_n is $\sigma(Y)$-measurable. Hence $\sigma(\{X_n\}) \subseteq \sigma(Y)$, i.e. $\mathscr{G} \subseteq \sigma(Y)$.

Combining both inclusions yields $\mathscr{G} = \sigma(Y)$. □

Proposition C.1.26 (HW2.5(A): Independence extends from algebras to generated σ-algebras).

Let $\{\mathscr{A}_n\}_{n\ge1}$ be a sequence of independent algebras in $(\Omega, \mathscr{F}, \mathbb{P})$, meaning: for every $k \ge 1$, every distinct indices $n_1, \dots, n_k$, and every choices $A_i \in \mathscr{A}_{n_i}$,

$$\mathbb{P}\Big(\bigcap_{i=1}^{k} A_i\Big) = \prod_{i=1}^{k} \mathbb{P}(A_i).$$

Then the σ-algebras $\sigma(\mathscr{A}_n)$ are independent.

Proof of Proposition C.1.26. It suffices to prove the extension for any fixed finite subcollection, say $n = 1, \dots, k$.

Fix $k \ge 2$ and fix sets $E_2 \in \sigma(\mathscr{A}_2), \dots, E_k \in \sigma(\mathscr{A}_k)$. Define

$$\mathscr{D}_1 := \left\{E \in \sigma(\mathscr{A}_1) : \ \mathbb{P}\Big(E \cap \bigcap_{i=2}^{k} E_i\Big) = \mathbb{P}(E) \prod_{i=2}^{k} \mathbb{P}(E_i)\right\}.$$

One checks $\mathscr{D}_1$ is a Dynkin (lambda) system: it contains Ω, is closed under complements (relative to Ω), and under countable disjoint unions (by countable additivity of $\mathbb{P}$). Moreover, $\mathscr{A}_1 \subseteq \mathscr{D}_1$ because the independence assumption holds whenever $E \in \mathscr{A}_1$ and $E_i \in \sigma(\mathscr{A}_i)$ are replaced by sets in $\mathscr{A}_i$, and then extended to $\sigma(\mathscr{A}_i)$ via the same Dynkin-system argument below.

To make that extension precise, proceed iteratively: for $j = 2, \ldots, k$, define $\mathscr{D}_j$ to be the class of sets in $\sigma(\mathscr{A}_j)$ for which the independence identity holds against all choices from the previously verified σ-algebras; each $\mathscr{D}_j$ is a Dynkin system containing the algebra $\mathscr{A}_j$, hence by the π-λ theorem $\mathscr{D}_j = \sigma(\mathscr{A}_j)$.

Consequently, for arbitrary $E_i \in \sigma(\mathscr{A}_i)$ we have

$$\mathbb{P}\Big(\bigcap_{i=1}^{k} E_i\Big) = \prod_{i=1}^{k} \mathbb{P}(E_i),$$

which is exactly independence of $\sigma(\mathscr{A}_1), \ldots, \sigma(\mathscr{A}_k)$. Since k was arbitrary, the full sequence $\{\sigma(\mathscr{A}_n)\}_{n\geq 1}$ is independent. $\square$

Proposition C.1.27 (HW2.5(B): Binary expansion gives i.i.d. Unif(0,1), and Lebesgue space supports them).

Let $\{\varepsilon_{m,n}\}_{m,n\geq 1}$ be independent Bernoulli(1/2) random variables. For each $m \geq 1$, set

$$U_m := \sum_{n=1}^{\infty} \frac{\varepsilon_{m,n}}{2^n}.$$

Then $\{U_m\}_{m\geq 1}$ are i.i.d. Unif$[0,1]$. Consequently, the Lebesgue space $([0,1], \mathscr{B}([0,1]), m)$ supports a sequence of independent Unif$[0,1]$ random variables.

Proof of Proposition C.1.27. Step 1 (Uniform distribution). Fix m. For $k \geq 1$ and $0 \leq j < 2^k$, let

$$I_{j,k} := \left[\frac{j}{2^k}, \frac{j+1}{2^k}\right).$$

Write j in binary with k digits: $j = \sum_{r=1}^{k} b_r 2^{k-r}$ with $b_r \in \{0,1\}$. Then

$$\{U_m \in I_{j,k}\} = \{\varepsilon_{m,1} = b_1, \ldots, \varepsilon_{m,k} = b_k\},$$

because the tail $\sum_{n>k} \varepsilon_{m,n}/2^n \in [0, 2^{-k})$. By independence and $\mathbb{P}(\varepsilon_{m,r} = b_r) = 1/2$, we get

$$\mathbb{P}(U_m \in I_{j,k}) = 2^{-k} = m(I_{j,k}).$$

The collection of dyadic half-open intervals $\{I_{j,k}\}$ is a π-system generating $\mathscr{B}([0,1])$, so the above identities imply $\mathcal{L}(U_m) = \text{Unif}[0,1]$.

Step 2 (Independence and identical distribution across m). For each m, let $\mathscr{H}_m := \sigma(\varepsilon_{m,n} : n \geq 1)$. Because the array is jointly independent, the algebras generated by finitely many $\varepsilon_{m,n}$ in distinct rows are independent, hence by HW2.5(A) the σ-algebras $\{\mathscr{H}_m\}$ are independent. Since each U_m is $\mathscr{H}_m$-measurable, the family $\{U_m\}$ is independent. All U_m have the same law by symmetry, so they are i.i.d. Unif$[0,1]$.

Step 3 (Realizing i.i.d. uniforms on Lebesgue space). On $([0,1], \mathscr{B}([0,1]), m)$, define the binary digits $(b_r(x))_{r\geq 1}$ of x by choosing the unique expansion not eventually equal to all 1 (this fails only on a null set of dyadic rationals), and set $b_r(x) \in \{0,1\}$ as the rth digit. Then for any fixed $(c_1, \ldots, c_k) \in \{0,1\}^k$,

$$m(\{x : \ b_1(x) = c_1, \ldots, b_k(x) = c_k\}) = 2^{-k}$$

because this event is exactly a dyadic interval of length 2^{-k}. Hence (b_r) are i.i.d. Bernoulli(1/2).

Choose a bijection $\phi : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ and define $\varepsilon_{m,n}(x) := b_{\phi(m,n)}(x)$. Then $\{\varepsilon_{m,n}\}$ is an i.i.d. Bernoulli(1/2) array under m, and defining

$$U_m(x) := \sum_{n=1}^{\infty} \frac{\varepsilon_{m,n}(x)}{2^n}$$

produces i.i.d. Unif$[0,1]$ random variables on Lebesgue space. □

Proposition C.1.28 (HW2.6(B): Biased random walk with $p > 1/2$ drifts to $+\infty$).

Let $X_1, X_2, \dots$ be i.i.d. with $\mathbb{P}(X_i = 1) = p$ and $\mathbb{P}(X_i = -1) = 1 - p$, and define $S_n := \sum_{i=1}^n X_i$. If $p > 1/2$, then $S_n \to +\infty$ almost surely.

Proof of Proposition C.1.28. By the strong law of large numbers,

$$\frac{S_n}{n} = \frac{1}{n}\sum_{i=1}^{n} X_i \longrightarrow \mathbb{E}[X_1] = 1 \cdot p + (-1) \cdot (1-p) = 2p - 1 > 0 \quad \text{a.s.}$$

Hence there exists (random) N such that for all $n \geq N$, $S_n/n > (2p-1)/2$, so $S_n > (2p-1)n/2 \to \infty$. Therefore $S_n \to +\infty$ almost surely. □

Proposition C.1.29 (HW2.6(C): Symmetric random walk is unbounded above and below).

In the setup of HW2.6, assume $p = 1/2$. Then

$$\mathbb{P}\Big(\limsup_{n\to\infty} S_n = +\infty\Big) = 1 \qquad \text{and} \qquad \mathbb{P}\Big(\liminf_{n\to\infty} S_n = -\infty\Big) = 1.$$

Proof of Proposition C.1.29. We show $\mathbb{P}(\sup_{n\geq 1} S_n \geq m) = 1$ for each fixed $m \geq 1$; then $\limsup S_n = +\infty$ follows by countable intersection. The $\liminf$ statement follows by symmetry (replace X_i with $-X_i$).

Fix integers $m \geq 1$ and $k \geq 1$. Define the hitting time

$$\tau_{m,k} := \inf\{n \geq 0 : \ S_n = m \text{ or } S_n = -k\}.$$

First, $\tau_{m,k} < \infty$ a.s.: indeed, set $\tau^N := \tau_{m,k} \wedge N$. Then $(S_n^2 - n)_{n\geq 0}$ is a martingale for $p = 1/2$, so optional stopping gives

$$0 = \mathbb{E}[S_{\tau^N}^2 - \tau^N] = \mathbb{E}[S_{\tau^N}^2] - \mathbb{E}[\tau^N] \quad \Rightarrow \quad \mathbb{E}[\tau^N] = \mathbb{E}[S_{\tau^N}^2] \leq \max\{m^2, k^2\}.$$

Letting $N \to \infty$ and using monotone convergence yields $\mathbb{E}[\tau_{m,k}] \leq \max\{m^2, k^2\} < \infty$, hence $\mathbb{P}(\tau_{m,k} < \infty) = 1$.

Now apply optional stopping to the bounded martingale $(S_{n\wedge\tau_{m,k}})$:

$$0 = \mathbb{E}[S_0] = \mathbb{E}[S_{\tau_{m,k}}] = m\,\mathbb{P}(S_{\tau_{m,k}} = m) + (-k)\,\mathbb{P}(S_{\tau_{m,k}} = -k).$$

Since the two probabilities sum to 1, we solve

$$\mathbb{P}(S_{\tau_{m,k}} = m) = \frac{k}{k+m}.$$

But the event $\{S_{\tau_{m,k}} = m\}$ is exactly $\{\text{hit } m \text{ before } -k\}$, and as $k \to \infty$ these events increase to $\{\exists n : \ S_n = m\}$. Therefore, by continuity from below,

$$\mathbb{P}(\exists n : \ S_n = m) = \lim_{k\to\infty} \frac{k}{k+m} = 1.$$

Since m was arbitrary, $\mathbb{P}(\forall m\ \exists n:\ S_n = m) = 1$, i.e. $\limsup_{n\to\infty} S_n = +\infty$ a.s. The $\liminf = -\infty$ statement follows similarly (or by symmetry). □

Proposition C.1.30 (HW2.6(A): The walk is unbounded in absolute value).

In the setup of HW2.6, for every $p \in [0,1]$,

$$\mathbb{P}\Big(\limsup_{n\to\infty} |S_n| = \infty\Big) = 1.$$

Proof of Proposition C.1.30. If $p > 1/2$, then by HW2.6(B) we have $S_n \to +\infty$ a.s., so $\limsup |S_n| = \infty$ a.s. If $p < 1/2$, apply HW2.6(B) to the walk $-S_n$ (which corresponds to swapping p and $1-p$) to get $S_n \to -\infty$ a.s., hence again $\limsup |S_n| = \infty$ a.s. If $p = 1/2$, HW2.6(C) gives $\limsup S_n = +\infty$ and $\liminf S_n = -\infty$ a.s., hence $\limsup |S_n| = \infty$ a.s. □

Proposition C.1.31 (HW2 Bonus: Limit of a convergent series of independent finite-valued r.v.'s is either purely atomic or non-atomic).

Let $X_1, X_2, \dots$ be independent random variables such that each X_n takes only finitely many values. Assume the partial sums $S_n := \sum_{k=1}^n X_k$ converge almost surely to some finite random variable S. Then exactly one of the following holds:

(i) There exists a countable set $A \subseteq \mathbb{R}$ such that $\mathbb{P}(S \in A) = 1$.

(ii) For every $\alpha \in \mathbb{R}$, we have $\mathbb{P}(S = \alpha) = 0$.

Proof of Proposition C.1.31. For each n, let μ_n be the law of X_n. Since X_n is finite-valued, μ_n is a discrete (purely atomic) probability measure. The law of the partial sum S_n is the convolution

$$\mu^{(n)} := \mu_1 * \mu_2 * \cdots * \mu_n.$$

Because $S_n \to S$ almost surely, we have $S_n \to S$ in distribution, hence $\mu^{(n)} \Rightarrow \mu$, where μ is the law of S. Thus μ is an *infinite convolution limit* of discrete measures.

A classical purity theorem of Jessen–Wintner for infinite convolutions of discrete measures implies that the limit law μ is of *pure type*: either (a) purely atomic, or (b) continuous with no atoms (more sharply: absolutely continuous or singular continuous). Consequently, either: (a) μ is purely atomic, in which case its set of atoms is countable and μ is supported on a countable set A, giving (i); or (b) μ has no atoms, i.e. $\mu(\{\alpha\}) = 0$ for all α, giving (ii). This yields the stated dichotomy. □

Proposition C.1.32 ((HW 3.1) Almost sure convergence implies convergence in probability; counterexample to the converse).

Let $(X_n)_{n\in\mathbb{N}}$ and X be real-valued random variables on a common probability space $(\Omega, \mathscr{F}, \mathbb{P})$. Recall:

- $X_n \to X$ *in probability* if for every $\epsilon > 0$,

$$\lim_{n\to\infty} \mathbb{P}\big(\,|X_n - X| > \epsilon\,\big) = 0.$$

- $X_n \to X$ *almost surely* if

$$\mathbb{P}\Big(\,\{\omega : \lim_{n\to\infty} X_n(\omega) = X(\omega)\}\,\Big) = 1.$$

(i) If $X_n \to X$ almost surely, then $X_n \to X$ in probability.

(ii) The converse need not hold: it is possible that $X_n \to X$ in probability but $\mathbb{P}(\{\lim_{n\to\infty} X_n = X\}) = 0$.

Proof of Proposition C.1.32. **(A)** Fix $\epsilon > 0$ and define events $A_n := \{|X_n - X| > \epsilon\} \in \mathscr{F}$. If $X_n \to X$ almost surely, then for $\mathbb{P}$-a.e. ω there exists $N(\omega)$ such that $|X_n(\omega) - X(\omega)| \le \epsilon$ for all $n \ge N(\omega)$. Equivalently, $\mathbb{1}_{A_n}(\omega) \to 0$ for $\mathbb{P}$-a.e. ω.

Since $0 \le \mathbb{1}_{A_n} \le 1$ and $\mathbb{1} \in L^1(\mathbb{P})$, the dominated convergence theorem yields

$$\mathbb{P}(A_n) \;=\; \mathbb{E}[\mathbb{1}_{A_n}] \;\longrightarrow\; 0.$$

Because $\epsilon > 0$ was arbitrary, this is exactly $X_n \to X$ in probability.

(B) (Typewriter sequence.) Let $\Omega = (0, 1]$ with $\mathscr{F}$ the Borel σ-algebra and $\mathbb{P}$ Lebesgue measure. For each integer $k \ge 1$ and $j \in \{1, \dots, 2^k\}$, define the dyadic interval

$$I_{k,j} := \Big(\frac{j-1}{2^k}, \frac{j}{2^k}\Big].$$

Enumerate these intervals by setting: for n with $2^k \le n \le 2^{k+1} - 1$, write $n = 2^k + (j - 1)$ with $j \in \{1, \dots, 2^k\}$ and define

$$X_n := \mathbb{1}_{I_{k,j}}, \qquad X := 0.$$

Convergence in probability. Fix $\epsilon \in (0, 1]$. Then $|X_n - 0| > \epsilon$ iff $X_n = 1$, hence

$$\mathbb{P}(|X_n - X| > \epsilon) = \mathbb{P}(X_n = 1) = \mathbb{P}(\mathbb{1}_{k,j}) = \frac{1}{2^k},$$

where k is determined by $2^k \le n \le 2^{k+1} - 1$. As $n \to \infty$ we have $k \to \infty$, so $2^{-k} \to 0$. Thus $\mathbb{P}(|X_n - X| > \epsilon) \to 0$ for every $\epsilon > 0$, i.e. $X_n \to 0$ in probability.

Failure of almost sure convergence (indeed, no pointwise limit a.e.). Fix $x \in (0, 1]$. For each $k \ge 1$ there is a *unique* $j(k) \in \{1, \dots, 2^k\}$ such that $x \in I_{k,j(k)}$. In the block of indices $n \in \{2^k, \dots, 2^{k+1} - 1\}$, exactly one n corresponds to $(k, j(k))$, so $X_n(x) = 1$ for that n. Hence $X_n(x) = 1$ for infinitely many n (one for each k), so $\limsup_n X_n(x) = 1$.

On the other hand, for each fixed k and the same block, there are many other $j \ne j(k)$, hence many n in that block with $x \notin I_{k,j}$, so $X_n(x) = 0$ for infinitely many n as well. Therefore $\liminf_n X_n(x) = 0$.

So for every $x \in (0, 1]$, $\liminf_n X_n(x) = 0$ and $\limsup_n X_n(x) = 1$, which implies $\lim_{n\to\infty} X_n(x)$ does *not* exist. Thus

$$\mathbb{P}\big(\{\lim_{n\to\infty} X_n = 0\}\big) = 0 \quad \text{and in particular} \quad \mathbb{P}\big(\{\lim_{n\to\infty} X_n = X\}\big) = 0.$$

□

Proposition C.1.33 ((HW 3.2) Indicator identity and inclusion–exclusion; derangements limit).

Let $(\Omega, \mathscr{F}, \mathbb{P})$ be a probability space and let $A_1, \dots, A_n \in \mathscr{F}$. Set $A := \bigcup_{i=1}^n A_i$.

(i) Show $\mathbb{1}_A = 1 - \prod_{i=1}^n (1 - \mathbb{1}_{A_i})$.

(ii) Deduce the inclusion–exclusion formula:

$$\mathbb{P}(A) = \sum_i \mathbb{P}(A_i) - \sum_{i<j} \mathbb{P}(A_i \cap A_j) + \sum_{i<j<k} \mathbb{P}(A_i \cap A_j \cap A_k) - \cdots + (-1)^{n+1} \mathbb{P}\Big(\bigcap_{i=1}^n A_i\Big).$$

(iii) If F_n is a uniform random permutation of $[n]$ and $Y_n := \sum_{i=1}^n \mathbb{1}_{\{F_n(i)=i\}}$ is the number of fixed points,

show $\lim_{n\to\infty} \mathbb{P}(Y_n = 0)$ exists and compute it.

Proof of Proposition C.1.33. **(A)** This is a pointwise identity. Fix $\omega \in \Omega$. If $\omega \notin A$, then $\mathbb{1}_{A_i}(\omega) = 0$ for all i, so $\prod_{i=1}^n (1 - \mathbb{1}_{A_i}(\omega)) = 1$ and the right-hand side equals $0 = \mathbb{1}_A(\omega)$. If $\omega \in A$, then $\omega \in A_{i_0}$ for some i_0, so $1 - \mathbb{1}_{A_{i_0}}(\omega) = 0$ and the product is 0, so the right-hand side equals $1 = \mathbb{1}_A(\omega)$.

(B) Expand the finite product using distributivity:

$$\prod_{i=1}^n (1 - \mathbb{1}_{A_i}) = 1 - \sum_i \mathbb{1}_{A_i} + \sum_{i<j} \mathbb{1}_{A_i}\mathbb{1}_{A_j} - \sum_{i<j<k} \mathbb{1}_{A_i}\mathbb{1}_{A_j}\mathbb{1}_{A_k} + \cdots + (-1)^n \prod_{i=1}^n \mathbb{1}_{A_i}.$$

Note $\prod_{i\in S} \mathbb{1}_{A_i} = \mathbb{1}_{\cap_{i\in S} A_i}$ for any finite set S. By part (A),

$$\mathbb{1}_A = 1 - \prod_{i=1}^n (1 - \mathbb{1}_{A_i}) = \sum_i \mathbb{1}_{A_i} - \sum_{i<j} \mathbb{1}_{A_i \cap A_j} + \sum_{i<j<k} \mathbb{1}_{A_i \cap A_j \cap A_k} - \cdots + (-1)^{n+1} \mathbb{1}_{\cap_{i=1}^n A_i}.$$

Taking expectations and using $\mathbb{E}[\mathbb{1}_B] = \mathbb{P}(B)$ gives the claimed inclusion–exclusion formula.

(C) Let $A_i := \{F_n(i) = i\}$, so $\{Y_n = 0\} = \bigcap_{i=1}^n A_i^c$ and

$$\mathbb{P}(Y_n = 0) = 1 - \mathbb{P}\Big(\bigcup_{i=1}^n A_i\Big).$$

Apply inclusion–exclusion from (B):

$$\mathbb{P}\Big(\bigcup_{i=1}^n A_i\Big) = \sum_{k=1}^n (-1)^{k+1} \sum_{1\le i_1<\cdots<i_k\le n} \mathbb{P}(A_{i_1} \cap \cdots \cap A_{i_k}).$$

Fix distinct indices $i_1, \ldots, i_k$. The event $A_{i_1} \cap \cdots \cap A_{i_k}$ means F_n fixes those k points. The number of permutations of $[n]$ fixing a prescribed set of k points is $(n-k)!$ (permute the remaining $n-k$ points arbitrarily). Since F_n is uniform on S_n,

$$\mathbb{P}(A_{i_1} \cap \cdots \cap A_{i_k}) = \frac{(n-k)!}{n!}.$$

There are $\binom{n}{k}$ choices of the k-tuple $\{i_1, \ldots, i_k\}$, hence

$$\sum_{1\le i_1<\cdots<i_k\le n} \mathbb{P}(A_{i_1} \cap \cdots \cap A_{i_k}) = \binom{n}{k} \frac{(n-k)!}{n!} = \frac{1}{k!}.$$

Therefore

$$\mathbb{P}\Big(\bigcup_{i=1}^n A_i\Big) = \sum_{k=1}^n (-1)^{k+1} \frac{1}{k!}, \qquad \text{so} \qquad \mathbb{P}(Y_n = 0) = \sum_{k=0}^n (-1)^k \frac{1}{k!}.$$

The right-hand side is the nth partial sum of the (absolutely convergent) power series for e^{-1}:

$$e^{-1} = \sum_{k=0}^\infty (-1)^k \frac{1}{k!}.$$

Hence the limit exists and equals e^{-1}. □

Proposition C.1.34 ((HW3: Bonus) Weierstrass approximation via Bernstein polynomials).

Let $f : [0,1] \to \mathbb{R}$ be continuous. Let $U_1, U_2, \dots$ be i.i.d. Unif$[0,1]$. For $n \geq 1$ and $x \in [0,1]$, define

$$p_n(x) := \mathbb{E}\Big[f\Big(\frac{1}{n}\sum_{i=1}^{n} \mathbb{1}_{[0,x]}(U_i)\Big)\Big].$$

(i) p_n is a polynomial in x of degree at most n.

(ii) For each fixed $x \in [0,1]$, $p_n(x) \to f(x)$ as $n \to \infty$.

(iii) In fact, $\sup_{x\in[0,1]} |p_n(x) - f(x)| \to 0$ as $n \to \infty$.

Proof of Proposition C.1.34. Fix $n \geq 1$.

(A) For fixed $x \in [0,1]$, the random variables $\mathbb{1}_{[0,x]}(U_i)$ are i.i.d. Bernoulli(x), since $\mathbb{P}(U_i \in [0,x]) = x$. Let $S_{n,x} := \sum_{i=1}^{n} \mathbb{1}_{[0,x]}(U_i) \sim \text{Bin}(n,x)$. Therefore

$$p_n(x) = \sum_{k=0}^{n} f\Big(\frac{k}{n}\Big)\mathbb{P}(S_{n,x} = k) = \sum_{k=0}^{n} f\Big(\frac{k}{n}\Big)\binom{n}{k} x^k (1-x)^{n-k}.$$

Each term $x^k(1-x)^{n-k}$ is a polynomial in x of degree n, hence p_n is a polynomial of degree at most n. (It can have degree $< n$ for special f, e.g. constant f.)

(B) Fix $x \in [0,1]$. By the strong law of large numbers applied to the i.i.d. Bernoulli(x) variables $\mathbb{1}_{[0,x]}(U_i)$,

$$\frac{1}{n}\sum_{i=1}^{n} \mathbb{1}_{[0,x]}(U_i) \;\longrightarrow\; \mathbb{E}[\mathbb{1}_{[0,x]}(U_1)] = x \quad \text{almost surely.}$$

By continuity of f, we have

$$f\Big(\frac{1}{n}\sum_{i=1}^{n} \mathbb{1}_{[0,x]}(U_i)\Big) \;\longrightarrow\; f(x) \quad \text{almost surely.}$$

Since f is continuous on a compact set, it is bounded: let $M := \sup_{t\in[0,1]} |f(t)| < \infty$. Then

$$\Big|f\Big(\frac{1}{n}\sum_{i=1}^{n} \mathbb{1}_{[0,x]}(U_i)\Big)\Big| \leq M \quad \text{for all } n.$$

Dominated convergence gives

$$p_n(x) = \mathbb{E}\Big[f\Big(\frac{1}{n}\sum_{i=1}^{n} \mathbb{1}_{[0,x]}(U_i)\Big)\Big] \;\longrightarrow\; f(x).$$

(C) Let $\omega_f(\delta) := \sup\{|f(s) - f(t)| : s,t \in [0,1],\, |s-t| \leq \delta\}$ be the modulus of continuity. Uniform continuity of f implies $\omega_f(\delta) \downarrow 0$ as $\delta \downarrow 0$.

Fix $\delta > 0$ and $x \in [0,1]$. Write $T_{n,x} := S_{n,x}/n$. Then

$$\begin{aligned}|p_n(x) - f(x)| &\leq \mathbb{E}\big[\,|f(T_{n,x}) - f(x)|\,\mathbb{1}_{\{|T_{n,x}-x|\leq\delta\}}\big] + \mathbb{E}\big[\,|f(T_{n,x}) - f(x)|\,\mathbb{1}_{\{|T_{n,x}-x|>\delta\}}\big] \\ &\leq \omega_f(\delta) + 2M\,\mathbb{P}(|T_{n,x} - x| > \delta).\end{aligned}$$

Now $\text{Var}(T_{n,x}) = \text{Var}(S_{n,x})/n^2 = (nx(1-x))/n^2 = x(1-x)/n \leq 1/(4n)$, so by Chebyshev,

$$\mathbb{P}(|T_{n,x} - x| > \delta) \leq \frac{\text{Var}(T_{n,x})}{\delta^2} \leq \frac{1}{4n\delta^2}.$$

This bound is uniform in x, hence

$$\sup_{x\in[0,1]} |p_n(x) - f(x)| \le \omega_f(\delta) + \frac{2M}{4n\delta^2} = \omega_f(\delta) + \frac{M}{2n\delta^2}.$$

Given $\epsilon > 0$, choose $\delta > 0$ so that $\omega_f(\delta) < \epsilon/2$, then choose n so large that $M/(2n\delta^2) < \epsilon/2$. Then $\sup_{x\in[0,1]} |p_n(x) - f(x)| < \epsilon$.

Thus $p_n \to f$ uniformly on $[0,1]$ by polynomials, proving the approximation claim. □

C.2 Independence

Proposition C.2.1 (Glivenko–Cantelli).

Let $X_1, X_2, \dots$ be independent, identically distributed real random variables with common cumulative distribution function $F(y) = \mathbb{P}(X_1 \le y)$. Define the empirical distribution function

$$F_n(y) := \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}_{\{X_i \le y\}}, \qquad y \in \mathbb{R}.$$

Then, with probability one,

$$\lim_{n\to\infty} \sup_{y\in\mathbb{R}} |F_n(y) - F(y)| = 0.$$

Proof of Proposition C.2.1. For each $y \in \mathbb{R}$, also define the left-empirical cdf

$$F_n(y-) := \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}_{\{X_i < y\}}.$$

Fix $y \in \mathbb{R}$. Then $\mathbb{1}_{\{X_i \le y\}}$ are i.i.d. Bernoulli with mean $F(y)$, so by the SLLN,

$$F_n(y) \to F(y) \quad \text{a.s.}$$

Likewise, $\mathbb{1}_{\{X_i < y\}}$ are i.i.d. Bernoulli with mean $F(y-) := \lim_{x\uparrow y} F(x)$, so

$$F_n(y-) \to F(y-) \quad \text{a.s.}$$

Now fix $\epsilon > 0$. Construct a finite grid $y_0 < y_1 < \cdots < y_m$ with $y_0 = -\infty$, $y_m = +\infty$ and

$$F(y_{k+1}-) - F(y_k) \le \epsilon \qquad \text{for all } k = 0, 1, \dots, m-1. \tag{C.1}$$

One way: set $y_0 = -\infty$ and recursively define

$$y_{k+1} := \inf\{y \in \mathbb{R} : F(y) \ge F(y_k) + \epsilon\}.$$

Because F increases from 0 to 1, this recursion terminates after at most $\lceil 1/\epsilon \rceil$ steps (once $F(y_k) \ge 1 - \epsilon$, the next step gives $y_{k+1} = +\infty$), so the grid is finite. By definition of an infimum, $F(y_{k+1}-) \le F(y_k) + \epsilon$, which is exactly (**??**).

Let

$$E := \bigcap_{k=0}^{m} \Big\{ \lim_{n\to\infty} F_n(y_k) = F(y_k) \Big\} \cap \bigcap_{k=0}^{m} \Big\{ \lim_{n\to\infty} F_n(y_k-) = F(y_k-) \Big\}.$$

Since there are only finitely many y_k's, $\mathbb{P}(E) = 1$.

Work on the event E. Then there exists N such that for all $n \geq N$,

$$\max_{0\leq k\leq m} |F_n(y_k) - F(y_k)| < \epsilon \qquad \text{and} \qquad \max_{0\leq k\leq m} |F_n(y_k-) - F(y_k-)| < \epsilon.$$

Fix any $y \in \mathbb{R}$. Choose k with $y_k \leq y < y_{k+1}$. Monotonicity of F_n and F gives

$$F_n(y_k) \leq F_n(y) \leq F_n(y_{k+1}-), \qquad F(y_k) \leq F(y) \leq F(y_{k+1}-).$$

Upper bound:

$$F_n(y) - F(y) \leq F_n(y_{k+1}-) - F(y_k) = \big(F_n(y_{k+1}-) - F(y_{k+1}-)\big) + \big(F(y_{k+1}-) - F(y_k)\big) < \epsilon + \epsilon = 2\epsilon,$$

using the choice of N and (**??**). Lower bound:

$$F(y) - F_n(y) \leq F(y_{k+1}-) - F_n(y_k) = \big(F(y_{k+1}-) - F(y_k)\big) + \big(F(y_k) - F_n(y_k)\big) < \epsilon + \epsilon = 2\epsilon.$$

Thus for all $n \geq N$,

$$|F_n(y) - F(y)| < 2\epsilon \quad \text{for every } y \in \mathbb{R},$$

so $\sup_{y\in\mathbb{R}} |F_n(y) - F(y)| \leq 2\epsilon$ for all $n \geq N$. Taking $n \to \infty$ and then $\epsilon \downarrow 0$ yields

$$\lim_{n\to\infty} \sup_{y\in\mathbb{R}} |F_n(y) - F(y)| = 0$$

on E, hence almost surely. □

Proposition C.2.2 (Exercise 2.17).

Let $X_1, X_2, \ldots$ be independent, identically distributed, each with the uniform distribution on $[0, 1]$. Explain why

$$\lim_{n\to\infty} \sup_{B\in\mathscr{B}_{[0,1]}} \left| \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}_B(X_i) - \lambda(B) \right| = 1.$$

Proof of Proposition C.2.2. Fix n. Consider the (random) set

$$B_n := \{X_1, \ldots, X_n\}.$$

This is a finite subset of $[0, 1]$, hence Borel and $\lambda(B_n) = 0$. On the other hand, $\mathbb{1}_{B_n}(X_i) = 1$ for every $i \leq n$, so

$$\frac{1}{n} \sum_{i=1}^{n} \mathbb{1}_{B_n}(X_i) = 1.$$

Therefore

$$\sup_{B\in\mathscr{B}_{[0,1]}} \left| \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}_B(X_i) - \lambda(B) \right| \geq |1 - 0| = 1.$$

But for any Borel B, both $\frac{1}{n}\sum_{i=1}^{n} \mathbb{1}_B(X_i)$ and $\lambda(B)$ lie in $[0, 1]$, so the absolute difference is always ≤ 1. Hence the supremum equals 1 for every n, and the limit is 1. □

C.3 Integration and Expectation

Proposition C.3.1 (Lemma 3.4).

If $f, g : \Omega \to \mathbb{R}$ are nonnegative simple functions, then for any two nonnegative scalars a, b,

$$\int (af + bg)\, d\mu = a \int f\, d\mu + b \int g\, d\mu.$$

Proof of Proposition C.3.1. Write $f = \sum_{i=1}^{m} \alpha_i \mathbb{1}_{A_i}$ and $g = \sum_{j=1}^{\ell} \beta_j \mathbb{1}_{B_j}$ where $\alpha_i, \beta_j \geq 0$ and $\{A_i\}_{i=1}^{m}$, $\{B_j\}_{j=1}^{\ell}$ are measurable partitions of Ω. Refine the partitions by $C_{ij} := A_i \cap B_j$. Then $\{C_{ij}\}_{i,j}$ is a measurable partition and

$$af + bg = \sum_{i=1}^{m} \sum_{j=1}^{\ell} (a\alpha_i + b\beta_j) \mathbb{1}_{C_{ij}}.$$

By the definition of the integral for nonnegative simple functions,

$$\int (af + bg)\, d\mu = \sum_{i,j} (a\alpha_i + b\beta_j)\mu(C_{ij}) = a \sum_{i,j} \alpha_i \mu(C_{ij}) + b \sum_{i,j} \beta_j \mu(C_{ij}).$$

Also,

$$\int f\, d\mu = \sum_i \alpha_i \mu(A_i) = \sum_i \alpha_i \sum_j \mu(C_{ij}) = \sum_{i,j} \alpha_i \mu(C_{ij}),$$

and similarly $\int g\, d\mu = \sum_{i,j} \beta_j \mu(C_{ij})$. Substituting gives the claim. □

Proposition C.3.2 (Monotone Convergence Theorem, Case (ii)).

Let $0 \leq f_1 \leq f_2 \leq \cdots$ be measurable and let $f = \lim_{n\to\infty} f_n$ pointwise. In the proof of the Monotone Convergence Theorem using the approximation-by-simple-functions strategy, complete the missing case where the comparison simple function

$$g = \sum_{i=1}^{m} a_i \mathbb{1}_{A_i} \leq f$$

satisfies $\mu(A_1) = \infty$ for some $A_1 > 0$ (so $\int g\, d\mu = \infty$).

Proof of Proposition C.3.2. In that strategy one sets $L := \lim_{n\to\infty} \int f_n\, d\mu$ (which exists in $[0, \infty]$ by monotonicity) and aims to show $L \geq \int g\, d\mu$ for every nonnegative simple $g \leq f$.

Assume $\mu(A_1) = \infty$ and $a_1 > 0$. Then $\int g\, d\mu = \infty$, so it suffices to show $L = \infty$.

Let $c := a_1/2 > 0$ and define

$$E_n := A_1 \cap \{f_n \geq c\}.$$

Because $f_n \uparrow f$ pointwise and $f \geq g \geq a_1 \mathbb{1}_{A_1}$, for every $\omega \in A_1$ we have $f_n(\omega) \uparrow f(\omega) \geq a_1$, hence $f_n(\omega) \geq c$ for all sufficiently large n. Therefore $E_n \uparrow A_1$, and by continuity from below,

$$\mu(E_n) \uparrow \mu(A_1) = \infty,$$

so $\mu(E_n) \to \infty$ in the extended-real sense. For each n,

$$\int f_n\, d\mu \;\geq\; \int_{E_n} f_n\, d\mu \;\geq\; \int_{E_n} c\, d\mu \;=\; c\,\mu(E_n).$$

Since $\mu(E_n) \to \infty$, the right-hand side becomes arbitrarily large, hence $\sup_n \int f_n \, d\mu = \infty$. But $\int f_n \, d\mu$ is nondecreasing in n, so its limit L must be ∞. This completes Case (ii). $\square$

Proposition C.3.3 (Exercise 3.15).

Verify that each of the following is an exponential family:

(a) $P_\mu =$ normal distribution with mean μ and variance 1.

(b) $P_\mu =$ Poisson distribution with mean e^μ.

(c) $P_\mu =$ Binomial(n, p) with $e^\mu = p/(1-p)$.

Proof of Proposition C.3.3. Recall the one-parameter exponential family form (with sufficient statistic $T(x) = x$): there is a base probability P_0 such that

$$\frac{dP_\mu}{dP_0}(x) = \exp(\mu x - \kappa(\mu)), \qquad \kappa(\mu) := \log \mathbb{E}_0[e^{\mu X}].$$

(a) Let $P_0 = \mathcal{N}(0,1)$ with density $\varphi_0(x) = (2\pi)^{-1/2} e^{-x^2/2}$. Then the $\mathcal{N}(\mu, 1)$ density is

$$\varphi_\mu(x) = (2\pi)^{-1/2} e^{-(x-\mu)^2/2} = (2\pi)^{-1/2} e^{-x^2/2} \, e^{\mu x - \mu^2/2}.$$

Thus $\frac{dP_\mu}{dP_0}(x) = e^{\mu x - \mu^2/2}$, so $\kappa(\mu) = \mu^2/2$.

(b) Work on $(\Omega, \mathscr{F}) = (\mathbb{N}, 2^{\mathbb{N}})$ with counting measure as reference. Let P_0 be Poisson(1): $P_0\{k\} = e^{-1}/k!$. For mean e^μ, the pmf is

$$P_\mu\{k\} = e^{-e^\mu} \frac{(e^\mu)^k}{k!} = \left(e^{-1} \frac{1}{k!}\right) \exp(\mu k - (e^\mu - 1)).$$

Hence $\frac{dP_\mu}{dP_0}(k) = \exp(\mu k - \kappa(\mu))$ with $\kappa(\mu) = e^\mu - 1$.

(c) Consider $X \in \{0, 1, \ldots, n\}$. Let P_0 be Binomial$(n, 1/2)$, so

$$P_0\{k\} = \binom{n}{k} \left(\frac{1}{2}\right)^n.$$

Let $p = \frac{e^\mu}{1+e^\mu}$ so that $\mu = \log \frac{p}{1-p}$, and

$$P_\mu\{k\} = \binom{n}{k} p^k (1-p)^{n-k} = \binom{n}{k} \left(\frac{e^\mu}{1+e^\mu}\right)^k \left(\frac{1}{1+e^\mu}\right)^{n-k} = \binom{n}{k} \exp(\mu k - n \log(1 + e^\mu)).$$

Dividing by $P_0\{k\} = \binom{n}{k} 2^{-n}$ gives

$$\frac{dP_\mu}{dP_0}(k) = \exp\Big(\mu k - n \log(1 + e^\mu) + n \log 2\Big) = \exp(\mu k - \kappa(\mu)),$$

where $\kappa(\mu) = n \log(1 + e^\mu) - n \log 2$. $\square$

Proposition C.3.4 (Corollary 3.17).

The mapping $\mu \mapsto \mu_\mu = \mathbb{E}_\mu X$ is a smooth, nondecreasing function of μ on the natural parameter space J. If $\mathrm{Var}_0(X) > 0$ then $\mu \mapsto \mu_\mu$ is strictly increasing, and consequently maps J homeomorphically onto an interval. *Note: To prove the second assertion, you must show that if* $\mathrm{Var}_0(X) > 0$ *then* $\varphi''(\mu) > 0$ *for all* $\mu \in J$.

Proof of Proposition C.3.4. Let $\varphi(\mu) := \mathbb{E}_0[e^{\mu X}]$ for μ in the interior of J, and let $\kappa(\mu) = \log \varphi(\mu)$. Under P_μ (the tilted measure), one has the standard identities

$$\mathbb{E}_\mu[X] = \kappa'(\mu) = \frac{\varphi'(\mu)}{\varphi(\mu)}, \qquad \mathrm{Var}_\mu(X) = \kappa''(\mu) = \frac{\varphi''(\mu)\varphi(\mu) - (\varphi'(\mu))^2}{\varphi(\mu)^2}.$$

In particular, $\mu \mapsto \mu_\mu = \mathbb{E}_\mu[X]$ is smooth on the interior of J.

Moreover,

$$\frac{d}{d\mu}\mu_\mu = \frac{d}{d\mu}\mathbb{E}_\mu[X] = \mathrm{Var}_\mu(X) \geq 0,$$

so $\mu \mapsto \mu_\mu$ is nondecreasing.

For strict monotonicity, it is enough to show $\mathrm{Var}_\mu(X) > 0$ for all $\mu \in J$. Suppose for contradiction that $\mathrm{Var}_{\mu^\star}(X) = 0$ for some $\mu^\star$ in the interior of J. Then X is $P_{\mu^\star}$-almost surely constant. But $P_{\mu^\star}$ is equivalent to P_0 because

$$\frac{dP_{\mu^\star}}{dP_0} = \exp(\mu^\star X - \kappa(\mu^\star))$$

is strictly positive P_0-a.s. Hence X would be P_0-a.s. constant, implying $\mathrm{Var}_0(X) = 0$, contradiction. Therefore $\mathrm{Var}_\mu(X) > 0$ for all $\mu \in J$, which implies $\mu \mapsto \mu_\mu$ is strictly increasing.

A strictly increasing continuous map from an interval J onto its image is a homeomorphism onto that image interval (the inverse exists and is continuous because strict monotonicity on an interval implies the inverse is also monotone). □

Proposition C.3.5 (Exercise 3.19).

If there exists $X \in L^1$ such that $|X_\lambda| \leq X$ pointwise for every $\lambda \in \Lambda$ then the collection $\{X_\lambda\}_{\lambda\in\Lambda}$ is uniformly integrable.

Proof of Proposition C.3.5. Fix $A > 0$ and any $\lambda \in \Lambda$. On the event $\{|X_\lambda| \geq A\}$ we also have $X \geq A$ because $|X_\lambda| \leq X$. Hence,

$$|X_\lambda|\mathbb{1}_{\{|X_\lambda|\geq A\}} \leq X\,\mathbb{1}_{\{X\geq A\}}.$$

Taking expectations and then $\sup_\lambda$ gives

$$\sup_{\lambda\in\Lambda} \mathbb{E}\big[|X_\lambda|\mathbb{1}_{\{|X_\lambda|\geq A\}}\big] \leq \mathbb{E}\big[X\,\mathbb{1}_{\{X\geq A\}}\big].$$

Because $X \in L^1$, the right-hand side tends to 0 as $A \to \infty$ (by dominated convergence applied to $X\mathbb{1}_{\{X\geq A\}} \downarrow 0$), so $\{X_\lambda\}$ is uniformly integrable. □

Proposition C.3.6 (Exercise 3.20).

If there is a function $\varphi : [0,\infty) \to [0,\infty]$ such that $\varphi(x)/x \to \infty$ as $x \to \infty$ and if

$$\sup_{\lambda\in\Lambda} \mathbb{E}\varphi(|X_\lambda|) < \infty$$

then the collection $\{X_\lambda\}_{\lambda\in\Lambda}$ is uniformly integrable. Thus, for example, if the second moments $\mathbb{E}X_\lambda^2$ are uniformly bounded then the collection $\{X_\lambda\}_{\lambda\in\Lambda}$ is uniformly integrable.

Proof of Proposition C.3.6. Let $C := \sup_{\lambda\in\Lambda} \mathbb{E}\varphi(|X_\lambda|) < \infty$. For $A > 0$, define

$$c(A) := \sup_{x\geq A} \frac{x}{\varphi(x)} \in [0,\infty].$$

The assumption $\varphi(x)/x \to \infty$ implies $c(A) \to 0$ as $A \to \infty$.

Fix $A > 0$ and $\lambda \in \Lambda$. On $\{|X_\lambda| \geq A\}$ we have

$$|X_\lambda| = \frac{|X_\lambda|}{\varphi(|X_\lambda|)}\varphi(|X_\lambda|) \leq c(A)\,\varphi(|X_\lambda|),$$

so

$$\mathbb{E}\big[|X_\lambda|\mathbb{1}_{\{|X_\lambda|\geq A\}}\big] \leq c(A)\mathbb{E}\varphi(|X_\lambda|) \leq c(A)\,C.$$

Taking $\sup_\lambda$ gives $\sup_\lambda \mathbb{E}[|X_\lambda|\mathbb{1}_{\{|X_\lambda|\geq A\}}] \leq c(A)C \to 0$ as $A \to \infty$, so the family is uniformly integrable.

For the "second moments" example, take $\varphi(x) = x^2$. Then $\varphi(x)/x = x \to \infty$, and $\sup_\lambda \mathbb{E}\varphi(|X_\lambda|) = \sup_\lambda \mathbb{E}[X_\lambda^2] < \infty$ implies uniform integrability by the first part. □

Proposition C.3.7 (Exercise 3.22).

Let X_n be random variables such that $X_n \to X$ almost surely. If $\{X_n\}_{n\geq 1}$ is uniformly integrable, prove:

(i) $\mathbb{E}|X| < \infty$.

(ii) The collection $\{X\} \cup \{X_n\}_{n\geq 1}$ is uniformly integrable.

Proof of Proposition C.3.7. (i) First show $\sup_n \mathbb{E}|X_n| < \infty$. Fix $A > 0$. Then for each n,

$$\mathbb{E}|X_n| = \mathbb{E}\big[|X_n|\mathbb{1}_{\{|X_n|<A\}}\big] + \mathbb{E}\big[|X_n|\mathbb{1}_{\{|X_n|\geq A\}}\big] \leq A + \mathbb{E}\big[|X_n|\mathbb{1}_{\{|X_n|\geq A\}}\big].$$

Taking $\sup_n$ and using uniform integrability, choose A so large that $\sup_n \mathbb{E}[|X_n|\mathbb{1}_{\{|X_n|\geq A\}}] \leq 1$. Then $\sup_n \mathbb{E}|X_n| \leq A+1 < \infty$

Now apply Fatou to the nonnegative random variables $|X_n|$: since $|X| = \liminf_{n\to\infty} |X_n|$ a.s., we have

$$\mathbb{E}|X| \leq \liminf_{n\to\infty} \mathbb{E}|X_n| \leq \sup_n \mathbb{E}|X_n| < \infty.$$

(ii) Since $X \in L^1$, the singleton family $\{X\}$ is uniformly integrable: indeed, $\mathbb{E}\big[|X|\mathbb{1}_{\{|X|\geq A\}}\big] \to 0$ as $A \to \infty$. Given $\epsilon > 0$, pick A so large that

$$\sup_{n\geq 1} \mathbb{E}\big[|X_n|\mathbb{1}_{\{|X_n|\geq A\}}\big] < \epsilon \quad \text{and} \quad \mathbb{E}\big[|X|\mathbb{1}_{\{|X|\geq A\}}\big] < \epsilon.$$

Then

$$\sup_{Z\in\{X\}\cup\{X_n\}_{n\geq 1}} \mathbb{E}\big[|Z|\mathbb{1}_{\{|Z|\geq A\}}\big] = \max\left\{\mathbb{E}\big[|X|\mathbb{1}_{\{|X|\geq A\}}\big],\ \sup_n \mathbb{E}\big[|X_n|\mathbb{1}_{\{|X_n|\geq A\}}\big]\right\} < \epsilon,$$

which is exactly uniform integrability of the union. □

Proposition C.3.8 (Exercise 3.28).

Show that if $f_n \to f$ almost surely then $f_n \to f$ in measure.

Proof of Proposition C.3.8. Assume $(\Omega, \mathscr{F}, \mu)$ has finite total mass $\mu(\Omega) < \infty$ (in particular this holds on probability spaces). Fix $\epsilon > 0$ and set $A_n := \{|f_n - f| > \epsilon\}$. Define

$$B_k := \bigcup_{n\geq k} A_n.$$

Then $B_k \downarrow \limsup_{n\to\infty} A_n$. If $f_n \to f$ a.s., then $\mathbb{1}_{A_n}(\omega) \to 0$ for a.e. ω, so ω belongs to only finitely many A_n;

equivalently, $\limsup_n A_n$ has measure 0. Hence

$$\mu(B_k) \downarrow \mu(\limsup_n A_n) = 0$$

by continuity from above (using $\mu(B_1) \leq \mu(\Omega) < \infty$). For $n \geq k$ we have $A_n \subseteq B_k$, so

$$\sup_{n \geq k} \mu(A_n) \leq \mu(B_k) \to 0.$$

Therefore $\mu(A_n) \to 0$, i.e. $\mu\{|f_n - f| > \epsilon\} \to 0$ for every $\epsilon > 0$, which is convergence in measure. □

Proposition C.3.9 (Exercise 3.29).

Show that if $f_n \to f$ in L^p then $f_n \to f$ in measure.

Proof of Proposition C.3.9. Fix $\epsilon > 0$. By Markov's inequality applied to the nonnegative function $|f_n - f|^p$,

$$\mu\{|f_n - f| > \epsilon\} = \mu\{|f_n - f|^p > \epsilon^p\} \leq \frac{1}{\epsilon^p} \int |f_n - f|^p \, d\mu.$$

If $f_n \to f$ in L^p, then the right-hand side tends to 0, hence $\mu\{|f_n - f| > \epsilon\} \to 0$ for all $\epsilon > 0$, which is convergence in measure. □

Proposition C.3.10 (Second Borel–Cantelli lemma).

Let $A_1, A_2, \ldots$ be independent events with

$$\sum_{i=1}^{\infty} \mathbb{P}(A_i) = \infty.$$

Then

$$\mathbb{P}(A_i \text{ i.o.}) = 1.$$

Proof of Proposition C.3.10. (a) By independence,

$$\begin{aligned}
\mathbb{P}\left(\bigcap_{i=1}^{n} A_i^c\right) &= \prod_{i=1}^{n} \mathbb{P}(A_i^c) \\
&= \prod_{i=1}^{n} (1 - \mathbb{P}(A_i)) \\
&= \exp\left(\sum_{i=1}^{n} \log(1 - \mathbb{P}(A_i))\right) \\
&\leq \exp\left(-\sum_{i=1}^{n} \mathbb{P}(A_i)\right) \to 0.
\end{aligned}$$

(b) Since $\bigcap_{i=1}^{n} A_i^c \downarrow \bigcap_{i=1}^{\infty} A_i^c$, continuity from above gives

$$\mathbb{P}\left(\bigcap_{i=1}^{\infty} A_i^c\right) = 0.$$

Hence

$$\mathbb{1}_{\cap_{i=1}^{\infty} A_i^c} = 0 \qquad \text{a.s..}$$

(c) Therefore, almost surely, there exists some i such that $\mathbb{1}_{A_i^c} = 0$, equivalently $\mathbb{1}_{A_i} = 1$. Thus

$$\sum_{i=1}^{\infty} \mathbb{1}_{A_i} \geq 1 \qquad \text{a.s..}$$

(d) Applying the same argument to the tail sequence $A_m, A_{m+1}, \ldots$, we get

$$\sum_{i=m}^{\infty} \mathbb{1}_{A_i} \geq 1 \qquad \text{a.s.}$$

for every $m \geq 1$.

(e) Hence

$$\mathbb{P}\left(\bigcup_{i=m}^{\infty} A_i\right) = 1$$

for every m. Since the events $\bigcup_{i=m}^{\infty} A_i$ decrease in m, continuity from above yields

$$\mathbb{P}\left(\bigcap_{m=1}^{\infty} \bigcup_{i=m}^{\infty} A_i\right) = \lim_{m\to\infty} \mathbb{P}\left(\bigcup_{i=m}^{\infty} A_i\right) = 1.$$

That is,

$$\mathbb{P}\left(A_i \text{ i.o.}\right) = 1.$$

□

C.4 Central Limit Theorem

Proposition C.4.1 (Almost sure convergence of averages implies integrability).

Let $X_1, X_2, \ldots$ be i.i.d. random variables, and set

$$S_n := \sum_{i=1}^{n} X_i.$$

If S_n/n converges almost surely, then $\mathbb{E}\left[|X_1|\right] < \infty$.

Proof of Proposition C.4.1. (a) Since

$$\frac{X_n}{n} = \frac{S_n - S_{n-1}}{n} = \frac{S_n}{n} - \frac{n-1}{n} \cdot \frac{S_{n-1}}{n-1},$$

we have $X_n/n \to 0$ almost surely.

(b) Suppose, toward a contradiction, that

$$\sum_{n=1}^{\infty} \mathbb{P}\left(|X_1| \geq n\right) = \infty.$$

Since the events $\{|X_n| \geq n\}$ are independent and

$$\mathbb{P}\left(|X_n| \geq n\right) = \mathbb{P}\left(|X_1| \geq n\right),$$

the second Borel–Cantelli lemma gives

$$\mathbb{P}\left(\frac{|X_n|}{n} \geq 1 \text{ i.o.}\right) = 1.$$

This contradicts $X_n/n \to 0$ almost surely. Therefore,

$$\sum_{n=1}^{\infty} \mathbb{P}\left(|X_1| \geq n\right) < \infty.$$

(c) Since $|X_1| \leq 1 + \sum_{n=1}^{\infty} \mathbb{1}_{|X_1| \geq n}$,

$$\mathbb{E}\left[|X_1|\right] = 1 + \sum_{n=1}^{\infty} \mathbb{P}\left(|X_1| \geq n\right) < \infty.$$

□

Proposition C.4.2 (Triangular array CLT with truncated centering).

Let $\{X_{n,i} : 1 \leq i \leq k(n)\}$ be independent within each row. Assume that for every $\epsilon > 0$,

(a)

$$\sum_{i=1}^{k(n)} \mathbb{P}\left(|X_{n,i}| > \epsilon\right) \to 0,$$

(b)

$$\sum_{i=1}^{k(n)} \operatorname{Var}\left(X_{n,i}\mathbb{1}_{\{|X_{n,i}| \leq \epsilon\}}\right) \to 1.$$

Define

$$A_n := \sum_{i=1}^{k(n)} \mathbb{E}\left[X_{n,i}\mathbb{1}_{\{|X_{n,i}| \leq 1\}}\right].$$

Then

$$\sum_{i=1}^{k(n)} X_{n,i} - A_n \rightsquigarrow N(0,1).$$

Proof of Proposition C.4.2. Choose $\epsilon_n \to 0$ such that both (a) and (b) hold with ϵ replaced by ϵ_n. Define

$$\xi_{n,i} := X_{n,i}\mathbb{1}_{\{|X_{n,i}| \leq \epsilon_n\}} - \mathbb{E}\left[X_{n,i}\mathbb{1}_{\{|X_{n,i}| \leq \epsilon_n\}}\right].$$

Then $|\xi_{n,i}| \leq 2\epsilon_n$ and $\mathbb{E}[\xi_{n,i}] = 0$.

Moreover,

$$\sum_i \operatorname{Var}(\xi_{n,i}) = \sum_i \operatorname{Var}\left(X_{n,i}\mathbb{1}_{\{|X_{n,i}| \leq \epsilon_n\}}\right) \to 1$$

by (b). Also,

$$\sum_i \mathbb{E}\left[|\xi_{n,i}|^3\right] \leq 2\epsilon_n \sum_i \mathbb{E}\left[\xi_{n,i}^2\right] = 2\epsilon_n \sum_i \operatorname{Var}(\xi_{n,i}) \to 0.$$

By the triangular array CLT,

$$\sum_i \xi_{n,i} \rightsquigarrow N(0,1).$$

Next, if $|X_{n,i}| \le \epsilon_n$ for all i, then $X_{n,i}\mathbb{1}_{\{|X_{n,i}|\le\epsilon_n\}} = X_{n,i}$ for every i, so the two sums coincide. Hence the event that they differ is contained in $\bigcup_i \{|X_{n,i}| > \epsilon_n\}$, and by the union bound,

$$\begin{aligned}
&\mathbb{P}\left(\sum_i \left(X_{n,i} - \mathbb{E}\left[X_{n,i}\mathbb{1}_{\{|X_{n,i}|\le\epsilon_n\}}\right]\right) \neq \sum_i \xi_{n,i}\right) \\
&\quad \le \mathbb{P}\left(\bigcup_i \{|X_{n,i}| > \epsilon_n\}\right) \\
&\quad \le \sum_i \mathbb{P}\left(|X_{n,i}| > \epsilon_n\right) \to 0.
\end{aligned}$$

Since $\mathbb{P}(A_n \neq B_n) \to 0$ implies $A_n - B_n \xrightarrow{p} 0$, writing

$$\underbrace{\sum_i X_{n,i} - \sum_i \mathbb{E}\left[X_{n,i}\mathbb{1}_{\{|X_{n,i}|\le\epsilon_n\}}\right]}_{A_n} = \underbrace{\sum_i \xi_{n,i}}_{B_n} + \underbrace{(A_n - B_n)}_{\xrightarrow{p} 0}$$

and applying Slutsky,

$$\sum_i X_{n,i} - \sum_i \mathbb{E}\left[X_{n,i}\mathbb{1}_{\{|X_{n,i}|\le\epsilon_n\}}\right] \rightsquigarrow N(0,1).$$

Finally,

$$\begin{aligned}
\left|\sum_i \mathbb{E}\left[X_{n,i}\mathbb{1}_{\{|X_{n,i}|\le\epsilon_n\}}\right] - A_n\right| &\le \sum_i \mathbb{E}\left[|X_{n,i}|\mathbb{1}_{\{\epsilon_n\le|X_{n,i}|\le 1\}}\right] \\
&\le \sum_i \mathbb{P}\left(\epsilon_n \le |X_{n,i}| \le 1\right) \to 0.
\end{aligned}$$

Therefore,

$$\sum_{i=1}^{k(n)} X_{n,i} - A_n \rightsquigarrow N(0,1).$$

□

Proposition C.4.3 (Triangular array CLT with prescribed mean and variance).

Let $\{X_{n,i} : 1 \le i \le k(n)\}$ be independent within each row. Suppose that for every $\epsilon > 0$,

(a)
$$\mathbb{P}\left(\max_{1\le i\le k(n)} |X_{n,i}| > \epsilon\right) \to 0,$$

(b)
$$\sum_{i=1}^{k(n)} \mathbb{E}\left[X_{n,i}\mathbb{1}_{\{|X_{n,i}|\le\epsilon\}}\right] \to \mu,$$

(c)
$$\sum_{i=1}^{k(n)} \mathrm{Var}\left(X_{n,i}\mathbb{1}_{\{|X_{n,i}|\le\epsilon\}}\right) \to \sigma^2.$$

Then
$$\sum_{i=1}^{k(n)} X_{n,i} \rightsquigarrow N(\mu, \sigma^2).$$

Proof of Proposition C.4.3. Define
$$\xi_{n,i} := X_{n,i}\mathbb{1}_{\{|X_{n,i}|\le\epsilon_n\}} - \mathbb{E}\left[X_{n,i}\mathbb{1}_{\{|X_{n,i}|\le\epsilon_n\}}\right].$$
Then $|\xi_{n,i}| \le 2\epsilon_n$ and $\mathbb{E}[\xi_{n,i}] = 0$.

Also,
$$\sum_i \mathrm{Var}(\xi_{n,i}) = \sum_i \mathrm{Var}\left(X_{n,i}\mathbb{1}_{\{|X_{n,i}|\le\epsilon_n\}}\right) \to \sigma^2$$
by (c), and
$$\sum_i \mathbb{E}\left[|\xi_{n,i}|^3\right] \le 2\epsilon_n \sum_i \mathbb{E}\left[\xi_{n,i}^2\right] = 2\epsilon_n \sum_i \mathrm{Var}(\xi_{n,i}) \to 0.$$
By the triangular array CLT,
$$\sum_i \xi_{n,i} \rightsquigarrow N(0, \sigma^2).$$

Since
$$\sum_i \mathbb{E}\left[X_{n,i}\mathbb{1}_{\{|X_{n,i}|\le\epsilon_n\}}\right] \to \mu$$
by (b), it follows that
$$\sum_i X_{n,i}\mathbb{1}_{\{|X_{n,i}|\le\epsilon_n\}} \rightsquigarrow N(\mu, \sigma^2).$$

Finally,
$$\begin{aligned}\mathbb{P}\left(\sum_i X_{n,i} \ne \sum_i X_{n,i}\mathbb{1}_{\{|X_{n,i}|\le\epsilon_n\}}\right) &\le \mathbb{P}\left(\max_i |X_{n,i}| > \epsilon_n\right)\\ &\to 0.\end{aligned}$$

Therefore,
$$\sum_{i=1}^{k(n)} X_{n,i} \rightsquigarrow N(\mu, \sigma^2).$$

□

Section D

Marcus Kuo

Appendix – Extensions

D.1 Law of Large Numbers

The main techniques are (1) direct Borel–Cantelli with a moment bound, (2) blocking + maximal inequality, and (3) truncation + Kolmogorov.

Theorem D.1.1 (SLLN via Kolmogorov when X_i are i.i.d. with $\mathbb{E}[X_1^2] < \infty$).

Let $X_1, X_2, \ldots$ be i.i.d. with $\mathbb{E}[X_1] = 0$ and $\mathbb{E}[X_1^2] = \sigma^2 < \infty$. Show $S_n/n \to 0$ almost surely.

Solution. Since the X_i are identically distributed,

$$\sum_{i=1}^{\infty} \frac{\mathbb{E}[X_i^2]}{i^2} = \sigma^2 \sum_{i=1}^{\infty} \frac{1}{i^2} < \infty.$$

Kolmogorov's LLN applies directly. □

Remark D.1.2. This is the key shortcut when you have *identical* distribution and a second moment: the Kolmogorov condition reduces to $\sum 1/i^2 < \infty$ and no truncation is needed. For the full SLLN with only $\mathbb{E}[|X_1|] < \infty$, the notes give the truncation argument.

Theorem D.1.3 (Non-i.i.d. SLLN; when does Kolmogorov apply?).

Let $X_1, X_2, \ldots$ be independent with $\mathbb{E}[X_i] = 0$ and $\mathbb{E}[X_i^2] = \sigma_i^2$.

(a) If $\sigma_i^2 = O(i^\alpha)$ for some $\alpha < 1$, show $S_n/n \to 0$ a.s.

(b) If $\sigma_i^2 = c \cdot i$ (linear growth), show $S_n/n^{3/4} \to 0$ a.s. for any $c > 0$.

Solution. **Part (a).** The Kolmogorov condition:

$$\sum_{i=1}^{\infty} \frac{\mathbb{E}[X_i^2]}{i^2} = \sum_{i=1}^{\infty} \frac{\sigma_i^2}{i^2} \le C \sum_{i=1}^{\infty} i^{\alpha-2},$$

which converges because $\alpha - 2 < -1$. Hence Kolmogorov's LLN gives $S_n/n \to 0$ a.s.

Part (b). We can no longer normalize by n; we need a larger denominator. Try the normalization $a_n = n^{3/4}$. Define $\xi_{n,i} = X_i/n^{3/4}$ and check the Kolmogorov condition for the sequence $\{X_i/n^{3/4}\}\ldots$ but this is a triangular array, not a single sequence.

Instead, use blocking with $n_k = 2^k$. Write $a_n = n^{3/4}$. For $n \in B_k$, $n \geq n_{k-1}$, so $a_n \geq n_{k-1}^{3/4}$. By the maximal inequality,

$$\sum_{k=1}^{\infty} \mathbb{P}\left[\max_{n \in B_k} \frac{|S_n|}{n^{3/4}} > \epsilon\right] \leq \sum_{k=1}^{\infty} \beta_k \mathbb{P}\left[|S_{n_k}| > \frac{n_{k-1}^{3/4}\epsilon}{2}\right].$$

Bound on β_k. $\mathbb{E}[(S_{n_k} - S_n)^2] \leq \mathbb{E}[S_{n_k}^2] = c\sum_{i=1}^{n_k} i \leq cn_k^2$. So

$$\frac{1}{\beta_k} \geq 1 - \frac{4cn_k^2}{n_{k-1}^{3/2}\epsilon^2} = 1 - \frac{4c \cdot 4^k}{2^{3(k-1)/2}\epsilon^2} = 1 - \frac{C}{2^{k/2}} \to 1,$$

giving $\beta_k \leq 2$ for $k \geq K$.

Tail sum. By Chebyshev,

$$\mathbb{P}\left[|S_{n_k}| > \frac{n_{k-1}^{3/4}\epsilon}{2}\right] \leq \frac{4cn_k^2}{n_{k-1}^{3/2}\epsilon^2} = \frac{4c \cdot 4^k}{2^{3(k-1)/2}\epsilon^2} = \frac{C'}{2^{k/2}},$$

so the sum over k converges. Borel–Cantelli gives $S_n/n^{3/4} \to 0$ a.s. □

Remark D.1.4. Part (b) illustrates the general principle: when $\sum \sigma_i^2/a_i^2 = \infty$ for the natural normalization $a_n = n$, try to find $a_n \gg n$ for which the blocking argument still works. Here $a_n = n^{3/4}$ succeeds because $n_k^2/n_{k-1}^{3/2} = 4^k/2^{3(k-1)/2} = O(2^{k/2})$ is summable.

Theorem D.1.5 (SLLN with polynomial blocks $n_k = k^2$).

Let $X_1, X_2, \ldots$ be independent with $\mathbb{E}[X_i] = 0$ and $\sup_i \mathbb{E}[X_i^2] \leq \sigma^2 < \infty$. Set $n_k = k^2$ and $B_k = \{n : (k-1)^2 < n \leq k^2\}$. Prove $\max_{n \in B_k} |S_n|/n \to 0$ almost surely.

Solution. Fix $\epsilon > 0$. For $n \in B_k$, $n \geq n_{k-1} = (k-1)^2$, so $|S_n|/n \leq |S_n|/n_{k-1}$. By the maximal inequality,

$$\mathbb{P}\left[\max_{n \leq n_k} |S_n| > n_{k-1}\epsilon\right] \leq \beta_k \, \mathbb{P}\left[|S_{n_k}| > \tfrac{n_{k-1}\epsilon}{2}\right], \qquad \frac{1}{\beta_k} = \min_{n \leq n_k} \mathbb{P}\left[|S_{n_k} - S_n| \leq \tfrac{n_{k-1}\epsilon}{2}\right].$$

Bound on β_k. By Chebyshev,

$$\mathbb{P}\left[|S_{n_k} - S_n| > \tfrac{n_{k-1}\epsilon}{2}\right] \leq \frac{4\,\mathbb{E}[S_{n_k}^2]}{n_{k-1}^2\epsilon^2} \leq \frac{4n_k\sigma^2}{n_{k-1}^2\epsilon^2} = \frac{4k^2\sigma^2}{(k-1)^4\epsilon^2} \xrightarrow{k\to\infty} 0,$$

so $\beta_k \leq 2$ for all $k \geq K$.

Tail sum. Chebyshev again:

$$\sum_{k=K}^{\infty} \mathbb{P}\left[|S_{n_k}| > \tfrac{n_{k-1}\epsilon}{2}\right] \leq \sum_{k=K}^{\infty} \frac{4k^2\sigma^2}{(k-1)^4\epsilon^2} \asymp \sum_{k=K}^{\infty} \frac{1}{k^2} < \infty.$$

Therefore $\sum_{k=1}^{\infty} \mathbb{P}[\max_{n \in B_k} |S_n|/n > \epsilon] < \infty$, and Borel–Cantelli gives the result. □

Remark D.1.6. The proof is identical in structure to the dyadic-block proof; only the ratio n_k/n_{k-1}^2 changes. For $n_k = k^2$, this ratio is $k^2/(k-1)^4 \sim 1/k^2$, which is summable. For $n_k = 2^k$, it is $2^k/2^{2(k-1)} = 2/2^k$, also summable. Any block sequence with $n_k/n_{k-1}^2 \to 0$ works.

Reference: If You Get Stuck on an LLN Problem

1. **Can you apply Borel–Cantelli directly?** Write down $\sum_n \mathbb{P}[|S_n|/a_n > \epsilon]$ and bound it with a moment inequality (Chebyshev, 4th-moment, Markov). If the sum converges, you are done.
2. **Does direct BC diverge?** Switch to blocking. Pick block endpoints n_k (try 2^k first, then k^2 if the ratio n_k/n_{k-1}^2 needs to be smaller). Show $\sum_k \mathbb{P}[\max_{n\in B_k} |S_n|/a_n > \epsilon] < \infty$ via the two-factor product $\beta_k \cdot \mathbb{P}[|S_{n_k}| > \cdots]$.
3. **Bounding** β_k: check whether $\mathbb{E}[S_{n_k}^2]/n_{k-1}^2 \to 0$ (uniformly bounded variances) or $\mathbb{E}[S_{n_k}^2]/n_k^2 \to 0$ (use the split at N from Kolmogorov's LLN). Once this ratio is less than $\frac{\epsilon^2}{16}$, we get $\beta_k \le 2$.
4. **Are moments too weak?** Truncate: $Y_i = X_i \mathbb{1}_{|X_i|\le b_i}$. Decompose into three parts and apply Kolmogorov to the centered truncated part.
5. **Key identity**: for i.i.d. X_i with $\mathbb{E}[X_i^2] < \infty$, $\sum \mathbb{E}[X_i^2]/i^2 = \mathbb{E}[X_1^2] \sum 1/i^2 < \infty$, so no truncation is needed.

D.2 Central Limit Theorem

Any CLT question reduces to verifying one of four sets of conditions: the triangular array CLT (third-moment condition $\sum \mathbb{E}[|\xi|^3] \to 0$), the Lindeberg condition ($L_n(\epsilon) \to 0$), the Lyapunov condition ($\sum \mathbb{E}[|X|^{2+\delta}]/S_n^{2+\delta} \to 0$), or Stein's inequality.

Theorem D.2.1 (Lindeberg condition for uniformly small summands).

Suppose $X_{n,1}, \dots, X_{n,k(n)}$ are independent within each row with $\mathbb{E}[X_{n,i}] = 0$, $\sum_{i=1}^{k(n)} \mathrm{Var}(X_{n,i}) = 1$, and $\max_{1\le i\le k(n)} |X_{n,i}| \le \delta_n$ almost surely where $\delta_n \to 0$. Prove $\sum_i X_{n,i} \rightsquigarrow N(0,1)$.

Solution. We verify the Lindeberg condition. Fix $\epsilon > 0$. For n large enough that $\delta_n < \epsilon$,

$$L_n(\epsilon) = \sum_{i=1}^{k(n)} \mathbb{E}\Big[X_{n,i}^2 \mathbb{1}_{|X_{n,i}|>\epsilon}\Big] = 0,$$

because $|X_{n,i}| \le \delta_n < \epsilon$ almost surely, so the indicator is identically 0. Hence $L_n(\epsilon) \to 0$, the Lindeberg CLT applies, and $\sum_i X_{n,i} \rightsquigarrow N(0,1)$. □

Remark D.2.2. This is the cleanest case of Lindeberg: the condition is automatically satisfied whenever the summands are uniformly negligible. "Uniform negligibility" ($\max_i |X_{n,i}| \to 0$) is the intuitive content of the Lindeberg condition; the condition itself weakens this to a second-moment version.

Theorem D.2.3 (Rademacher triangular array).

For each n, let $\epsilon_{n,1}, \dots, \epsilon_{n,n}$ be i.i.d. ± 1 with equal probability (Rademacher), and let $\sigma_{n,1}, \dots, \sigma_{n,n} > 0$. Set $X_{n,i} = \sigma_{n,i}\epsilon_{n,i}$ and $S_n^2 = \sum_{i=1}^n \sigma_{n,i}^2$. Suppose $\max_{i\le n} \sigma_{n,i} = o(S_n)$. Prove $\frac{1}{S_n}\sum_{i=1}^n X_{n,i} \rightsquigarrow N(0,1)$.

Solution. We verify the Lindeberg condition for $Y_{n,i} = X_{n,i}/S_n$. Since $Y_{n,i}^2 = \sigma_{n,i}^2/S_n^2$ is deterministic,

$$L_n(\epsilon) = \sum_{i=1}^n \mathbb{E}\Big[Y_{n,i}^2 \mathbb{1}_{|Y_{n,i}|>\epsilon}\Big] = \sum_{i=1}^n \frac{\sigma_{n,i}^2}{S_n^2} \mathbb{1}_{\sigma_{n,i}/S_n>\epsilon}.$$

Let $m_n = \max_{i\le n} \sigma_{n,i}$. Then $\sigma_{n,i}/S_n > \epsilon$ implies $\sigma_{n,i} > \epsilon S_n$, so $m_n > \epsilon S_n$. Therefore, for n large enough that $m_n < \epsilon S_n$ (which holds since $m_n = o(S_n)$), every indicator is 0 and $L_n(\epsilon) = 0$. Hence $L_n(\epsilon) \to 0$.

Since $\sum_i \mathrm{Var}(Y_{n,i}) = 1$, all hypotheses of the Lindeberg CLT (with $\mathbb{E}[Y_{n,i}] = 0$ because $\epsilon_{n,i}$ are symmetric) are satisfied. $\square$

Remark D.2.4. This shows that for Rademacher arrays, $\max_i \sigma_{n,i} = o(S_n)$ is *necessary and sufficient* for the Lindeberg condition. For general independent summands, $\max_i \sigma_{n,i} = o(S_n)$ plus some moment control suffices but is not always necessary.

Theorem D.2.5 (Lyapunov CLT for non-i.i.d. with uniformly bounded third moments).

Let $X_1, X_2, \ldots$ be independent with $\mathbb{E}[X_i] = 0$, $\mathbb{E}[X_i^2] = \sigma^2 > 0$ for all i, and $\sup_i \mathbb{E}[|X_i|^3] \leq M < \infty$. Show $\frac{S_n}{\sigma\sqrt{n}} \rightsquigarrow N(0,1)$.

Solution. Set $S_n^* = \sigma\sqrt{n}$ so that $S_n^{*2} = \sum_i \mathbb{E}[X_i^2]$. Apply the Lyapunov CLT with $\delta = 1$: it suffices to check

$$\frac{1}{S_n^{*3}} \sum_{i=1}^n \mathbb{E}[|X_i|^3] \leq \frac{nM}{(\sigma\sqrt{n})^3} = \frac{M}{\sigma^3\sqrt{n}} \to 0.$$

The Lyapunov condition is satisfied, so $S_n/(\sigma\sqrt{n}) \rightsquigarrow N(0,1)$. $\square$

Remark D.2.6. This is the "non-i.i.d. but identically second-momentedn" CLT. The classical CLT assumes i.i.d.; this result shows that identical distribution can be replaced by equal second moments plus a uniform third-moment bound. Contrast with Problem C.2.2: there we dropped the third-moment assumption entirely but needed the Lindeberg condition to hold for a different structural reason.

Theorem D.2.7 (Lyapunov CLT for growing variances).

Let $X_1, X_2, \ldots$ be independent with $\mathbb{E}[X_i] = 0$, $\mathbb{E}[X_i^2] = i$, and $\mathbb{E}[|X_i|^3] \leq C\, i^{3/2}$ for some constant $C > 0$. Show

$$\frac{S_n}{\sqrt{n(n+1)/2}} \rightsquigarrow N(0,1).$$

Solution. Let $S_n^2 = \sum_{i=1}^n i = n(n+1)/2$. Apply the Lyapunov CLT with $\delta = 1$: verify

$$\frac{1}{S_n^3} \sum_{i=1}^n \mathbb{E}[|X_i|^3] \leq \frac{C \sum_{i=1}^n i^{3/2}}{\big(n(n+1)/2\big)^{3/2}}.$$

We bound the numerator: $\sum_{i=1}^n i^{3/2} \leq \int_0^n x^{3/2}\,dx + n^{3/2} = \frac{2}{5}n^{5/2} + n^{3/2} \leq n^{5/2}$ for $n \geq 1$. The denominator satisfies $\big(n(n+1)/2\big)^{3/2} \geq (n^2/4)^{3/2} = n^3/8$. Hence

$$\frac{1}{S_n^3} \sum_{i=1}^n \mathbb{E}[|X_i|^3] \leq \frac{C\, n^{5/2}}{n^3/8} = \frac{8C}{n^{1/2}} \to 0.$$

The Lyapunov condition holds, and $S_n/\sqrt{n(n+1)/2} \rightsquigarrow N(0,1)$. $\square$

Remark D.2.8. The key computation is always the same: put a rough bound on $\sum \mathbb{E}[|X_i|^{2+\delta}]$ (here $O(n^{5/2})$ for $\delta = 1$) and compare it to $S_n^{2+\delta}$ (here $O(n^3)$). If the ratio is $o(1)$, Lyapunov applies.

Theorem D.2.9 (Wasserstein CLT via Stein's method).

Let $X_1, \ldots, X_n$ be independent with $\mathbb{E}[X_i] = 0$, $\mathbb{E}[X_i^2] = 1$, and $\mathbb{E}[|X_i|^3] < \infty$. Set $W = \frac{X_1+\cdots+X_n}{\sqrt{n}}$ and $W_i = W - \frac{X_i}{\sqrt{n}}$. Show

$$\mathscr{W}(W,\, N(0,1)) \leq \frac{3}{\sqrt{n}} \frac{1}{n} \sum_{i=1}^{n} \mathbb{E}\Big[|X_i|^3\Big].$$

Solution. By definition of Wasserstein distance and the Stein-equation bound $\|f_h\|_\infty, \|f_h'\|_\infty, \|f_h''\|_\infty \leq 2\,\mathrm{Lip}(h)$,

$$\mathscr{W}(W,Z) = \sup_{\mathrm{Lip}(h)\leq 1} |\mathbb{E}[h(W)] - \mathbb{E}[h(Z)]| \leq \sup_{\substack{\|f\|_\infty \leq 2 \\ \|f'\|_\infty \leq 2 \\ \|f''\|_\infty \leq 2}} \big|\mathbb{E}[f'(W)] - \mathbb{E}[Wf(W)]\big|.$$

It suffices to bound the Stein discrepancy $|\mathbb{E}[f'(W)] - \mathbb{E}[Wf(W)]|$ for any such f.

Since $W = \sum_i X_i/\sqrt{n}$, write $\mathbb{E}[Wf(W)] = \frac{1}{\sqrt{n}} \sum_i \mathbb{E}[X_i f(W)]$. Because $X_i \perp\!\!\!\perp W_i$, the term $\mathbb{E}[X_i f(W_i)] = \mathbb{E}[X_i]\mathbb{E}[f(W_i)]$ Hence

$$\mathbb{E}[X_i f(W)] = \mathbb{E}[X_i(f(W) - f(W_i))].$$

Adding and subtracting the first-order Taylor term $f'(W_i)(W - W_i) = f'(W_i) \cdot X_i/\sqrt{n}$:

$$\mathbb{E}[X_i f(W)] = \underbrace{\mathbb{E}\big[X_i(f(W) - f(W_i) - f'(W_i)(W - W_i))\big]}_{(\star)} + \frac{1}{\sqrt{n}} \mathbb{E}\Big[X_i^2 f'(W_i)\Big].$$

By Taylor's theorem, $f(W) - f(W_i) - f'(W_i)(W - W_i) = \frac{1}{2} f''(\xi_i)(W - W_i)^2$ for some ξ_i between W_i and W. Since $W - W_i = X_i/\sqrt{n}$,

$$(\star) = \frac{1}{2n} \mathbb{E}\Big[X_i^3 f''(\xi_i)\Big].$$

Summing over i and dividing by $\sqrt{n}$:

$$\mathbb{E}[Wf(W)] = \underbrace{\frac{1}{2n\sqrt{n}} \sum_i \mathbb{E}[X_i^3 f''(\xi_i)]}_{A_n} + \underbrace{\frac{1}{n} \sum_i \mathbb{E}[f'(W_i)]}_{B_n}.$$

Now $B_n \approx \mathbb{E}[f'(W)]$ because W_i and W are close; more precisely,

$$|\mathbb{E}[f'(W)] - \mathbb{E}[f'(W_i)]| \leq \|f''\|_\infty \mathbb{E}[|W - W_i|] = \|f''\|_\infty \mathbb{E}[|X_i|]/\sqrt{n},$$

so

$$\big|\mathbb{E}[f'(W)] - B_n\big| \leq \frac{\|f''\|_\infty}{\sqrt{n}} \cdot \frac{1}{n} \sum_i \mathbb{E}[|X_i|] \leq \frac{2m_1}{\sqrt{n}},$$

where $m_1 = \frac{1}{n} \sum_i \mathbb{E}[|X_i|]$. Since $\|f''\|_\infty \leq 2$. Also $|A_n| \leq \frac{\|f''\|_\infty}{2\sqrt{n}} \cdot \frac{1}{n} \sum_i \mathbb{E}[|X_i|^3] \leq \frac{m_3}{\sqrt{n}}$ where $m_3 = \frac{1}{n} \sum_i \mathbb{E}[|X_i|^3]$. Thus

$$\big|\mathbb{E}[f'(W)] - \mathbb{E}[Wf(W)]\big| \leq |A_n| + \big|\mathbb{E}[f'(W)] - B_n\big| \leq \frac{m_3 + 2m_1}{\sqrt{n}} \leq \frac{3m_3}{\sqrt{n}},$$

where the last step uses $m_1 \leq (m_3)^{1/3} \leq m_3$ (by Jensen, since $\mathbb{E}[X_i^2] = 1$ gives $m_1 \leq 1 \leq (m_3)^{2/3}$, so $m_1 \leq m_3$). Substituting the sup bound:

$$\mathscr{W}(W, N(0,1)) \leq \frac{3m_3}{\sqrt{n}} = \frac{3}{\sqrt{n}} \cdot \frac{1}{n} \sum_{i=1}^{n} \mathbb{E}[|X_i|^3]. \qquad \square$$

Remark D.2.10. The three-step pattern in Stein's method is always: (1) bound $\mathscr{W}$ by the Stein discrepancy via the Stein-equation solution f_h, (2) expand $\mathbb{E}[Wf(W)]$ using the decomposition $W = W_i + X_i/\sqrt{n}$, and (3) kill lower-order terms using independence $X_i \perp\!\!\!\perp W_i$. The result is non-asymptotic: it holds for all n, not just in the limit.

Reference: If You Get Stuck on a CLT Problem

1. **Identify the theorem to apply.** If you are given a finite third moment or Lyapunov-type condition, aim for the triangular array CLT or Lyapunov. If you only have second moments, aim for Lindeberg.

2. **Setting up the triangular array.** Rescale: if X_i are i.i.d. with $\mathbb{E}[X_i^2] = \sigma^2$ and you want $S_n/(\sigma\sqrt{n}) \rightsquigarrow N(0,1)$, define $\xi_{n,i} = X_i/(\sigma\sqrt{n})$ and verify (i) mean $\to \mu$, (ii) variance $\to \sigma^2$, (iii) $\sum \mathbb{E}[|\xi_{n,i}|^3] \to 0$.

3. **Verifying the Lindeberg condition.** The target is $L_n(\epsilon) = \sum_i \mathbb{E}[X_{n,i}^2 \mathbb{1}_{|X_{n,i}|>\epsilon}] \to 0$. Two shortcuts:

   - If $\max_i |X_{n,i}| \le \delta_n \to 0$, then for large n the indicator is identically 0 and $L_n(\epsilon) = 0$ (Problem C.2.1).
   - If the $X_{n,i}$ are Rademacher ($\pm\sigma_{n,i}$), then the indicator is $\mathbb{1}_{\sigma_{n,i}/S_n>\epsilon}$, and $\max_i \sigma_{n,i} = o(S_n)$ kills it (Problem C.2.2).

4. **Applying Lyapunov.** Compute $\sum_i \mathbb{E}[|X_{n,i}|^{2+\delta}]$ (try $\delta = 1$ first) and $S_n^{2+\delta}$. If their ratio $\to 0$, you are done. Typically one is $O(n^\alpha)$ and the other $O(n^\beta)$ with $\beta > \alpha$; verify $\beta - \alpha > 0$ (Problem C.2.3–4).

5. **Truncation.** If $\mathbb{E}[|X_i|^3]$ is not given, truncate X_i at $\sqrt{n}$ (i.e., $\xi_{n,i} = (X_i/\sqrt{n})\mathbb{1}_{|X_i|\le\sqrt{n}}$), verify the three conditions, then bound

$$|\mathbb{E}[f(S_n/\sqrt{n})] - \mathbb{E}[f(\sum_i \xi_{n,i})]| \le 2\|f\|_\infty \sum_i \mathbb{P}[|X_i| > \sqrt{n}] \le 2\|f\|_\infty \mathbb{E}[X_1^2 \mathbb{1}_{|X_1|>\sqrt{n}}] \to 0.$$

6. **Stein's method (alternative).** Show $|\mathbb{E}[f'(W_n)] - \mathbb{E}[W_n f(W_n)]| \to 0$ for all admissible f. Use the bound $\|f_h\|_\infty, \|f_h'\|_\infty, \|f_h''\|_\infty \le 2\|h'\|_\infty$ to control the Stein discrepancy term.

D.3 Stein Characterizations Beyond Gaussian and Poisson

Every distribution admits a *Stein operator* L such that $\mathbb{E}[Lf(X)] = 0$ for all admissible f characterizes the distribution of X. The three examples below follow the same two-step template as Chapters 5–6: (i) prove the identity by integration by parts (continuous) or index-shifting (discrete) against the density/pmf, and (ii) establish the converse by exhibiting a solution to the Stein equation and arguing that the identity forces distributional agreement.

D.3.1 Exponential distribution

Lemma D.3.1 (Stein identity for $\mathrm{Exp}(\lambda)$).

Let $\lambda > 0$. We have

$$X \sim \mathrm{Exp}(\lambda) \iff \mathbb{E}[f'(X)] = \lambda\,\mathbb{E}[f(X)]$$

for every absolutely continuous f with $f(0) = 0$ and $\mathbb{E}[|f'(X)|] < \infty$.

[1]

[1] The restriction $f(0) = 0$ is needed to kill the boundary term in integration by parts; without it the identity picks up an extra $-\lambda f(0)$ on the right.

Proof of Lemma D.3.1. ($\Longrightarrow$) Let $p(x) = \lambda e^{-\lambda x}$ denote the $\mathrm{Exp}(\lambda)$ density on $(0, \infty)$. Since $p'(x) = -\lambda p(x)$, integration by parts gives

$$\begin{aligned}\mathbb{E}\left[f'(X)\right] &= \int_0^\infty f'(x)\, p(x)\, dx \\ &= \left[f(x)\, p(x)\right]_0^\infty - \int_0^\infty f(x)\, p'(x)\, dx\end{aligned}$$

The boundary term equals $0 - \lambda f(0)$ (it vanishes at ∞ by integrability, and equals $\lambda e^0 f(0) = \lambda f(0)$ at 0 with a minus sign from the lower limit). Since $p'(x) = -\lambda p(x)$:

$$\begin{aligned}&= -\lambda f(0) + \lambda \int_0^\infty f(x)\, p(x)\, dx \\ &= -\lambda f(0) + \lambda\, \mathbb{E}\left[f(X)\right].\end{aligned}$$

Setting $f(0) = 0$ yields $\mathbb{E}\left[f'(X)\right] = \lambda\, \mathbb{E}\left[f(X)\right]$.

($\Longleftarrow$) Fix $t \geq 0$ and consider the *Stein equation*

$$f'(x) - \lambda f(x) = \mathbb{1}_{x \leq t} - \mathbb{P}\left[X \leq t\right], \quad f(0) = 0.$$

Multiplying by the integrating factor $e^{-\lambda x}$,

$$\frac{d}{dx}\left[e^{-\lambda x} f(x)\right] = e^{-\lambda x}(\mathbb{1}_{x \leq t} - \mathbb{P}\left[X \leq t\right]).$$

Integrating from 0 to x with $f(0) = 0$:

$$f_t(x) = e^{\lambda x} \int_0^x e^{-\lambda s}(\mathbb{1}_{s \leq t} - \mathbb{P}\left[X \leq t\right])\, ds.$$

Note $f_t(0) = 0$, so f_t is admissible. For any random variable W satisfying the hypothesis, take expectations of the Stein equation evaluated at W:

$$\underbrace{\mathbb{E}\left[f_t'(W)\right] - \lambda\, \mathbb{E}\left[f_t(W)\right]}_{=0 \text{ by hypothesis}} = \mathbb{P}\left[W \leq t\right] - \mathbb{P}\left[X \leq t\right].$$

Hence $\mathbb{P}\left[W \leq t\right] = \mathbb{P}\left[X \leq t\right]$ for all $t \geq 0$, so $W \sim \mathrm{Exp}(\lambda)$. □

Remark D.3.2. The structural parallel with the Gaussian is exact: for $N(0,1)$, $\phi'(x) = -x\phi(x)$ drives the identity $\mathbb{E}\left[f'(Z)\right] = \mathbb{E}\left[Zf(Z)\right]$; for $\mathrm{Exp}(\lambda)$, $p'(x) = -\lambda p(x)$ drives $\mathbb{E}\left[f'(X)\right] = \lambda\, \mathbb{E}\left[f(X)\right]$. Both Stein equations are first-order linear ODEs solved by an integrating factor.

D.3.2 Gamma distribution

Lemma D.3.3 (Stein identity for $\mathrm{Gamma}(\alpha, 1)$).

Let $\alpha > 0$. We have

$$X \sim \mathrm{Gamma}(\alpha, 1) \iff \mathbb{E}\left[Xf'(X)\right] = \mathbb{E}\left[(X - \alpha)\, f(X)\right]$$

for every absolutely continuous f with $\mathbb{E}\left[|Xf'(X)|\right] < \infty$ and $\mathbb{E}\left[|Xf(X)|\right] < \infty$.

Proof of Lemma D.3.3. ($\Longrightarrow$) Let $p(x) = x^{\alpha-1}e^{-x}/\Gamma(\alpha)$ on $(0,\infty)$. Then

$$p'(x) = \frac{(\alpha-1-x)}{x}\,p(x), \quad \text{so} \quad xp'(x) = (\alpha-1-x)\,p(x).$$

Therefore $\frac{d}{dx}[xp(x)] = p(x) + xp'(x) = p(x) + (\alpha-1-x)p(x) = (\alpha-x)p(x)$. Integration by parts with $u = xp(x)$ and $dv = f'(x)\,dx$ gives

$$\begin{aligned}\mathbb{E}\left[Xf'(X)\right] &= \int_0^\infty f'(x)\,xp(x)\,dx \\ &= \left[xp(x)f(x)\right]_0^\infty - \int_0^\infty f(x)\,(\alpha-x)\,p(x)\,dx.\end{aligned}$$

The boundary term vanishes:[2] therefore

$$\mathbb{E}\left[Xf'(X)\right] = -\mathbb{E}\left[(\alpha-X)f(X)\right] = \mathbb{E}\left[(X-\alpha)f(X)\right].$$

($\Longleftarrow$) The *Stein equation* is

$$xf'(x) + (\alpha-x)f(x) = h(x) - \mathbb{E}_\alpha\left[h\right],$$

where $\mathbb{E}_\alpha\left[h\right] := \mathbb{E}\left[h(X)\right]$ for $X \sim \text{Gamma}(\alpha,1)$. Rewriting as $f'(x) + \frac{\alpha-x}{x}f(x) = \frac{h(x)-\mathbb{E}_\alpha[h]}{x}$, the integrating factor is

$$\mu(x) = \exp\left(\int \frac{\alpha-x}{x}\,dx\right) = e^{\alpha\ln x - x} = x^\alpha e^{-x},$$

so $\frac{d}{dx}[\mu(x)f(x)] = x^{\alpha-1}e^{-x}(h(x) - \mathbb{E}_\alpha[h])$. Integrating with the boundary condition $f(0^+)$ finite:[3]

$$f_h(x) = \frac{e^x}{x^\alpha}\int_0^x t^{\alpha-1}e^{-t}(h(t) - \mathbb{E}_\alpha\left[h\right])\,dt.$$

For any W satisfying the hypothesis, take expectations of the Stein equation at W:

$$\underbrace{\mathbb{E}\left[Wf_h'(W)\right] - \mathbb{E}\left[(W-\alpha)f_h(W)\right]}_{=0 \text{ by hypothesis}} = \mathbb{E}\left[h(W)\right] - \mathbb{E}_\alpha\left[h\right].$$

Hence $\mathbb{E}\left[h(W)\right] = \mathbb{E}_\alpha\left[h\right]$ for all admissible h, so $W \sim \text{Gamma}(\alpha,1)$. □

Remark D.3.4. The Stein operators Lf for the three continuous distributions are: $f' - xf$ for $N(0,1)$, $f' - \lambda f$ for $\text{Exp}(\lambda)$, and $xf' + (\alpha-x)f$ for $\text{Gamma}(\alpha,1)$. The factor of x on f' in the Gamma case reflects the support $(0,\infty)$: the natural weight $xp(x)$ appears as the u in integration by parts, whereas for distributions on $\mathbb{R}$ or with constant density shape, no such weight is needed.

[2] $xp(x) = x^\alpha e^{-x}/\Gamma(\alpha) \to 0$ as $x \to 0^+$ (since $\alpha > 0$) and as $x \to \infty$ (exponential decay dominates).

[3] Near $x = 0$,

$$\begin{aligned}&\int_0^x t^{\alpha-1}e^{-t}(h - \mathbb{E}_\alpha h)dt \\ &\approx (h(0) - \mathbb{E}_\alpha h)\frac{x^\alpha}{\alpha},\end{aligned}$$

so $f_h(x) \approx \frac{h(0)-\mathbb{E}_\alpha h}{\alpha}$, which is finite.

D.3.3 Binomial distribution

Lemma D.3.5 (Stein identity for $\mathrm{Bin}(n,p)$).

Let $n \in \mathbb{N}$, $p \in (0,1)$, $q = 1-p$. We have

$$X \sim \mathrm{Bin}(n,p) \iff p\,\mathbb{E}\left[(n-X)f(X+1)\right] = q\,\mathbb{E}\left[Xf(X)\right]$$

for every bounded $f : \{0,\dots,n\} \to \mathbb{R}$.

Proof of Lemma D.3.5. ($\Longrightarrow$) Let $\pi(k) = \binom{n}{k}p^k q^{n-k}$. We use the combinatorial identity[4] $(n-k)\binom{n}{k} = (k+1)\binom{n}{k+1}$ to reindex the left-hand side. Substituting $j = k+1$:

$$\begin{aligned}
p\,\mathbb{E}\left[(n-X)f(X+1)\right] &= \sum_{k=0}^{n-1}(n-k)\binom{n}{k}p^{k+1}q^{n-k}f(k+1) \\
&= \sum_{k=0}^{n-1}(k+1)\binom{n}{k+1}p^{k+1}q^{n-k}f(k+1) \\
&= \sum_{j=1}^{n} j\binom{n}{j}p^{j}q^{n-j+1}f(j) \\
&= q\sum_{j=1}^{n} j\binom{n}{j}p^{j}q^{n-j}f(j) \\
&= q\,\mathbb{E}\left[Xf(X)\right].
\end{aligned}$$

($\Longleftarrow$) For each $A \subseteq \{0,\dots,n\}$, consider the *Stein–Binomial difference equation*

$$(n-k)p\,f(k+1) - kq\,f(k) = \mathbb{1}_{k\in A} - \mathbb{P}\left[X \in A\right], \quad k = 0,\dots,n.$$

Multiplying by $\pi(k)$ and using the *detailed balance* $(n-k)p\,\pi(k) = (k+1)q\,\pi(k+1)$:[5]

$$(k+1)q\,\pi(k+1)f(k+1) - kq\,\pi(k)f(k) = \pi(k)(\mathbb{1}_{k\in A} - \mathbb{P}\left[X \in A\right]).$$

Setting $G(k) = kq\,\pi(k)f(k)$ with $G(0) = 0$, this is a telescoping relation

$$G(k+1) - G(k) = \pi(k)(\mathbb{1}_{k\in A} - \mathbb{P}\left[X \in A\right]),$$

which sums to the explicit solution

$$f_A(k) = \frac{1}{kq\,\pi(k)}\sum_{j=0}^{k-1}\pi(j)(\mathbb{1}_{j\in A} - \mathbb{P}\left[X \in A\right]), \quad k = 1,\dots,n.$$

Consistency at $k = n$: $G(n+1) = \sum_{j=0}^{n}\pi(j)(\mathbb{1}_{j\in A} - \mathbb{P}[X \in A]) = 0$ since π is a probability mass function, and $(n+1)q\,\pi(n+1) = 0$.[6]

[4] Both $(n-k)\binom{n}{k}$ and $(k+1)\binom{n}{k+1}$ equal $\frac{n!}{k!(n-k-1)!}$.
[5] Both sides equal $(k+1)\binom{n}{k+1}p^{k+1}q^{n-k}$; this is the discrete analogue of $p'(x) = -\lambda p(x)$.
[6] $\pi(n+1) = \binom{n}{n+1}p^{n+1}q^{-1} = 0$, so $G(n+1) = 0$ automatically.

For any W satisfying the hypothesis, take expectations of the difference equation at W:

$$\underbrace{p\,\mathbb{E}\left[(n-W)f_A(W+1)\right] - q\,\mathbb{E}\left[W f_A(W)\right]}_{=0 \text{ by hypothesis}} = \mathbb{P}\left[W \in A\right] - \mathbb{P}\left[X \in A\right].$$

Since A is arbitrary, $W \sim \mathrm{Bin}(n,p)$. $\square$

Remark D.3.6. Comparing the discrete Stein operators: $\mathrm{Poi}(\lambda)$ has the identity $Lf(k) = \lambda f(k+1) - kf(k)$; $\mathrm{Bin}(n,p)$ has $Lf(k) = (n-k)p\,f(k+1) - kq\,f(k)$. Both are solved by multiplying by the pmf and telescoping. As $n \to \infty$ with $np \to \lambda$, $(n-k)p \approx \lambda$ and $kq \approx k$, so the Binomial operator converges to the Poisson operator, reflecting $\mathrm{Bin}(n, \lambda/n) \rightsquigarrow \mathrm{Poi}(\lambda)$.